

LASERTEC CORPORATION (TOKYO: 6920)

Japan's Most Actively Traded Stock Is A Colossal Fraud And Ticking Time Bomb, Reminiscent of Olympus, Toshiba, And Other Infamous Accounting Scandals

- One of the largest corporate frauds in Japanese history, a textbook accounting scheme faking revenue, margins, and earnings, with a looming impairment charge that will wipe out most of its earnings since the launch of its flagship ACTIS EUV inspection machine – the driver of its \$23B market cap and improbable rise from a penny stock to one of the best performing and most actively traded stocks on earth
- The ACTIS system is defective with a fatally flawed EUV light source and other insurmountable problems, which it has concealed from investors - the absurdity of an EUV concept stock which has no functioning EUV technology
- An alarming level of inventory that is the highest of any semicap tools supplier in the world, a sign of fraudulent accounting and the core of its accounting charade
- The “new” ACTIS A300, recently announced as its next-generation EUV product and driver of growth, is a hoax – a cover story to conceal flaws in its flagship A150 EUV tool and its desperate recall and replacement effort
- The company's new Innovation Park in Yokohama is a brazen fraud, a purported R&D and production campus with two supposed “fabs” that is 5X the size of its headquarters and the next phase of its expansion, announced in 2022 as the largest capital expenditure in its history
- Our investigators visited ~20 times from Dec 2023 to May 2024 and found a generally deserted facility with no R&D or production activity - contradicting the company's claims and suggesting its promotional pics of the “fab” are fake, based on extensive surveillance including site interviews, weeks of 24/7 time-lapse camera footage, and aerial drones
- Its largest customers – TSMC, Intel, and Samsung which are ~80% of revenue – are exasperated and furious to the point that TSMC has ceased buying its EUV machines, and the key Intel executive in charge of Lasertec's tools has been allegedly banned from meetings for “blowing up” on Lasertec due to his anger at the tool's failure
- KLA's imminent EUV market entry will be “catastrophic” for Lasertec, with customers “crying” and “begging” KLA to hurry – “devastating”; “game over”; “everything could turn around in a second...they're out of business, basically”

“To hide a lie, a thousand lies are needed”

嘘を隠すには、もっと大きな嘘が必要になる

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<i>Lasertec was a small, obscure semiconductor equipment supplier that <u>struggled for a decade as a penny stock</u>, trading at 1-2X revenue with a <u>\$50-100MM market cap</u> and no growth – and now suddenly sports a \$23B valuation, hailed as “Japan’s ASML.” and one of the most actively traded stocks on earth. It spent almost nothing on R&D yet claims to have a monopoly with its actinic EUV mask inspection tool, magically replicating a technology that took ASML 20 years, \$10B in investment, and 1,000 engineers to develop. We conducted an <u>intensive 5-month investigation</u> encompassing over 20 research interviews and ~20 investigator field visits to its new R&D campus in Yokohama, leading us to conclude that Lasertec is one of the <u>largest frauds in Japanese corporate history</u>.</i>	
2. Lasertec is an <u>extreme, textbook accounting fraud</u> that checks off every box: claiming to have the highest margins in the industry; but alarmingly little cash flow and the lowest cash conversion among its peers; accompanied by the highest inventory level of any semicap equipment supplier in the world.	60-101
<i>Overstated revenue and earnings are at the core of the fraud, offset by <u>improperly valued inventory</u>, with a looming, inevitable 65-70% <u>write-down and impairment charge</u> – a loss that will wipe out ~70% of its cumulative earnings since the launch of its flagship ACTIS EUV system and <u>85% of its retained earnings</u>. The hit will far <u>exceed what Olympus incurred</u> and is considerable versus Toshiba, when their frauds were exposed.</i>	
3. The origin of the accounting fraud: Lasertec’s <u>flagship ACTIS actinic EUV mask inspection machine is defective</u> with a fatally flawed EUV light source – the core of any EUV system – in addition to insurmountable problems with other key components like the optics and mask stage. Shockingly, it has a \$23B market cap as an EUV story without an EUV technology that works.	102-140
<i>Its inspection tool is an <u>unmitigated disaster due to</u> a) contamination and debris from tin splatter; b) instability and flickering which leads to false positives and other mask inspection failures; c) onerous cleaning and maintenance that leads to unacceptably low reliability, uptime, and throughput; and d) a short lifespan that drives frequent replacement and a burdensome cost of ownership. The <u>failures have wiped out its margin</u> – 3-4 full-time field service engineers per each machine for repairs and troubleshooting; frequent, costly replacement of optics components; a <u>covered-up recall effort</u> to swap out the source; discounts and other concessions to angry customers - thus <u>forcing it to fabricate its earnings and the value of \$1.1B of worthless inventory</u>.</i>	

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4. The “new” ACTIS A300, announced with fanfare in Nov 2023 as Lasertec’s next-generation EUV product and driver of growth, is a hoax – a cover story to conceal fatal flaws with the EUV light source in its flagship tool, the A150, and the resulting, costly replacement and recall crisis. The A300 is identical to the A150 and the substitute source is just as defective. 141-167
- The company’s EUV story - and its stock - would have collapsed had it revealed that the source – the foundation of an EUV system - was defective and had to be hastily, secretly swapped with one from a small, 12-person Russian R&D project called Isteq, which it misrepresents as in-house Japanese technology under the name “Urashima.” However, the Isteq/Urashima source is just as defective as the previous Ushio one. The A300 is simply an attempt to mislead investors and spin a damage-control and recall effort as a “new” product. In addition, the much-hyped claim that the A300 is High NA-compatible – a term for ASML’s next-generation EUV lithography tool – is false as it has no such capability; field engineers revealed to us that the A300 is the same as the A150. Four different TSMC sources told us that the A300 recently failed their inspection process and that they have no intention of buying any, similar to color from other customers like Global Foundries and Marvell.*
5. Lasertec’s EUV ambitions are a delusion given the impossibility of developing an EUV light source without investing USD billions. It is not - and will never be - a serious, credible player in EUV lithography/inspection, and investors who perceive it as “Japan’s ASML” are misguided. 168-184
- We spoke with two of the most knowledgeable EUV light source experts in the world: a leading figure in ASML/Cymer’s source development, and a senior R&D individual at Isteq. They elaborated on the history and origin of ASML’s source; and Lasertec’s struggles and failures with 2 different source suppliers, Ushio and Isteq. Even Isteq stated that the “new” source technology they licensed to Lasertec for its A300 “Urashima” is a pipe dream that is as fatally flawed as Ushio’s; that Lasertec’s EUV tools are a failure; and that key customers they speak to like TSMC and Intel are angry.*
6. Lasertec’s largest customers – notably TSMC, Intel, and Samsung, which are 77% of its revenue – are struggling, exasperated, and/or furious with its EUV mask inspection system, as well as its previous-generation DUV tools which we believe to still comprise the vast majority of its revenue. Key customers are “really struggling about what to do”; “it’s painful, extremely painful” to use Lasertec’s tool; “it would be stupid” to buy more machines. 185-191

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- 6A. **Struggles at Intel**. Lasertec's EUV mask inspection effort is Intel's baby, which it sponsored with \$300-500MM in advances >5 years ago as its "founding" customer after failing to convince KLA. Intel's mask group is trying to save face and cover up that its investment has been a debacle, giving it supplier prizes for public PR while "blowing up" on Lasertec behind the scenes, according to multiple research sources. 192-203
- A Lasertec field engineer – based on-site at their Santa Clara, CA mask shop – revealed that Intel's mask shop head was so enraged at the tool's failure that he was banned from attending meetings for blowing up at staff. One of Intel's most senior and tenured executives, with a leadership role in EUV, confirmed that Lasertec's EUV source "is not a well-controlled process...it was painful...if you choose a bad technology you just can't get there"; Intel's VP mask operations VP confirmed the struggles.*
- 6B. **Struggles at TSMC**. The world's largest chipmaker's experience with Lasertec has been a disaster, not only with its ACTIS EUV systems but also with its outdated, previous-generation DUV tools. 204-225
- Multiple sources including TSMC's ex-head of R&D, a procurement executive who first brought Lasertec into TSMC, and the head of TSMC's mask shop – one of whom laughed at Lasertec's tool - indicated that TSMC stopped buying its ACTIS machines years ago; that they are so unreliable that masks require parallel, redundant inspection with older tools; that at best it is a niche, laboratory R&D tool unsuitable for fabs; that KLA's older DUV tools are superior for EUV mask inspection and that KLA has already taken 2/3 of their mask inspection share. Lasertec's status is so dire that TSMC has forced them to attend a standing, weekly crisis meeting: "it's just not stable...it doesn't work...it just dies"; "TSMC is not happy with Lasertec"; "TSMC will not consider the A300 or A150 for production." Critically, four different TSMC sources indicated to us that the "new" ACTIS A300 with the Urashima light source recently failed their evaluation process.*

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- 6C. Struggles at other customers, such as Samsung, Marvell, and Global Foundries, the latter ridiculing the quality of Lasertec's EUV tool and ceasing new purchases in 2021, calling it "just unacceptable" with a "problematic light source." 226-236

Global Foundries blasted the tool as "not as sophisticated or as capable as the classic [DUV] KLA mask inspection tools; mocked their A300 announcement by saying "no question...it sounds sketchy"; and confirmed that other customers are equally "frustrated and disappointed" and are refusing to take shipment. As a sign of Lasertec's urgent situation, they stated that "senior technical people" from Intel and Samsung recently flew to Japan to express their "concerns with the equipment." Marvell stated it also has no interest in further ACTIS purchases; that ASML "echoed the same problems that we are seeing"; expressed skepticism of the A300; revealed that Lasertec offered them ~75% discounts in a desperate attempt to offload inventory; and that they recently saw high levels of inventory piling up at Lasertec, leading them to question their inventory valuation and accounting.

7. Lasertec customers who have recently visited its headquarters confirmed seeing highly unusual levels of inventory - "pretty scary...pretty weird" - and other red flags: their "factory floor for finished goods is full.. a substantial amount of finished goods inventory that's either "not passing [evaluation]" or "customers have deferred shipment"; "literally no space left on their final inspection bays before the equipment ships." 237-251

Research sources indicated that customers are refusing to accept shipment of tools; that Lasertec "recognized revenue for prepayments of systems that aren't yet shipped"; that most of their initial advances and purchase orders were from Intel and are being "pushed, pushed, pushed out" and will never be filled, with Intel allegedly trying and failing to pawn off PO's to TSMC; are misclassifying inventory using "tricks that any supplier can play"; inflating the value of inventory - "they're going to have to significantly mark down"; "haven't taken any impairments"; "they don't want to record a hit on their gross margin"; and detailed aggressive, questionable discounts, trade-ins, inducements, and concessions Lasertec is quietly offering customers, as well as "reimbursement...for the lost time of use" due to the ACTIS tool's defects and lack of uptime.

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8. Lasertec's small, one-time EUV market opportunity is already exhausted, with little to no growth left in its flagship ACTIS line. Aside from product defects, irate customers, and the lack of a viable EUV source, an analysis of its growth drivers in the last 5 years indicates they were ephemeral. 252-265

We analyze a) the industry transition to the 5nm node, which is now complete and during which there was a momentary window for its EUV system; b) the ratio/correlation of EUV mask inspection tools per ASML EUV scanner, a canary in the coal mine as ASML's shipments have hit the wall; and c) the installed base and trend of ACTIS tools inside each use case at TSMC, Intel, and Samsung, whether mask shop or fab. As Intel and TSMC's mask shops only have 5 ACTIS tools in total and ceased new installs, and TSMC views it as a failure in the fab, Lasertec's only remaining hope is Intel adding tools in its fabs – a fantasy, as most of its recent revenue "growth" has come from movements in its deferred revenue and advances account, with therefore little cash flow.

9. KLA's impending launch of an EUV actinic patterned mask inspection tool will be catastrophic for Lasertec. We spoke with 3 KLA sources, as well as senior individuals at Intel, TSMC, Samsung, Global Foundries and others – all indicated that KLA's tool is likely coming in 2025, has a superior technology, and will destroy Lasertec's market share – hardly a surprise, given previous short-lived "monopolies" in semicap equipment that collapsed overnight when leapfrogged by a new entrant. 266-285

Customers irate with Lasertec are "begging" KLA and "crying" for it to hurry its launch; KLA's entry will be "devastating," "catastrophic," and "game over" for Lasertec which "will disappear" – "everything could turn around in a second...they're out of business, basically." KLA's tool uses a superior EUV source technology and is specifically timed for the transition to High NA, for which Lasertec has no capability. KLA is in pole position given Biden's CHIPS Act, which is placing pressure on Intel, TSMC, and Samsung to buy from KLA, given the tens of billions in US subsidies they were just granted, which sources call a seismic event that spells major trouble for a Japanese supplier like Lasertec.

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10. Lasertec's EUV tool is so defective, unreliable, and lacking in throughput that TSMC, Intel, and Samsung have all quietly substituted non-EUV alternatives for inspecting EUV masks, improvising with KLA's legacy Teron DUV tool plus electron beam inspection - indicating that actinic EUV is useful if and when it works but is not essential and certainly not a "monopoly" that justifies Lasertec's absurd valuation. 286-296

Sources indicate that customers – presumably Intel - who claim to use Lasertec's EUV tool are secretly using its old DUV system instead; that TSMC is running parallel, redundant inspections with non-EUV tools and has reverted to the DUV-based inspection process it used prior to Lasertec; that the market window for actinic EUV mask inspection is rapidly closing, as the next phase in EUV is High NA, which requires an electron beam solution, as OPC patterns and sub-resolution assist features (SRAF) are smaller than 13.5nm and cannot be detected with actinic EUV light – leading ASML, KLA, and others to develop multi-beam inspection tools for High NA masks that render the ACTIS tool obsolete.

11. Lasertec's new "Innovation Park" in Yokohama is a fraud – a purported R&D and production campus with two supposed "fabs" that is 5X the size of its headquarters and its next driver of growth, announced in 2022 as the largest capital expenditure and expansion in its history. Multiple investigators visited the campus ~20 times between Dec 2023 and May 2024 and found a generally deserted and empty facility with no evidence of R&D, production, nor any indication that Fab 4A matches its promotional photos, leading us to question if the pics are fake. 297-334

We conclude that the company's statements - in press releases, earnings Q&A, presentations, and an in-person IR meeting - are false and an epic scam, based on extensive surveillance including field visits/interviews; 24/7 time-lapse and motion-detector camera footage over several weeks; and aerial drone surveillance on two occasions. We also hired a local IR consultant who met with the company in March – Lasertec IR agreed to a brief tour of their clean room at headquarters, but strongly refused multiple requests for a tour of the clean room (Fab 4A) at the Park. We then asked for a brief tour of the Park without entering Fab 4A – also refused. Finally, we asked for simply a video clip of Fab 4A, which was also denied.

Lasertec has a \$23B (¥3,630B) market cap and is the most actively traded stock in Japan, trading ~\$2B a day despite ranking only 64th in market capitalization – a sign of extreme speculation. Depending on the week, its daily trading volume is in the top ten globally, in proximity with Nvidia, Tesla, Microsoft, Google, Amazon, and Meta, making it one of biggest bubble stocks on earth. It is also the most actively traded semiconductor equipment stock in the world, surpassing the five largest players ASML (\$377B market cap), Applied Materials (\$177B), Lam (\$122B), Tokyo Electron (\$101B), and KLA (\$102B). On some days it trades 10X the dollar volume of ASML, the world's largest semicap player. Its multiple is equally preposterous at 17X sales, eclipsing all of the leading semicap suppliers.

Lasertec 10-year stock chart



The longtime CEO Osamu Okabayashi abruptly announced his resignation a few weeks ago, having run the company for the last 15 years, on the same day as earnings with the stock near all-time highs – typically an ominous sign and one of our most reliable indicators that the end is near. In one major corporate fraud after another, we see the CEO and/or CFO depart when the scheme becomes too big to sustain. Investors may recall the **surprise departure of the CEO of Enron – one of the largest frauds in US history** – a few months before its collapse and bankruptcy, baffling unsuspecting investors at the time. Okabayashi provided **no explanation for the unexpected timing.** The stock jumped anyway despite the news, given investor complacency and speculative fervor.

Lasertec outgoing CEO on Nikkei CNBC



Founded in the 1960's, Lasertec was a small, obscure supplier of basic measurement and inspection equipment, with revenue stuck between \$30-100MM for most of the 2000's. It struggled the entire decade prior to its EUV product launch in 2017, with mediocre margins and no growth over the period, trading at 1-2X revenue as a penny stock around \$0.50 to \$2 (¥75 to ¥300), unable to exceed \$50-200MM market cap. Out of nowhere, the stock has spiked 100-200X since it became an EUV story, depending on the baseline date, making it the best performing stock in Japan in the last 5 years (1,700%) and 10 years (16,000%); and the 6th best in the world >\$5B market cap over the last decade.

Lasertec media coverage

Japan's Hottest Stock Is Tiny Maker of \$40 Million Machines

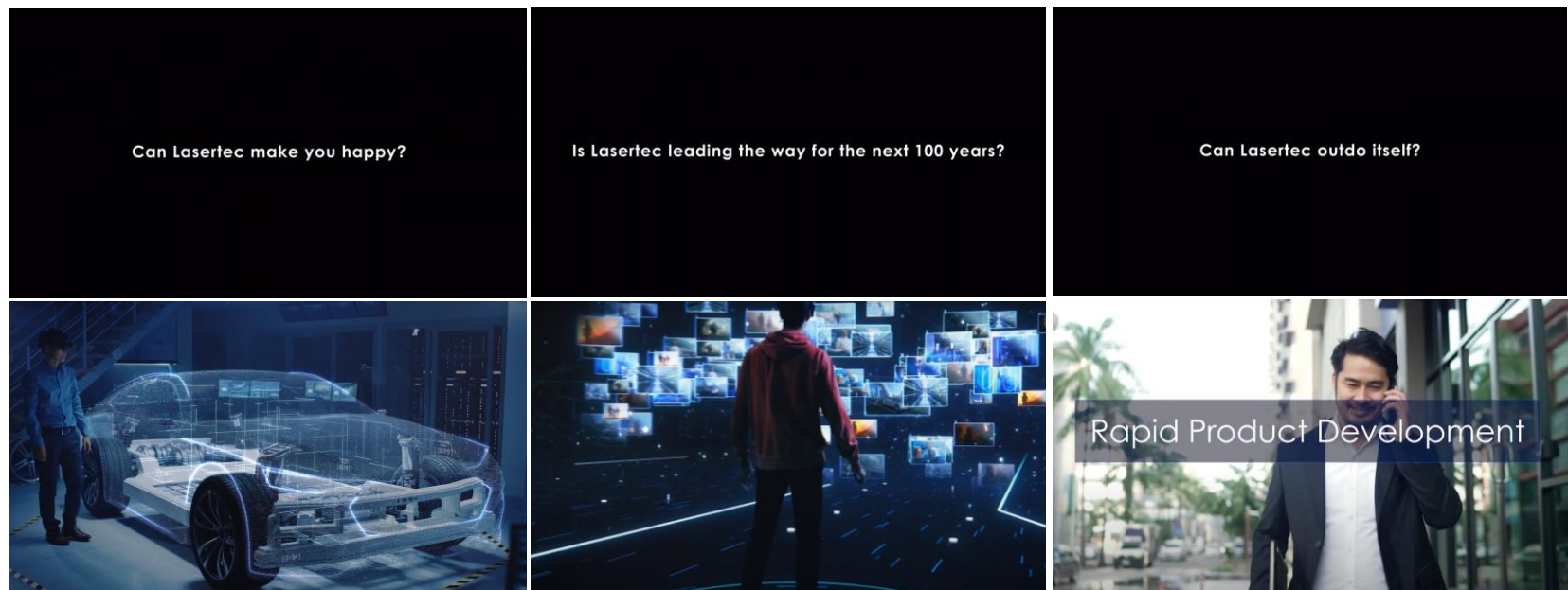
This little-known Japanese company is a key player in the leading-edge chip industry— iPhones Are Improving Every Year Thanks to an Obscure Japanese Company

**The obscure firm making iPhones better:
Japan's Lasertec masters extreme
ultraviolet lithography**

Source: <https://www.bloomberg.com/news/articles/2020-05-25/japan-s-hottest-stock-is-tiny-maker-of-40-million-chip-machines?embedded-checkout=true>; <https://finance.yahoo.com/news/little-known-japanese-company-key-210000435.html>; <https://www.japantimes.co.jp/news/2019/03/12/business/corporate-business/obscure-firm-making-iphones-better-japans-lasertec-masters-extreme-ultraviolet-lithography/>; www.bloomberg.com/news/articles/2019-03-11/iphones-improve-every-year-thanks-to-an-obscure-japanese-company

Lasertec is an EUV concept stock – a concept most investors couldn't explain, much less what the letters stand for. We shortly provide a tutorial. Amazingly, Lasertec refuses to reveal how much of its revenue, orders, or backlog actually come from EUV, nor the trend – because, we think, the contribution is now likely close to zero. It is nothing more than an outdated supplier of DUV tools from a prior era, masquerading as EUV. We remind that its market cap during the peak period of DUV lithography in the late 1990's to early 2010's was ~\$100MM with a \$1 stock price. Lasertec's site is too technical for most, so for them it has a slick corporate video – as fluffy and light as cotton candy on a sunny day, the perfect sugar high for the retail day trader: "Can Lasertec make you happy?"; "Is Lasertec leading the way for the next 100 years?"; "Can Lasertec outdo itself?"

Lasertec corporate video



Lasertec is a textbook fraud – a marginal company that found itself in the right place at the right time, rode a buzzword called EUV, with no way to live up to the hype and expectations for earnings and growth, inevitably succumbing to the temptation to maintain its stock price – first with little lies, snowballing into a giant lie that brings the whole thing crashing down. The Japanese have a concept similar to 15 minutes of fame: *isshun no eikou* - 瞬の栄光“ - momentary glory before fading back to obscurity, to be remembered only as just another accounting scandal. Lasertec’s accounting scheme bears similarities to Olympus and Toshiba, as we shall explain, with a looming impairment that will exceed the former’s. We note that a key part of the Toshiba scandal involved fraudulent inventory valuation at its semiconductor business unit, as we shall also detail.

Japanese corporate frauds – media coverage

Japan Resists Audit Reforms Despite Toshiba, Olympus Scandals

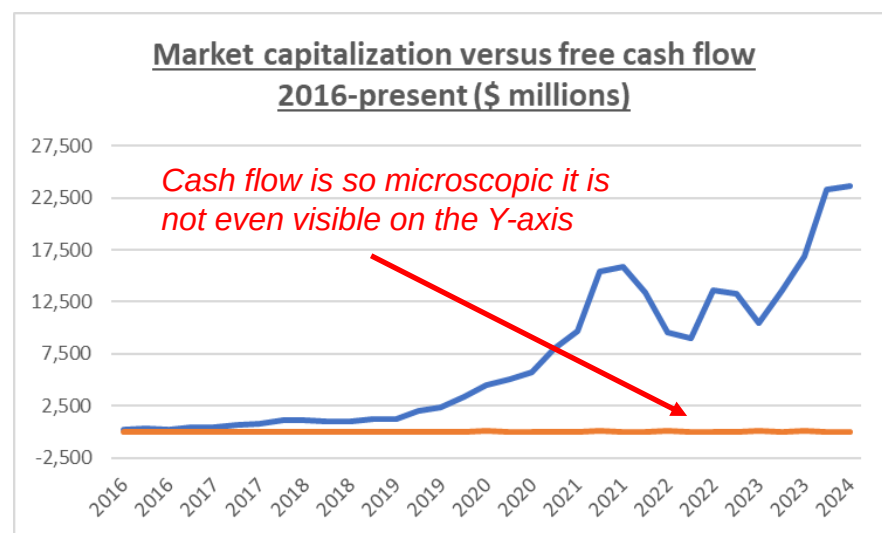
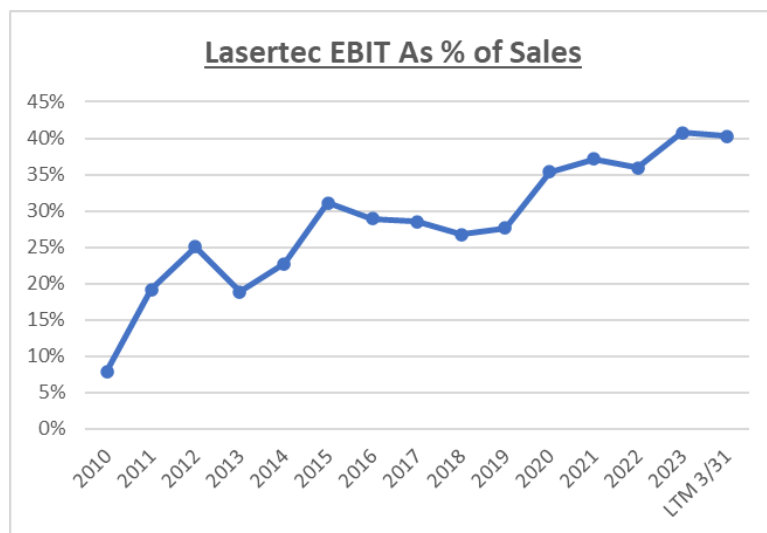
Toshiba Accounting Scandal Highlights Issues in Corporate Governance

Olympus admits it hid losses for decades

How Olympus Became Japan's Biggest Fraud

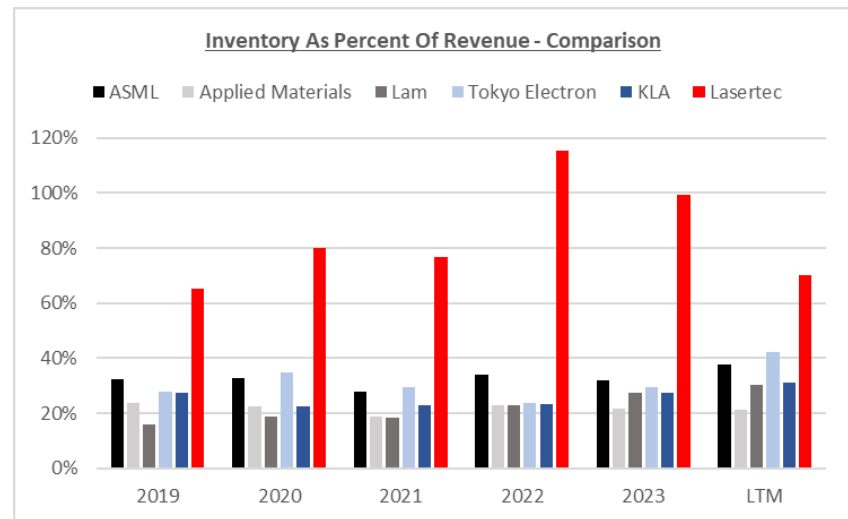
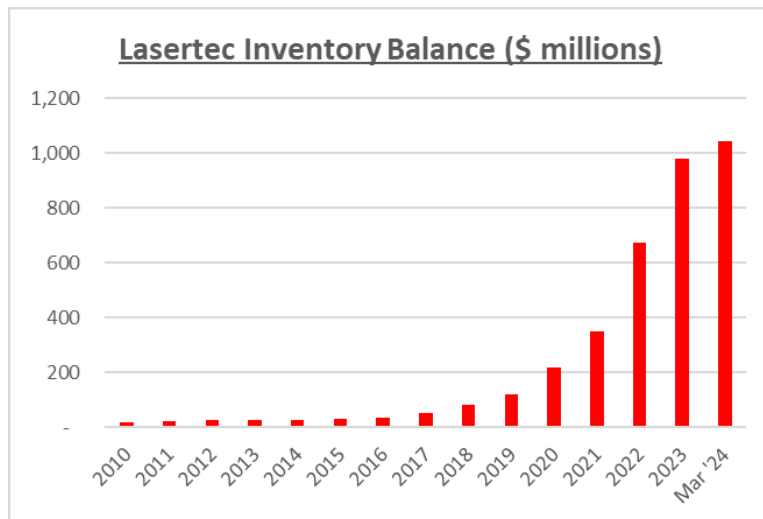
Lasertec sparked our curiosity for the same reason many frauds do: the story is simply too good to be true, and anomalies in its financial statements are extreme - identical to those that preceded the largest blow-ups in history, in Japan and globally. The irregularities are hiding in plain sight but easily missed – a complex industry with terms like EUV lithography and mask inspection that the public barely grasps; speculative frenzy; and the usual gaslighting and deceit by management. Yet, the red flags check off almost every box: a microcap supplier with a mediocre history and non-existent R&D and capex comes out of nowhere to replicate an EUV source that took ASML two decades and \$10B to develop; claims the highest margins not only in its industry but exceeding all but one of the largest tech companies in the world; yet has almost no cash flow nor cash on its balance sheet, relative to its revenue or market cap. The next chapter analyzes the accounting fraud in great detail, but for now we offer just a simplistic preview:

Lasertec margins over time; market capitalization versus free cash flow



The warning signs are classic: firing their longtime auditor right when the anomalies first appeared; secrecy as evidenced by vague disclosures in the financial statements; and terse, evasive replies by the CEO and CFO during quarterly Q&A. Aside from the unusual margins and missing cash flow, the anomaly that drew our focus was the alarming level of inventory - \$1.1B, a figure that has barely fluctuated in 6 quarters, itself a sign of a fabricated number. The absolute level of inventory has gone parabolic; is highly abnormal as a percent of revenue versus its historical trend; began to spike right when it entered the EUV market; hit a preposterous 120-135% of revenue in some recent quarters; hit its highest level in the last 2 quarters as a percent of assets; is the most inflated inventory level of any semicap supplier in the world; and has no explanation versus the 5 largest players - triple or quadruple ASML, Applied Materials, Lam, Tokyo Electron, KLA - even though they all make complex, long lead-time tools.

Lasertec inventory over time and versus the 5 largest semicap players



The CFO's comments a few weeks ago during the Apr 30, 2024, earnings Q&A should alarm Lasertec investors. A BofA analyst questioned why inventories are so high and when they “expect inventories to drop.” The CFO offered no explanation, stating only that “the current situation is a stalemate” and that “we do not necessarily have an accurate grasp of the situation” – and shockingly, that inventories would rise even higher: “we believe that this will increase”; “I think we will see...or rather, an increase in inventories”; and affirmed that the current \$1.1B is the “minimum level forecast.”

Quarterly earnings Q&A, Apr 30, 2024 – transcript excerpts

Continuing on, Mr. Hirakawa of BofA Securities, please go ahead.

Hirakawa [Q]: Thank you very much. Inventories, even after the end of Q3, were probably around JPY160 billion and have remained high for the past three or four quarters. I wonder if you could tell me the level you are targeting in the medium term, and when you expect inventories to drop.

Misawa [A]: Regarding inventories, Misawa will answer. Inventories vary depending on the progress of orders received, anticipated production, and the acceptance of products by customers. As of today, the current situation is a stalemate, or about JPY160 billion.

We believe that this will increase as market conditions recover and orders become more active, and also as products with long lead times increase. We do not necessarily have an accurate grasp of the situation, but we are proceeding with production while properly controlling risks.

Hirakawa [Q]: So you are saying that if market conditions recover, there will be another increase, and that you are currently holding this amount as a minimum level forecast.

Misawa [A]: Yes. We are of course still engaged in anticipated production, but once market conditions recover, I think we will see more active production, or rather, an increase in inventories.

Lasertec's charade is not only brazen but hiding in plain sight – it claims that 100% of its \$1.1 billion in inventory is raw materials and work-in-process with 0% finished goods. Incredibly, its financial statements over the past 5 years record **zero finished goods in any period** – its balance sheet doesn't even have a line item for it. **Misclassifying and improperly valuing inventory** to evade impairment charges that would wipe out historical earnings is a tactic **common to some of the most significant fraud cases over time**. Lasertec has no option but to claim it has no finished goods – its stock would have crashed if investors saw machines piling up, while work-in-process fools them into thinking that components/parts will shortly convert to new sales; and that finished machines ship so fast that they never remain on the balance sheet. The **lack of finished goods lacks credibility versus the five largest semicap suppliers, for whom finished goods has averaged 30%** as a percent of total over the last 5 years – each one discloses raw materials, work-in-process, and finished goods, plus other detail lacking here.

Lasertec financial filings contain no line item for finished goods – most recent filing as example, Apr 30, 2024

	Total Inventory By Category (JPY)					
	2019	2020	2021	2022	2023	Mar'24
Raw materials	15,479	30,083	48,712	94,308	131,056	126,755
Work-in-process	3,278	3,978	5,176	9,774	21,017	34,872
Total inventory	18,757	34,061	53,888	104,082	152,073	161,627

	Total Inventory By Category (USD)					
	2019	2020	2021	2022	2023	Mar'24
Raw materials	101	197	318	616	857	828
Work-in-process	21	26	34	64	137	228
Total inventory	123	223	352	680	994	1,056

2. Consolidated financial statements and notes

(1) Consolidated balance sheet

	Previous (As at June 30, 2023)	Current (As at March 31, 2024)
(In millions of yen)		
Assets		
Current assets		
Cash and deposits	29,773	29,744
Notes and accounts receivable-trade, and contract assets	21,611	20,494
Work in process	131,056	126,755
Raw materials and supplies	21,017	34,872
Accounts receivable - other	19,640	263
Other	8,029	7,872
Allowance for doubtful accounts	(38)	(21)
Total current assets	231,090	219,981

To hide a lie as flagrant as no finished goods, of course, means that other lies are needed. It appears to be an open secret among Lasertec's investor relations team that inventory is being misclassified. We hired a local consultant to meet with IR in March 2024 at Lasertec headquarters, who at one point asked “And no finished goods in there, right?” The IR rep replied “No. All work in process.” while smiling with a knowing wink. In a later section, we share photographic evidence that Lasertec's new “Innovation Park” and its so-called “fabs” are essentially a fraud with no production or R&D. During ~20 visits to the campus, our investigators found an empty and deserted facility. Its only purpose appears to be storage for finished goods – we show below a sample of photos of machines being loaded into storage, all hermetically sealed in protecting wrapping and clearly finished goods. Given that some of these machines sell for \$50MM each, Lasertec's claim of 100% raw materials and work-in-process and 0% finished goods is clearly false.

Investigator photos of finished goods being moved into storage at “Fab 4A” at Lasertec Innovation Park



The elephant in the room – aside from Lasertec’s lack of cash – is, of course, why the inventory is so high. Every accounting anomaly has an explanation. To find the answer, we undertook an intensive 5-month investigation encompassing over 20 in-depth research interviews, typically 60-90 minutes each; ~20 investigator field visits to Lasertec’s new Innovation Park; and an extensive review of the scientific literature on EUV light sources and actinic mask inspection, including key papers and presentations at SPIE - the industry’s society for photonics and optics - by Lasertec, TSMC, Intel, KLA, and other players. Our sources included 6 contacts at TSMC, including the ex-head of R&D and the purchasing executive who first brought in Lasertec; one of Intel’s most senior and tenured engineering executives, with leadership responsibility in EUV mask inspection; a longtime Samsung executive in a leadership role in EUV and a decision-maker for Lasertec tools; and other customers like Global Foundries and Marvell.

Sample of experts we consulted as part of our research (cont’d)

TSMC - 6 sources; including a Lasertec field engineer based inside its fab in Taiwan; plus other experts in touch with TSMC including the head of its mask shop

- Former head of TSMC R&D; played key role in 5nm transition and vendor/tool selection
- Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC
- Others we cannot detail due to confidentiality agreements

Intel – 1 senior executive; several Lasertec field engineers based full-time inside Intel’s mask shop or Oregon fab; other sources in touch with Intel and familiar with its experience with Lasertec’s technology

- One of Intel’s most senior and tenured engineering executives, leadership responsibility for EUV/mask inspection

Samsung – 1 senior executive

- Longtime Samsung executive in leadership role in EUV; worked closely with Lasertec; decision-maker for inspection tools

Other customers

- Global Foundries: Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection
- Marvell: Senior employee at Marvell Technology, directly involved with Lasertec’s systems

Additional research sources included 3 Lasertec field engineers working full-time on its ACTIS EUV systems, including one based inside Intel's mask shop in Santa Clara, CA and one at TSMC's fab in Taiwan, and an Ushio engineer inside Intel's fab responsible for Lasertec's EUV source; two of the leading experts in EUV source development, including one who played a key role in ASML/Cymer's source and one at Isteq; multiple industry executives in Japan, including two in touch with Lasertec's CEO; a leading EUV lithography expert close to Lasertec and its key customers; and several sources at KLA, such as an ex-executive in a leadership position in its EUV mask inspection group.

Sample of experts we consulted as part of our research (cont'd)

Lasertec and Ushio field service engineers working full-time on its ACTIS EUV systems, and one for its DUV tools

- *Former Lasertec field service engineer who worked in the Intel mask shop*
- *Former Lasertec field service engineer at TSMC fab in Taiwan*
- *Former Lasertec product engineer; 20 years in the industry and previously part of the EUV task force at one of the top 3 semiconductor companies in the world*
- *Ushio field engineer who maintains light sources on Lasertec ACTIS machines at Intel's Oregon fabs*

Leading figures in EUV light source development

- *Key figure in the development of Cymer/ASML's EUV light source; headed KLA's EUV source effort; one of the top scientist/experts in the field*
- *Key R&D employee at Isteq, the Russian R&D project from which Lasertec licensed their "new" Urashima light source technology*

KLA; 2 sources plus other industry sources in touch with KLA regarding its imminent EUV mask inspection system

- *Former executive in a leadership position in their EUV mask inspection group, described recent interactions directly with Lasertec*
- *Ex-KLA director of engineering focused on EUV mask inspection*

Other industry executives and experts

- *Leading EUV lithography expert close to Lasertec, Intel, KLA, and other key players*
- *Longtime semiconductor consultant recently working for Lasertec; close to KLA management*
- *Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO*

The balance sheet already told us where the bodies are buried, but didn't foretell the stench when we found them. Lasertec's inventory is inflated is because 1) its flagship EUV product line is defective and fatally flawed; 2) it has no margin because of costly component replacements, recalls, discounts, and inducements; and 3) worthless inventory is hence fraudulently overstated as an offset to understated expenses and fake earnings, the accounting mechanics of which we shall explain. Lasertec has never had an EUV technology that actually works: key customers have stopped buying new ACTIS systems; are refusing to accept shipment with ones like Intel "pushing, pushing, pushing out" purchase orders; inventory is piling up; the head of Intel's EUV mask shop is allegedly so angry that he has been banned from meetings for blowing up at Lasertec; TSMC, Global Foundries, and Marvell laugh at the product; the "new" ACTIS A300 failed inspection at TSMC, per 4 sources; and key customers told us that Lasertec is quietly offering them ~75% discounts, trade-in's, and other inducements and concessions – proof that Lasertec's reported inventory is worthless.

Lasertec desperate to offload ACTIS inventory; offering up to ~75% discounts on A150; offering various inducements

Q: "Do you remember any color around the inventory build-up?"

A: "When our [Marvell senior executive – title redacted] went to go visit, **they were trying to also offer additional A150's that they had on the books that weren't, let's say, being taken ownership of by other competitors that had presumably forecasted that they needed them. And so, we were getting a reduced price as well if we wanted to purchase additional A150s and that price was closer to about, in the range of \$15 to 30 million.**"

Q: "So, they were willing to try to sell those machines at half or a third of their price?"

A: "Yeah, and also, again, that's why the comment that I gave earlier, which is that if we were to buy an A150—"

Q: "Yeah, they've got to write down all that inventory."

A: "Yeah."

Q: **"My question is, did that really happen? They offered you guys \$15 to 30 million to take A150's? This literally just happened a few weeks ago?"**

A: **"Yeah."** – Senior employee at Marvell Technology

Source: Scorpion Capital consultation calls with experts

Lasertec is a relatively simple accounting fraud - one can easily estimate what its normalized inventory should be, and therefore the level by which both inventory and earnings are inflated and the size of the inevitable impairment charge and retained earnings hit – we explain the accounting math and interrelationships in the next chapter. Normalized inventory may be calculated by using 5 methods: 1) the 3-year average of inventory as a percent of sales across ASML, Applied Materials, Lam, Tokyo Electron, and KLA – 26% on average with little dispersion, and hence a robust measure, given similar product and business characteristics; 2) 3-year average of inventory as a percent of assets across the same peer group; 3) Lasertec’s historical, “normal” inventory as a percent of sales, from 2010 to 2017, immediately prior to its EUV product launch and before it spiked to unusual levels; 4) as a “normal” percent of its assets with the same methodology; and 5) by marking down its inventory by the 50-75% discount it is quietly offering to customers, per our research interviews, thereby adjusting it to “market value.”

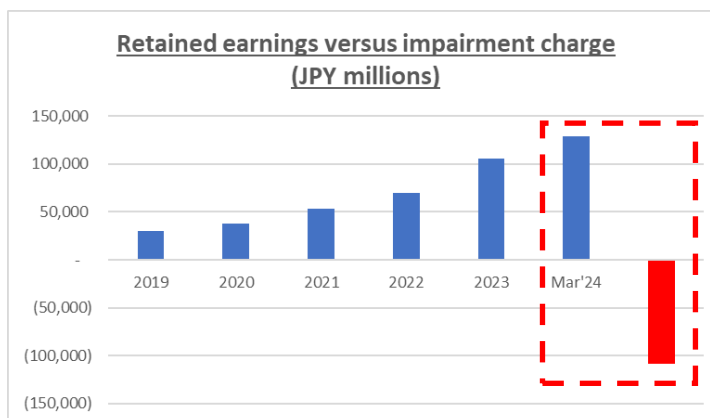
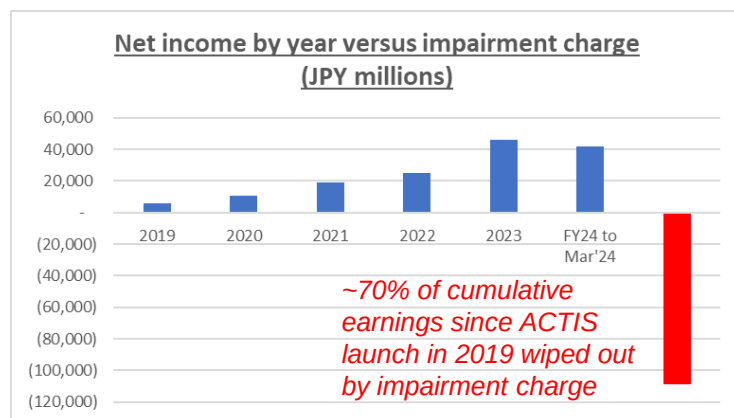
First two methodologies – normalizing inventory to peer group % of sales; and to % of assets; other methods shown in next chapter

Method 1: Normalize inventory to peer group % of sales	USD	JPY
Lasertec inventory as % of sales, 3-year avg	97%	97%
Peer group 3-year avg	26%	26%
Lasertec LTM revenue	1,525	230,604
Implied normalized inventory (peer group avg x LTM revenue)	397	59,957
Current Lasertec inventory, Mar '24	1,069	161,627
Implied impairment charge (current inventory less normalized #)	673	101,670
Implied impairment as % of current inventory	63%	63%

Method 2: Normalize inventory to peer group % of assets	USD	JPY
Current Lasertec inventory as % of assets	62%	62%
Peer group 3-year avg	21%	21%
Lasertec total assets Mar '24	1,713	259,146
Implied impairment charge	709	107,206
Implied impairment as % of current inventory	66%	66%

All five methods for estimating the amount by which Lasertec is fraudulently valuing inventory – and therefore earnings, as well as the magnitude of the impairment charge it is trying to avoid – arrive at roughly the same figure, one that is devastating and indicates the degree to which its stock price has been based on an illusion: an inevitable 65-70% write-down of its \$1.1B / ¥162B of inventory as of Mar '24 – a ~\$700-750MM / ¥105-113B loss that would wipe out ~70% of its cumulative earnings since the launch of its ACTIS product in late 2019 through the most recent quarter. The charge would also wipe out ~85% of its retained earnings of \$832MM / ¥129B, as of the most recent quarter. The loss and retained earnings hit we think is imminent far exceeds what Olympus incurred in 2012 in the immediate aftermath of its scandal (¥49B/\$300MM and ¥53B/\$460MM, respectively); and is considerable relative to Toshiba's loss and retained earnings hit of ¥400B/\$2.6B and ¥378B/\$2.4B in 2014/2016 – a sobering fact given Lasertec is only 3.5% of Toshiba's revenue of ¥6,300B/\$40B at the time

Net income and retained earnings versus looming impairment charge



As a preview of the fiasco that Lasertec is trying to conceal with fake accounting, TSMC's experience with Lasertec's EUV and DUV systems has been a debacle, the specific reasons for which we detail in a separate chapter. We did two research calls with a former longtime procurement executive at TSMC, who first brought Lasertec into TSMC; the second call was after he checked with current colleagues there for recent color. He chuckled when stating that the "new" ACTIS A300 EUV system failed their evaluation, which we confirmed with a total of four TSMC sources; was blunt in stating that "TSMC will not consider the A300 or A150 for production"; and stated "no, unfortunately, no" when asked if TSMC is still buying Lasertec's EUV tools.

Lasertec's new A300 actinic tool "failed" evaluation at TSMC

Q: "I heard the light source in the new Lasertec tool, the A300, doesn't work, and TSMC evaluated it, but I heard that it failed at TSMC. Is that true?"

A: "[chuckles] Yes. It failed." – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

TSMC "will not consider" Lasertec's EUV tool for production; only a niche lab/R&D tool in the mask shop; using legacy DUV inspection machines instead as EUV inspection "is not necessary"; only purchased six A150 machines

A: "Both the LasertecX8 Ultra and X9 Ultra are the DUV light source, but they are inspecting the EUV mask. The main reason is for the throughput. And it is not necessary..."

Q: "You're saying TSMC uses the X8 and X9 Ultra in the fab instead of the ACTIS?"

A: "Yes. Because of the throughput. TSMC will not consider the A300 or A150 for production. They will only exist in the EPO mask house." – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

TSMC stopped buying Lasertec EUV machines as it's only a niche mask shop took; "not a tool to support production"

Q: "Has TSMC basically stopped buying A150 machines from Lasertec? Are they still buying Lasertec A150 machines or no?"

A: "No, unfortunately, no."

Q: "When did they stop buying them?"

A: "I will say something – even if TSMC continues to work with Lasertec, now it's mainly in the mask house, instead of the production house, production area. So, for the EPL, for the mask house..it's not a tool to support production [...] the Lasertec tool cannot meet demands in a production line." – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

Industry sources outside of TSMC corroborated our findings, such as two different experts from KLA, one who recently spoke with Lasertec: “TSMC is not buying the High-NA Lasertec tool,” the ACTIS A300; “no confidence...it will be reliable product”; “I hear offline it’s not very reliable”; “the A150...needs a lot of support, and it has a lot of issues”; “what I know is that TSMC is upset because the source is not working,” referring to the Urashima source in the A300; “TSMC is upset...the source does not have enough uptime...cost of service is too high...they are flickering”; “I heard that they failed” evaluation at TSMC and that “Lasertec probably must write off those tools.”

TSMC is “not buying” the A300; “no confidence” that it’s a “reliable product”; A150 is also “not very reliable,

Q: “Is anybody buying less Lasertec, or have they essentially idled the Lasertec machines because of these frustrations?”

A: “Well, a recent example would be that TSMC is not buying the High-NA Lasertec tool, to my knowledge. And they're the company that does high volume, high-performance chips...”

Q: “You’re talking about the recent ACTIS A300 that they announced in November, which sort of tied into the Intel High-NA announcement. Is that what we’re talking about—the recent product announcement?”

A: “Yes, I’m talking about that...there’s no confidence in my opinion that it will be a reliable product...”

Q: “You said you don’t think this A300, this High-NA tool, is going to be reliable.”

A: “I reviewed again the papers presented in the past year at the various conferences—there was the advanced lithography conference and a photomask conference. I reviewed the papers...reliability, the big thing I hear offline is it’s not very reliable. I’m talking about the A150. A300 is just a product announcement. It’s not shipped. The A150...needs a lot of support, and it has a lot of issues.” – Ex-KLA director of engineering focused on EUV mask inspection

TSMC is upset with Lasertec; problems with the source; A300 with new light source failed in the qualification period

A: “Lasertec convinced TSMC to take a new upgrade of the source, which is an Urashima source. There was a lot of wow around it. And when they released that source to TSMC [...] What I know is that TSMC is upset because the source is not working.”

Q: “So, TSMC evaluated the Isteq source for six months and it failed.”

A: “Yes. Intel doesn’t return the tool. TSMC is upset. The source does not have enough uptime, cost of service is too high, they are flickering [...] Mid last year to the end of last year. At the beginning of the year, I heard that they failed.” [...]

Q: “The A150 used the Ushio source. The A350, the Urashima, is the Isteq source, which already failed after six months with TSMC.”

A: “Correct. That’s my speculation...now I know exactly what happened because I thought that the tools were shipped back. Lasertec probably must write off those tools.” - Former KLA executive in a leadership role in their EUV mask inspection group

Global Foundries, also one of the world's largest chipmakers, dismissed the A300 as “just a press release” to match Intel's acceptance of ASML's first High NA scanner, saying “no question, it sounds sketchy”; “if the A300 remains only on paper...it could be pure PowerPoint...that the A300 never really materializes...they would be in a lot of trouble at Lasertec...would end up having to impair...would probably require an asset write-down.”

ACTIS A300 is just a press release timed to coincide with Intel's High NA EUV announcement

*“I have not even heard of a single A300 system, a 300 system that shipped even. I don't know if it's pure vaporware, and it's **just a press release to match the shipment of the first High-NA system to Intel, and they want to match that press release.**” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection*

New Urashima is a light source is a key risk and “sounds sketchy”

*“Urashima is the **risk for them, no question...it sounds sketchy.**” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection*

Global Foundries sees KLA as “huge risk to Lasertec” that “obviously will” take share; Lasertec would have “significantly mark down” and “impair” its inventory; “would probably require an asset write-down”

Q: “How many more Lasertec machines are people going to buy in 2024? Because if finished goods inventory is already piling up, 3-nanometer delays, TSMC--?”

*A: “I think **there's a huge risk here to Lasertec, if machines are piling up and KLA is going around reassuring people that if you can hold off until mid-2025,** we will have a system commercially available to you. And in fact, we'll give early access in the second quarter of 2025, you can come and evaluate...High-NA is going to accelerate. People are making these massive capital investments in inspection equipment. I do think, by the way, even if KLA came out with, let's say, an A150 competitor, yes, **they will take share from Lasertec. They obviously will. And Lasertec could be caught with inventory they're going to have to significantly mark down** even to move on the A150 side. But **if the A300 remains only on paper...it could be pure PowerPoint capability that the A300 never really materializes. And then they would be in a lot of trouble at Lasertec...**”*

Q: “This is a scenario where their sales completely collapse.”

*A: “Yes, for EUV, correct. Correct. And they **would end up having to impair** I don't know price point-wise. **KLA, if they wanted to...could put a lot of pressure on them, which would probably require an asset write-down on the part of Lasertec.**” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection*

Marvell, another major player, was just as skeptical, sharing color which was damning and indicates that Lasertec's inventory is worthless and that the A300 is simply a ruse to swap out defective A150's while disguising it as "growth": the A300 "doesn't make sense for us right now"; "there was no data that they had"; "no real-world case studies from independently verified semiconductor industry associations"; "no real way to discern that this actually has a value-add"; "we haven't seen real proof of concept." Marvell stated that a senior executive recently met with Lasertec in Japan, and that they were offered aggressive trade-in's and discounts - a shocking sign of desperation - to take back the ACTIS A150 and "give us an A300 at a reduced price"; "effectively companies would take the A300 and not use their A150's anymore...that would be the idea...so, the value of the A150 that we have on hand wouldn't be worth that much."

Lasertec had "no data" to show for the A300; "no real world case studies" or independent evaluation; "no real way to discern" any value

Q: "The A300 sounded like vaporware to me. Did they even have any evaluation data or benchmarks? What was the status?"

A: **"No, there was no data that they had.** They essentially just had value-add of how this is different, namely in their light source...there's **no real-world case studies from independently verified semiconductor industry associations** or a semiconductor consortium. And so, there's **no real way to discern that this actually has a value-add** that would offset the overall cost of taking possession of one of these machines." – Senior employee at Marvell Technology

Swap for A150 implies existing inventory of A150's has almost no value

Q: "When they said swap, what did they mean? You don't have to pay for it? Or you trade it in?"

A: "So, **they would take back the A150 and then they would give us an A300 at a reduced price** from the retail price."

Q: "And did they say what the price is and what the discounted price might be?"

A: "That I can't speak on because I think those are confidential terms. I can just speak to the fact that what **they were offering was to take the A150 back and give us an A300 and essentially give us a discounted value based on us returning the A150."**

Q: "The swap thing is weird. How is there any trade-in value from that? It just makes no sense to me."

A: "...**Effectively, companies would take the A300 and not use their A150s anymore; that would be the idea. So, the value of the A150 that we have on hand wouldn't be worth that much anyhow.**" – Senior employee at Marvell Technology

A leading figure in ASML's EUV source development expressed skepticism of Lasertec's reported margins: "whatever they shipped so far, they basically lost money...they basically are spending constantly on cleaning, constantly replacing tools, and keep losing money." Other sources stated that Lasertec is well aware their tool is defective and quietly admit that it doesn't work, such as an industry executive in Japan who stated he is friends with the CEO: "they are very secretive but if you talk to the right people, they will tell you...the light source only lasts for three months and it's only usable for two months, and replacement is very time-consuming and very costly." When we asked if Lasertec is worried, he stated that "they don't show it, but I think they are." He compared it to another Japanese supplier that is "also struggling because tools are not moving out"; "they have a cash flow problem...I would assume that Lasertec is in a similar situation."

Lasertec executives quietly admit their product doesn't work

Q: "Does Lasertec admit that they have these quality and reliability problems? Do they admit it, or are they secretive about it?"

A: "They are **very secretive, but if you talk to the right people, they will tell you.**"

Q: "And what do those people say? It sounds like you know some people there. What do they say?"

A: "One of the people I know, he is [redacted] of Lasertec and he told [redacted] about these problems, **the light source only lasts for three months, and it's only useable for two months, and replacement is very time-consuming and very costly.** That's what I heard from [redacted]." - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

Executive who knows Lasertec CEO believes they are worried; suggests struggling with moving inventory and cash flow

Q: "**Do you hear any worry from the people at Lasertec? Any concern?**"

A: "**They don't show it, but I think they are.** I've heard rumors from [redacted] that they're also struggling - because tools are not moving out as quickly as they want, that they have a cash flow problem. **I would assume that Lasertec is in a similar situation.**"
- Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

High skepticism of Lasertec's reported margins given need for "spending constantly" to keep tools running

"Last [quarter], Lasertec said that they believed they shipped the last low-profit versions of the mask inspectors - **whatever they shipped so far, they basically lost money. They basically are spending constantly on cleaning, constantly replacing tools, and keep losing money.**" – Leading figure in the development of Cymer/ASML's EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

The executive continued that he is “concerned about their future” and that their product problems are essentially unsolvable: “it’s not so easy for them to move forward...it was big hype for them to come out with an actinic inspection tool but when it comes to adopting the tool...there are many unforeseen problems...and I would be a bit concerned about the capability of Lasertec to really overcome these,” indicating “they will be in serious trouble” the second there is a viable competitor. The TSMC procurement executive we quoted stated he was recently in Japan and met with a Lasertec executive, who confided that things are “not well.” The TSMC exec derided their EUV system as just an irrelevant niche: “we call it a lab tool, R&D tool...not a production tool” that is useful inside a fab. He added that TSMC has a weekly, standing crisis meeting with Lasertec in Taiwan “to push Lasertec to improve,” as we detail later.

Senior executive who’s friendly with Lasertec CEO is concerned about their future; “big hype” quickly followed by “unforeseen problems”; difficult to solve and “in serious trouble” with upcoming electron beam inspection tools
“I’m a little bit concerned about their future. If they cannot come up with a very disruptive light source, it’s **not so easy for them to move forward. It was big hype** for them to come out with an actinic inspection tool, being the only supplier, but when it comes to adopting the tool and the process in the fab, that there are **many unforeseen problems that need to be solved, and I would be a little bit concerned about the capability of Lasertec to really overcome these problems timely**. And if somebody comes out with an EB inspection tool that really works, **they will be in serious trouble**.” - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec’s CEO

Lasertec executive recently told TSMC contact that things are “not well”; TSMC views it as a niche “lab tool, R&D tool”

Q: **“When you talked to the Lasertec executive, what else did they say? Are things going well or not going well for them?”**

A: **“Not well.”**

Q: “What did he say?”

A: **“Not well. I think Lasertec will be kind of—we call it a lab tool, R&D tool.”**

Q: “It’s an R&D tool? Lab tool, not a production tool.”

A: “Yes, **a lab tool. Yes, not a production tool**. Because you have a lot of limitations for productivity...even today, TSMC still has a weekly meeting with Lasertec. **My colleagues in procurement and from the fab told me they have a weekly meeting to push Lasertec to improve.**” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

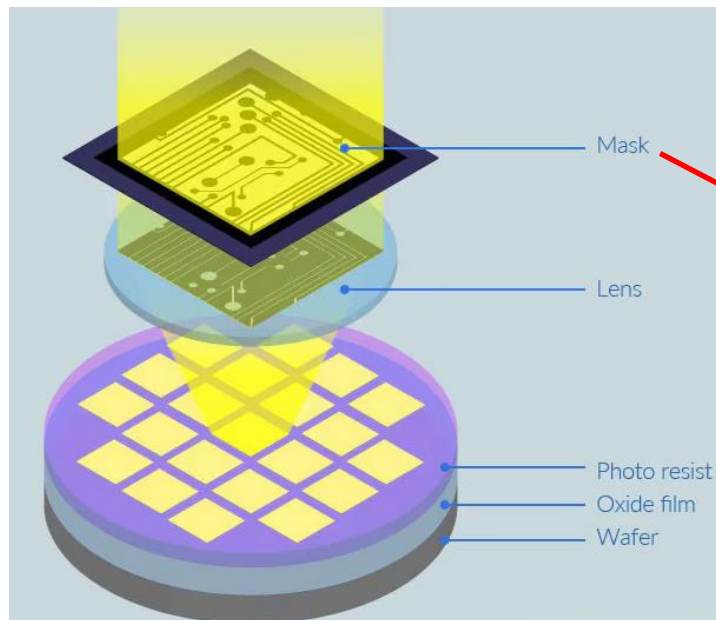
Former Lasertec engineers corroborated the company's secrecy and concealment of the various defects that plague its EUV inspection tool, such as one who prior to Lasertec was part of the EUV task force at one of the largest semis players: "there were too many secrets and there were too many things that were hidden by the company"; "you will never get an answer from the company...Lasertec was not open about anything that could be a problem...the company was not willing to be open about...minor problems let alone any major problems with the EUV tool...utilization rate or more major issues"; "I never received any answer and all I was told was to wait...nothing ever gets solved." He alleged that Lasertec hides product defects and data from customers. As the engineer was primarily focused on servicing machines at Intel, we presume that it is one of the customers he meant: "...tried to hide some data or technical things...the company will not tell its customer...." He added that Ushio didn't care that its source was defective: "their caliber of caring wasn't as high...they're more like, we'll just replace it."

Lasertec is secretive and conceals product problems even from its engineers and employees

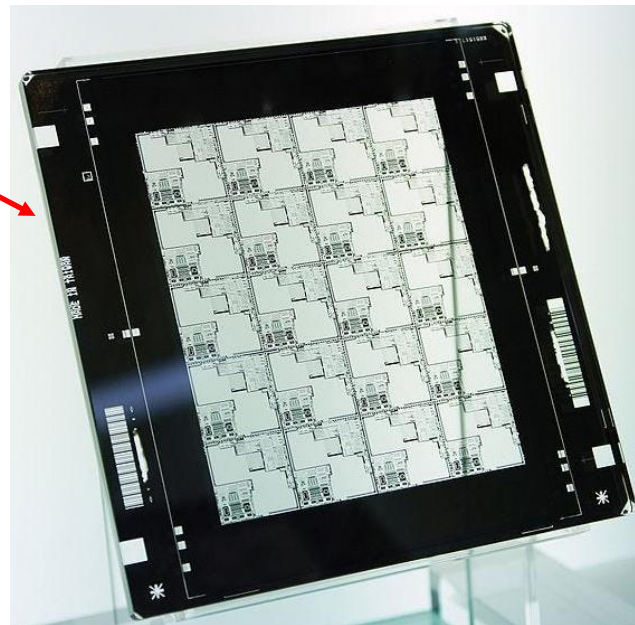
"There were too many secrets and there were too many things that were hidden by the company...it is a Japanese company, so even if you are one of the company's employees, you will never get an answer from the company. Lasertec was not open about anything that could be a problem, and that could be a target of attack from its competitors. The company was not willing to be open about any of these minor problems, let alone any major problems, with the EUV tool. Recently, there could be some utilization rate or more major issues. When I asked something or when I tried to raise something, I never received any answer, and all I was told was to wait...Lasertec USA does not even have the technical background, so they are just waiting. Nothing ever gets solved." [...] The company could have tried to hide some data or technical things...for example, if there was breakage, if some sample masks were damaged while installing them or if some parts were broken with the optics part of the machine, the company will not tell its customer about that mistake, and it will just request replacement parts and do the work. It will just tell the customers that the process is being delayed. So, when these kinds of mistakes happen, Lasertec would rather address all of the issues by itself without telling the customer exactly what happened.... In terms of Ushio, I would say their caliber of caring wasn't as high. They're more like, we'll just replace it when it needs to be replaced." – Former Lasertec product engineer; 20 years in the industry and previously part of the EUV task force at one of the top 3 semiconductor companies in the world

In order to understand what EUV is and what Lasertec actually sells, we first explain how semiconductors such as memory chips and microprocessors are made. The process is called photolithography, akin to traditional photography where an image is projected onto a film. A pattern is projected onto a silicon wafer coated with a light-sensitive material called photoresist, similar to how light exposes photographic film. The light causes chemical changes where it hits the photoresist, and this pattern guides where the wafer will be etched or layered to form the intricate circuits of a microchip. Light is projected onto the wafer through a photomask with the desired pattern or image of a chip layer. The photomask is typically called a “reticle” and resembles a slide in a projector.

Semiconductor photolithography

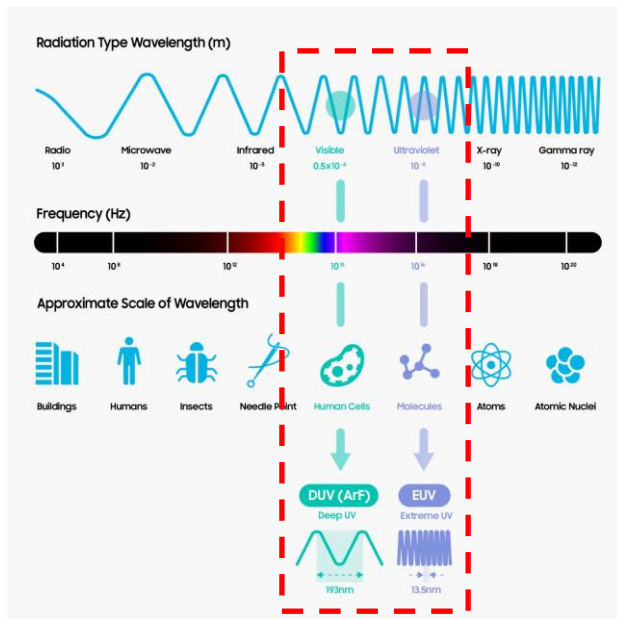


Photomask, also known as a “reticle”

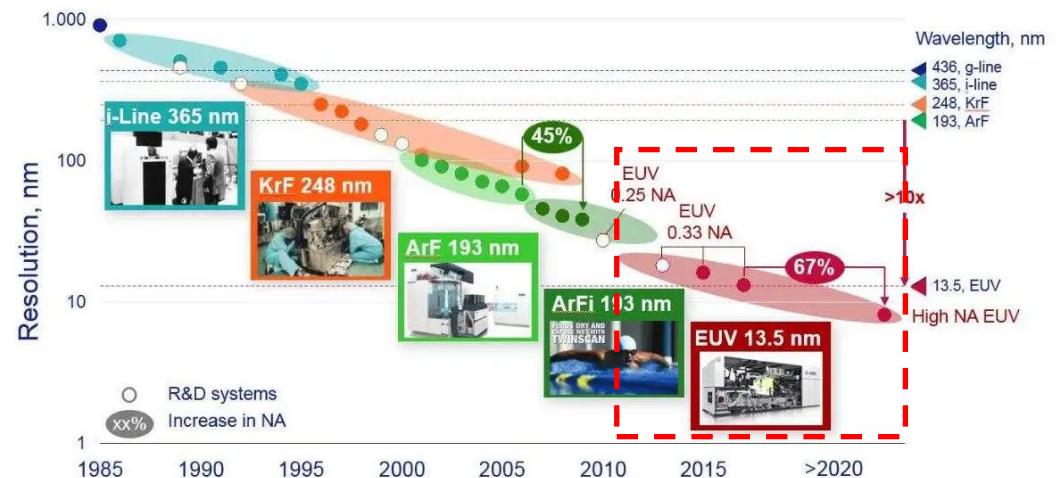


Photolithography is classified by the wavelength of light used, such as ultraviolet lithography (UV), deep ultraviolet (DUV), extreme ultraviolet (EUV), or X-ray. The wavelength determines the minimum feature size that can be printed on a wafer. The 1970's and 80's were the UV lithography era, using mercury vapor lamps that emitted wavelengths of 365nm and 436 nm. DUV followed in the 1990's with excimer lasers and shorter wavelengths of 248nm and 193nm and became the industry standard, allowing smaller features for more compact, powerful chips that pushed the edge of Moore's Law. EUV is the next-generation of lithography, with a far shorter wavelength of 13.5nm, and became a reality in the late 2010's.

DUV/EUV wavelength comparison

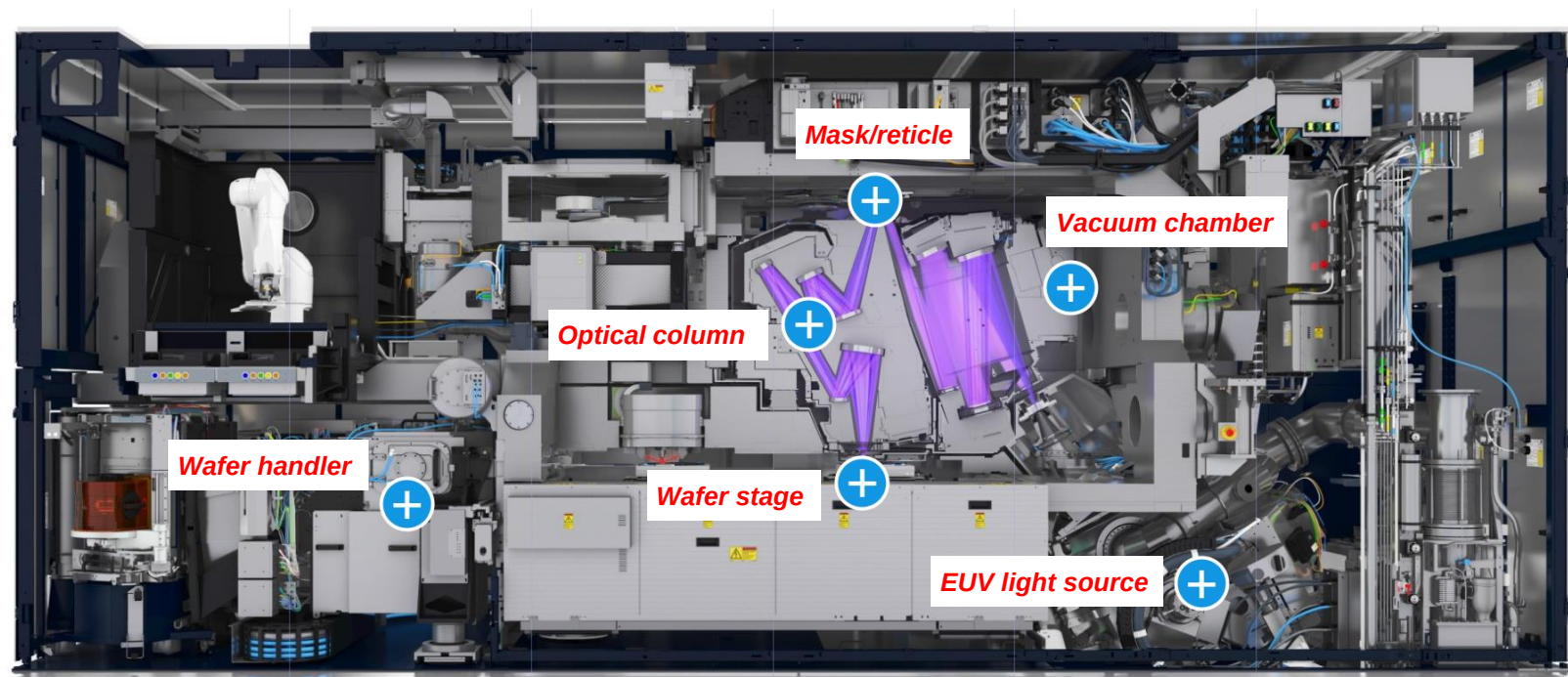


Timeline for transition from UV to DUV to EUV lithography



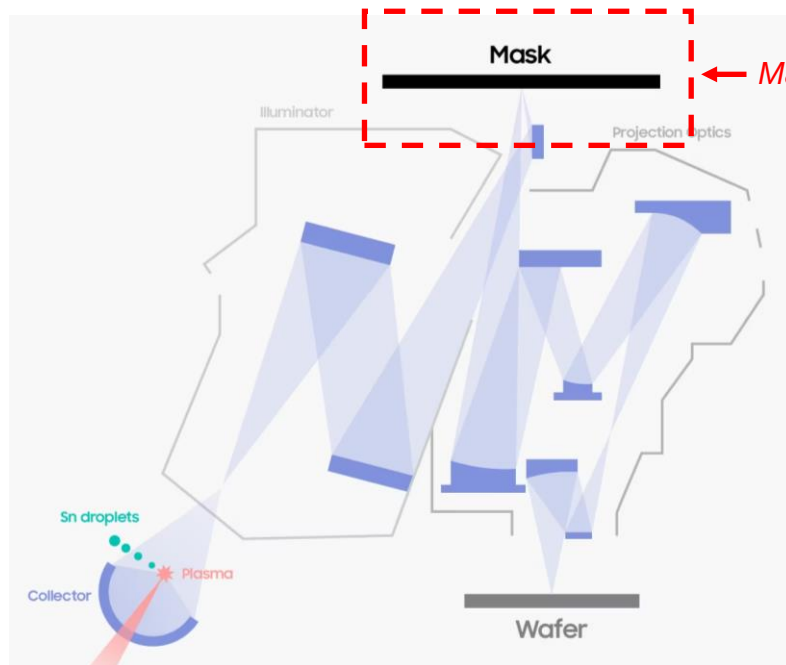
Netherlands-based ASML has a monopoly as the only company that produces and sells EUV systems for chip production. The technology took decades to develop and faced enormous struggles and delays, with the first EUV scanner – the machine that “writes” onto a wafer - shipping in 2013 and costing upwards of \$200MM each. EUV scanners are the size of a truck and weigh 180 tons, requiring three Boeing 747’s to ship - among the most complex systems developed by man. The most critical and challenging component is the EUV light source; other key sub-systems are the reticle (mask), optical column, and wafer stage.

ASML EUV scanner – interior



The optics in an EUV scanner are completely different than for DUV, as EUV light is absorbed by glass and air. It therefore requires a vacuum and uses a series of reflective mirrors, instead of using lenses to focus light. In a DUV system, light is projected onto a wafer through a mask and series of lenses, whereas in EUV it bounces off highly complex mirrors. Masks for EUV also work by reflecting light.

EUV mask reflects light vs. DUV light going through



ASML EUV scanner optical column

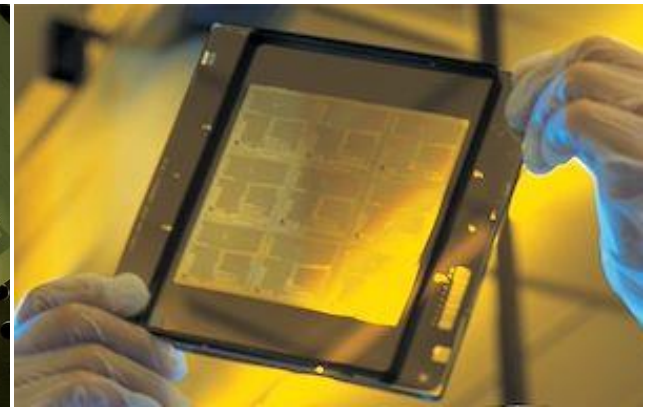
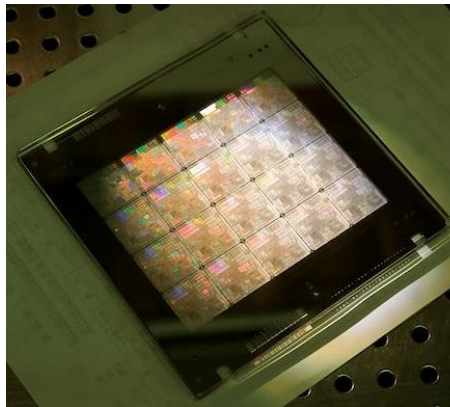


Machines made specifically for inspecting photomasks have been Lasertec's bread and butter since the 1970's – a small niche versus other segments of the tools market like wafer inspection, defect review, or metrology, hence its historical market cap of \$100-200MM. If the pattern on a mask is defective or its surface is contaminated with a microscopic particle, every wafer printed with it will be defective and yields will be low. In photolithography, several masks are used in turn, each one reproducing a layer of the completed circuit design, known together as a mask set. Masks are opaque plates made of multiple layers of various materials, with EUV masks using 40-50 alternating layers of molybdenum and silicon. The process begins with a mask blank, which is then coated with photoresist, exposed to a laser, and developed to etch the final pattern.

EUV mask blank prior to patterning

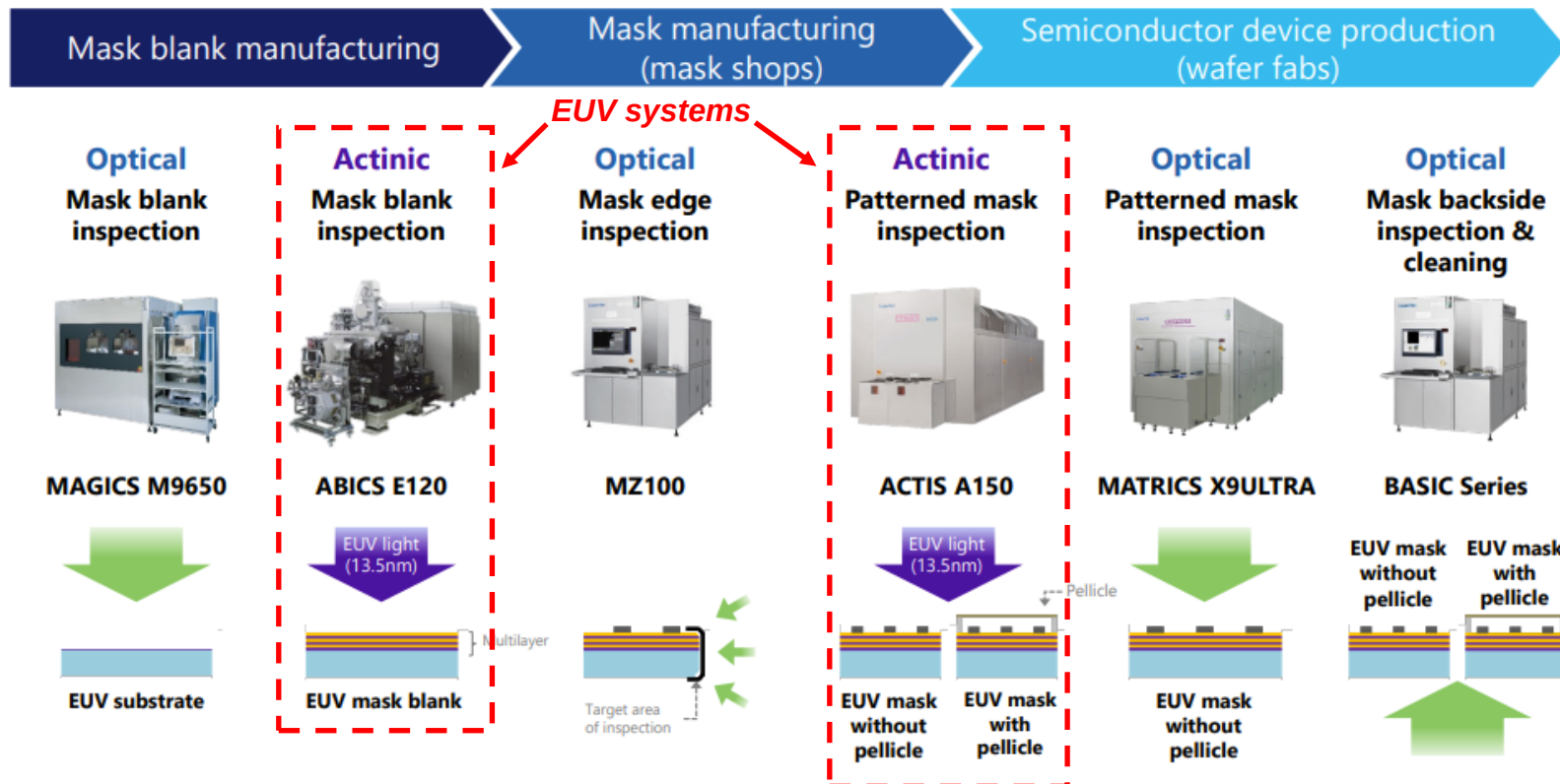


EUV masks, known as “reticles”



Lasertec sells two types of mask inspection systems: one for mask blanks, and one for finished masks. In the 1970's/80's, its tools were UV-based. Today, each of these two types of machines comes in an old DUV (~200nm source) and new EUV version (13.5nm). The DUV product line is called **MAGICS** (blanks) and **MATRICES** (masks); the EUV line is called **ABICS** (blanks) and **ACTIS** (masks). Lasertec sells a few other tools for inspecting wafers or FPD (flat panel display) blanks/masks, but these are old, inexpensive commodity systems with less sensitivity.

Lasertec product lines – slide from corporate presentation



Lasertec's revenue and stock went nowhere for a decade until the launch of its first EUV product in 2017, the ABICS system for inspecting blanks, with a 13.5nm light source. The stock came to life, and then exploded in 2019 with the launch of its first EUV system for inspecting masks, the ACTIS A150 – its flagship product and the driver of its stock. The ACTIS tool's claim-to-fame is that it is the first and only “actinic” EUV mask inspection machine.” Actinic simply means that it uses a 13.5nm EUV light source, which theoretically detects smaller mask defects at EUV feature sizes versus older ~200nm DUV tools. ACTIS tools sell for \$40-60MM each; we estimate ABICS at \$25MM each. Its DUV systems sell for much less, with MAGICS (for blanks) at ~\$6MM and MATRICS (for masks) at ~\$9-12MM, per our estimates.

Flagship EUV mask inspection machine ACTIS A150, launched Sep 2019



EUV blanks inspection machine ABICS E120, launched Apr 2017



For all intents and purposes, Lasertec and the ACTIS A150 are Intel's baby - giving Intel's mask group the same incentive to cover up its epic failure. When ASML's EUV scanners began to gain traction in the late 2010's, the industry faced a problem: pre-occupied with a decade of delays to bring the technology to market, ASML and key customers like Intel and TSMC neglected to develop a system to inspect EUV masks. Intel in particular was caught flat-footed, having made a massive bet on EUV via a \$4B investment in ASML in 2012. Intel hence pushed KLA to develop an EUV mask inspection tool, but our research indicates that KLA demanded too much money. Intel then turned to Lasertec, sponsoring its EUV mask inspection effort as its "founding customer," investing \$300-\$500mm, by our estimate, via advance purchase orders. Multiple sources told us that the timeframe Intel pushed was unrealistic, forcing Lasertec to pick the only EUV source available - from a Japanese supplier called Ushio, which was defective and the start of its EUV fiasco.

Ushio EUV light source announcement – July 2019, the key component of Lasertec's ACTIS A150

USHIO Launch Lithography Mass Production EUV Mask Inspection Device

Intel pushed Lasertec to develop an actinic tool in an unrealistic timeframe; forced them to hastily pick a source supplier

"At the early stage Intel is the one that pushed them. They heavily invested, and instead of having a program that will take five, six years for development, they laid it down and said, you need to ship it in three years. Lasertec said there is no way that we can develop stuff that fast. So they used a Japanese connection. They went to Ushio, which was the only company that had a source that has enough brightness." –Former KLA executive in a leadership role in their EUV mask inspection group

As the only “actinic” 13.5nm EUV mask inspection machine currently on the market, naïve investors perceive and value it as a “monopoly” akin to “Japan’s ASML.” Sell-side broker reports and articles use the label freely, such as a recent UBS report: “Lasertec should continue to enjoy monopoly profits as long as it does not lose its technology lead over KLA”; “will maintain its monopoly because EUV is exceedingly difficult”; and has “a dominant share in the EUV mask/blank inspection market.” A longtime industry consultant who recently worked for Lasertec stated that the Japanese have bid it up as Europeans have bid up ASML: “...in Japan Lasertec has a similar story...we’re a monopoly blah-blah...they’re kind of nuts...people tend to view them as...the Japanese version of ASML...so it’s got everything.”

UBS initiation report Jan 18, 2024 - excerpts

“The company is the only one with mask inspection technology using an EUV light source [...] We think Lasertec should continue to enjoy monopoly profits as long as it does not lose its technology lead over KLA...”

“Our base scenario assumes Lasertec will maintain its monopoly simply because the company’s EUV technology is exceedingly difficult”

“Thus, EUV light source technology has had an unusually high barrier to entry. In the field of mask defect inspection, no other company has the same level of technology as Lasertec”

Lasertec valuation is “kind of nuts”; viewed as the “junior brother” to ASML or “the Japanese version of ASML”

“Lasertec is in a similar unique monopolistic situation. Much like you have a lot of European investors going after ASML stock and bidding it up to ridiculously high P/E ratios, I think in Asia and particularly in Japan, Lasertec has a similar story: we’re a unique EUV semiconductor driven. We’re the only game in town. We’re a monopoly, blah-blah-blah. So, you’re going to see similar nosebleed valuations out of them. I can’t suggest that I agree with the valuations. I think they’re kind of nuts. But when you’re the only game in town, investors may decide to bid it up to some stupid P/E...” So, I think a lot of people tend to view them as sort of the junior brother to ASML or the Japanese version of ASML. Whether it’s deserved or not, it is what it is...I don’t know of any other company in Japan that’s had the cache of being a monopoly and having its name associated with EUV. So, it’s got everything. ” – Longtime semiconductor consultant recently working for Lasertec

Source: UBS Initiation of Coverage “Lasertec Corporation Aiming to Protect the Dominance of ACTIS,” Jan 18, 2024; <https://www.smartkarma.com/insights/lasertec-6920-a-monopoly-with-a-long-runway>; Scorpion Capital consultation calls with experts

Lasertec's EUV mask inspection tools are a niche – most of the industry's value is captured by ASML's EUV scanners that produce wafers. The use cases and total addressable market for Lasertec is far smaller than investors realize. Two suppliers (Hoya and AGC) dominate the market for EUV blanks, which caps the number of customers for its ABICS EUV blank inspection tool. Its flagship ACTIS system is dependent on 3 customers – TSMC, Samsung, and Intel – who drove 77% of total company revenue in both 2022 and 2023. Within these 3 customers, ACTIS has only two use cases: in the EUV mask shops at Intel and TSMC that produce masks; and in the fab to periodically check masks during wafer production. However, Intel and TSMC's mask shops only installed about 6 ACTIS systems in total over the last few years - and don't need more. That leaves only the fabs. Unfortunately, as detailed in a later section, TSMC views ACTIS as a colossal failure in the fab – thus leaving only Intel's fabs as a potential use case.

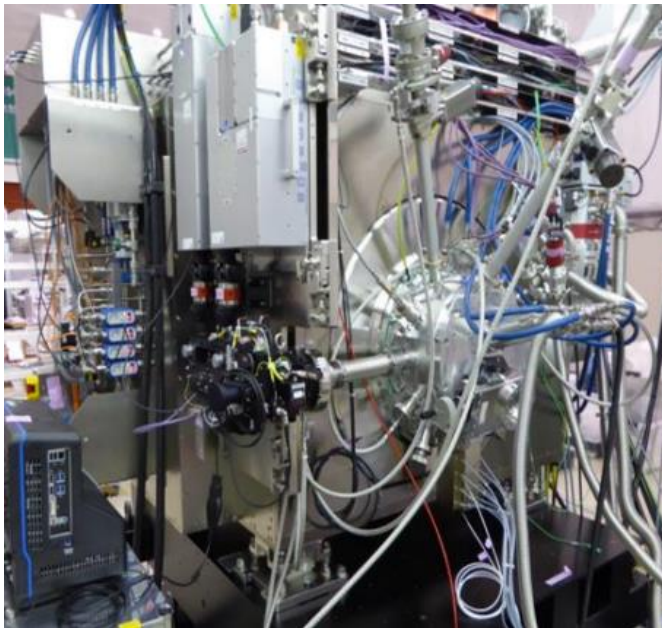
Sales by customer – TSMC, Intel, Samsung, in JPY and USD

	Sales by Key Customer (JPY millions)				
<i>JPY millions</i>	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>
TSMC	6,869	17,891	17,625	18,833	51,782
Intel	-	6,337	10,331	28,568	48,003
Samsung	4,666	4,745	14,042	21,750	17,871
Total	11,535	28,973	41,998	69,151	117,656
TSMC	24%	42%	25%	21%	34%
Intel	0%	15%	15%	32%	31%
Samsung	16%	11%	20%	24%	12%
Total	40%	68%	60%	77%	77%

	Sales by Key Customer (USD millions)				
<i>USD millions</i>	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>
TSMC	44	115	114	122	334
Intel	-	41	67	184	310
Samsung	30	31	91	140	115
Total	74	187	271	446	759
TSMC	24%	42%	25%	21%	34%
Intel	0%	15%	15%	32%	31%
Samsung	16%	11%	20%	24%	12%
Total	40%	68%	60%	77%	77%

Lasertec is a marginal, outdated DUV systems supplier masquerading as an EUV player. The reason is simple: unlike ASML, it never developed a 13.5nm EUV light source, and the source is the foundation and critical technology in any EUV system, whether for wafer production or inspection. All other components are secondary. In contrast to ASML, Lasertec is just an over-hyped assembler of third-party components with no EUV capability or intellectual property, first licensing an EUV source from Ushio, which is defective and a failure, as we shall document in great detail. It then quietly licensed one from a tiny Russian R&D project called Isteq, giving it a Japanese name - “Urashima” - to misrepresent it as “in house” technology. That source is also a debacle with the same fatal flaws.

Ushio LDP-based EUV source



Isteq LPP-based EUV source



In order to understand why ASML is a monopoly in EUV and has a \$350B market cap, one must grasp the extreme difficulty and cost of developing of a reliable, high-power EUV light source. EUV light can only be generated by ionizing tin into plasma, which emits a 13.5nm wavelength. ASML spent \$7B on EUV source R&D, and then had to acquire Cymer for \$4B in 2013, which had 300 R&D staff just on source development; it took ASML and Cymer a combined 20 years to make it work. We spoke to a leading figure in Cymer's source effort, who said ASML currently has 1,000 people still working on the source – “that's the scale.” LinkedIn suggests the actual number is 1,300.

Press regarding ASML closing of Cymer acquisition, May 2013

ASML Holding NV Completes Acquisition of Cymer for \$3.7B

Impassible for Lasertec to replicate ASML's source; took Cymer/ASML 20 years and required hundreds of people; required “several billion dollars on the source only”; ASML currently has ~1,000 staff working on their EUV source; “that's the scale”
“Now, the ASML EUV source is tin-based. That's what I'm kind of telling, that yes, it's possible, but it would require hundreds of people, which are like 100 people, 10 years, and a lot of money. I don't think that, for example, Lasertec is not in a position to throw 100 people working on a source. **ASML and Cymer spent 20 years, and, on average, it was probably 300 people working per year, and then 20 years. ASML spent several billion dollars on the source only - maybe a couple of billion dollars in development. That's the scale.** I joined ASML at Cymer in 1999 or maybe 2000, and I worked there for 11 years. And it went into production basically in 2017. That tells you the scope. When I left Cymer, there were 250 or 300 people in the source development team engineering. **Right now, if you go to San Diego, I'm sure that there are 600-700, maybe 1,000 people working on it. It's a huge development.**” – Leading figure in the development of Cymer/ASML's EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

EUV source technology is so challenging that only ASML has managed to produce one that actually works. The history is instructive: R&D efforts began in the early 1990's and took several decades of US government-funded research by Lawrence Livermore National Laboratories, Lawrence Berkeley, and Sandia. Japanese companies were denied the ability to license the technology as it was US-taxpayer funded. ASML eventually licensed the technology in the late 1990's, making it – not Lasertec – a “monopolist” with the only working EUV source and “the sole benefactor of the critical technology.”

Wikipedia on the history of EUV lithography

“In 1991, scientists at Bell Labs published a paper demonstrating the possibility of using a wavelength of 13.8 nm for the so-called Soft X-Ray Projection Lithography. To address the challenge of EUV lithography, researchers at Lawrence Livermore National Laboratory, Lawrence Berkeley National Laboratory, and Sandia National Laboratories were funded in the 1990s to perform basic research into the technical obstacles. The results of this successful effort were disseminated via a public/private partnership Cooperative R&D Agreement (CRADA) with the invention and rights wholly owned by the US government, but licensed and distributed under approval by DOE and Congress. The CRADA consisted of a consortium of private companies and the Labs, manifested as an entity called the Extreme Ultraviolet Limited Liability Company (EUV LLC). Intel, Canon, and Nikon (leaders in the field at the time), as well as the Dutch company ASML and Silicon Valley Group (SVG) all sought licensing. **Congress denied the Japanese companies the necessary permission** as they were perceived as strong technical competitors at the time, and should not benefit from taxpayer-funded research at the expense of American companies. **In 2001 SVG was acquired by ASML, leaving ASML as the sole benefactor of the critical technology...**this made the once small company ASML the world leader in the production of scanners and **monopolist in this cutting edge technology.”**

In contrast to ASML, Lasertec is a small company with only 859 employees and historically negligible R&D investment – it is simply unrealistic that it could have developed a reliable EUV source with 1% of the billions that ASML and Cymer spent. Lasertec spent a mere \$9 to \$15MM per year in R&D in the years preceding its EUV market entry, struggling as an irrelevant supplier with \$30-100MM in revenue while ASML and Cymer were making unprecedented investments – hardly the profile of a cutting-edge innovator. Ushio – its original EUV source supplier – spends \$80-90MM annually on R&D, but EUV is only a small piece of its revenue and R&D. Isteq, which developed Lasertec’s “new” Urashima EUV source is little more than a Russia/Netherlands-based science project with only 12 employees on LinkedIn.

Lasertec and Ushio R&D expenditure was minimal prior to 2019 launch

\$ millions	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Lasertec										
Revenue	134	124	149	155	192	267	395	633	665	1,057
R&D expenditure	10	9	15	13	25	33	31	51	63	76
Ushio										
Revenue	1,533	1,328	1,594	1,550	1,634	1,490	1,478	1,072	1,226	1,317
R&D expenditure	87	91	100	88	101	91	95	91	87	86

“New” Urashima source supplier only has 12 employees on LinkedIn

ISTEQ B.V.
 We make light brighter
 Nanotechnology Research · Eindhoven, North Brabant · 225 followers ·

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Home About Posts Jobs People **Insights**

PREMIUM
 Total employee count 
 Based on LinkedIn data.

12
 total employees

Lasertec's current annual R&D is so laughably small that it is less than the price of one of its ACTIS tools, which sell for \$40-60MM. Low R&D to market cap is one of our favorite indicators for fraudulent or vaporeous bubble stocks – it signals the valuation is driven by hype rather than substance. Legitimate, innovative companies worth billions typically make proportional investments in R&D, product development, and intellectual property. In the last 5 years, Lasertec's R&D spend was a mere \$33MM (2019), \$31MM (2020), \$51MM (2021), \$64MM (2022) and \$76MM (2023) – reminiscent of medieval alchemy to conjure \$25B in market value from only \$250MM in cumulative R&D. Its valuation relative to 5-year R&D is the 3rd most extreme among all large-cap (>\$10B) tech stocks globally (~200 total) – investors are valuing its “R&D” even higher than Nvidia, at least per this metric; 3X higher than Microsoft and TSMC, and 4X higher than ASML and Apple.

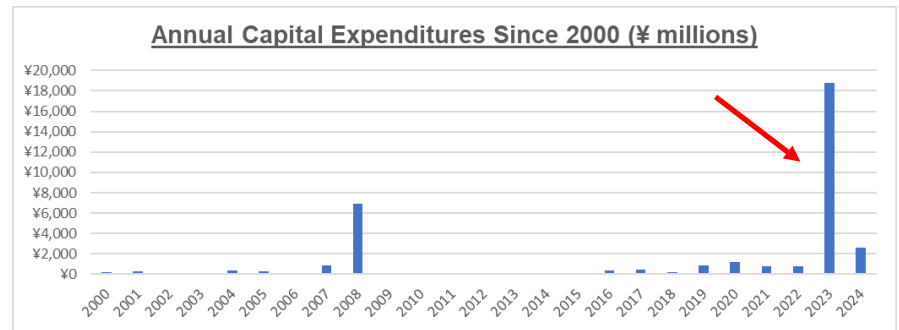
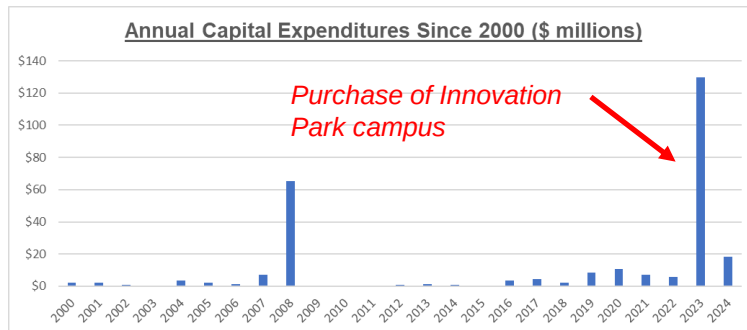
Capital IQ screen of market cap to 5-year cumulative R&D, tech companies >\$10B market cap

Company Name	Market cap	Market cap to cumulative 5 yr R&D
Keyence Corporation (TSE:6861)	110,911	182
HANMI Semiconductor Co., Ltd. (KOSE:A042700)	10,431	164
Lasertec Corporation (TSE:6920)	24,958	98
NVIDIA Corporation (NasdaqGS:NVDA)	2,618,961	93
Jabil Inc. (NYSE:JBL)	14,423	77
Disco Corporation (TSE:6146)	42,461	66
Amphenol Corporation (NYSE:APH)	81,826	55
First Solar, Inc. (NasdaqGS:FSLR)	29,623	53
MicroStrategy Incorporated (NasdaqGS:MSTR)	29,885	52
Delta Electronics (Thailand) Public Company Limited (SET:DELTA)	25,614	52
BE Semiconductor Industries N.V. (ENXTAM:BESI)	11,737	49
WiseTech Global Limited (ASX:WTC)	21,644	47
HCL Technologies Limited (NSE:HCLTECH)	43,856	45
Tyler Technologies, Inc. (NYSE:TYL)	20,890	44
VeriSign, Inc. (NasdaqGS:VRSN)	17,049	43
Fair Isaac Corporation (NYSE:FICO)	34,216	43
Super Micro Computer, Inc. (NasdaqGS:SMCI)	51,757	43
Zhongji Innolight Co., Ltd. (SZSE:300308)	18,020	42

Lasertec valued at a higher premium to R&D than Nvidia

The semiconductor industry - including semicap equipment - is one of the most capital intensive on earth. Lasertec is once again an anomaly – it spends almost nothing on R&D and even less on capital expenditures. Lasertec claims it is “fab lite,” a meaningless term as it sells equipment vs. designing chips, and simply a euphemism for small, sub-scale assembler with negligible investment in facilities, machinery, technology, or intellectual property. Its only meaningful capex was in 2023, spending \$120MM for a new R&D campus called “Innovation Park” in Yokohama, which as we detail later is a hoax. From 2010 to 2022 - just prior to the park - Lasertec invested a cumulative \$45MM in capex – less than \$1MM in some years, and again less than the price of one ACTIS machine. In the last 5 years from 2019 to 2023, its total capex was only \$30MM plus the amount it paid for the park – a nonsensical amount versus the 5 largest semicap players ASML (\$6.8B), Applied Materials (\$3.4B), Lam (\$1.9B), Tokyo Electron (\$2.3B), and KLA (\$1.2B).

Lasertec capital expenditures over time – JPY and USD millions



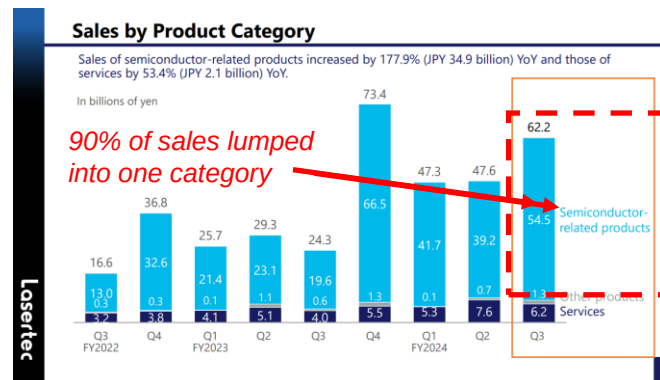
As yet another stark contrast to its insignificant R&D and capex, we note the massive sums KLA is spending to develop its upcoming actinic EUV mask inspection tool – which as we detail in a later section will be catastrophic for Lasertec. For anyone who doubts that KLA will enter the market, various statements by its management team underscore their seriousness. At a 2023 investment conference, KLA’s CFO stated they have already invested “over \$1 billion” specifically for EUV mask inspection: “If you look at the reticle space, for EUV...at the end of this year, I was adding up the numbers as CFO, you kind of roll your eyes you go holy – wow, over \$1 billion across multiple products to solve the reticle challenge in terms of R&D investment.” He noted their dominance and effective monopoly in inspecting EUV masks with their Teron DUV systems: “in the EUV space, 90%-plus of reticles are running through KLA systems” – once again illustrating the naivete of those who mistakenly think Lasertec is the monopoly.

KLA CFO Bren Higgins at Morgan Stanley, March 2023

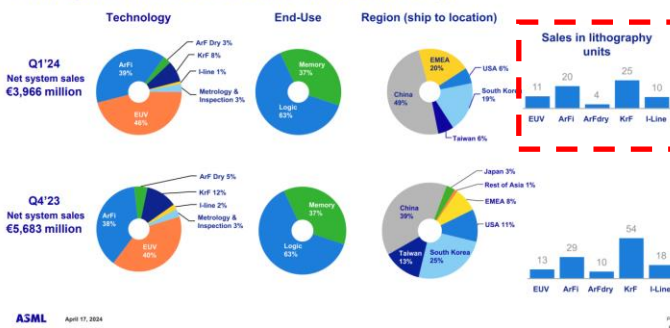
*“And the economics of running the fab is playing to our optical solutions and our wafer inspection solutions to validate reticles in the fab environment. So clearly, that’s playing out **in the EUV space, 90%-plus reticles are running through KLA systems**. We have new capability in the market that as they start to shrink feature sizes, that’s an e-beam, electron beam-based capability that should meet those higher end post writing inspection requirements. And we feel pretty good about the strategy and the introduction of new capability, similar to what our competitor is offering and trying to engineer today when you move into a high NA environment [...] and so a lot of investment in the company to meet that timing. **If you look at the reticle space, for EUV, the company. I think at the end of this year, I was adding up the numbers as CFO, you kind of roll your eyes you go holly -- wow, over \$1 billion across multiple products to solve the reticle challenge in terms of R&D investment.**” – KLA CFO Bren Higgins*

Investors have naively turned Lasertec into an EUV meme stock without realizing that it remains an antiquated seller of last-generation DUV machines. The lack of a reliable EUV source, given no credible R&D and capital expenditure to date, is one reason the euphoria is misguided. The other is that systems with a 13.5nm EUV source are a relatively insignificant percent of its revenue, by our estimate, with most of its business dependent on older machines with a ~200nm DUV source. Lasertec is secretive - typical of frauds – failing to provide basic disclosures typical of ASML, KLA, and others such as the number of tools sold by EUV vs. DUV, ASP's, or segment mix. ASML, for example, lists the exact number of EUV and DUV systems sold each quarter – while the title of Lasertec's "Sales by Product Category" disclosure is a joke in itself, as no categories are shown – 90% of sales are lumped into one bucket called "Semiconductor-related products."

Lasertec (left) fails to break out EUV revenue, while ASML (right) discloses exact number of EUV vs. DUV tools sold quarterly as well as other detailed information



Net system sales breakdown (Quarterly)



ASML breaks out number of tools sold every quarter by source wavelength

Investors may naturally wonder why Lasertec is afraid to disclose how much of its revenue is from EUV systems, and what the quarterly or annual trend is for the number of tools sold. The CEO and CFO are among the most evasive we have ever seen across any company during quarterly analyst Q&A's, with a pattern of vague, non-committal answers and tersely dismissing even the most basic requests for details typical on other semicap earnings Q&A. Analysts have repeatedly tried and failed over the last 5 years to get any color on ACTIC EUV as a percentage of total sales, orders, or backlog versus its older MATRICS DUV.

Quarterly Q&A sessions with analysts – CEO replying in each exchange below

Feb 1, 2022, Jefferies analyst

A: "In this sense, we expect that there will be a demand for EUV-related inspection equipment."

Q: "Does that include ACTIS?"

A: **"Well, I don't know if it will be for MATRICS X8ULTRA or ACTIS**, but in any case, we are expecting to receive inquiries for inspection systems for EUV masks.

Feb 1, 2022, IwaiCosmo analyst – CEO replying

Q: "In terms of orders, you mentioned that the X8ULTRA is selling well, but are you talking about a volume basis? If we look at it on a monetary basis, for example, **is it correct to say that ACTIS has a larger presence in your orders?**"

A: "Orders, yes. **We have not explained the breakdown of orders received** but looking at the orders received in the first half, both MATRICS X8ULTRA and ACTIS orders are recorded."

Feb 1, 2022, Macquarie analyst – CEO replying

Q: "For Intel, it is common knowledge that your company's ACTIS is included. The X8ULTRA will probably bring out this kind of demand, so will there be more opportunities here?"

A: **"Sorry. We cannot disclose which systems are being used** by our customers."

Aug 5, 2022, Nomura analyst

Q: "I understand. So, is this a situation where the full-blown explosion of ACTIS is not yet in sight, or is it already exploding and there is no need to worry about it?"

A: **"Talking about our EUV-related products, we continue to receive orders for both MATRICS and ACTIS."**

The level of stonewalling by the CEO – when asked for even the tiniest crumbs on EUV contribution – would alarm any investor who reviews each quarterly Q&A in the 5 years since its ACTIS EUV tool launched: “we have not explained the breakdown of orders received”; “sorry...we cannot disclose which systems are being used”; “I hope you understand that we do not disclose information by product type”; “we do not disclose the ratio between ACTIS and MATRICS.” In the recent Q&A on April 30, Lasertec talked up demand for its purported ACTIS A300 machines with High NA capability – actively pushed by sell-side brokers as key to the next phase of growth - but refused to provide even vague, qualitative color regarding its contribution to orders. Analysts thus beg politely for just a hint, to no avail: “Is it a little difficult for you to tell your direction as well [within EUV]?”

Quarterly Q&A sessions with analysts - CEO replying in each exchange below

Aug 5, 2022, Jefferies analyst

Q: “I know there is not much disclosure with respect to the orders, but...can you tell us which of the two, ACTIS and ULTRA, grew more in the fiscal year that has just finished and also which will grow more in the current fiscal year based on the forecast?”

A: “I’m sorry. The breakdown within EUV-related systems is not disclosed.”

Q: “Is it a little difficult for you to tell us your direction as well?”

A: “In that sense, ACTIS will be used for 5 nm and 3 nm logic devices, and it is also to be used for memory in the future.”

Jan 31, 2023, CLSA analyst

Q: “In terms of orders, I believe ACTIS is used more frequently for mask manufacturing processes. In that sense, if the orders from the fab side are lower, I understood that they are for the ULTRA series.”

A: “ACTIS itself is beginning to see demand from fabs. However, I hope you understand that we do not disclose information by product type, such as the destination of each product.”

Jan 31, 2024, CEO reply to CLSA analyst

“In that sense, although we do not disclose the ratio between ACTIS and MATRICS, I think investment overall will...recover...”

Apr 30, 2024, CLSA analyst

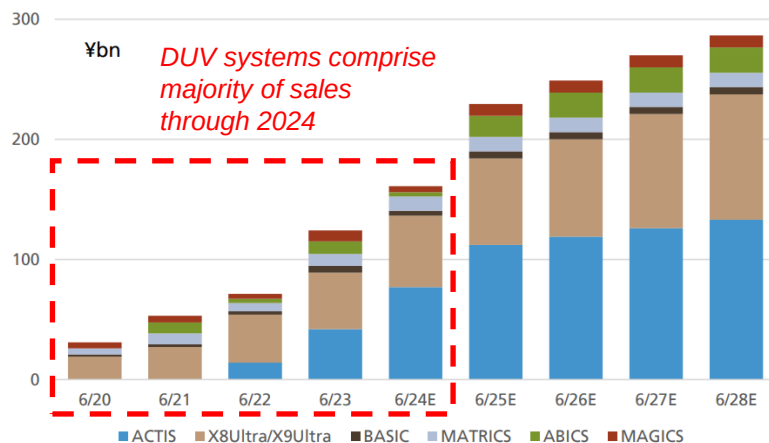
Q: “It’s about orders in Q3. You mentioned that ACTIS was strong, but what was the percentage of High-NA A300 among them?”

A: “Yes, unfortunately we do not disclose the ratio between High-NA and conventional ACTIS, but we have continued to receive orders for High-NA ACTIS this fiscal year, and we expect more orders to come in...”

In the absence of any information from the company on EUV sales, the Japanese investment analysts who cover the stock are guessing and bewildered, with their estimates of EUV percentage diverging wildly – a stunning fact as Lasertec has a \$23B market cap due only because of its EUV hype. For example, a UBS initiation report in Jan 2024 suggests - per a chart for sales by product - that Lasertec is still mainly a DUV supplier, with EUV comprising almost none of its revenue through 2021, ~20% in 2022, and only ~40-45% in 2023/2024. A Citigroup report published 3 weeks before this one, however, pegged EUV at 80% of sales by 2022 – 4X the UBS estimate.

UBS initiation report Jan 18, 2024 - excerpt

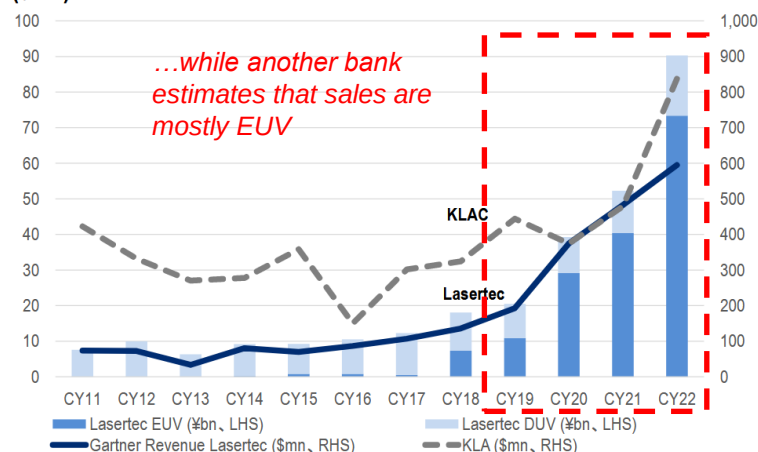
Figure 14: Sales of semiconductor-related equipment by type



Source: UBS estimates based on company data

Citi Research report Dec 23, 2023 - excerpt

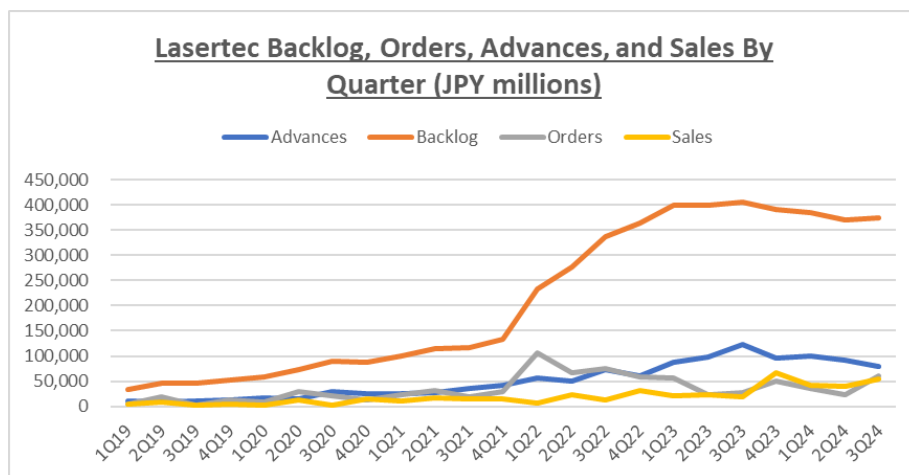
Figure 11. Lasertec sales (¥bn) and Gartner equipment value estimate (\$mn)



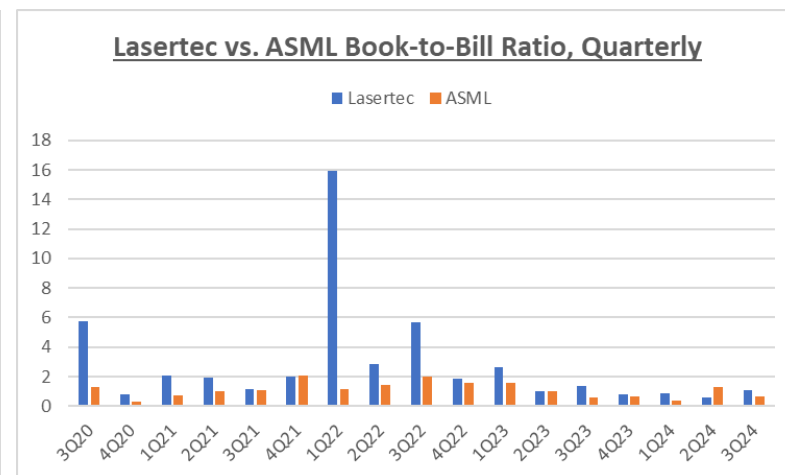
Source: UBS Initiation of Coverage "Lasertec Corporation Aiming to Protect the Dominance of ACTIS," Jan 18, 2024, p.11; Citi Research "Lasertec (6920.T) Expecting High Growth to Continue But Downgrading to Neutral," Dec 21, 2023, p.8

Brokers who cover the stock seem equally enamored of Lasertec's orders and bookings – even though the figures strain credibility and common sense. Orders in the most recent quarter were the highest in almost two years, and appeared to be the reason the stock jumped after the earnings release. Lasertec's book-to-bill ratio – the ratio of orders to sales, a common measure in the industry – was regularly above 2 in recent years, hitting a preposterous and unheard of 3, 6, and 16 in some quarters. ASML, for example, has generally hovered around a more realistic ratio of 1, never exceeding 2 even in some heady quarters in 2022. The parabolic spike in Lasertec's backlog strikes us similarly too good to be true, particularly relative to advances which one would expect to follow as customers provide partial funds up front, yet the gap has widened astronomically.

Backlog vs. orders vs. advances vs. sales



Orders vs. sales comparison



Source: ASML and Lasertec quarterly earnings presentations and filings; Lasertec backlog, sales, and orders are for "semiconductor-related products" per the quarterly IR Excel spreadsheet with financial results and key metrics

Our research indicates that the vast majority of Lasertec's sales remain its older DUV mask inspection systems – not EUV. Former employees told us that its business is still dependent on the MATRICS X8 ULTRA , launched in 2018 – back when it traded at a fraction of today's market cap and multiple. Even its DUV technology is so outdated that the X8 uses the same 213nm DUV light source as its predecessor the X810 launched in 2012. The X9 ULTRA was announced in 2022 with a slightly upgraded 193nm DUV source, finally getting with the times as KLA introduced 193nm in the 1990's. Comically, the “new” 193nm source in the X9 is as unreliable as its EUV source, as we detail later, illustrating why Lasertec was irrelevant even in the previous DUV era with only a \$100-200MM market cap.

213nm DUV system – MATRICS X8 ULTRA, X9 ULTRA announced in April 2022, 193nm launched in 2017

Lasertec releases a New Mask Inspection System, the MATRICS X9ULTRA Series

2022.04.26



A mask inspection system designed for pellicle-less EUV mask inspection in the technology nodes of 3nm and beyond

Lasertec announced the release of the MATRICS X9ULTRA series, a mask inspection system designed to inspect the photomasks used in extreme ultraviolet (EUV) lithography while their pellicles are not attached.

Given its EUV hype, investors may be shocked to realize how few ACTIS tools Lasertec has sold since the 2019 launch – which explains its silence on the actual number, revenue percentage, or the trend. Based on over 20 research interviews – including multiple sources from TSMC, Intel, and Samsung - we conclude that only ~30 systems have been installed, merely 6 per year on average, with new installs dropping off a cliff in recent years. We estimate Intel has 13 tools: 3 in its mask shop in Santa Clara, with new installs hitting the wall years ago, and ~10 in its Oregon fab. TSMC has 6, with no plans to purchase more, of which 2 are in its mask shop. Samsung has 9; Global Foundries has 2 and hasn't bought a new one since 2021; and Marvell has one. We thus estimate total 5-year ACTIS sales at \$1.36B, at best - <\$300M per year, on average - at a \$40MM price for Intel/Samsung; \$50MM for TSMC; and \$60MM for others, based on our data points. ACTIS is hence 40% of revenue, since its 2019 launch to March 2024 (¥513B, or \$3.3B at 155 JPY/USD). The actual percentage is likely lower, as we suspect Intel's price is even less; and as the average is driven by earlier years ~2020/2021, we think ACTIS today is <10% of sales – if not below.

Cumulative ACTIS units and sales - estimate

	Cumulative ACTIS Units and Sales - Estimate						
	<u>Mask shop</u>	<u>Fabs</u>	<u>Total</u>	x	<u>Price (\$MM)</u>	=	<u>Sales</u>
Intel	3	10	13		\$40		\$520
TSMC	2	4	6		\$50		\$300
Samsung	NA	NA	9		\$40		\$360
Global Foundries	NA	NA	2		\$60		\$120
Marvell	NA	NA	1		\$60		\$60
Total			31				\$1,360

Lasertec cleverly tries to spin its outdated DUV systems as EUV – a deceptive attempt to sprinkle itself with ASML’s glitter. Its site and materials go to great lengths to dress up and insert its DUV products under “EUV” or “EUV-related systems” – the word “related” being the tell. Most investors are unaware that EUV masks can be inspected with DUV or EUV wavelengths – DUV is lower resolution and requires an extra step where defect locations are then analyzed with a SEM (scanning electron microscope) tool; however, DUV tools are much faster than EUV, which means higher throughput overall despite the extra step. Thus, while Lasertec pushes its DUV tools as part of its EUV story, it undermines the entire bull case – it shows that the ACTIS EUV tool is unnecessary for EUV mask inspection, isn’t worth \$50MM and the huge price premium vs. DUV, and is not a monopoly.

Lasertec website/materials misrepresent its outdated DUV tools as “EUV”

Only two of the tools shown (circled) even have a 13.5nm EUV light source; rest are older DUV or scanning systems – yet are misrepresented as “EUV”

Lineup of **EUV-related systems** from Lasertec



The use of DUV tools for EUV mask inspection allows Lasertec to masquerade as an “EUV lithography” player, but it raises an obvious question – why is an older DUV technology used at all, given the ACTIS EUV system and Lasertec’s purported “actinic patterned mask inspection” monopoly? The answer is key to understanding the fraud, and detailed in a later section: the ACTIS EUV system is so unreliable and slow that TSMC, Intel, and Samsung have simply gone back to using DUV and homegrown systems to inspect EUV masks. Even if ACTIS weren’t defective, it would have almost no use case or market - EUV masks in production in the fab only need periodic, quick, low-resolution checks, for which DUV is sufficient. That leaves only one use case for ACTIS: R&D in the two mask shops at TSMC and Intel, where a negligible ~6 tools have been installed in total – recall that EUV masks are inspected in only two settings, the mask shop and the fab.

TSMC relying on legacy DUV and scanning electron microscope tools

Q: “You’re saying the fact that they’re using optical means that they’re not getting the results from the Lasertec machine by itself. They have to use a second tool?”

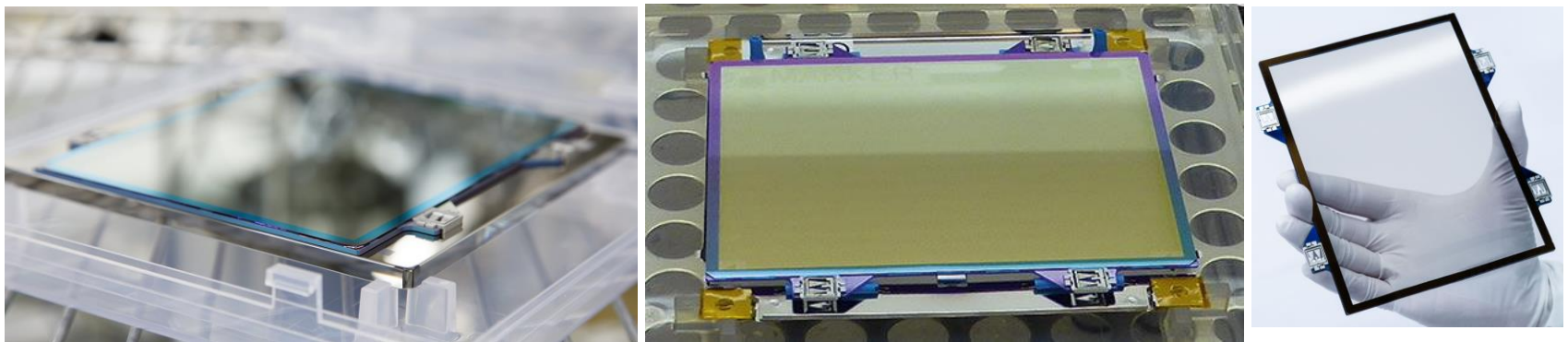
A: “Correct. But the second tool—before Lasertec came out with their actinic tool, they needed to establish some way of inspecting the EUV masks. **What they did was use an optical tool from KLA – the Teron product line. And this is an optical tool with a 193-nanometer light source.** So, there’s a limitation to what defects it can detect. Very small defects, it cannot clearly distinguish. For something that looks like a defect, **they record the coordinates, and then these coordinates are transferred to an electron beam inspection tool,** which is only looking at these suspicious-looking defect locations. Is the defect on print or not? So, they need a **combination of optical DUV and electron beam inspection to inspect the mask.**” - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec’s CEO

KLA non-actinic DUV alternatives are “work really well” and are “well-engineered”; EUV wavelength is less important

“The **current KLA systems are pretty good,** but they’re not as good as actinic. Having said that, **a well-engineered system is more important than the wavelength**...they’re using the current DUV tools, and it was announced at the most recent conference that people who are appreciating all the work that both KLA did on supporting EUV with the DUV tools. **EUV and DUV are different but for supporting EUV inspection, the DUV tools actually do work...Those tools work really well and have been well-engineered for a long period of time.** That’s what people use. I think today, I think people are using the Lasertec actinic pattern mask inspection tool - they use that when it works...” – Ex-KLA director of engineering focused on EUV mask inspection

Investors with a keen understanding of EUV lithography may dispute the previous page and argue that the ACTIS EUV system has a “monopoly” because it is the only tool capable of “actinic” inspection through a pellicle. Pellicles are a much-hyped concept in the EUV mask space and key to the Lasertec story among sell-side analysts pushing the stock. A pellicle is a thin membrane to protect the mask from particle contamination that would cause defects during wafer exposure; a mask must be inspected again in case pellicle placement caused new particles. In theory, only an “actinic” 13.5nm EUV source can inspect through a pellicle with sufficient sensitivity. Sadly for bulls, a pellicle-based thesis is a fantasy: TSMC – the largest player – doesn’t use pellicles for EUV masks and is content with non-actinic DUV tools; and for others like Intel, the ACTIS is highly contamination-prone and spews particles during inspection, defeating its purpose and requiring redundant inspection with KLA’s Teron DUV system, even if a pellicle is used.

Examples of EUV mask pellicles – two left from ASML, far right from Imec



Lasertec may try to spin its archaic DUV tools as “EUV-related,” but its DUV business is just as irrelevant as its EUV product. KLA locked up the mask inspection space long ago with its Teron DUV systems – giving it a well-entrenched monopoly unlike the imaginary one Lasertec claims in EUV. At a 2023 conference, KLA’s CFO stated that “in the EUV space, 90%-plus reticles are running through KLA systems” – reticle being another name for mask, and illustrating the dominance of KLA DUV tools for EUV mask inspection. Those who think Lasertec’s MATRICS DUV system can support its story and valuation in the EUV era may wish to review Lasertec’s market cap during the peak period of DUV lithography in the late 1990’s to early 2010’s: about \$100MM with a \$1 stock price.

KLA CFO Bren Higgins at Morgan Stanley, March 2023; KLA website excerpts for Teron DUV systems

*“And the economics of running the fab is playing to our optical solutions and our wafer inspection solutions to validate reticles in the fab environment. So clearly, that’s playing out **in the EUV space, 90%-plus reticles are running through KLA systems**. We have new capability in the market that as they start to shrink feature sizes, that’s an e-beam, electron beam-based capability that should meet those higher end post writing inspection requirements. And we feel pretty good about the strategy and the introduction of new capability, similar to what our competitor is offering and trying to engineer today when you move into a high NA environment [...] and so a lot of investment in the company to meet that timing. **If you look at the reticle space, for EUV, the company. I think at the end of this year, I was adding up the numbers as CFO, you kind of roll your eyes you go holly -- wow, over \$1 billion across multiple products to solve the reticle challenge in terms of R&D investment.**” – KLA CFO Bren Higgins*



**Teron™ SL670e and Teron™ SL670e XP:
EUV Reticle Quality Control**

2. Lasertec is an extreme, textbook accounting fraud that checks off every box: claiming to have the highest margins in the industry; but alarmingly little cash flow and the lowest cash conversion among its peers; accompanied by the highest inventory level of any semicap equipment supplier in the world.

A simple accounting framework explains the fraud at Lasertec. The most common type of fraud is 1) fictitious revenue; and/or 2) fake, overstated earnings. Many companies go to extremes and do both, which inflates net income on the income statement and leads to a corresponding overstatement of retained earnings on the balance sheet, a component of owner's equity. In order for the balance sheet to balance (assets = liabilities + equity), the company must therefore: a) overstate assets; b) understate liabilities; or c) understate an equity account. Hence, a fraudster is trapped by double-entry bookkeeping and the interdependence between the income statement, balance sheet, and cash flow statement – they cannot overstate profits on the income statement without also overstating assets or understating liabilities.

Income statement

$$\text{Revenues} - \text{Expenses} + \text{Gains} - \text{Losses} = \text{Net Income}$$

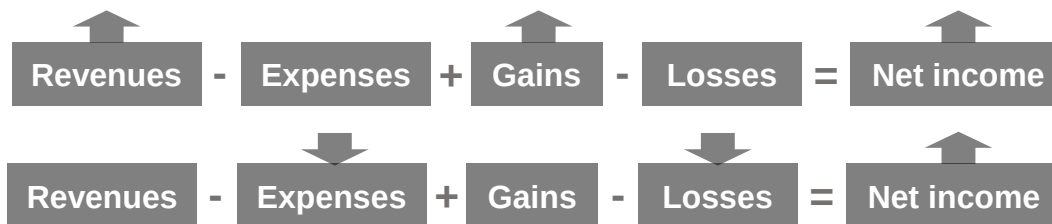

Balance sheet

$$\text{Assets} = \text{Liabilities} + \text{Owner's Equity}$$


*Balance sheet impact
of increase in net
income > leads to an
increase in retained
earnings, a component
of owner's equity*

The critical point is that aggressive or fraudulent revenue recognition or understated expenses results in an overstatement of some asset, which helps to identify the source of the problems. The typical warning sign that a forensic accountant therefore looks for: are there any unusual assets or unexplained large increases in assets such as inventory, particularly relative to revenues? The most likely overstatements of assets in order of likelihood are accounts receivable, cash, inventory, investments, or intangible or other assets. In the case of fake revenue, the most common offset is to overstate assets through an increase in accounts receivable; in the case of understated expenses and overstated margins, the offset is often highly elevated inventory.

Impact of overstating revenue/gains or understating expenses/losses



Potential balance sheet impacts



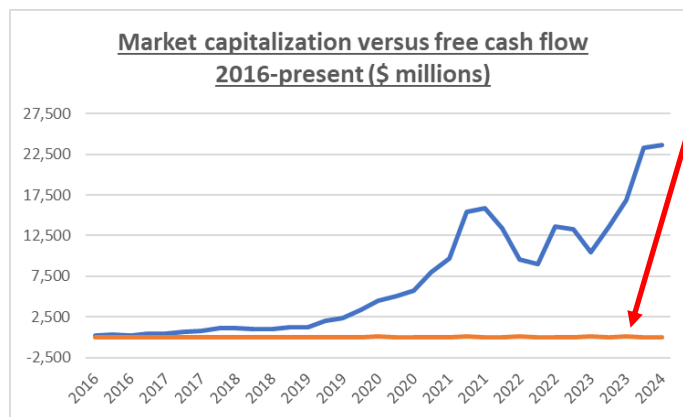
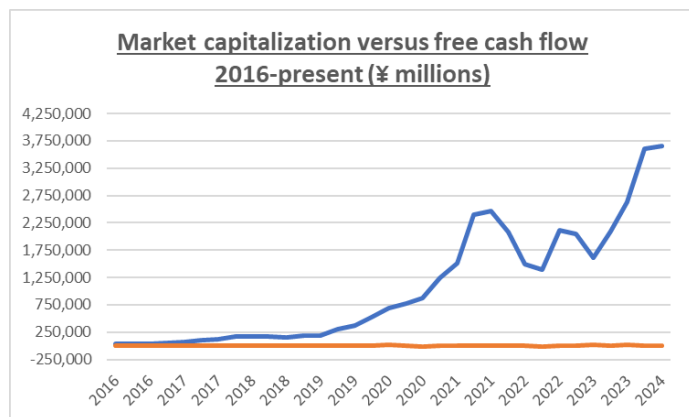
For example, Olympus was exposed in 2011 as one of Japan's largest accounting frauds. It hid massive losses and liabilities via a *tobashi* scheme using off-balance sheet entities. When forced to consolidate them after accounting rules changed, it hatched a scheme to buy them at prices that far exceeded their value, given the embedded losses. The difference was recorded as goodwill and long-term investments – which suddenly spiked and reached 35-45% of total assets in the years prior to 2011. Thus, Olympus kept losses off its books (understating liabilities and overstating owner's equity). When forced to buy the entities and record the liability, it overstated assets (goodwill) as an offset. The scheme collapsed when it was forced to recognize the loss via an impairment charge for the acquisitions – exactly as Lasertec will when it admits that earnings and inventory are fraudulently inflated and takes a similar, massive impairment.

Olympus total assets by category (¥ billions) – goodwill and long-term investments spiked to ~45% of assets in the years prior to 2011 when the fraud was exposed

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Cash & equivalents	149,836	121,626	166,930	224,933	158,284	137,076	206,783	213,561	200,088	229,610
Total receivables	114,323	149,250	164,059	189,474	190,446	167,744	163,902	155,817	166,384	143,711
Other current assets	143,890	163,963	158,536	158,431	194,575	169,947	162,191	164,156	160,086	167,716
Net pp&e	101,104	120,053	130,810	140,089	150,036	163,113	144,494	141,341	127,808	129,802
Long-term investments	119,643	177,696	209,420	217,690	193,762	127,080	140,233	117,646	103,740	104,179
Goodwill and other intangibles	0	57,737	102,666	100,351	404,030	264,620	265,646	205,979	197,145	174,606
Loans receivable	0	0	668	561	3,277	3,811	3,988	5,734	3,108	1,040
Deferred tax assets	7,788	9,839	9,324	9,606	9,665	15,661	9,492	14,926	8,167	9,418
Other long-term assets	46,089	57,919	33,719	50,665	54,274	57,266	55,498	0	0	0
Total assets	682,673	858,083	976,132	1,091,800	1,358,349	1,106,318	1,152,227	1,019,160	966,526	960,082
Goodwill/investments % of total assets	18%	27%	32%	29%	44%	35%	35%	32%	31%	29%

The second critical warning sign, aside from unusual and large increases in balance sheet accounts, is abnormally low cash conversion - particularly if reported margins are high. Cash flow is a check on how real revenues or earnings are. The income statement summarizes the changes in the retained earnings section of the balance sheet, while the cash flow statement summarizes the changes in cash accounts during the period. If high or rising net income is accompanied by low or declining cash flow, the net income is suspect and warrants investigation – earnings are reported on an accrual basis on the income statement, dependent on accounting estimates, assumptions, and plugs in a spreadsheet versus cash in a bank account. We will shortly analyze Lasertec's cash conversion ratios, but we first simply show that it has essentially never had any meaningful cash flow, especially relative to its market capitalization.

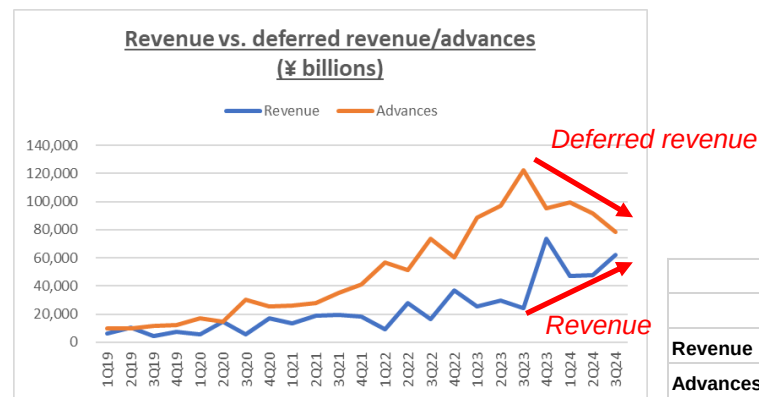
Lasertec market capitalization vs. free cash flow 2016 to Q1 2024 (JPY and USD)



Free cash flow is so microscopic it is not even visible if graphed next to market capitalization

Even if a company isn't booking fictitious revenue, it may engage in aggressive revenue recognition and report a sale before it has occurred or before the revenue is earned – for example, recording revenue when a contract is signed even though the delivery date is far into the future, leading to a spike in accounts receivable. Deferred revenue is a similar high-risk category, and is one of the largest line items in Lasertec's balance sheet, which it calls "advances" – a liability representing funds that have been purportedly collected but not yet earned as revenue. A fraudster can manipulate deferred revenue to create current period revenue to meet expectations. Recent, unusual swings in Lasertec's deferred revenue are a major red flag, showing that basically all of its incremental revenue in the last 5 quarters – since Mar '23 – have come from siphoning the account, which is of course consistent with the lack of cash flow.

Lasertec revenue vs. deferred revenue/advances

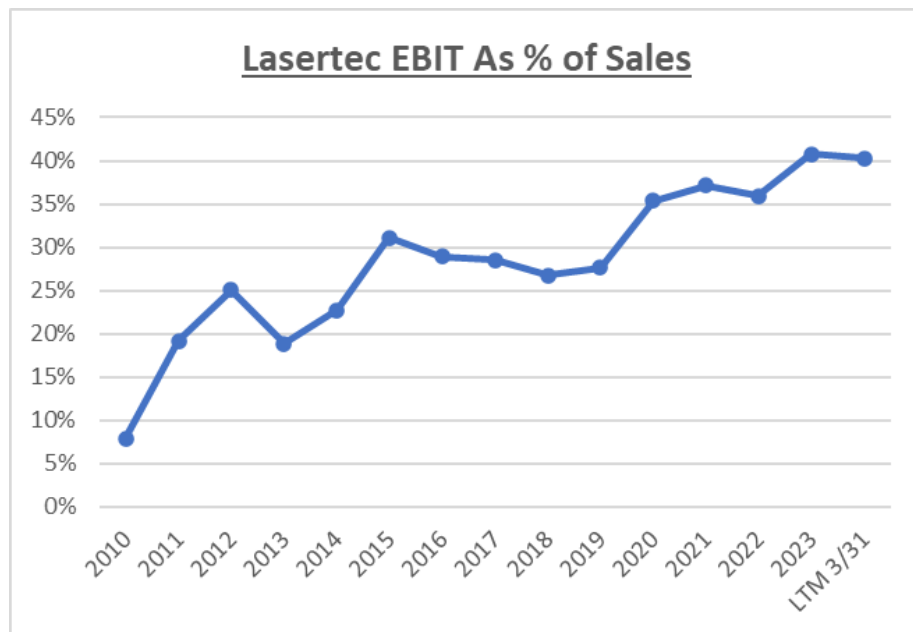


Basically all revenue growth since Mar '23 has come from siphoning deferred revenue – an abrupt change of trend and the first major, sustained decline in “advances.” Increase in revenue of ¥38B/\$245MM is similar to ¥44B/\$283MM decline in deferred revenue.

	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	Change from 3Q23 to 3Q24
Revenue	36,812	25,723	29,376	24,329	73,401	47,305	47,684	62,213	37,884
Advances	60,092	88,615	97,202	122,416	95,155	99,406	91,758	78,556	43,860

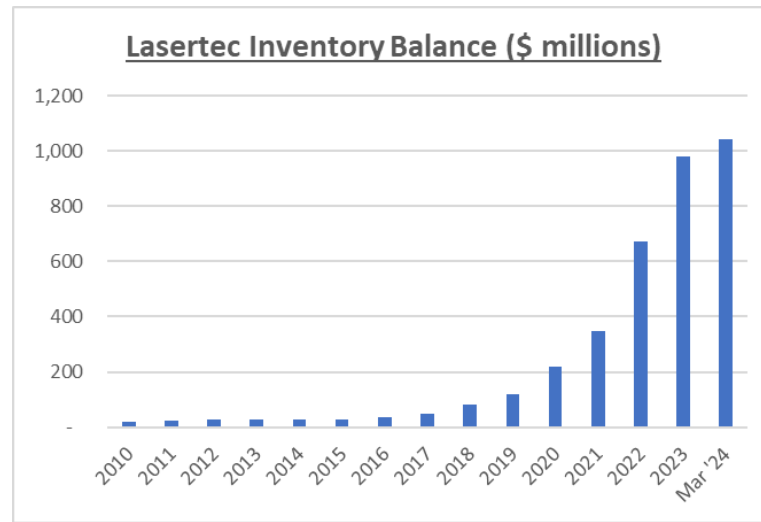
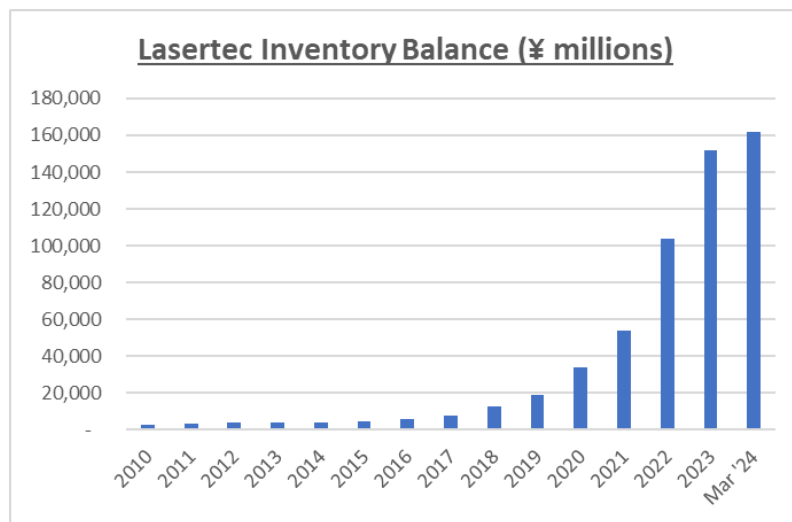
Similar to overstating revenues, which inflates net income, understating expenses will result in overstated retained earnings and must be offset by an overstatement of assets, understatement of liabilities, or understatement of another component of equity. For example, if a company sells goods (inventory) and reports the revenue on the income statement but fails to transfer the cost of that inventory (an asset on the balance sheet) to cost of goods sold (an expense on the income statement), earnings and assets in the form of inventory both become overstated. Thus, spiking inventory is associated with an overstatement of income by understating cost of goods sold – leading to margins that are unusually high. As preview of Lasertec's unusually lofty margins, we show the annual trend below.

Lasertec margins over time 2010-present – suspiciously high with a suspiciously consistent trend



Over time, this results in a large increase in the inventory account on the balance sheet. Effectively, the company is deferring reporting inventory expenses to a later period. The inventory account will increase rapidly over time and keep getting larger and larger. In frauds there is a limit and eventually the bubble must burst – one cannot outrun the fundamental accounting equation between the income statement, balance sheet, and cash flow statement. Eventually everything must balance. The moment of truth typically involves an impairment charge and restatement that wipes out the many previous years of fake revenue and/or profits, thus exposing the details of the fraud. We shall shortly analyze Lasertec's various inventory ratios, but here we simply show its inventory balance over time.

Lasertec inventory over time 2010-present (using 155 JYP/USD)



Similarly, if an asset - such as finished goods inventory - has declined in value, the company can continue to show it on the balance sheet at the older, higher cost or value. An additional trick in the repertoire – clumsily used by Lasertec – is to misclassify finished goods inventory as raw materials and/or work-in-progress. A company thus overstates earnings, equity, and assets – by improperly valuing the inventory and deferring an inevitable write-down and impairment charge. We note that a major part of that Toshiba fraud exposed in 2015 involved over-valuation of inventory in its semiconductor business, thereby inflating profits.

Investigation Report published by Toshiba independent committee, July 2015 – excerpt from the chapter on inventory fraud in its semiconductor division (pages 272-306)

2. Scope of the Investigation

The scope of the Investigation consisted of the appropriateness of accounting treatments related to the valuation of inventory in the Semiconductor Business. Specifically, the following three points were reviewed based on the specific matters delegated by Toshiba.

Summary of inventory fraud at Toshiba in a separate paper analyzing the investigation report

ACCOUNTING FRAUD AND ACCOUNTING STANDARDS: THE CASE OF TOSHIBA'S FRAUDULENT ACCOUNTING

related to cost accounting. In the treatments with no relation to cost accounting, inventory was classified as finished goods inventories which is under the management of the Sales department, and raw materials or works in process which are under the management of Operations (p. 249) and thus, outside the scope of revaluation (p. 253). Furthermore, a part of the finished goods inventories did not devalue in conformity with the method of devaluing book value to the expected disposal value (p. 255). In the investigation report, it was said that for most of the inventory, losses were recognized for accounting purposes for the first time at the time of disposal (p. 245); to begin with, neither was there an accounting rule nor were there similar de

Lasertec's valuation is a function of revenue growth and rapid margin expansion – an explosive combination that often leads investors to pay an extreme multiple. As shown below, its EBIT margin has spiked from 28% in 2019 when it launched its ACTIS EUV machine to 41% in 2023. Surprisingly, Lasertec's reported margin is significantly higher than every one of the five largest semiconductor equipment companies in the world by revenue – even eclipsing ASML's, a \$350 billion market cap giant which invented EUV lithography systems and has a real monopoly unlike Lasertec's fictional one. Lasertec's margins are so astonishing that they even exceed Apple's (30%), Google's (29%), Meta/Facebook's (37%), and are on par with Microsoft's (42%).

Lasertec EBIT margin versus 5 largest semicap companies by revenue

	Revenue (2023)		Operating Margin (EBIT % of Sales)					
	<u>USD</u>	<u>JPY</u>	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>LTM</u>
ASML	27,559	4,216,451	24%	29%	35%	31%	33%	32%
Applied Materials	26,517	4,057,101	23%	26%	31%	30%	29%	29%
Lam	17,429	2,666,563	26%	27%	31%	31%	30%	29%
Tokyo Electron	14,438	2,209,025	24%	21%	23%	30%	28%	25%
KLA	10,496	1,605,897	30%	30%	36%	40%	38%	36%
Lasertec	999	152,832	28%	35%	37%	36%	41%	40%

Lasertec reports the highest EBIT margin of any semicap equipment company in the world with more than one billion USD of revenue. A CapitalIQ screen shows 45 such companies globally in the semiconductor equipment and materials industry. A broader screen of all companies greater than \$500MM market cap shows about 150 companies globally, and Lasertec's margins are the fifth highest in that list, exceeded only by a few small, questionable publicly-traded companies in Malaysia, China, and South Korea.

Capital IQ screen of all semicap companies >\$1B USD revenue, sorted by EBIT % - top 25 of 45

Company Name	EBIT Margin %	Revenue (LTM)
Lasertec Corporation (TSE:6920)	40%	1,525
Disco Corporation (TSE:6146)	40%	2,033
KLA Corporation (NasdaqGS:KLAC)	36%	9,599
Zhejiang Jingsheng Mechanical & Electrical Co., Ltd. (SZSE:300316)	33%	2,617
ASML Holding N.V. (ENXTAM:ASML)	32%	28,161
Applied Materials, Inc. (NasdaqGS:AMAT)	29%	26,485
Lam Research Corporation (NasdaqGS:LRCX)	29%	14,241
GlobalWafers Co., Ltd. (TPEX:6488)	28%	2,306
Tokyo Electron Limited (TSE:8035)	25%	13,064
ASM International NV (ENXTAM:ASM)	25%	2,766
Axcelis Technologies, Inc. (NasdaqGS:ACLS)	24%	1,129
Sino-American Silicon Products Inc. (TPEX:5483)	24%	2,675
Soitec SA (ENXTPA:SOI)	24%	1,079
Xinyi Solar Holdings Limited (SEHK:968)	23%	3,409
Wuxi Autowell Technology Co., Ltd. (SHSE:688516)	23%	1,001
GCL Technology Holdings Limited (SEHK:3800)	22%	4,753
Xinjiang Dago New Energy Co., Ltd. (SHSE:688303)	21%	2,002
Shenzhen S.C New Energy Technology Corporation (SZSE:300724)	20%	1,299
NAURA Technology Group Co., Ltd. (SZSE:002371)	20%	3,333
Teradyne, Inc. (NasdaqGS:TER)	19%	2,659
Kokusai Electric Corporation (TSE:6525)	18%	1,360
SCREEN Holdings Co., Ltd. (TSE:7735)	18%	3,359
Dago New Energy Corp. (NYSE:DQ)	17%	2,013
Sumco Corporation (TSE:3436)	17%	3,022
Flat Glass Group Co., Ltd. (SEHK:6865)	17%	3,031

Lasertec's gross margin and net income margin are just as vertigo-inducing as its operating margin. In each of last 5 years, its gross margin exceeded ASML's and each of the 5 largest semicap companies except for KLA. Lasertec claims its net income margin has expanded significantly over the last five years, and in the last 12 months now exceeds even KLA's.

Lasertec gross margin and net income margin versus largest semicap equipment companies

	Gross Margin					
	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>LTM</u>
ASML	45%	49%	53%	51%	51%	51%
Applied Materials	44%	45%	47%	47%	47%	47%
Lam	45%	46%	47%	46%	45%	47%
Tokyo Electron	41%	40%	40%	46%	45%	45%
KLA	59%	58%	60%	61%	60%	60%
Lasertec	55%	54%	53%	53%	55%	51%

	Net Income Margin					
	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>LTM</u>
ASML	22%	25%	32%	27%	28%	27%
Applied Materials	19%	21%	26%	25%	26%	27%
Lam	23%	22%	27%	27%	26%	25%
Tokyo Electron	19%	16%	17%	22%	21%	19%
KLA	26%	21%	30%	36%	32%	27%
Lasertec	21%	25%	27%	27%	30%	29%

With margins far higher than the five largest semicap companies, one would expect Lasertec's net income – the earnings reported on an accrual basis on its income statement – to convert to copious amounts of operating cash flow – and at a higher rate given its higher margins. But therein lies the key to the fraud – while Lasertec claims to have the highest profitability, its cash conversion is the lowest – a striking anomaly at the center of the largest corporate frauds in history. As shown below, cash conversion has plummeted since the launch of its ACTIS EUV line in 2019/2020 – over the last 5 years, one dollar of earnings has produced only 65 cents of cash, in contrast to its peers which have extraordinary cash conversion, such as ASML with 130-180% conversion despite margins far lower than the ones Lasertec claims; and KLA at >100% in all but one of the last 5 years.

Lasertec operating cash flow divided by net income versus largest semicap players

	Operating Cash Flow To Net Income							
	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>LTM</u>		<u>5 year</u>
ASML	126%	130%	184%	151%	69%	63%		128%
Applied Materials	120%	105%	92%	83%	127%	122%		104%
Lam	145%	94%	92%	67%	115%	136%		98%
Tokyo Electron	76%	137%	60%	65%	90%	NA		82%
KLA	98%	146%	105%	100%	108%	129%		108%
Lasertec	98%	152%	54%	-14%	88%	46%		65%

Lasertec's cash conversion has dropped sharply over time, particularly over the last two years – collapsing even further in the last two quarters (30% Dec'23 and 28% Mar'24), in addition to exhibiting troubling volatility. This is a typical progression in frauds with a high-flying stock and pressure to show rising earnings, which requires the accounting fraud to get larger every period until it becomes unsustainable and collapses. Over time, the ratio of operating cash flow to net income should exceed 1, due to the fact that noncash charges such as depreciation are deducted from net income but are not an operating cash outflow. A ratio consistently below 1 indicates that earnings are suspect. In the 5 year period preceding its ACTIS EUV line (2014-2018), its cash conversion was 90% - versus only 16% and 52% in 2022 and 2023 using a rolling average. In 3 of the last 4 quarters, the conversion was 26-30% - indicating that 70% of reported net income is just accrual estimates and assumptions in a spreadsheet versus cash.

Lasertec cash flow conversion over time

Operating Cash Flow to Net Income - Annual											Last four quarters			
<u>2014</u>	<u>2015</u>	<u>2016</u>	<u>2017</u>	<u>2018</u>	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>LTM 3/31</u>	<u>Jun'23</u>	<u>Sep'23</u>	<u>Dec'23</u>	<u>Mar'24</u>
199%	35%	96%	100%	67%	98%	152%	54%	-14%	88%	46%	26%	189%	30%	28%

Operating Cash Flow to Net Income - Two Year Average									
<u>2014</u>	<u>2015</u>	<u>2016</u>	<u>2017</u>	<u>2018</u>	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>
126%	100%	67%	98%	82%	85%	133%	90%	16%	52%

It's extraordinary that Lasertec has a \$23B market cap yet only \$197MM of cash on its balance sheet – flat over the 5 years since the launch of its ACTIS EUV product, despite 40% margins. Over the same period, peers like ASML, Applied Materials, Lam, and KLA have seen their cash reserves surge – despite lower margins compared to Lasertec – even after repurchasing billions in stock and funding growth and inventory. Shockingly, Lasertec generates so little cash that it can't seem to fund its dividend without debt – putting it at risk. In the Sep 2023 quarter, it had to borrow \$67MM (¥10B) to fund a semiannual dividend of \$77MM (¥7.8B) – highly unusual. Its dividend payout has averaged ~25% of net income. Earnings grew to \$443MM in the last 12 months, implying a \$111MM dividend, while free cash flow was only \$192MM. The dividend alone is thus ~60% of free cash flow (and of cash balance). As the discrepancy between accounting income and cash flow widens, funding the dividend becomes tricky – consensus earnings estimates for 2025/2026 imply a dividend that is unrealistic vs. cash or cash flow.

Dividend vs. free cash flow and cash balance – JPY and USD millions

¥ millions	2019	2020	2021	2022	2023	LTM Mar '24	2025E	2026E
Operating cash flow	5,800	16,486	10,488	(3,464)	40,548	30,981		
Capital expenditures	(897)	(1,178)	(771)	(760)	(18,752)	(1,956)		
Free cash flow	4,903	15,308	9,717	(4,224)	21,796	29,025		
Net income	5,933	10,823	19,250	24,850	46,164	66,969	84,064	106,847
Cash dividends	(1,713)	(2,795)	(4,238)	(7,845)	(10,551)	(18,127)	(22,697)	(28,849)
Dividend payout ratio	29%	26%	22%	32%	23%	27%	27%	27%
Cash balance	13,185	24,723	27,918	23,494	29,773	29,744		

\$ millions	2019	2020	2021	2022	2023	LTM Mar '24	2025E	2026E
Operating cash flow	37	106	68	(22)	262	200		
Capital expenditures	(6)	(8)	(5)	(5)	(121)	(13)		
Free cash flow	32	99	63	(27)	141	187		
Net income	38	70	124	160	298	432	542	689
Cash dividends	(11)	(18)	(27)	(51)	(68)	(117)	(146)	(186)
Dividend payout ratio	29%	26%	22%	32%	23%	27%	27%	27%
Cash balance	85	160	180	152	192	192		

Other cash flow-based ratios are also useful in assessing quality of earnings – and confirm the problem at Lasertec. We compute an operating cash margin to show the percentage of revenue reflected in operating cash flow; as well as a cash return on assets. Lasertec spins to investors that it is an “asset light” or “fab-light” company: “most of our production is outsourced to third parties under [a] fab-lite strategy.” Its cash return on assets should therefore be far higher, but it is only about half that of ASML and KLA – and has also collapsed over time. Each dollar of revenue at Lasertec similarly produces far less cash than its peers.

Lasertec operating cash margin and cash return on assets versus largest semicap equipment companies

	Operating Cash Flow as Percent of Revenue						
	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>LTM</u>	<u>5 year</u>
ASML	28%	33%	58%	40%	20%	17%	35%
Applied Materials	22%	22%	24%	21%	33%	33%	25%
Lam	33%	21%	25%	18%	30%	34%	25%
Tokyo Electron	15%	22%	10%	14%	19%	NA	16%
KLA	25%	31%	32%	36%	35%	35%	33%
Lasertec	20%	39%	15%	-4%	27%	13%	18%

	Operating Cash Flow as Percent of Assets					
	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>5 year</u>
ASML	14%	19%	38%	26%	14%	22%
Applied Materials	18%	18%	23%	21%	30%	22%
Lam	26%	16%	24%	19%	29%	23%
Tokyo Electron	15%	20%	11%	17%	20%	17%
KLA	16%	19%	22%	29%	28%	23%
Lasertec	13%	25%	10%	-2%	18%	13%

Lasertec has been sitting on over \$1 billion of inventory for the last six quarters. Aside from being alarmingly elevated, the inventory balance has barely fluctuated during each of these quarters, also highly unusual. In addition, inventory as a percent of revenue is highly abnormal versus its historical trend – increasing in an out-of-control manner. It is critical to note that inventory began to spike in 2019/2020 when Lasertec launched its ACTIS EUV line and has become even more unhinged in recent quarters. In some quarters in 2022 and 2023, inventory hit 120-135% of revenue, compared to a baseline of 30%. Inventory as a percent of assets shows the same pattern – hitting its highest level ever in the last two quarters.

Lasertec inventory as a percent of revenue and assets over time

Inventory As Percent Of Revenue - Annual														
<u>2010</u>	<u>2011</u>	<u>2012</u>	<u>2013</u>	<u>2014</u>	<u>2015</u>	<u>2016</u>	<u>2017</u>	<u>2018</u>	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>LTM 3/31</u>
34%	27%	33%	37%	31%	29%	37%	45%	58%	65%	80%	77%	115%	100%	70%

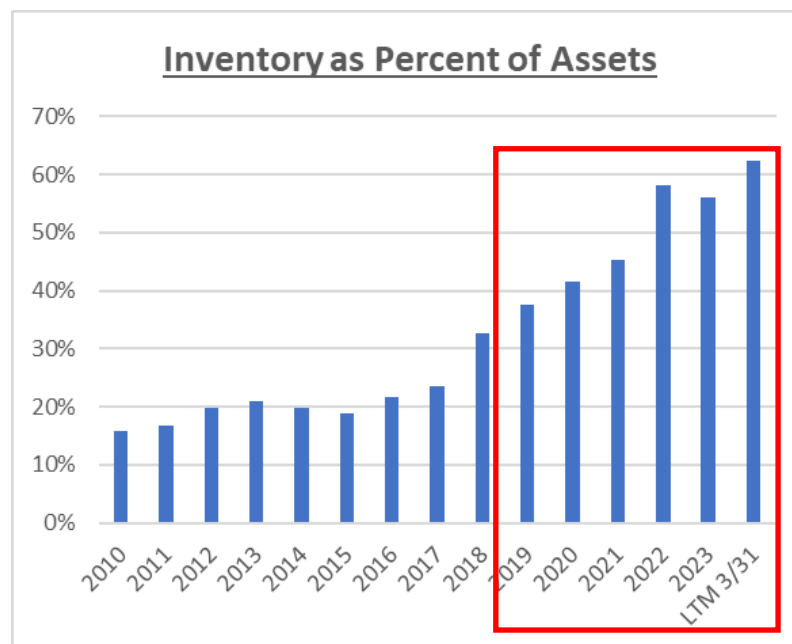
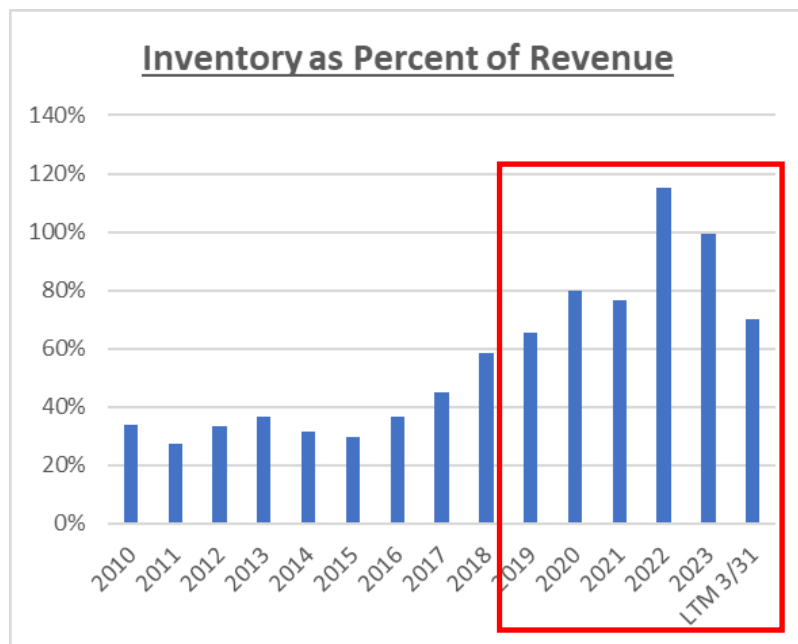
Inventory As Percent Of Revenue - Last Four Quarters							
<u>Jun'22</u>	<u>Sep'22</u>	<u>Dec'22</u>	<u>Mar'23</u>	<u>Jun'23</u>	<u>Sep'23</u>	<u>Dec'23</u>	<u>Mar'24</u>
115%	112%	123%	135%	100%	86%	84%	70%

Inventory As Percent Of Assets - Annual														
<u>2010</u>	<u>2011</u>	<u>2012</u>	<u>2013</u>	<u>2014</u>	<u>2015</u>	<u>2016</u>	<u>2017</u>	<u>2018</u>	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>LTM 3/31</u>
16%	17%	20%	21%	20%	19%	22%	24%	33%	37%	42%	45%	58%	56%	62%

Inventory As Percent Of Assets - Last Four Quarters							
<u>Jun'22</u>	<u>Sep'22</u>	<u>Dec'22</u>	<u>Mar'23</u>	<u>Jun'23</u>	<u>Sep'23</u>	<u>Dec'23</u>	<u>Mar'24</u>
58%	52%	55%	60%	56%	59%	64%	62%

The alarming spike is apparent visually – we show inventory as a percent of both revenue and assets annually from 2010 to the last 12 months. We again highlight the key fact that inventory began to spike in 2019/2020 when Lasertec launched its ACTIS EUV line – the period when its reported revenue and purported margins both began to rapidly expand, fueling a 2,000% rise in its stock price.

Lasertec inventory as a percent of revenue and assets over time



Shockingly, Lasertec has the most inflated inventory level of any public semiconductor equipment company in the world. A CapitalIQ screen returns ~150 such companies globally >\$500MM market cap. Lasertec tops the list on inventory as a percent of assets, and is near the top of the list on inventory as a percent of revenue, with only a few non-comparable China-listed companies that are worse – each of which is dubious, small, or in the solar industry. We excerpt the top ~30.

Inventory as percent of revenue – CapitalIQ screen

Company Name	Revenue	Inventory/Revenue
Shenzhen S.C New Energy Technology Corporation (SZSE:300724)	1,232	244%
Jiangsu Leadmicro Nano-Equipment Technology Ltd (SHSE:688147)	237	192%
Piotech Inc. (SHSE:688072)	382	168%
Suzhou Maxwell Technologies Co., Ltd. (SZSE:300751)	1,141	133%
Yingkou Jinchen Machinery Co., Ltd. (SHSE:603396)	317	129%
Beijing Jingyi Automation Equipment Co., Ltd. (SHSE:688652)	105	129%
Hangzhou Changchuan Technology Co.,Ltd (SZSE:300604)	250	122%
Wuxi Autowell Technology Co.,Ltd. (SHSE:688516)	889	121%
Wuhan DR Laser Technology Corp.,Ltd (SZSE:300776)	227	119%
ACM Research (Shanghai) Inc. (SHSE:688082)	548	101%
Lasertec Corporation (TSE:6920)	1,057	100%
ACM Research, Inc. (NasdaqGM:ACMR)	558	98%
Hwatsing Technology Co., Ltd. (SHSE:688120)	354	96%
JHT Design Co.,Ltd. (SHSE:603061)	49	96%
KINGSEMI Co., Ltd. (SHSE:688037)	242	95%
Shanghai GenTech Co., Ltd. (SHSE:688596)	541	88%
Zhejiang Jingsheng Mechanical & Electrical Co., Ltd. (SZSE:300316)	2,536	86%
NAURA Technology Group Co., Ltd. (SZSE:002371)	3,114	77%
Crystal Growth & Energy Equipment Co.,Ltd. (SHSE:688478)	57	69%
AEM Holdings Ltd. (SGX:AWX)	365	68%
Advanced Micro-Fabrication Equipment Inc. China (SHSE:688012)	883	68%
SICC Co., Ltd. (SHSE:688234)	176	67%
AIXTRON SE (XTRA:AIXA)	696	63%
Techwing, Inc. (KOSDAQ:A089030)	103	61%
HANMI Semiconductor Co., Ltd. (KOSE:A042700)	123	60%
Shenzhen Xinyichang Technology Co., Ltd. (SHSE:688383)	147	58%
Rorze Corporation (TSE:6323)	622	57%
SÜSS MicroTec SE (XTRA:SMHN)	336	55%
Sumco Corporation (TSE:3436)	3,022	51%
Grand Process Technology Corporation (TPEX:3131)	116	50%
Wonik Ips Co., Ltd (KOSDAQ:A240810)	532	49%

Inventory as percent of assets

Company Name	Revenue	Inventory/Assets
Lasertec Corporation (TSE:6920)	1,057	56%
AEM Holdings Ltd. (SGX:AWX)	365	46%
SÜSS MicroTec SE (XTRA:SMHN)	336	45%
AIXTRON SE (XTRA:AIXA)	696	38%
ACM Research, Inc. (NasdaqGM:ACMR)	558	37%
SMA Solar Technology AG (XTRA:S92)	2,105	34%
Rorze Corporation (TSE:6323)	622	34%
SolarEdge Technologies, Inc. (NasdaqGS:SEDG)	2,977	31%
Wonik Ips Co., Ltd (KOSDAQ:A240810)	532	31%
Advantest Corporation (TSE:6857)	3,217	30%
SDI Corporation (TWSE:2351)	354	30%
Tokyo Electron Limited (TSE:8035)	16,618	28%
Philoptics Co., Ltd. (KOSDAQ:A161580)	231	26%
ASMP Limited (SEHK:522)	1,881	26%
Ichor Holdings, Ltd. (NasdaqGS:ICHR)	811	26%
Lam Research Corporation (NasdaqGS:LRCX)	17,429	26%
Tokyo Seimitsu Co., Ltd. (TSE:7729)	1,104	26%
Axcelis Technologies, Inc. (NasdaqGS:ACLS)	1,131	24%
PSK Inc. (KOSDAQ:A319660)	271	24%
Eugene Technology Co.,Ltd. (KOSDAQ:A084370)	213	23%
Grand Process Technology Corporation (TPEX:3131)	116	23%
Oporun Co.,Ltd. (TSE:6235)	261	23%
UMS Holdings Limited (SGX:558)	227	23%
ASML Holding N.V. (ENXTAM:ASML)	30,462	22%
MPI Corporation (TPEX:6223)	266	22%
EO Technics Co., Ltd. (KOSDAQ:A039030)	244	22%
SCREEN Holdings Co., Ltd. (TSE:7735)	3,467	22%
TOWA Corporation (TSE:6315)	405	21%
Disco Corporation (TSE:6146)	2,033	21%
KLA Corporation (NasdaqGS:KLAC)	10,496	20%
Sumco Corporation (TSE:3436)	3,022	20%

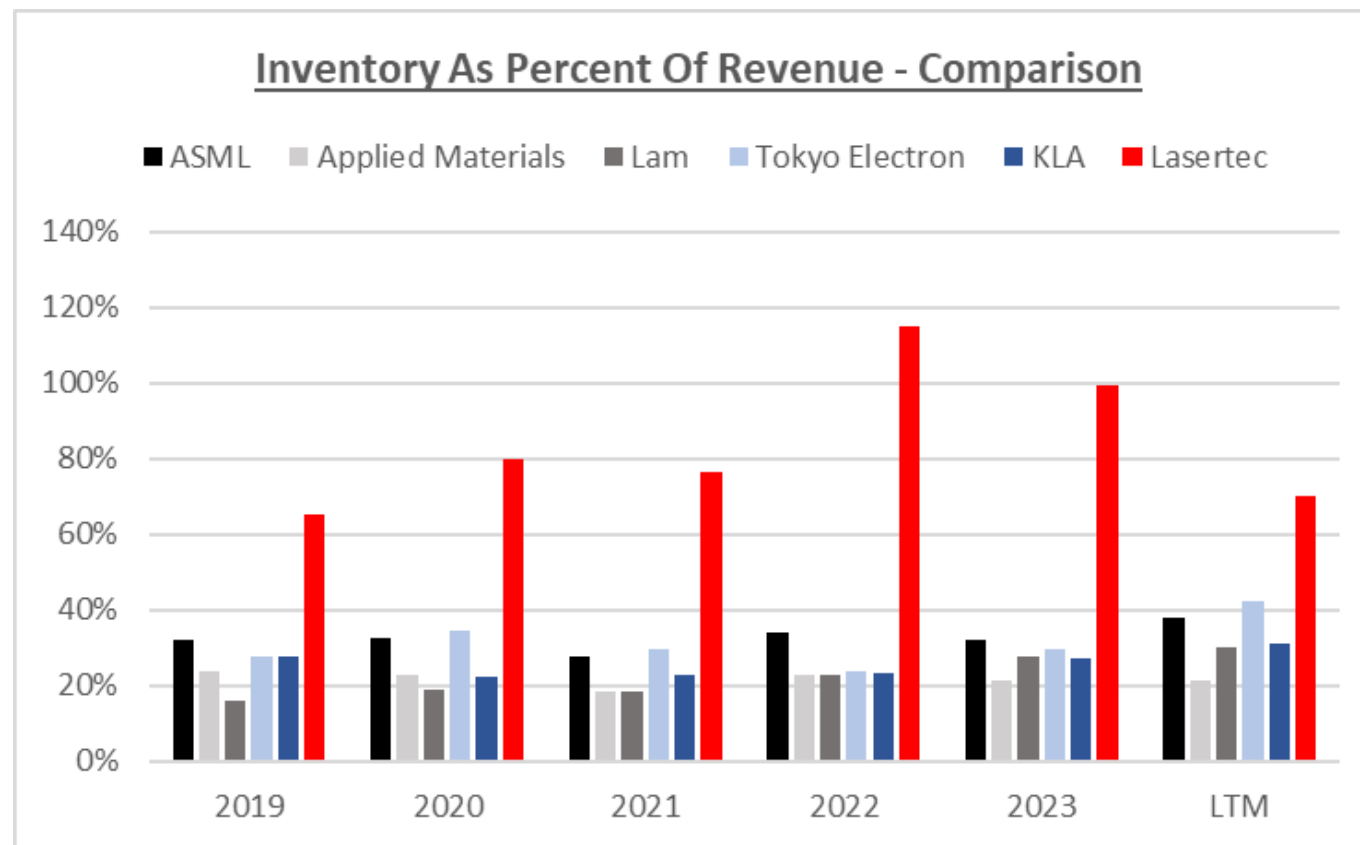
Lasertec's inventory balance has no plausible explanation when compared to the five largest semicap equipment players – it is triple or quadruple as a percent of revenue in the last 2 to 3 years. Each of them – ASML, Applied Materials, Lam Research, Tokyo Electron, and KLA – makes complex, expensive equipment with long lead times. They all share similar balance sheet, income statement, and cash flow characteristics and hence have comparable inventory – except for Lasertec.

Inventory as percent of revenue – comparison

	Inventory As Percent Of Revenue						
	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>LTM</u>	<u>3 YR AVG</u>
ASML	32%	33%	28%	34%	32%	38%	31%
Applied Materials	24%	23%	19%	23%	22%	21%	21%
Lam	16%	19%	18%	23%	28%	30%	23%
Tokyo Electron	28%	35%	30%	24%	30%	42%	28%
KLA	28%	23%	23%	23%	27%	31%	24%
Lasertec	65%	80%	77%	115%	100%	70%	97%

A visual representation of inventory as percent of revenue versus the five largest semicap equipment companies makes clear how unusual Lasertec's reported inventory balance is.

Lasertec inventory versus five largest semicap equipment companies

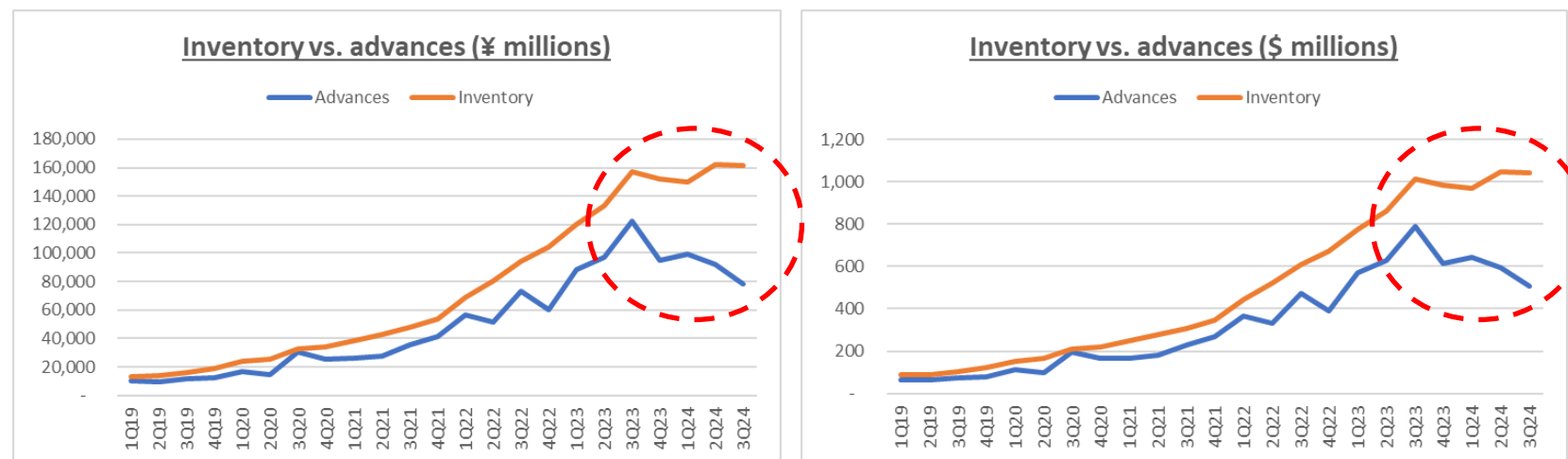


As an additional indication that the inventory account makes no sense, we note a striking breakdown in the historical relationship between inventory and advances – Lasertec’s term for deferred revenue. We note that the CEO stated in a quarterly Q&A that “in terms of ACTIS, we have reached agreements with customers to receive advance payments...at the time of order” – clearly indicating that advances should grow in lockstep with inventory. At this point, the inventory balance is little more than a plug in an Excel spreadsheet to balance the accounts, spiraling out of control as is typical when an accounting fraud becomes difficult to sustain.

Lasertec CEO, Feb 2021 Q&A

“Furthermore, as you stated earlier, we receive advance payments from customers at the time of order. **Also, in terms of ACTIS, we have reached agreements with customers to receive advance payments from customers at the time of order.”**

Lasertec inventory versus advances (JPY and USD) by quarter 2019 to Mar ‘24



The CFO's comments a few weeks ago during the Apr 30, 2024, earnings Q&A should alarm Lasertec investors. A BofA analyst questioned why inventories are so high and when they “expect inventories to drop.” The CFO offered no explanation, stating only that “the current situation is a stalemate” and that “we do not necessarily have an accurate grasp of the situation” – and shockingly, that inventories would rise even higher: “we believe that this will increase”; “I think we will see...or rather, an increase in inventories”; and affirmed that the current \$1.1B is the “minimum level forecast.”

Quarterly earnings Q&A, Apr 30, 2024 – transcript excerpts

Continuing on, Mr. Hirakawa of BofA Securities, please go ahead.

Hirakawa [Q]: Thank you very much. Inventories, even after the end of Q3, were probably around JPY160 billion and have remained high for the past three or four quarters. I wonder if you could tell me the level you are targeting in the medium term, and when you expect inventories to drop.

Misawa [A]: Regarding inventories, Misawa will answer. Inventories vary depending on the progress of orders received, anticipated production, and the acceptance of products by customers. As of today, the current situation is a stalemate, or about JPY160 billion.

We believe that this will increase as market conditions recover and orders become more active, and also as products with long lead times increase. We do not necessarily have an accurate grasp of the situation, but we are proceeding with production while properly controlling risks.

Hirakawa [Q]: So you are saying that if market conditions recover, there will be another increase, and that you are currently holding this amount as a minimum level forecast.

Misawa [A]: Yes. We are of course still engaged in anticipated production, but once market conditions recover, I think we will see more active production, or rather, an increase in inventories.

Lasertec's inventory charade is not only brazen but hiding in plain sight – it claims that 100% of its \$1+ billion in inventory is raw materials and work-in-process with 0% finished goods. Incredibly, its financial statements over the past 5 years record zero finished goods in any period – its balance sheet doesn't even include a line item for it. Misclassifying and improperly valuing inventory in an attempt to evade impairment charges that would wipe out historical earnings is a tactic common to some of the most significant fraud cases in history. Lasertec has no option but to claim it has no finished goods – its stock would have imploded if investors saw machines piling up. Work-in-process fools investors into thinking that components/parts will shortly convert to new sales; and that finished machines ship so fast that they never remain on the balance sheet.

Lasertec financial filings contain no line item for finished goods – most recent filing as example, Apr 30, 2024

	Total Inventory By Category (JPY)					
	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>Mar'24</u>
Raw materials	15,479	30,083	48,712	94,308	131,056	126,755
Work-in-process	3,278	3,978	5,176	9,774	21,017	34,872
Total inventory	18,757	34,061	53,888	104,082	152,073	161,627

	Total Inventory By Category (USD)					
	<u>2019</u>	<u>2020</u>	<u>2021</u>	<u>2022</u>	<u>2023</u>	<u>Mar'24</u>
Raw materials	101	197	318	616	857	828
Work-in-process	21	26	34	64	137	228
Total inventory	123	223	352	680	994	1,056

2. Consolidated financial statements and notes

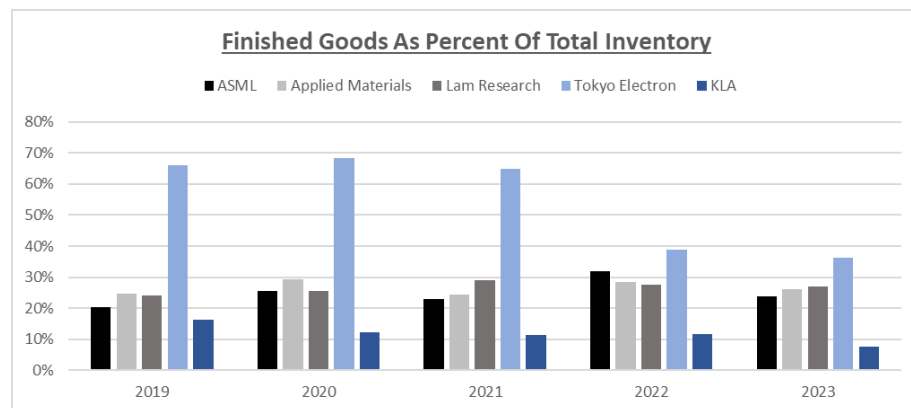
(1) Consolidated balance sheet

	(In millions of yen)	
	Previous (As at June 30, 2023)	Current (As at March 31, 2024)
Assets		
Current assets		
Cash and deposits	29,773	29,744
Notes and accounts receivable-trade, and contract assets	21,611	20,494
Work in process	131,056	126,755
Raw materials and supplies	21,017	34,872
Accounts receivable - other	19,640	263
Other	8,029	7,872
Allowance for doubtful accounts	(38)	(21)
Total current assets	231,090	219,981

Lasertec's purported lack of finished goods inventory and reporting methodology is highly abnormal and lacks credibility versus the five largest semicap equipment players, for whom finished goods has averaged 30% as a percent of total over the last 5 years versus 0% for Lasertec – each one breaks out inventory by raw materials, work-in-process, and finished goods; several include additional disclosure for allowances and progress payments.

Finished goods as percent of total inventory

	Finished Goods As Percent Of Total Inventory				
	2019	2020	2021	2022	2023
ASML	20%	25%	23%	32%	24%
Applied Materials	25%	29%	24%	28%	26%
Lam Research	24%	26%	29%	28%	27%
Tokyo Electron	66%	68%	65%	39%	36%
KLA	16%	12%	11%	12%	7%
Lasertec	0%	0%	0%	0%	0%



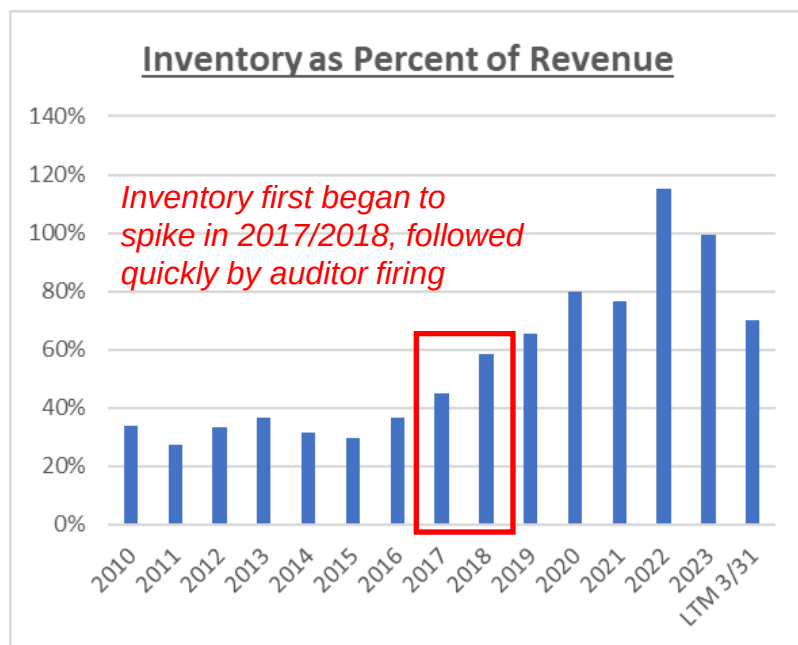
It appears to be an open secret among Lasertec's investor relations team that inventory is being misclassified. We hired a local consultant to meet with IR in March 2024 at Lasertec headquarters, who at one point asked “And no finished goods in there, right?” The IR rep replied “No. All work in process.” while smiling with a knowing wink. In a later section, we share photographic evidence that Lasertec's new “Innovation Park” and its so-called “fabs” are essentially a fraud with no production or R&D. During ~20 visits to the campus, our investigators found an empty and deserted facility. Its only purpose appears to be storage for finished goods – we show below a sample of photos of machines being loaded into storage, all hermetically sealed in protecting wrapping and clearly finished goods. Given that some of these machines sell for \$50MM each, Lasertec's claim of 100% raw materials and work-in-process and 0% finished goods is clearly false.

Investigator photos of finished goods being moved into storage at “Fab 4A” at Lasertec Innovation Park



Lasertec suddenly fired their longtime auditor Deloitte Touche Tohmatsu on Aug 8, 2018, after using them for at least the previous 15 years – a red flag as auditors are typically only fired after they start sniffing around and asking too many questions. The timing is suspicious as the firing occurred right after the close of the 2018 fiscal year in June – and 2017/2018 were the first years that inventory as a percent of revenue began to spike. Lasertec provided no explanation for the firing in its press release.

Lasertec inventory as a percent of revenue



Auditor firing, press release Aug 8, 2018

FOR IMMEDIATE RELEASE

Lasertec Corporation
2-10-1 Shin-yokohama, Kohoku-ku, Yokohama
(Code 6920 / Tokyo Stock Exchange 1st Section)

Announcement of Financial Auditor Change

Yokohama, August 8, 2018 - Lasertec Corporation announced today that the Audit and Supervisory Board resolved on August 8, 2018, to change the certified public accountants who perform audit certification in accordance with Article 193-2, Paragraphs 1 and 2 of the Financial Instruments and Exchange Act, and that the Board of Directors resolved on the same day to propose the appointment of a new accounting firm as financial auditor at the 56th Ordinary General Meeting of Shareholders scheduled for September 26, 2018.

Manipulating inventory to falsify gross margins and earnings is an easy temptation for a semiconductor company trying to meet lofty market expectations. Inventory games in the semiconductor unit at Toshiba were a key part of the fraud and scandal that blew up the company in 2014. The potential for abuse is highlighted in the audit opinions at several of the largest semicap players. Auditors for both KLA and Lam recently noted inventory valuation as the one and only Critical Audit Matter in their last audit opinions, a designation reserved for the highest-risk and most material areas in the financial reports.

KLA 10K Aug 2023 – PriceWaterhouse Coopers audit opinion excerpt

“The estimate of net realizable value of inventory is impacted by assumptions regarding general semiconductor market conditions, manufacturing schedules, technology changes, new product introductions and possible alternative uses, and **require management to use significant judgment that may include uncertain elements**....The principal considerations for our determination that performing procedures relating to the **valuation of product inventory is a critical audit matter** are (i) the significant judgment by management when developing the estimated obsolescence used for determining the net realizable value of product inventory and (ii) a high degree of auditor judgment, subjectivity, and effort in performing procedures and evaluating management’s assumption related to the future demand for product inventory.”

Lam Research 10K Aug 2023 – Ernst& Young audit opinion excerpt

“Inventory – Valuation [...] Auditing **management’s estimates for excess and obsolete inventory involved subjective auditor judgment** because management’s assessment of whether a write down is required and the measurement of any excess of cost over net realizable value is **judgmental and considers a number of qualitative factors** that are affected by market and economic conditions outside the Company’s control.”

Although KLA and Lam’s auditors flag inventory accounting as a risk, their cash flows don’t indicate a major issue – at similar companies, however, inventory has been at the core of large-scale fraud, such as at Toshiba’s semiconductor unit. MagnaChip, a Korean company which trades on the New York Stock Exchange, settled fraud charges with the Securities and Exchange Commission in 2017, and shows the potential for accounting tricks: “MagnaChip recognized revenue on sales of incomplete or unshipped products...and delayed booking obsolete or aged inventory to manipulate its gross margin.” The SEC detailed how Magnachip “overstated revenue for nearly two years in response to immense pressure...each quarter to meet revenue and gross margin targets that had been communicated to the public...a panoply of accounting tricks to artificially meet financial targets.”

SEC press release, May 2017

Semiconductor Company and Former CFO Settle Accounting Fraud Charges

The SEC’s order finds that MagnaChip Semiconductor Corp. overstated revenues for nearly two years in response to immense pressure placed on employees each quarter to meet revenue and gross margin targets that had been communicated to the public. Then-CFO Margaret Sakai directed or approved several fraudulent accounting practices to make it falsely appear the company had met those targets. For example, MagnaChip recognized revenue on sales of incomplete or unshipped products, and the company delayed booking obsolete or aged inventory to manipulate its reported gross margin. MagnaChip also engaged in roundtrip transactions to manipulate accounts receivable balances, and concealed from auditors that there were side agreements with distributors to induce them to accept products early.

The 16-page SEC complaint reads like a manual for how to cook the books if you're a semiconductor company, detailing the accounting tricks and levers applicable to the industry: 1) “fraudulent revenue recognition practices” such as “pull in” sales by offering “undisclosed concessions...to order products earlier than wanted or needed so Magnachip would hit revenue targets”; 2) “recognizing revenue on ‘sales’ of non-existent or unfinished products...falsified books and records”; 3) “manipulated gross margin...delayed scrapping obsolete inventory by several months or years...inflated its gross margin...records did not accurately reflect inventory values...made improper entries...to increase the inventory value when no manufacturing process had occurred.”

SEC order issued against Magnachip Semiconductor, May 2017 - excerpts

“Sun Ip Go” – Recognizing Revenue on Sales of Unfinished Products

15. MagnaChip also improperly recognized revenue on “sales” of non-existent or unfinished products in order to meet revenue targets. In 2011, some MagnaChip Korean sales

A. MagnaChip’s Fraudulent Revenue Recognition Practices

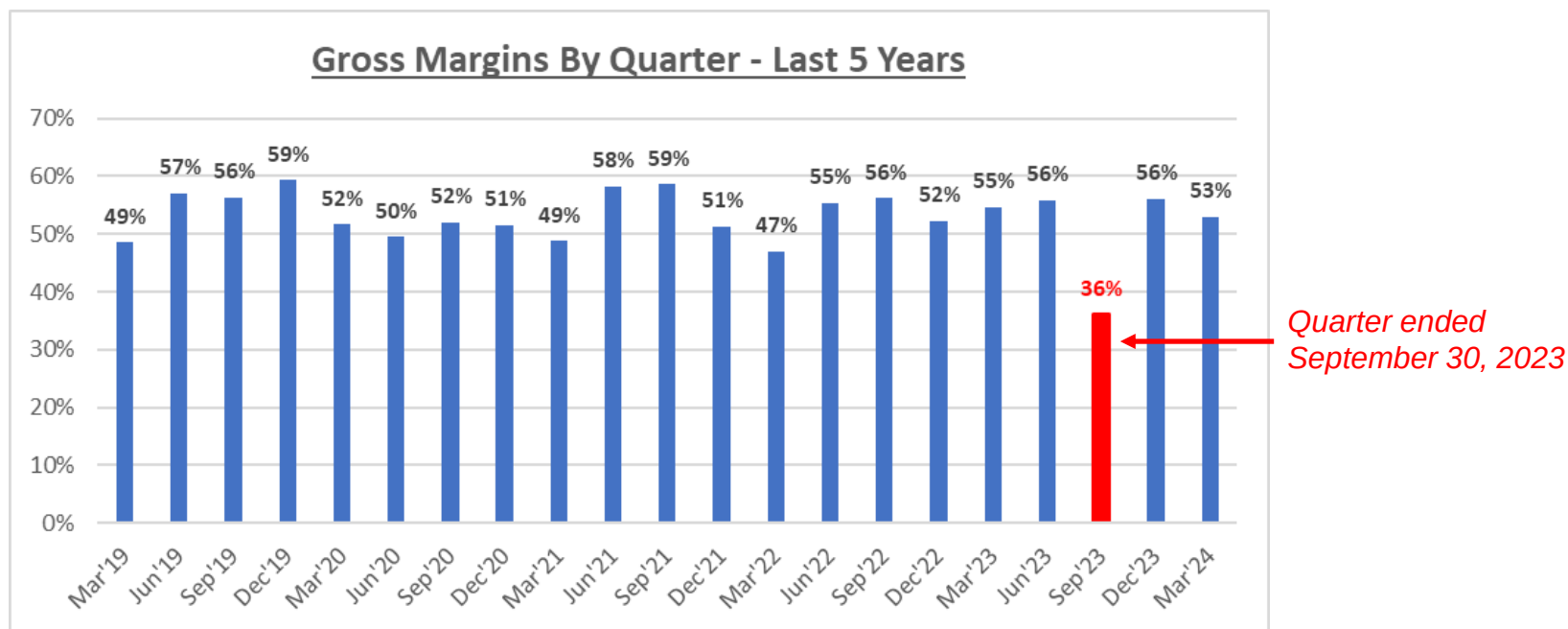
8. To help ensure that MagnaChip hit its targets, certain employees in Korea engaged in two primary improper practices to inflate its revenue. The first, which certain employees

B. MagnaChip Manipulated Gross Margin

27. Sakai either knew or was reckless in not knowing that MagnaChip sometimes engaged in efforts to manipulate its gross margins. For example, certain MagnaChip

The final stage of a large-scale fraud is usually preceded by unusual tremors – ignored or misunderstood by most – as pressure builds before the inevitable earthquake. A major tremor occurred in the quarter ended Sep 30, 2023, when Lasertec's gross margin mysteriously plunged from 56% to 36%, magically recovering to 56% the next quarter. The company's explanations were evasive. Unexpected, dramatic anomalies in the financial reports are a tell-tale sign that a company is struggling to hold a scheme together.

Lasertec quarterly gross margins



The quarterly earnings presentation on Oct 31, 2023, was vague regarding the gross margin collapse, saying only that “profit margins fell due to an unfavorable product mix in the short term.” In his prepared comments, the CFO added a critical hint: “this is the result of the recognition of some low-profit ACTIS sales” – referring to Lasertec’s flagship EUV system. A Citigroup analyst asked how many ACTIS machines were recognized, given their \$40-60MM price, and if “this low-profit ACTIS has zero gross profit?” The CEO evaded the question: “I cannot give you any information on gross margin.”

Earnings presentation, comments, and analyst Q&A on Oct 31, 2023, for the quarter ended Sep 30

Results in Perspective

Sales increased significantly YoY. Profit margins fell due to an unfavorable product mix in the short term.

In millions of yen

	FY2023				FY2024	YoY (vs Q1) FY2023	QoQ (vs Q4) FY2023
	Q1	Q2	Q3	Q4	Q1		
Net Sales	25,723	29,376	24,329	73,401	47,305	+83.9%	▲35.6%

“This is the result of the **recognition of some of the low-profit ACTIS sales.**” – Lasertec CFO, Oct 31, 2023, earnings presentation

“This is the result of the **recognition of some of the low-profit ACTIS sales.**” – Lasertec CFO, Oct 31, 2023, earnings presentation

Shibano [Q]: Thank you. My name is Shibano from Citigroup Securities...in the president's explanation...he explained that the **acceptance of the low-profit ACTIS** will start from Q1 and was completed. I think what this means is that there were **several units, although you cannot disclose how many**, and that the acceptance had begun in July. **I am not sure how many units there were, but I understand that all of the low-profit ACTIS units finished recording sales in Q1, is that correct?**

Okabayashi [A]: “Yes. Mr. Shibano is correct.”

Shibano [Q]: “By the way, **am I right in understanding that this low-profit ACTIS has zero gross profit?**”

Okabayashi [A]: “**I cannot give you any information on gross profit.**”

When asked again later in the same Q&A, the CEO stated that “it took time for this low-profit ACTIS to be accepted because we had to change parts and add resources, but I think it is safe to say that we have already gotten out of that situation.” In the next quarter’s Q&A, the CEO stated that “in Q1 there were ACTIS units with very low gross margin from the first lots” – without ever explaining what “first lots” means; and that “we have finished recording the sales of ACTIS with poor gross profit margins.”

Quarterly Q&A on Oct 31, 2023, for the quarter ended Sep 30

Yamamoto [Q]: “So, you are saying that all low-margin ACTIS sales have been completed, and the negative impact on the gross margin of the entire company will gradually disappear, and we will not hear much about how the margins are still a little low...is my understanding correct?”

Okabayashi [A]: “Yes, you are right. Yes. It took time for this low-profit ACTIS to be accepted because we had to change parts and add resources, but I think it is safe to say that we have already gotten out of that situation.”

Quarterly Q&A on Jan 31, 2024, for the quarter ended Dec 30

Nakamura [Q]: “The gross margin in Q1 was 36%, and in Q2 it was 56%, which is quite a big jump. I am aware that sales for less profitable ACTIS units were recorded in Q1, but what factors, if any, boosted the gross margin in Q2?...”

Okabayashi [A]: “...in Q1, there were ACTIS units with very low gross margins from the first lots...Q2 saw an increase in gross margin because there were no such items. Future gross profit margins will depend on the product mix. However, singling out ACTIS, the gross profit margin of ACTIS sales that will be recorded from Q2 onward has improved because we have finished recording the sales of ACTIS with poor gross profit margins.”

The elephant in the room and the question no analyst asked is key to Lasertec's fraud: why did the ACTIS tools "sold" in Q1 – its flagship EUV product and driver of its stocks – have "very low gross margin"? And why only those sold in Q1, and not all systems sold since launch in 2019? The CEO provided a clue, albeit vague: "because we had to change parts and add resources," without specifying which parts or why they had to be changed. The answer, as we detail shortly, is that the ACTIS system is defective and unreliable to the point that Lasertec has no functioning EUV technology. Its EUV light source – the key component – is fatally flawed; has a short lifespan of only a few weeks before failing, requiring replacement; and the optics system comprising the collector and mirrors needs costly refurbishment every few weeks, due to contamination from the source. An expert shared the bill of materials and staggering cost for key components: source (\$8MM), stage (>\$3MM), optics (\$6MM), collector (\$4MM) – before millions more for enclosure, assembly, fixed costs – all prior to adding gross margin.

ACTIS bill of materials is ~\$25MM; "it doesn't" add up"

A: "...the current source cost, together with the current stage cost, doesn't make sense. I don't think they can shave a lot compared to KLA and, in a 2x-led price war, it doesn't add up to me. By adding up all the major components, I can project how much they cost [...] the **Sumitomo stage will not cost less than \$3 million...that complex source with all the discounts, \$8 million. The optics are at least, \$4 million is the mirror. The optics--\$6 million.** And then you put another \$1 million for the enclosure, the imaging computing, everything. Do the math and the source, isolation system maybe together is \$3 million. All together, we are almost \$20 million or \$15. Where am I now? \$3 million for the stage and we say \$8 million for the source and then we have another \$3 million that combines all the rest, and the rest is image acquisition, the whole enclosure and isolation system....that's the raw material. To turn it to the cost, you have to add their fixed costs on their labor plus clean rooms."

Q: "They report a gross margin of 50% to 55%."

A: "No-no. Gross margin is after. I gave you building materials. To turn it to cost, you need to add another - the type of tool is at least \$3-4 million just to assemble. It's complex. Now **we are around—we are \$23-24 million.**" –Former KLA executive in a leadership role in their EUV mask inspection group

As a result, our research indicates that key customers like TSMC and Intel are exasperated, angry, and refusing or delaying shipment and/or acceptance of Lasertec's tools. We suspect that they – in particular TSMC, according to several interviews - refused to pay or demanded compensation that wiped out the margin; and/or were induced to take tools at sharp discounts – similar to the Magnachip fraud. In addition to the highlighted replacement costs, experts indicated that it costs \$2.5MM/year just to clean the collector; plus 3-4 full-time, dedicated field service engineers per tool for troubleshooting and repairs – eating up the margin and driving a burdensome cost of ownership. Global Foundries, for example, complained that the cost of each field engineer is \$400,000/year – “it’s very expensive” – and \$3-4MM per machine just for maintenance. They stated that Lasertec has to eat the service cost for the first 18 months – “just there, they lose around \$4MM”- with at least \$6-7MM thereafter in annual service after mark-up: “I’m sure customers are not happy with that...the cost has skyrocketed.” Our research indicates Lasertec is now offering to assume that cost as a concession.

Maintenance for each machine adds up to \$3-4MM annually, costing unhappy customers \$6-7MM in service contracts

Q: “If you add up having to clean the collector, clean the lenses, what do you think the **annual maintenance cost** is of parts?”

A: **“Between three to four million, definitely. It’s true that the service contract is also a few million dollars a year.** But they have to offer a one-year warranty for every tool. That’s part of the contract. That’s the minimum they have to give. And together with that, they have six months until they get acceptance. So, they install the tool. Between three to six months, you can only then get acceptance, and then you have a one year warranty. So, it’s 18 months that they need to sponsor it, and it’s part of the cost of the sale. **Just there, they lose around \$4 million. After that—that’s exactly the problem—how much you can sell service cost—at least they need to double these numbers. So, it has to cost \$6-7 million a year in service costs, and I’m sure the customers are not happy with that...the cost [of service contracts] has skyrocketed...\$400,000 a year [per technician]...it’s very expensive.**” — Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

Lasertec never explained what was different about the ACTIS tools sold in Q1 such that they had no margin – each tool is the same, and the obvious implication is that if those are worthless, then so is the \$1.2B of inventory on its balance sheet, which is therefore mis-valued to avoid an impairment charge. As a preview of a later section, we quote a Marvell senior employee who shared color from an executive's recent visit to Lasertec's headquarters: they were desperate to offload ACTIS inventory, offering Marvell up to ~75% discounts on the A150, as well as other inducements and concessions that acknowledged uptime and reliability problems: "they were trying to also offer additional A150's that they had on the books that weren't, let's say, being taken ownership of...we were getting a reduced price as well if we wanted to purchase additional A150's...in the range of \$15-30MM." Marvell stated they previously paid \$62MM for the A150.

Lasertec desperate to offload ACTIS inventory; offering up to ~75% discounts on A150; offering various inducements

Q: "Do you remember any color around the inventory build-up color?"

A: "When our [Marvell senior executive – title redacted] went to go visit, **they were trying to also offer additional A150's that they had on the books that weren't, let's say, being taken ownership of by other competitors that had presumably forecasted that they needed them. And so, we were getting a reduced price as well if we wanted to purchase additional A150s and that price was closer to about, in the range of \$15 to 30 million."**

Q: "So, they were willing to try to sell those machines at half or a third of their price?"

A: "Yeah, and also, again, that's why the comment that I gave earlier, which is that if we were to buy an A150—"

Q: "Yeah, they've got to write down all that inventory."

A: "Yeah."

Q: **"My question is, did that really happen? They offered you guys \$15 to 30 million to take A150's? This literally just happened a few weeks ago?"**

A: **"Yeah."** – Senior employee at Marvell Technology

The details that Marvell provided suggest that Lasertec is desperate to offload bloated EUV inventory: “they made a multitude of offers to try to get us to basically purchase more equipment.” Notably, Lasertec offered terms that indicate an attempt at damage control given customer frustration with product defects and low uptime: “other concessions were made like extended warranty service...at no additional cost...even guaranteeing service time like 60-70%...if not then we could take some kind of reimbursement...for the lost time of use” – making us question how Lasertec could retain any margin given the three field service engineers required to maintain each machine plus the millions to service, clean, and replace the light source and optics.

Lasertec desperation evident in contract concessions and inducements

A: “But we also declined it because we just don't need it. We didn't need that, and then we also weren't looking for an evaluation on their A300. So, **they made a multitude of offers to try to get us to basically purchase more equipment.**”

Q: “Wow, I'm shocked.”

A: “What **they were also offering was that if we purchased an additional machine, we would get—there were other concessions that were made like extended warranty service that would be at no additional cost** or discounted warranty services or discounted repair costs or **even guaranteeing service time, like 60-70% time, for example, and if not, then we could take some kind of reimbursement from them for the lost time of use.**” – Senior employee at Marvell Technology

Marvell told us that when the senior executive visited Lasertec in December, Lasertec offered an arrangement where “they would actually swap out the A150 that we have with an A300 at a discounted price...because they wanted to try to get early purchase volume.” The offer suggests not only desperation but an attempt to mislead investors by essentially fabricating sales of the “new” A300, which is simply the A150 with a different light source and a new name in an attempt to cover up a defective light source. Marvell indicated that they declined the offer and that “we haven’t seen real proof of concept and enough case studies to actually show the real-world application of it versus the A150.”

Lasertec offered to swap A300 for A150 “at a discounted price” so that they could show “early purchase volume”; no “real proof of concept”

Q: “What have you heard about the A300? Have you actually had a conversation or engaged with Lasertec on it?”

A: “We’ve had some introductory conversations with them about it, and actually, insofar as if we were to purchase—one of the conversations we had was that **they would actually swap out the A150 that we have with an A300 at a discounted price.** Again, it wasn’t something that we were looking to entertain just because the overall quote that we got on it is fairly significant. But I have heard that they used a new light source.”

Q: “Yeah, Urashima.”

A: “Urashima, yeah. The only thing is that **we haven’t seen real proof of concept and enough case studies to actually show the real-world application** of it versus the A150.”

Q: “When were these conversations? They announced it in November. When did you have these conversations with Lasertec about the A300?”

A: “This was about a few weeks ago, around December, because they did a press release and **[an executive at Marvell – title redacted to preserve anonymity] who actually went to visit their headquarters in Japan and had a conversation with them about it and one of the things that they had stated was that because they wanted to try to get early purchase volume, they were willing to potentially entertain a swap of the A150 for an A300.** We respectfully just said that we wouldn’t be looking at evaluating something like this until probably closer to 2025.” – Senior employee at Marvell Technology

Marvell was one of several customers we interviewed who recently visited Lasertec's headquarters and reported seeing an unusual level of machine inventory – the Marvell executive “mention[ed] a build-up of inventory when they visited the Lasertec headquarters – and indicated skepticism of Lasertec's inventory valuation: “I can tell you right off the bat...they haven't taken any impairments against their balance sheet, so they're probably over-valuing their inventory far greater than it's actually worth...a lot of these companies are sitting on so much inventory....”

Marvell witnessed “a build-up of inventory” during Lasertec visit; “lot of these companies are sitting on so much inventory” due to 5N slowdown

Q: “What was the overall feedback from the [Marvell executive – title redacted] who visited Lasertec?”

A: “[T]hey did mention **a build-up of inventory when they visited the Lasertec headquarters**...I can tell you right off the bat that the reason why that is inducements probably because they **haven't taken any impairments against their balance sheet, so they're probably over-valuing their inventory far greater than what it's actually worth.**”

Q: **“What was the inventory comment that [Marvel executive – title redacted] made?** It's very interesting because that's a second data point, so I'm just trying to see how credible it is.”

A: **“There's credibility to it because that's what I've also heard as well. A lot of these companies are sitting on so much inventory** because the 5 nanometer, the market size of it, has basically softened to a significant degree from 2022 to currently. I can give you some additional context. In 2021, when we forecasted a 5-nanometer wafer throughput for our needs to, let's say, TSMC, we were forecasting that by 2023, we would need about 15,000 to 18,000 5-nanometer wafers. And our total throughput was only about 4,000-5,000. Now, when we did that forecast in 2021, that's when—you're not just forecasting TSMC but you're forecasting to all your hardware needs, all of your hardware manufacturers that you work with...**everybody is sitting on additional inventory...that's an issue across all 5-nanometer architecture.**” – Senior employee at Marvell Technology

Lasertec is a relatively simple accounting fraud - one can easily estimate what its normalized inventory should be, and therefore the level by which both inventory and earnings are inflated and the size of the inevitable impairment charge and retained earnings hit. Normalized inventory may be calculated by using 5 methods: 1) the 3-year average of inventory as a percent of sales across ASML, Applied Materials, Lam, Tokyo Electron, and KLA – 26% on average with little dispersion, and hence a robust measure, given similar product and business characteristics; 2) 3-year average of inventory as a percent of assets across the same peer group; 3) Lasertec’s historical, “normal” inventory as a percent of sales, from 2010 to 2017, immediately prior to its EUV product launch and before it spiked to unusual levels; 4) as a “normal” percent of its assets with the same methodology; and 5) by marking down its inventory by the 50-75% discount it is quietly offering to customers, per our research interviews, thereby adjusting it to “market value.”

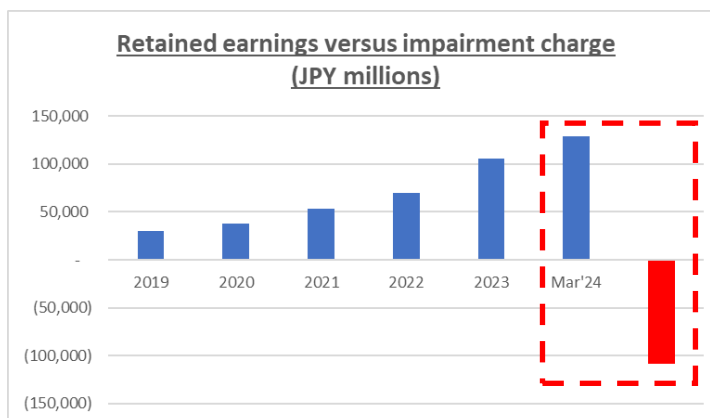
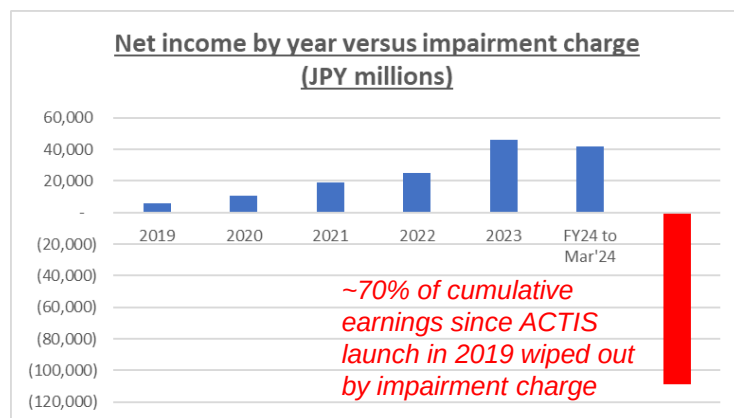
First two methodologies – normalizing inventory to peer group % of sales; and to % of assets; other methods shown in next chapter

Method 1: Normalize inventory to peer group % of sales	USD	JPY
Lasertec inventory as % of sales, 3-year avg	97%	97%
Peer group 3-year avg	26%	26%
Lasertec LTM revenue	1,525	230,604
Implied normalized inventory (peer group avg x LTM revenue)	397	59,957
Current Lasertec inventory, Mar '24	1,069	161,627
Implied impairment charge (current inventory less normalized #)	673	101,670
Implied impairment as % of current inventory	63%	63%

Method 2: Normalize inventory to peer group % of assets	USD	JPY
Current Lasertec inventory as % of assets	62%	62%
Peer group 3-year avg	21%	21%
Lasertec total assets Mar '24	1,713	259,146
Implied impairment charge	709	107,206
Implied impairment as % of current inventory	66%	66%

All five methods for estimating the amount by which Lasertec is fraudulently valuing inventory – and therefore earnings, as well as the magnitude of the impairment charge it is trying to avoid – arrive at roughly the same figure, one that is devastating and indicates the degree to which its stock price has been based on an illusion: an inevitable 65-70% write-down of its \$1.1B / ¥162B of inventory as of Mar '24 – a ~\$700-750MM / ¥105-113B loss that would wipe out ~70% of its cumulative earnings since the launch of its ACTIS product in late 2019 through the most recent quarter. The charge would also wipe out ~85% of its retained earnings of \$832MM / ¥129B, as of the most recent quarter. The loss and retained earnings hit we think is imminent far exceeds what Olympus incurred in 2012 in the immediate aftermath of its scandal (¥49B/\$300MM and ¥53B/\$460MM, respectively); and is considerable relative to Toshiba's loss and retained earnings hit of ¥400B/\$2.6B and ¥378B/\$2.4B in 2014/2016 – a sobering fact given Lasertec is only 3.5% of Toshiba's revenue of ¥6,300B/\$40B at the time

Net income and retained earnings versus looming impairment charge



The first two calculations for normalizing inventory - to peer group percent of sales and assets - were shown two pages ago. We show below the remaining three methodologies: normalizing inventory to Lasertec's historical, "normal" inventory as a percent of sales and assets; and by adjusting it to market value by applying the 50-75% discount quietly being shopped to key customers.

Additional three methodologies – normalizing inventory to peer group % of sales; and to % of assets

Method 3: Normalize inventory to Lasertec average pre-EUV % of sales	USD	JPY
Lasertec inventory as % of sales, 3-year avg	97%	97%
Lasertec pre-EUV average % of sales (2010-2016)	33%	33%
Implied impairment charge	566	85,528
Implied impairment as % of current inventory	53%	53%

Method 4: Normalize inventory to Lasertec average pre-EUV % of assets	USD	JPY
Current Lasertec inventory as % of assets	62%	62%
Lasertec pre-EUV average % of assets (2010-2016)	19%	19%
Implied impairment charge	744	112,389
Implied impairment as % of current inventory	70%	70%

Method 5: Reduce inventory value by Lasertec's 50-75% discount	USD	JPY
Discount, assume 62.5% as average of 50-75% discount	62.5%	62.5%
Current Lasertec inventory, Mar '24	1,069	161,627
Implied impairment charge	668	101,017
Implied impairment as % of current inventory	63%	63%

3. The origin of the accounting fraud: Lasertec's flagship ACTIS actinic EUV mask inspection system is defective with a fatally flawed EUV light source – the core of any EUV system – in addition to insurmountable problems with other key components like the optics and mask stage. Shockingly, it has a \$23B market cap as an EUV story but lacks an EUV technology that actually works.

Lasertec's Ushio-supplied EUV source is plagued by a number of fatal problems: 1) contamination and debris from tin splatter; 2) instability and flickering which leads to false positives and other mask inspection failures; 3) onerous cleaning and maintenance that leads to a pitifully low uptime and throughput; and 4) a short lifespan that drives frequent replacement and a burdensome cost of ownership. We begin with contamination, which results from the ignition of molten tin into plasma prior to the production and collection of EUV light. An ex-KLA executive in a leadership role in their EUV mask inspection group – who has extensively interacted with Lasertec and is intimately familiar with their technology - described “molten tin that splashes everywhere” with “puddles in the air” that “coat everything,” with “no way out” of the problem via attempts at “debris mitigation.”

Lasertec EUV light source ignites tin and splashes it everywhere, creating puddles of tin splatter, debris, and contamination; debris mitigation approaches don't work; “no way out”

Q: “You also mentioned a problem with the stability of their light source and other issues. What's the problem with their source? You said they're trying to switch the source to somebody else now.”

A: “Yes. This source is based on electrical discharge between two electrodes. The discharged light between them and then, if you think about two propellers, like two fans facing each other. I don't know if you know how hydraulic torque converters in pneumatic ear boxes are made, but think of two fans, and there is **molten tin that splashes in between those scans and creates a sustaining puddle in the air, almost like a supersonic jet, you have those puddles** - a three-dimensional donut - the same thing. And now, you have these two electrodes that discharge, almost like lightning, and this ignites the tin and turns it to plasma, and from there, you harvest the EUV, all shine with EUV light. But the problem is that not just EUV is emitting, all the **molten tin is splashing everywhere. Of course, they have a lot of - they call it “tin mitigation” or “debris mitigation,” but it's not enough. The amount of tin that splashes coats everything and propagates upstream, and there is no way out. It's too messy. That's why there are now—there have been in the last two years doing this project, Urashima.**” –Former KLA executive in a leadership role in their EUV mask inspection group

Contamination from tin-produced plasma is a well-known and long-standing obstacle in the scientific literature on EUV light source development, as igniting tin is the only way to produce the correct wavelength at the right power. The problem mainly afflicts the laser-assisted discharge-produced-plasma (LDP) approach employed by Lasertec, in contrast to ASML's laser-produced plasma (LPP) where the plasma is better controlled. In 2021, Ushio and Intel presented an SPIE paper describing the Ushio EUV source technology used inside Lasertec's machine, describing “several technical challenges and infrastructure gaps that need to be resolved to make EUVL suitable for high volume manufacturing...these gaps relate to...a stable and reliable high power EUV source...”

March 2021 conference paper jointly authored by Ushio and Intel employees

23 March 2021

Laser-assisted discharge produced plasma (LDP) EUV source for actinic patterned mask inspection (APMI)

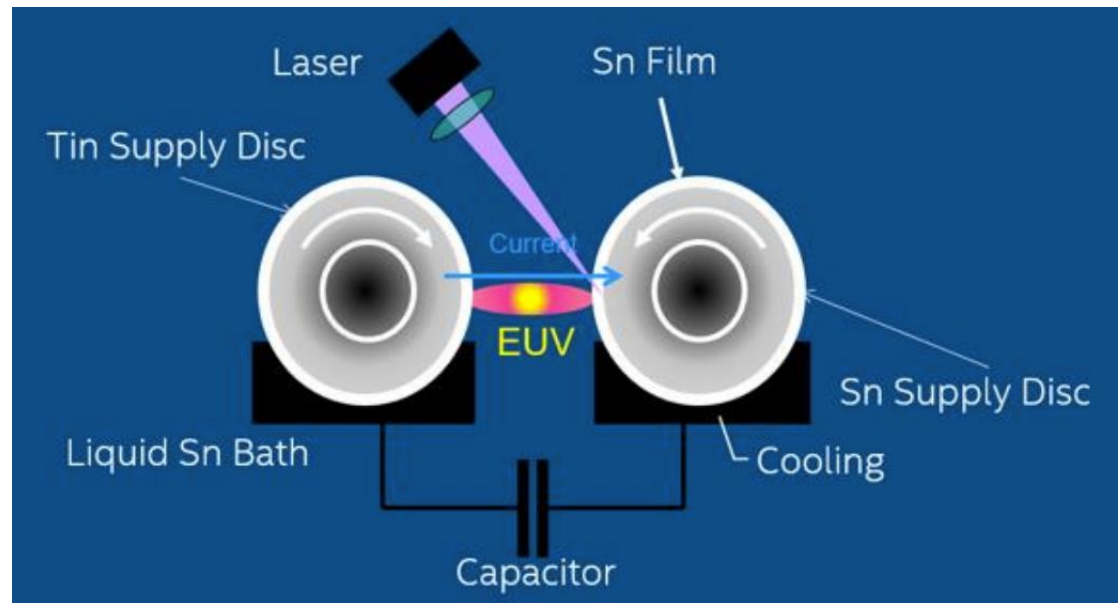
Safak Sayan, Kishore Chakravorty, Yusuke Teramoto, Takahiro Shirai, Shunichi Morimoto, Hidenori Watanabe, Yoshihiko Sato, Kazuya Aoki, Ted Liang, Yoshihiro Tezuka, Marieke Jager, Firoz Ghadiali, Frank Abboud, Steven L. Carson

Abstract

Improved lithography resolution provided by EUVL simplifies the patterning process and makes it possible to use less restrictive design rules. This in turn enables cost effective scaling with extendibility. There are several technical challenges and infrastructure gaps that need to be resolved to make EUVL suitable for high volume manufacturing (HVM). These gaps relate to development of a stable and reliable high power EUV source, EUV resist and EUV compatible photomask infrastructure. Realization of Actinic patterned mask

The paper describes the key components for the EUV source: a liquid “hot tin bath”; a “source head” consisting of two rotating metal discs submerged in liquid tin; and an electrode/capacitor bank that ignites the tin mixture similar to a spark plug in a car, creating plasma that emits EUV light. It is self-evident that the process would create tin splatter and clouds of debris inside the Lasertec tool.

March 2021 Ushio/Intel paper illustrates the key components of the EUV light source, excerpts below



“LDP EUV source is the most critical enabler of actinic patterned mask inspection technology”

“Laser irradiation not only vaporizes tin and forms a tin vapor cloud...”

A CAD schematic and actual photograph of the Ushio EUV light source that powers Lasertec's tool illustrates its electrical/mechanical complexity and its vulnerability to clouds of molten tin, gas, and splatter.

March 2021 Ushio/Intel paper shows the key components and an actual EUV source

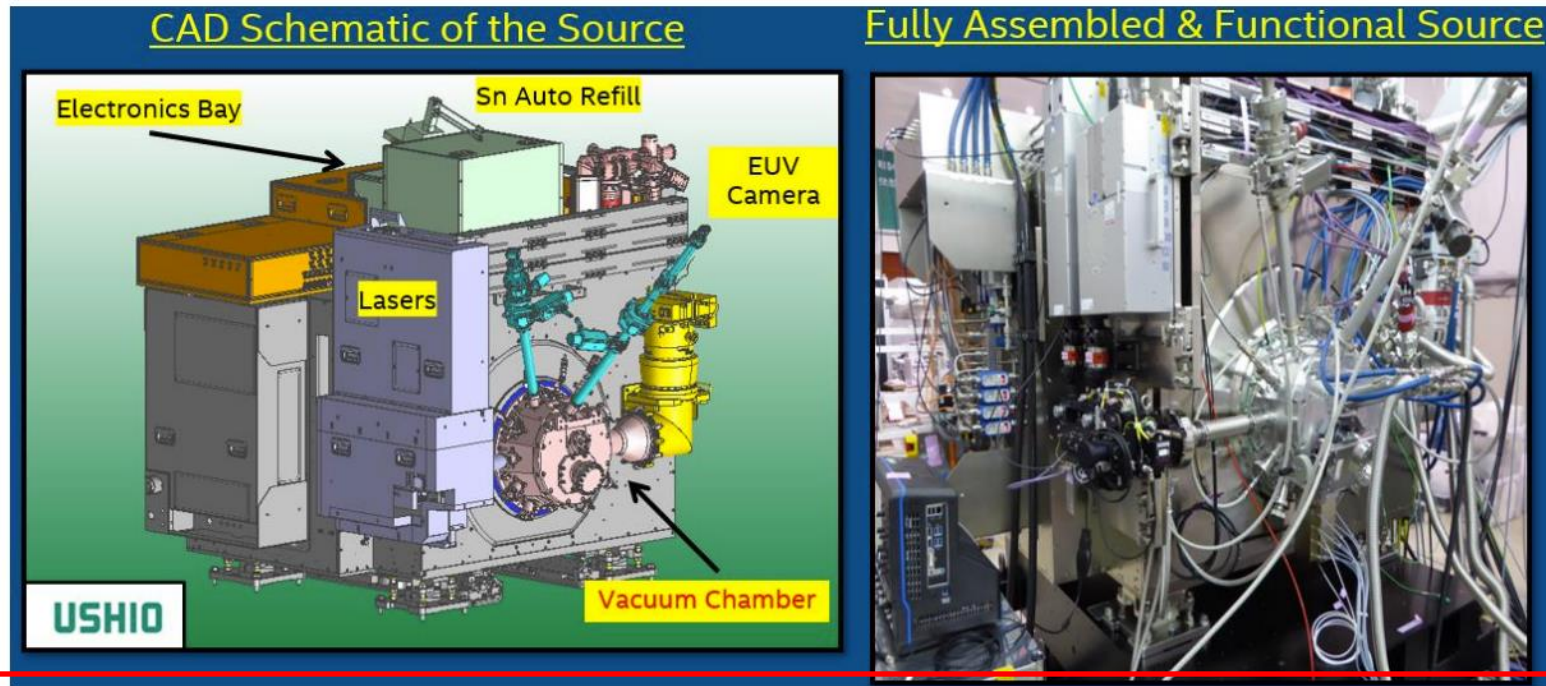


Figure.2 CAD schematics of the Ushio LDP EUV source (left panel), and fully assembled and functional source picture (right panel)

The tin contamination problem that plagues Lasertec's EUV light source is detailed in various papers. We quote three representative ones below. One describes three ways that tin debris "can damage the collector optic": "sputtering, implantation, and deposition" – two of which are "irreversible." A second further classifies the debris into three different types: "micro-particles or liquid Sn [tin] droplets, the slow atomic-ionic debris, and fast ionic debris."

2017 paper by plasma physicists and others at U.C. Berkeley, Applied Materials et al

Removal of Tin from Extreme Ultraviolet Collector Optics by *In-Situ* Hydrogen Plasma Etching

Abstract Extreme ultraviolet (EUV) lithography produces 13.5 nm light by irradiating a droplet of molten Sn with a laser, creating a dense, hot laser-produced plasma and ionizing the Sn to the + 8 through + 12 states. An unwanted by-product is deposition of Sn debris on the collector optic, which focuses the EUV light emitting from the plasma. Consequently, collector reflectivity is degraded. Reflectivity restoration can be accomplished by

This Sn-based plasma emits Sn ion and neutral debris, which can damage the collector optic in three ways: sputtering, implantation, and deposition. The first two damage processes are irreversible and are caused by the high energies (1–10 keV) of the ion debris [8].

2022 paper by Chinese scientists

The development of laser-produced plasma EUV light source

occur, caused by large heat load from the discharge plasma⁸³. The corroded electrode materials are introduced onto the collector as debris contamination, which reduces the lifetime of the collector mirror. In order to of two rotating electrodes was employed^{39,40,43}. Generally, the debris generated by Sn-based DPP sources can be divided into three different types in addition to electrode erosion: the micro-particles or liquid Sn droplets, the slow atomic-ionic debris and the fast ionic debris⁸³. Furthermore, by using

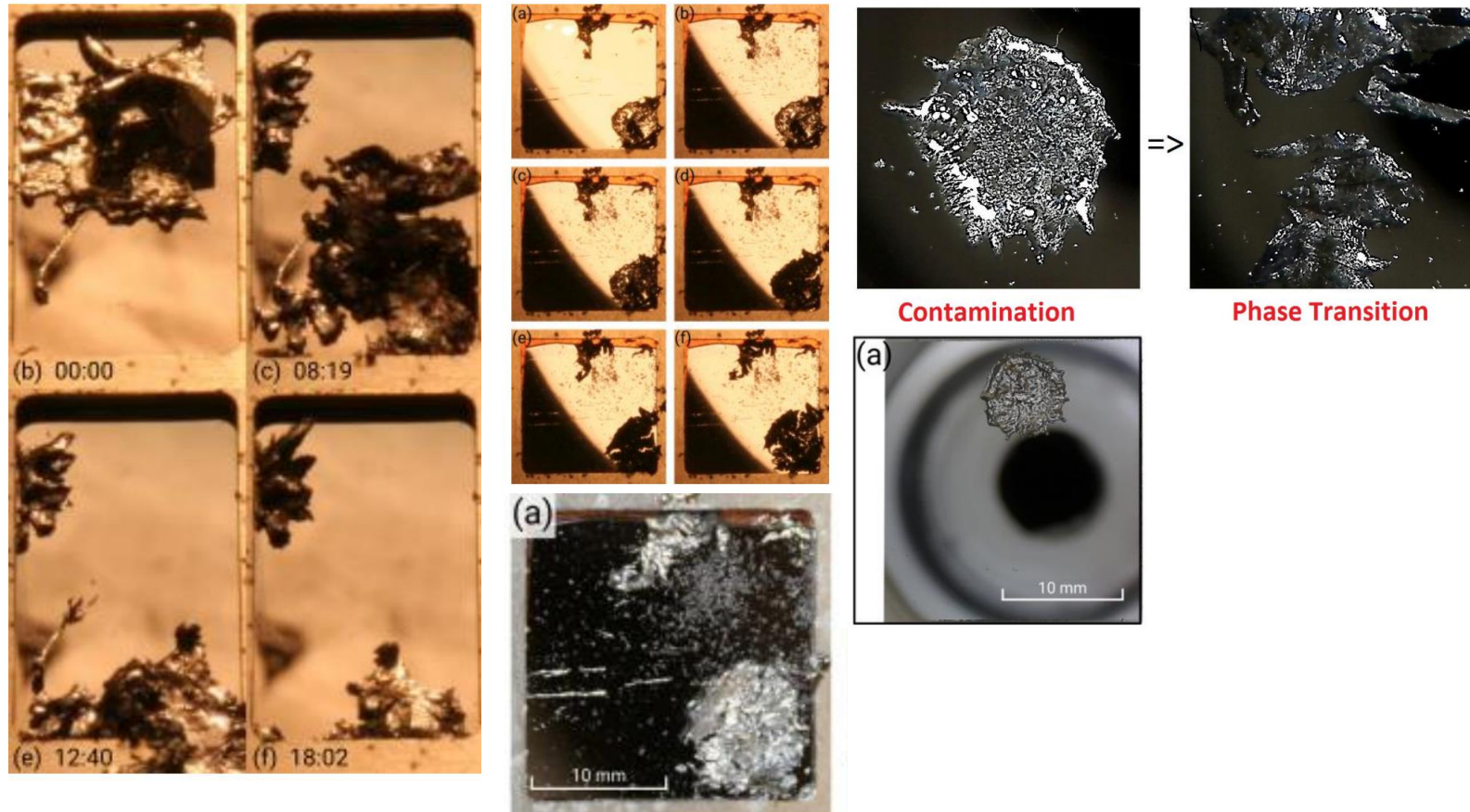
2019 presentation by German physicists

In-situ Cryogenic Cleaning Techniques for Tin-Contaminated EUV Mirrors

- Considerable contamination of the collector mirror occurs by tin debris
- Chip manufacturing needs increased source availability and productivity
- Increase of collector lifetime needed: more than several months, > 400 Gp
- Internal tin-cleaning by plasma etching is too slow for thick deposits

The papers provide representative images of what molten tin splatter and debris looks like when it coats, contaminates, and damages the numerous mirrors and optics inside Lasertec's EUV mask inspection machine.

Representative images of contamination from tin plasma used to produce EUV light, from several papers



A KLA presentation on EUV mask inspection stressed the dangers of particle contamination – the tools “must find defects but not contaminate!” and must be “the cleanest machines among all tools used in a fab/mask shop.” Unfortunately, as the photos of tin splatter show, “plasma based sources are the dirtiest parts in EUV tools...in inspectors, the problem of isolating the mask from source debris is more severe than in scanners” – “bounced/released micro-particles, liquid tin splashing, and accumulation (and sputtering) of deposited tin at the walls.”

KLA presentation “Source performance metrics for EUV mask inspection”



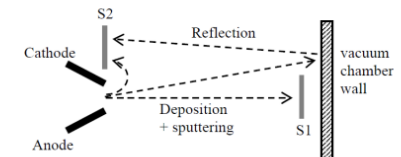
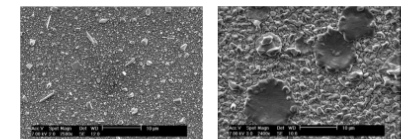
Source performance
metrics for EUV mask
inspection
Larissa Juschkin, Dan Wack, and
James Westphal

KLA

System and source cleanliness requirements

- EUV mask inspection, specifically final one at mask shop, must find defects but **not contaminate!**
⇒ These inspectors are the cleanest machines among all tools used in a fab / mask shop.
- Example Teron 640e: less than **0.1 PRP** (particles per reticle pass). *Requirement will tighten for future nodes!*
- Plasma based sources are the dirtiest parts in EUV tools. In inspectors, the problem of isolating the mask from source debris is more severe than in scanners.
 - Scanner best values: 1 adder per 10k wafers, 140 wafers/hr => ~70 hours (status 2016: 120 adders).
 - ⇒ Assuming 5 hrs per mask inspection, it corresponds to at least 1 adder per ~15 inspections! (2016-equivalent: ~10 adders per 1 inspection).
 - Based on scanner research, tin may reach the mask due to secondary effects of bounced/released micro-particles, liquid tin splashing, and accumulation (and sputtering) of deposited tin at the walls.

I. V. Fomenkov, “EUV sources for high-volume manufacturing”,
from *EUV Lithography*, ed. V. Bakshi (2018)



K. Gielissen, “The nature and characteristics of particles produced by EUV sources: exploration, prevention and mitigation”, PhD thesis TU Eindhoven, 2009

The seriousness of the contamination problem was confirmed by key Lasertec customers such as TSMC and Intel; multiple Lasertec field service engineers; Lasertec's EUV source technology suppliers such as Ushio and Isteq; and numerous other experts. The pages that follow will detail our research interviews with these experts, but first we share further color from an ex-KLA executive in a leadership role in their EUV mask inspection group, given his recent interactions with Lasertec and detailed knowledge of their product. He described Lasertec's Ushio-supplied source as "ridiculous" and inherently flawed and stated that they "made an awful mistake" as they were desperate to launch a product given pressure from Intel: "they took a source without understanding the complexity...and the problems associated with damage from debris...they did this because it was the only available source."

Lasertec's "ridiculous" light source is inherently flawed with excess power; "made an awful mistake" as were desperate for a solution; "damage from debris"; KLA source is more appropriately powered

"They have a tool for blank inspection but blank inspection needs a fraction of the light. It doesn't need the brightness. There's a difference between brightness and power, and the mistake that they made was that they took an Ushio source with the right brightness but it has 400x more power than actually needed. **That's how ridiculous it is.** Two things are important for inspection: one is the brightness and the second is the power, which means that with the same brightness, I can illuminate the spot 1 mm by 1 mm or I can illuminate 25x25. The area difference is 400x, 25x25. But the brightness is still tiny; each spot has the same plug or the same brightness. For inspection, what is important, it's important to have the brightness but the spoke is 1 mm x 1 mm, and **they made an awful mistake. They took a source without understanding the complexity of the source and the problems associated with damage from debris.** So, instead of going with something lighter like KLA that will create much less of a problem - **they did this because it was the only available source.**" –Former KLA executive in a leadership role in their EUV mask inspection group

The expert elaborated that the EUV source “is terrible” and “creates a mess” given its use of discharge electrodes - similar to car spark plugs – and propellers splashing in molten tin, in contrast to ASML’s superior laser-based solution as well as the xenon-based source KLA is developing. He stated that off-gassing from tin plasma and volatile organic compounds are additional sources of mask contamination, undermining the machine’s sole purpose of mask inspection: “one is particles...two is VOC...off-gassing...which coats the optics and produces damage to the optical surfaces...the particles propagate and eventually find themselves on the mask and you find these specks.”

Ushio source “is terrible” and “creates a mess”

A: **“The KLA Xenon works.** The Ushio tin source failed. ASML is different. They're using tin and they're using IR laser to ignite it. That's ASML. The KLA technology is published, It uses a frozen Xenon gas with IR laser as the originator. **Ushio uses molten tin like ASML, but instead of using IR, they're using discharge electrodes, which is terrible. And the concept of maintaining two propellers creates a mess.**”

Q: “It sounds like it takes ten years to figure it out. It sounds very complex.”

A: “Correct...I think **that Isteq source will not be done in a professional way and it cannot work. You need to understand Isteq was done for research. Research is done in bursts, not continuously. For inspection, you need something to work 24/7, and that's it. And it's really hard. It's totally different.**” –Former KLA executive in a leadership role in their EUV mask inspection group

Off-gassing is an additional source of contamination from the EUV light source

“It's not just particles, there is another aspect coming from the source in terms of contamination, you need to remember two things: **One is particles. Two is VOC, volatile organic compounds, basically, off-gassing.** Off-gassing is like, let's say, a particle machine has oil and you heat it up, it will propagate and release gases - alcohol or oil – as VOC's, especially with a reaction with EUV light, which **coats the optics and produces damage to the optical surfaces.** When you deal with deep UVs, we learned this, and EUV is ten-fold harder. They have two problems: one, the particles propagate and **eventually find themselves on the mask and you find these specks.** And second, they propagate in the system and they coat all the optics, including the image in the pristine optics.” –Former KLA executive in a leadership role in their EUV mask inspection group

Particles generated by Lasertec's machine leads to "degradation of the imaging optics...and particulation on the actual mask," which "you cannot clean" as it coats the source of the EUV light": "gradually you lose it and you have to replace it." The KLA expert indicated the problem degrades each of the key optical components, including the EUV collector and four mirrors inside the tool.

Numerous problems such as "degradation of imaging optics" and "particulation" onto the mask; debris coating requires imaging optics to be replaced annually as well as numerous mirrors at a cost of millions per year

Q: "You said the optics are getting off-gassing and particles from the stage. So, you said they have to replace the optics or they have to clean those as well?"

A: "Together with the problem with the source of off-gassing, also **the stage has a problem with particulation and off-gassing, all of them contribute now to the degradation of the imaging optics and also the particulation on the actual mask** of the inspection. And this one, you cannot clean. It coats the source of the EUV light until it develops a few **mono-layers of coating of this organic composition, and gradually, you lose it, and you have to replace it."**

Q: "So, you have to replace the optics. How often do they have to replace the optics, and how much does that cost?"

A: "I heard **every year that you need to replace the imaging optics right now...the coating and the amount of VOC are such** that every year they have to replace the optics, and this costs—the **refurbishing of those optics costs at least a million dollars or more. They need to replace mirror number two, mirror number four, and that costs at least half a million dollars each."**

—Former KLA executive in a leadership role in their EUV mask inspection group

Off-gassing coats the collector and other key optics components; cost millions to replace; KLA executive witnessed Lasertec optics having to be "repeatedly replaced"

Q: "The off-gassing is coating the optics?"

A: "There is one **big collector next to the source. They have to frequently replace it in maintenance.** Then after the collector, Lasertec has four mirrors. And those **four mirrors are coating the imaging optics."**

Q: "When you say collector, you're referring to a mirror?"

A: "Yes, everything in EUV is mirrors. There is no lens, a different curvature. So, the collector is number one. And there are another four surfaces or four mirrors that we call **imaging mirrors. Each of them costs around a million dollars to produce.** That's how much it costs. If you start and gradually, those are maybe once a year or once a half a year they need to replace it because they have problems. **They have problems there as well. I know that as a fact. I was in the shop and I saw them repeatedly replacing the optics."** —Former KLA executive in a leadership role in their EUV mask inspection group

Lasertec's particle contamination problem makes it highly problematic for P2 mask inspection – the only step in the inspection process where its EUV tool can be used. Mask inspection involves four steps, each with distinct inspection tools: P0, P1, P2, and P3. The first step P0 refers to inspection of the mask blank in the blank shop, before it's etched with a chip pattern. Customers typically use Lasertec's relatively inexpensive ABICS blank inspection tool. **P1 inspection occurs after mask writing, where KLA's Teron inspection machines dominate and have a monopoly** due to their unique "Die to Database" technology. Defects discovered by the Teron in P1 are then fixed with a Carl Zeiss tool, **after which P2 inspection occurs – the primary use case for Lasertec's ACTIS EUV tool.**

KLA owns the P1 inspection in the mask shop with their Teron product line and no one else can compete

A: **"P0 is done in the blank shop. That's a Lasertec tool that will remain a Lasertec tool. KLA is not there."**

Q: "But that's a pretty cheap tool, right?"

A: "Yes, the source, buying from Energetiq, and that costs \$1 million. I think Ushio bought that company as well. From there, it goes to the mask shop. In the mask shop, the first tool is the EUV writer that writes the mask. **Then you have P1, which is post-writing inspection.** After that, it goes to Carl Zeiss repair tool. You can think about it as a multimillion-dollar tweezer that you can see all the problems that were discovered by KLA, and the **KLA P1 tool is Teron 640.**"

Q: "You're saying KLA owns the P1 market, and nobody's going to beat them. They have a monopoly in the P1 market."

A: **"There's a monopoly there because KLA is the only tool that has Die to Database.** You can compare the right chip to the left chip, and you know that if three are the same, there is no problem."

Q: "So, KLA is going to keep the P1 market. Then P2, that's where Lasertec is used?"

A: **"P2 is before you put the pellicle after you repair. P3 is after you put the pellicle.** Let's start with P3 and go back to P2. After you put the pellicle, the concern is something will fold them up and you have to inspect P3 before you chip it with a stamp. However, in order to know—we talked about die-to-database, which is important to understand because Lasertec will claim that they are working on it. KLA developed this die-to-database. You compensate all the optical distortion and build a synthetic image. That is a huge task—more than 10 years of development. So, **that's perfected and that's what KLA is good at, so they dominate the P1.**" –Former KLA executive in a leadership role in their EUV mask inspection group

However, since Lasertec's EUV mask inspection tool introduces new particle contamination in P2, key customers like Intel and TSMC are forced to then do a redundant inspection with the same KLA Teron tool used in P1 – defeating Lasertec's entire purpose and creating issues prior to the final step, called P3 which refers to inspection after a thin protective membrane called a "pellicle" is placed on the mask. As a result, Lasertec "customers are furious because they need to maintain another KLA tool as a backup on P2": "the problem that the customer has...because Lasertec introduces particles as you clean and put the pellicle on...so you defeat the purpose."

Lasertec is used for P2 but "defeats the purpose" as it "introduces particles" and debris and requires duplicate work with a KLA tool; "customers are furious"

A: "P2, P3 is after pelliclization, and the problem is that in order to know that something happened during the pelliclization. You need to compare P2 and P3. Basically, there is a map to do the P2 and P3 in the same tool. Because you inspect the change before and after. And now **the problem that the customer has is they cannot put a Lasertec tool there because Lasertec introduces particles as you clean and put the pellicle on. So, you defeat the purpose.** So, what they do is inspect the Lasertec, they clean it and then they inspect it with KLA Teron again and then send it to pelliclization. And after that, again, the Lasertec because after pelliclization, it doesn't matter—"

Q: "Where does Lasertec play? P1 is KLA, so Lasertec is P2 and P3?"

A: "Currently, P2 and P3, but the **customers are furious because they need to maintain another KLA tool as a backup on P2. And look what's happening in the mask right now.**"

Q: "And that's because the Lasertec tool introduces debris?"

A: **"Correct. Particles, yes. Absolutely."** - Former KLA executive in a leadership role in their EUV mask inspection group

After particle contamination, the second fatal problem with Lasertec's EUV light source is instability and flickering, which leads to false positives and other mask inspection failures. The light source is inherently unstable for various reasons, such as electrode erosion that plagues their discharge-produced plasma approach. We interviewed four on-site field service engineers tasked with addressing problems with Lasertec's actinic mask inspection (ACTIS) machines and the EUV light source in each one. Three of the field engineers were recently employed by Lasertec and one is currently at Ushio to specifically address EUV source failures. All four confirmed that the machines were so instable that each one required 3 full-time Lasertec field service engineers, for a total of 9 just in Intel's mask shop alone, which they stated has 3 Lasertec actinic inspection tools: "nine of us for the San Jose location; Ushio separately also "had a whole team that would work on it"; a "whole system of ejecting tin droplets...was probably the size of half" a car.

Lasertec needed 3 full time service technicians per tool, and Ushio had a separate team to fix problems with the light source

A: "The basic concept for creating EUV is you have to transmute tin to a plasma state. The traditional method is using a high-powered CO2 laser. There is a whole laser module. There'd be a chamber it's running in and the cooling system, and the dispensation of the tin itself. That whole system of **ejecting tin droplets and then shooting with a laser and directing it within the tool was probably the size of half a Mini-Cooper.**"

Q: "Did Ushio have a person on site like Lasertec did, just to work on the light source?"

A: **"Yes. They had a whole team that would work on it.** We had some of the guys on speed dial if we needed their direct support.

Q: "Were you the only Lasertec person on site, or were there other people onsite who were basically full-time along with you?"

A: "We had a team of three people per tool, so **nine of us for the San Jose location.**" – Former Lasertec field service engineer who worked in the Intel mask shop

The field service engineers provided color that was consistent and across all key customers such as Intel and TSMC. One engineer we interviewed was based full-time at Intel's mask shop in California; a second at their fab in Oregon, along with the current Ushio engineer; and one was based at TSMC's fab in Taiwan. The one at Intel's mask shop stated that "across the three machines, we'd have daily issues" and that they were forced to "sacrifice" throughput and functionality "for maintaining uptime." He confirmed that Intel's fabs in Oregon experienced the same problems with Lasertec's machine as the mask shop in California.

All three Intel machines had "daily issues"; had to sacrifice throughput to keep them operational

A: "If it got into a state where they weren't able to utilize it, then we would get it back into a state where they could utilize it. We helped them troubleshoot certain issues."

Q: "How often was the machine having issues?"

A: "**Across the three machines, we'd have daily issues**. Some consumable parts within, such as mirrors and the like, that when they were approaching their end of life, we'd see an uptake in issues, but it wouldn't warrant shutting the tool down just because the end goal was trying to maintain 24/7 operation based on customer needs. So, there was a balance between **sacrificing a little bit of speed and functionality for maintaining uptime.**" – Former Lasertec field service engineer who worked in the Intel mask shop

Similar product issues in Oregon fabs as in the Santa Clara mask shop

Q: "Did you hear anything from other field service colleagues about what was going on at the Intel fabs in Oregon with the Lasertec machines? Were there the same issues?"

A: "Yeah, we would cross-reference information. Like if they struggled with something, then they would usually contact us for support. **I'd say similar issues**..." – Former Lasertec field service engineer who worked in the Intel mask shop

The engineers corroborated information we received from a KLA mask inspection executive that Lasertec's EUV source is highly instable due to electrode erosion: "a problem because the tin when it splashes between those two fans and the discharge has electrodes" which "have erosion and therefore it's not stable and it's changing all the time." An ex-field engineer stated he "was hands-on the ACTIS line, particularly the ACTIS A150" and that "it was basically like a flickering candle": "having high repeatability became a big challenge"; "the light drifted out of focus or too far out that Intel engineers couldn't step in and resolve it...."

ACTIS light source "main problems" – "flickering candle"; "high repeatability" was a "big challenge"; "drifted out of focus"

"I was a field service engineer at the home office location in San Jose for Lasertec USA. I was hands-on on the ACTIS line, particularly the ACTIS A150. These were for mask inspection exclusively. There were three machines...I'd say there were definitely a lot of challenges with the tool, and the light source in and of itself was out of Lasertec's hand at the time with the A150. Ushio would do the light source replacements and maintenance. There were engineering differences to try to resolve some of the main problems that we experienced with Ushio - it was basically like a flickering candle when you were trying to utilize it like a laser. Having high repeatability became a big challenge. They needed someone on the back end, like someone in my position, to be able to call, worst case, if the light drifted out of focus or too far out that Intel engineers couldn't step in and resolve it themselves." – Former Lasertec field service engineer who worked in the Intel mask shop

Lasertec light source is highly instable due to electrode erosion where the tin is ignited to create plasma

Q: "You said aside from the debris, you said there was an issue with the stability of the light and it's flickering."

A: "Yeah. Because the steady state, how much you can bond that tin, is not perfect. If you think about ASML, how they manage the tin, they are using a droplet generator. They are making sure that every drop is the same, and then they knock that with IR, and you have the EUV emitting for each drop the same way, unlike the **Ushio source, which has a problem because the tin, when it splashes between those two fans and the discharge has electrodes, the electrodes have erosion and, therefore, it's not stable, and it's changing all the time.**" –Former KLA executive in a leadership role in their EUV mask inspection group

Another former Lasertec application engineer, who was primarily allocated to Intel, stated that “the ACTIS A150...has a lot of problems” especially “regarding the chamber structure and the source” which “tends to be unstable”: “customers want better maintenance and stability...the problem was the light source...it requires more time for maintenance once the machine breaks down.” The Ushio engineer added many important details, stating that Ushio has 20 field engineers in the US to service the light sources in the small number of ACTIS machines installed in total, plus another 10 for managing parts: “I’m on the Ushio team that maintains the light sources”; Intel’s “flagship cleanroom is in Oregon, D1X.”

ACTIS A150 “has a lot of problems” especially with “the chamber structure and the source”; source is “unstable”; customers want “better maintenance and stability”

“The ACTIS A150 is an earlier version, and it has a lot of problems. When the company switched to EUV, it had to change the structure of the chamber, and there were some problems regarding the chamber structure and the source. Initially, they used the source of a company called Ushio, and later, they upgraded that source...The laser source tends to be unstable, and it’s hard to make it. Customers want better maintenance and stability...the problem was the light source...it requires more time for maintenance once the machine breaks down.” – Former Lasertec product engineer; 20 years in the industry and previously part of the EUV task force at one of the top 3 semiconductor companies in the world

Ushio has about 20 field engineers in the US to service the light sources in just a handful of A150 machines; plus another 10 staff

A: “I’m on the Ushio team that maintains the light sources in Hillsboro. Their flagship cleanroom is in Oregon, D1X. On that site, there are multiple fabs, but D1X is the biggest and the newest and is kind of like their skunkworks.”

Q: “Ushio is there primarily to service the Ushio components in Lasertec machines for EUV mask inspection - is that correct?”

A: “That is correct.”

Q: “How many Ushio people are there in the U.S. servicing these Lasertec machines?”

A: “In the U.S., I think about 20...And then we have a team of about 10 that just helps do the work, exchanging parts mostly” – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

The Ushio engineer - we emphasize that Ushio is Lasertec's EUV source supplier - stated that "Lasertec causes a lot of headaches for their end user" and that each machine requires "a lot of work" in order "to keep it running": "on the light source side we have some issues...things break." He stated that "the struggle is to reduce" the light source downtime, and parts that need to be replaced;" that the source fails rapidly and has to be replaced every 2 to 4 weeks; that the light source is compromised to run "dimmer" to extend its lifetime by a week or two; that it breaks down faster in the mask shop because it has to run "at a much higher brightness"; and that Lasertec is "lucky that the competitors didn't jump on making pattern inspection machines right off the bat."

Lasertec "causes a lot of headaches" its customers; "lot of work...to keep it running"; the light source "breaks"

"There is a lot of work that the Lasertec engineers have to put into it to keep it running. And then, on the light source side, we have some issues. Things break, and we repair them...it does seem that the product was rushed and that Lasertec was out of their element when they got thrown into this...Lasertec causes a lot of headaches for their end user. They're lucky that the competitors didn't jump on making pattern inspection machines right off the bat. Lasertec is not nearly as well established in the fab as KLA or obviously, ASML." – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

"Struggle" to "reduce light source downtime"; source has to be replaced every 2 to 4 weeks; brightness is compromised

A: "The struggle is to reduce the amount of light source downtime, and parts need to be replaced. It takes a finite amount of time to replace them, and the customer and end-user want it to be done as quickly as possible."

Q: "How often does the light source need servicing? How often does it break down and need to be cleaned and whatever?"

A: "For Ushio, we keep to a pretty regular one-month schedule in production."

Q: "Does that mean it needs to be taken offline and fixed, or the light source needs to be replaced?"

A: "Yeah, it's got to be taken offline, and parts have to be replaced. In Hillsboro, they run the light sources dimmer so the light source can last about a month. At Santa Clara, they run it at a much higher brightness, a higher power, and they are changing parts every two weeks or two to three weeks." – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

The Ushio engineer elaborated on the mask inspection problems and failures caused by the EUV light source's instability and lack of brightness and power: “they get a dark frame or they get a false defect...it's light source fluctuation...the brightness is just going to fluctuate just as a function of the physics of the plasma.” He stated that the tin-based plasma reaction is poorly controlled: “the plasma has a lot of freedom to move...that causes fluctuation...a difficult thing to control”; “it's hard to get enough power” to “get the brightness up...as a result the plasma is smaller...that can create problems with not getting enough illumination into the tool.” With insufficient light source stability and brightness, “you'll get a frame where there's just not enough light and uniformity...you're going to have dark area and that's going to cause a false defect to show up.”

Ushio light source causes inspection problems such as dark frames and false defects due to light source fluctuation; problems to due to difficulties in controlling plasma; Urashima doesn't have enough power and illumination

A: “It's different issues, but the **same results happen on their inspections. They get a dark frame or they get a false defect...it's light source fluctuation. In these light sources, the brightness is just going to fluctuate** just as a function of the physics of the plasma. In the Ushio source there's a discharge between two electrodes. That means the **plasma has a lot of freedom to move. And so, that causes fluctuation and brightness. That's a difficult thing to control.** The other light source is laser-produced plasma. There's no discharge between electrodes. It's just the laser creates the plasma. That tends to be much more stable, but **it's hard to get enough power into that to get the brightness up. As a result, the plasma is usually smaller. That can create problems with not getting enough illumination into the tool.**”

Q: “What happens when you don't get enough illumination in the tool?”

A: “It's similar. **If the plasma is moving, every once in a while, you'll get a frame where there's just not enough light and uniformity. If you're not getting full brightness all the way to the edge of the frame, then you're going to have dark areas, and that's going to cause a false defect to show up**...multiple things can combine, so if a light source is starting to dim and there's a component on the tool side that's dimming, a mirror or lens or something, then they combine and cause problems.”—
Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

LinkedIn profiles of current and former field service engineers hint at the challenges: “tackled challenging problems regarding losing EUV light transmission through EUV optics”; “we needed to work with...troubles with our existing equipment”; “troubleshooting...equipment problem”; “cooperated with design engineers in Japan...to resolve complex maintenance issues”: “providing service support to customers encompassing...system breakdown.”

Lasertec/Ushio field service engineer bios on LinkedIn



Sr. Field Service Engineer

Lasertec USA · Full-time

Jul 2020 - Sep 2021 · 1 yr 3 mos

- Tackled challenging problems regarding losing EUV light transmission through EUV optics.

Lasertec

Field Service Engineer

Lasertec Corporation

Jul 2018 - Oct 2020 · 2 yrs 4 mos

Malta, New York, United States

• Cooperated with Design Engineers in Japan as well as other US Field Service Engineer to resolve complex maintenance issues and to conduct intricate recovery work

Lasertec

Field Service Engineer

Lasertec Corporation · Full-time

Jan 2021 - Oct 2021 · 10 mos

대한민국 경기도 화성시

EUV Photomask inspection 설비의 설치, 유지보수, Troubleshooting

a Japanese semiconductor equipment company. I have always achieved my goals by maximizing team work. As the company has grown in the EUV mask defect inspection field, the number of equipment orders has also rapidly increased with it. We needed to work with the new orders as well as continue to deal with PMs and troubles with our existing equipment. Although there were many responsibilities, as a team, we had made sure that not one single person was

Lasertec

Service Engineer

Lasertec Corporation · Full-time

Feb 2019 - Jun 2022 · 3 yrs 5 mos

新市區, 台灣

Customer service in mass production and RD FAB (TSMC)

• Trouble shooting in both process and equipment problem in photo mask product.(SPC/available time)



Field Services Engineer

Ushio America, Inc. · Full-time

Aug 2021 - Present · 2 yrs 11 mos

Santa Clara County, California, United States

Support Photo Masks inspection system at IMO (Intel Mask Operation). Performed daily to collect data and PM for AIP 01/02/03 (Automation Inspection Process). Mechanical, electrical, and leak check tool after PM, High Voltage Test 5000V for shortage circuit for source head Cathode and Anode.

Providing service support to customers encompassing system, system breakdown and preventative maintenance.

A leading EUV lithography expert close to all major players including Lasertec, ASML, and KLA - as well as the industry's three key customers TSMC, Intel, and Samsung – described Lasertec's source as an unmitigated disaster. He stated that Ushio was secretive about the defects, but that he received some data and modeled it, which revealed it was fatally flawed: "I asked Ushio what was wrong with their source, and they wouldn't say...if your source is not stable, then the intensity of light coming in through it fluctuates...so you have a stability issue...they gave us the data, and we had about six to seven different models of Ushio data...the bottom line is that there is something wrong."

Ushio source "is not stable" and "fluctuates"; Ushio is trying to hide their source problems

"I asked Ushio what was wrong with their source, and they wouldn't say. I can tell what it is because what happens in a mask inspection system is that your NA – numerical aperture - is very, very small, which means that you don't need only a very small source or a large source that you need for a scanner. But the problem with that is if your source is not stable, then the intensity of light coming in through it fluctuates. So, you have a stability issue." – Leading EUV lithography expert close to Lasertec's customers

Ushio light source data shows that "the bottom line is there is something wrong"

"We asked Ushio, please to share their experimental data in terms of plasma emissions. They gave us the data, and we had about six to seven different models of Ushio data...the bottom line is that there is something wrong." – Leading EUV lithography expert close to Lasertec's customers

The EUV lithography expert was highly critical of Intel's credibility with respect to Lasertec, noting their large investment and need to save face as its sponsor. He stated "the Ushio source [had] degradation," and criticized "people" who we presume to be Intel's mask inspection group: "...the same people who came to me and said, wow, we won the war and we got Lasertec...it's all good with Ushio...the same people two years later said yeah by the way everybody in the field has taken the source out..."; "Intel came in and did their song and dance a couple of years ago, saying we've got Lasertec working...we're the winner...the champ...but three years later we're hearing the Ushio source doesn't work."

Ushio source had "degradation"; Lasertec/Intel did a premature victory lap and then had to rip out the light sources

"The **Ushio source [had] degradation and they took it out**. So, the same people who came to me and said, wow, we won the war, and we got Lasertec; it's all good with Ushio. There was this celebration. The **same people came to me two years later and said, yeah, by the way, everybody in the field has taken the sources out, and we're replacing it with Isteq. Well, Isteq has not quite done their job**. I don't think they have their sources in the field—if they have one or two already, there's not enough data. So, if you ask my opinion, **Lasertec and Intel will come and do a song and dance and say, well, now we've got this new source supplier**, and everything is good, it's all wonderful and all that stuff. My response will be, well, **how much data do you have to say that because you were the same guys who came and said the same about Ushio? And I have not seen long-term performance data at all.**" – Leading EUV lithography expert close to Lasertec's customers

Intel outsourced actinic development to Lasertec and proclaimed themselves as "the winner" only to have the tool fail

"Intel said we're going to go ahead and play with Lasertec. The reason they went with Lasertec is that KLA wanted too much money upfront for investment...KLA also dropped the ball and they essentially laid off the whole team. So, **Intel said, we're going to develop this thing ourselves, and they went with Lasertec**. This is a strange business because everybody has to come in and show that they succeeded in doing a little song and dance. So, when **Intel came in and did their song and dance a couple of years ago, saying we've got Lasertec working. We're the first one; we're the winner; we're the champ. But three years later, we're hearing the Ushio source doesn't work, and now we have another source from a Russian company called Isteq.**" – Leading EUV lithography expert close to Lasertec's customers

The third fatal problem with Lasertec's EUV-based inspection tool, in addition to contamination/debris and light source instability/flickering, is pitifully low reliability that leads to low uptime and throughput, compounded by onerous cleaning and maintenance. We previously quoted an ex-Lasertec field service engineer who stated that each single ACTIS actinic inspection machine requires 3 full-time engineers for troubleshooting. A second ex-product engineer stated that the service requirements are so exorbitant that **each tool actually requires “at least 4 skilled field service engineers.”** Multiple engineers indicated that the tools only work a bit more than half the time, with **60-65% uptime, angering both Intel and TSMC,** as we detail in a later section. When we remarked that the utilization is poor, one engineer replied “Yeah. That kind of came with the territory.”

Each ACTIS actinic inspection tool requires “at least 4 skilled field service engineers”

“The actinic tool requires **at least four skilled field service engineers** for the laser source, chamber, hardware, and others.” – Former Lasertec product engineer; 20 years in the industry and previously part of the EUV task force at one of the top 3 semiconductor companies in the world

ACTIS actinic inspection tools have only 60% uptime

Q: “What was the uptime on the Lasertec actinic machine?”

A: **“[T]he actinic and EUV tools remain at 60%.”** – Former Lasertec product engineer; 20 years in the industry and previously part of the EUV task force at one of the top 3 semiconductor companies in the world

ACTIS tool only had 65% uptime at Intel mask shop

Q: “When you added it up, how much usage was Intel able to get out of the machine? How often was it down?”

A: **“Utilization, we were probably at 65% based on the 24/7 goal.”**

Q: “That’s pretty poor.”

A: “Yeah. That kind of came with the territory.” – Former Lasertec field service engineer who worked in the Intel mask shop

We contrast Lasertec's low ~60% uptime with 96% for ASML's EUV source, according to data they presented at an SPIE lithography and patterning conference in 2023. In addition, ASML's EUV collector lasts for over a year, versus only a couple of weeks for the same critical component in Lasertec's machine.

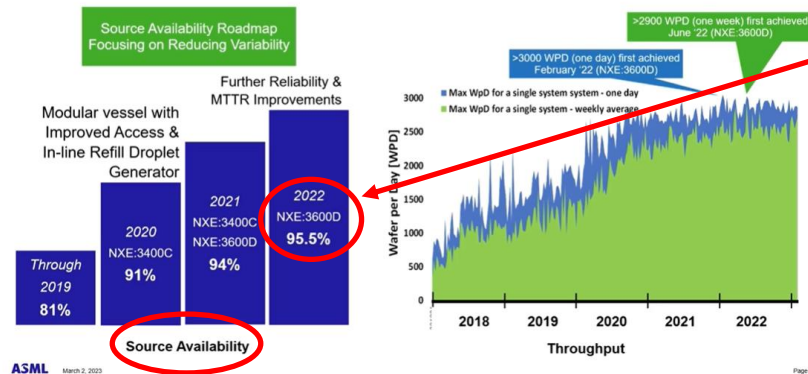
ASML SPIE paper, April 2023 - excerpt

EUV light source for high-NA and low-NA lithography

Karl Umstadter, Matthew Graham, Michael Purvis, Alex Schafgans, Jayson Stewart, Peter Mayer, Daniel Brown

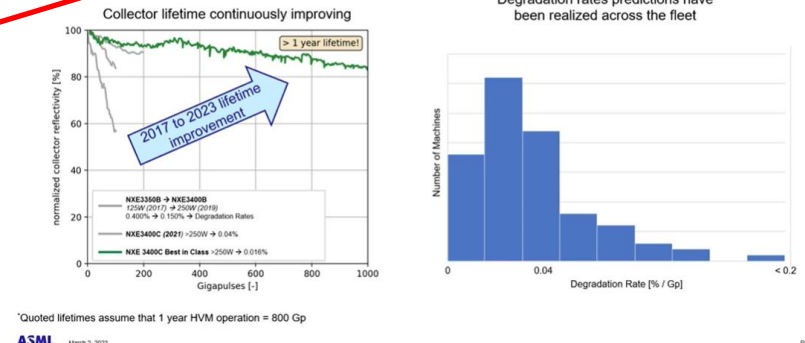
HVM with EUV enabled by ASML's NXE platform

Availability & throughput have matured and are continuously improving



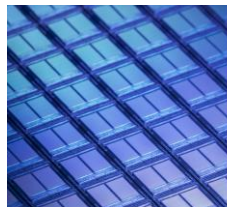
Collector Lifetime Continuously Improving

Median degradation rate 0.04% / Gp with some systems ≤ 0.02% / Gp (**>1 year lifetime**)



A KLA presentation on EUV mask inspection highlighted the absolute requirement for light source uptime, which Lasertec's pitiful ~60% metric obviously fails to meet: "Every % of tool availability is critical for mask shops that must run 24/7...anything less than 85% is not acceptable"; "the source has to be highly reliable to avoid impact to overall system availability." They emphasized that plasma EUV is challenging as tin matter "must be ablated >5000 times per second...life-time is the biggest challenge"; "replacement of components (e.g. electrodes) every week is not an option...maintenance time of 5-10% represents a serious problem for the entire tool availability."

KLA presentation "Source performance metrics for EUV mask inspection"



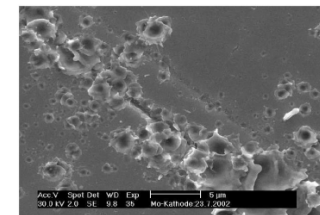
KLA

Source performance
metrics for EUV mask
inspection

Larissa Juschkin, Dan Wack, and
James Westphal

Production systems need high availability and long life-time

- Every % of tool availability is critical for mask shops that run 24/7. Anything less than 85% is not acceptable.
- In APML, the source has to be highly reliable to avoid impact to overall system availability.
- Plasmas are efficient EUV sources, but to produce one the matter must be ablated > 5000 times per second. The problem of target recovery and life-time is the biggest challenge. This includes life-time of any component facing plasma and/or being in close proximity to it.
- Example of electrode erosion in discharges:
H. Verbraak et al., J. Appl. Phys. 108, 093304 (2010)
J. Pankert et al., Proc. SPIE 6151, 61510Q (2006)
 - Electrode mass loss is about 1 g per Mshot (corresponds to 5 kg per 10 days).
 - Electrode life-time of 2 billion shots is reported for LDP concept (corresponds to ~5 days @ 5 kHz).
- For an inspection tool, replacement of components (e.g. electrodes) every week is not an option. Maintenance time of 5-10% represents a serious problem for the entire tool availability.



Electrode surface after one shot in a HCT discharge showing cathode spots

L. Juschkin, G. Derra, and K. Bergmann, "EUV light sources", from *Low temperature plasmas*, ed. R. Hippler et al. (2008)

Several long-time industry executives close to and/or in contact with Lasertec's CEO and/or management team provided information identical to that shared with us by field service engineers. One stated that his color on reliability issues comes directly from "close connections at Lasertec" but he was apprehensive about sharing more: "...but I really can't go into more than that"; "some of the stuff I really can't share." Lasertec's secrecy is a recurring theme echoed by its own engineers, which we shall return to. He continued that "Lasertec system reliability is pretty low...60% reliable...so it's not reliable enough...making chips work requires every step to be 99.9% reliable...the mask is super critical because if one mask is bad, then every chip you make after that bad mask is bad."

KLA expert confirms information on Lasertec reliability issues as coming from "close connections at Lasertec"

Q: "You made a lot of very interesting comments about reliability, which you're very consistent with what I've heard from different customers or from former employees of Lasertec. That all lines up. What is the basis for your saying it's not reliable? Is that coming from one of these particularly large customers who you maybe visited or interacted with?"

A: "Sorry, some of the stuff I really can't share. **I have some connections which are very close with Lasertec and that's what I hear from some close connections at Lasertec. But I really can't go into more than that.**" – Ex-KLA director of engineering focused on EUV mask inspection

Lasertec "system reliability is pretty low" and is "not reliable enough"; reliability needs to be 99.9% else a bad mask leads to defective chips

"The **Lasertec system reliability is pretty low**, similar to what you have heard. That's what I have heard as well. Systems need to be 95%, 99% reliable, and the Lasertec system is but **60% reliable. So, it's not reliable enough**. It has the form that's needed, but it's not reliable enough. **Making chips work requires every step to be 99.9% reliable**. If one step is not good, we're f-cked. The mask is super critical because if one mask is bad, then every chip you make after that bad mask is bad. That's why making a super reliable, effective mask system is important. Do you write the mask correctly? Do you check it? That's why that's so important." – Ex-KLA director of engineering focused on EUV mask inspection

The reliability is so low that Lasertec's tool is simply not “production-worthy”: “their tool doesn't work reliably”; “doesn't work well enough for production”; doesn't work well enough overall,” which leads customers to beg KLA to hurry its competing actinic inspection tool – “and that's why customers want KLA to make a reliable tool.” The expert stated that “the reliability is bad...they don't meet the high 90%'s reliability at all,” which then undermines the economics of buying ASML's expensive EUV scanners: “it needs to be reliable...otherwise the economics don't work...the scanners are getting very, very expensive.”

Lasertec “reliability is bad”; “very difficult to work with”; “don't meet high 90's reliability at all”

Q: “Who have you heard from TSMC, Intel, and Samsung?”

A: “With the Lasertec system, **the reliability is very difficult to work with...the reliability is bad. So, they don't meet the high 90%'s reliability at all.** That's the general thing that I hear that I can share overall. It matters. People spend lots of money based on what's going to make their production work.” – Ex-KLA director of engineering focused on EUV mask inspection

Lasertec tool “doesn't work well enough for production” and is not “production worthy”; “doesn't work well enough overall”; hence “customers want KLA to make a reliable tool”

“Their tool doesn't work reliably...the tool doesn't work well enough for production. It has the capabilities that no one else has. So, I would say on the plus side, the capabilities are great. **Production worthy, not great.** So, you have to mix those things together...reliability is not so good. It's actinic. It should do what needs to be done and that's what customers think is that when it works. **It doesn't work well enough overall, and that's why customers want KLA to make a reliable tool.”** – Ex-KLA director of engineering focused on EUV mask inspection

Lack of Lasertec mask inspection reliability harms the economics of EUV scanners

“There's a huge value in EUV because we make more devices per nanometer. We need EUV to move forward, but **it needs to be reliable. Otherwise, the economics don't work out. Bottom line, it comes down to economics. And it goes to the scanners. The scanners are getting very, very expensive and the inspection system should support the scanners** and they're getting very expensive, so it doesn't make sense.” – Ex-KLA director of engineering focused on EUV mask inspection

Lasertec's lack of production reliability makes it a non-starter for widespread use inside wafer fabs and radically shrinks its TAM – total addressable market. Actinic mask inspection tools have only two use cases: 1) inspection inside the mask shop where each new EUV mask is prepared; and 2) inside fabs next to ASML scanners, where masks are periodically inspected during production use. However, the **TAM inside mask shops is miniscule and already exhausted.** As we detail later, we believe that Lasertec's two largest customers **Intel and TSMC have only 3 and 2 ACTIS machines each, respectively, in their mask shops and appear to have ceased any new installs,** making any future tool purchases dependent on production use in fabs: "the mask shop...to be blunt...that's where more of the money is"; "**spending money on EUV actinic inspection in a wafer fab seems unlikely...it would be a lower priority...**"

Not a "production-worthy" tool; tool purchases for wafer fabs are less likely as the value is mostly in the mask shop

A: "Production-worthy, not so quick."

Q: "So, **you're saying it's not production-worthy?**"

A: "**Yeah, definitely. The question is, is it production-worthy? Is it reliable? People want a 95% reliable tool. The Lasertec tool is not 95% reliable...the mask shop, and that's where—to be blunt—that's where more of the money is.** The volume is in the fab. Making sure that the mask is good is super important because every mask is making so many chips. Inspecting in process in the wafer fab - the importance is lower than the mask shop, and the market value is lower. And that's why **spending money on an EUV actinic inspection in a wafer fab seems unlikely to me...it would be a lower priority overall.**" – Ex-KLA director of engineering focused on EUV mask inspection

As a preview of damning customer color detailed in a later section – from the company’s largest customers - we note that TSMC doesn’t appear interested in purchasing new Lasertec EUV inspection tools: the ACTIS “A150 needs a lot of support and it has a lot of issues”; “reliability, the big thing I hear offline is it’s not very reliable.” Reliability is equally poor among Lasertec’s next-generation ACTIS A300, as we document in a subsequent chapter: “there’s no confidence in my opinion that it will be a reliable product.”

TSMC is “not buying” the A300; “no confidence” that it’s a “reliable product”; A150 is also “not very reliable”; “has a lot of issues”

Q: “Is anybody buying less Lasertec, or have they essentially idled or just reduced their utilization of the Lasertec machines because of these frustrations?”

A: “Well, a recent example would be that **TSMC is not buying the High-NA Lasertec tool**, to my knowledge. And they're the company that does high volume, high-performance chips...”

Q: “You’re talking about the recent ACTIS A300 that they announced in November, which sort of tied into the Intel High-NA announcement. Is that what we’re talking about—the recent product announcement?”

A: “Yes, I’m talking about that...**there’s no confidence in my opinion that it will be a reliable product...**”

Q: “You said you don’t think this A300, this High-NA tool, is going to be reliable.”

A: “I reviewed again the papers presented in the past year at the various conferences—there was the advanced lithography conference and a photomask conference. I reviewed the papers...**reliability, the big thing I hear offline is it’s not very reliable.** I’m talking about the A150. A300 is just a product announcement. It’s not shipped. **The A150...needs a lot of support, and it has a lot of issues.**” – Ex-KLA director of engineering focused on EUV mask inspection

Lasertec's EUV light source is so plagued by the fatal problems just described - 1) contamination and debris from tin splatter and off-gassing; 2) instability and flickering; 3) low reliability, uptime, and throughput along with onerous cleaning and maintenance - that it has a short lifespan of only a couple of weeks before needing replacement and/or being sent to Germany for refurbishment, driving a burdensome cost of ownership. A former field service engineer stated "they were replacing them every, give or take, two weeks...being able to work on the system became rather challenging." He highlighted additional downtime from de-pressurizing and pressurizing the chamber, as EUV light is absorbed by air and glass and has to be enclosed in a vacuum: "...pumping up and pumping down was a significant amount of time...every time the chamber was opened or the light source was taken off, we had to de-vac...then pump it back up to vacuum."

Light source had to be replaced every two weeks; "challenging" to use the machine; cycling vacuum pressure up and down caused significant downtime

Q: "How much downtime did the machines have? How long would it take to replace the optics, or how long would it be out of commission when the light source went bad?"

A: "Light source replacement would probably be a one-day cycle. **They were replacing them every, give or take, two weeks. Being able to work on the system became rather challenging** because one of the things with EUV is you can't really have it travel through air; you have to make a vacuum. So, pumping up and pumping down was a significant amount of time. So, some of those downtimes could be—**every time the chamber was opened, or the light source was taken off, we had to de-vac, like get out a vacuum, and then pump it back up to vacuum before we could make it operational again.**"

Q: "How long would that have to happen? Every two weeks?"

A: "Yes, about every two weeks."

Q: **"The light source had to be replaced every two weeks or fixed every two weeks?"**

A: **"Replaced."** – Former Lasertec field service engineer who worked in the Intel mask shop

An industry executive in touch with Lasertec detailed the vast amount of cleaning and maintenance required to service and maintain the other optics besides the light source, such as the collector and mirrors that direct EUV light. He described seeing piles of Lasertec EUV collectors in Germany awaiting cleaning at a specialized vendor: “you replace the collector and send it to Germany”; “I’ve seen a bunch of these unbelievable amounts of collectors there from Lasertec”; “it costs hundreds of thousands of dollars” to clean each collector every month; “takes almost two days to chisel all this debris.” The expert suggested the annual cost to clean each collector is \$2.5 million: “I’m telling you, the problem...started two years ago and they said that we cannot ship...they are not shipping because it’s a liability...it’s consuming all the margin.”

Light source and optics needs to be sent to Germany monthly for cleaning; takes days and costs millions a year to make them usable again; expert has seen “unbelievable amounts” of Lasertec optics components piling up in Germany

Q: “Do they have to replace their source or they just have to clean the mirrors and the optics?”

A: “They don’t replace the source. The source is the size of a mini-bus and weighs like - it’s massive. You cannot replace it. Therefore, the only way you do it open the entire chamber of the source. **You replace the collector and send it to Germany** and I’ve been visiting my favorite vendors in Germany for optics, and **I’ve seen a bunch of these unbelievable amounts of collectors there from Lasertec**. They strip it, and then they polish it again and then they coat it. And it **costs hundreds of thousands of dollars to do this operation for each collector. And you need to do it for every month...it takes almost two days to chisel all this debris and make the source workable.**”

Q: “And they have to do this every month, and it costs a couple of hundred thousand dollars?”

A: “Correct.”

Q: “That’s **like \$2.5 million a year just to clean the collector.**”

A: “That was the beginning. How much it improves the tin, that’s exactly the reason why they went to the Urashima light source...I’m telling you, that’s where the **problem started two years ago and they said that we cannot ship - and that was why they are not shipping because it’s a liability. They ship something and they know that it’s consuming all the margin they have.**” – Former KLA executive in a leadership role in their EUV mask inspection group

Our research interviews indicate that the net result of the contamination and frequent maintenance and refurbishing is that the entire optical system inside Lasertec's EUV inspection tool needs to be replaced at least every year: "...it coats the source of the EUV light until it develops a few mono-layers of coating...and gradually you lose it...and have to replace it": "I heard every year that they have to replace the optics...the refurbishing of those optics costs at least a million dollars or more...they need to replace mirror number two, mirror number four, and that could cost at least half a million dollars each."

Entire optical system needs to be replaced annually; costs millions to refurbish and repair

A: "...and this one, you cannot clean. It coats the source of the EUV light until it develops a few mono-layers of coating of this organic composition, and gradually, you lose it, and you have to replace it."

Q: "So, you have to replace the optics. How often do they have to replace the optics, and how much does that cost?"

A: "I heard every year that you need to replace the imaging optics right now...the coating and the amount of VOC are such that every year they have to replace the optics, and this costs—the refurbishing of those optics costs at least a million dollars or more. They need to replace mirror number two, mirror number four, and that would cost at least half a million dollars each." —Former KLA executive in a leadership role in their EUV mask inspection group

Another longtime semiconductor industry executive based in Japan – who stated he is friendly with Lasertec’s CEO – corroborated that their EUV source is defective with a short lifespan and low uptime. As a result, customers have to waste money buying duplicate back-up machines: “at the early stage of their product, they had big trouble with the light source”; “they were struggling” because of the short lifespan “and then it needed replacement and after replacement they needed one month to tune the light source...so the tool was only available for maybe 60% or 70%...customers needed to buy more equipment than they originally planned to cover their requirements.”

Extensive troubles with light source; short lifespan, can only use for two months before replacement; low uptime; TSMC still using alternative legacy optical and e-beam tools due to problems; failed TSMC qualification

A: “They were the only company that came out with a so-called actinic inspection tool. They had bought out a new one for High-NA exposure equipment. **At the early stage of their product, they had big trouble with the light source.**”

Q: “This is the ACTIS A150?”

A: “Yes. **They were struggling because the light source had a lifespan of three months, and then it needed replacement, and after replacement, they needed one month to tune the light source. So, the tool was only available for maybe 60% or 70%.** So, customers needed to buy more equipment than they originally planned to cover their requirements. So, the switch from optical inspection to actinic inspection was kind of slow, slower than everybody anticipated. From what I hear from others who are involved in the inspection and quality assurance flow at **TSMC, I hear that they are still using optical inspection tools from KLA and combined with an electron beam kind of tool to review potential defects. The current setup is based on an optical and electron beam inspection scheme, not actinic.**”

Q: “And is that because of these problems with the A150 as far as the optics?”

A: “Yes. The KLA leading-edge optical inspection tool costs maybe \$40 or \$50 million. The SEM tool costs maybe \$5 or \$7 million. Scanning electron microscope. This costs maybe \$5 to 7 million, so the **financial setup of optical and the CD-SEM is less money.**” - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec’s CEO

The intelligence provided by the executive is particularly damning, given his leadership role at a Japanese semiconductor equipment company and his relationship with Lasertec's CEO. He elaborated that the EUV light source is "very low efficiency...in order to provide sufficient capacity, customers need to purchase more than one tool where they only would need one."

Light source fails quickly and has low efficiency, forcing customers to buy back-up machines in excess of their needs

Q: "What have you heard about the light source problem, and what the status is with this Urashima product?"

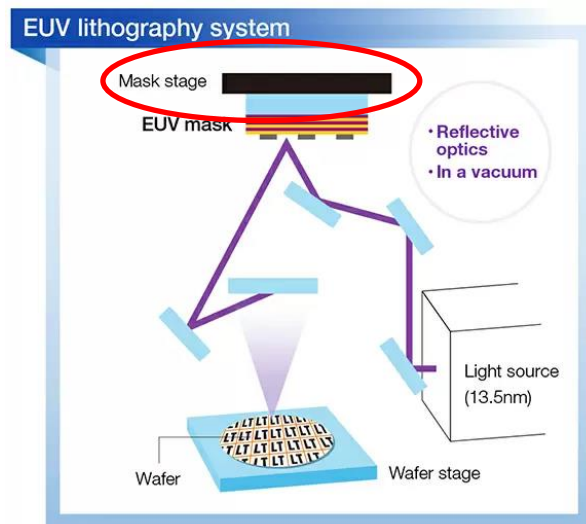
A: "I heard it's made by Ushio, and the problem is the lifetime is only three months. And out of the three months, tuning is one month, so they **can only use it for two months until they have to replace it again. So, it's very low efficiency. In order to provide sufficient capacity, customers need to purchase more than one tool where they only would need one.**"

Q: "They must be completely pissed off."

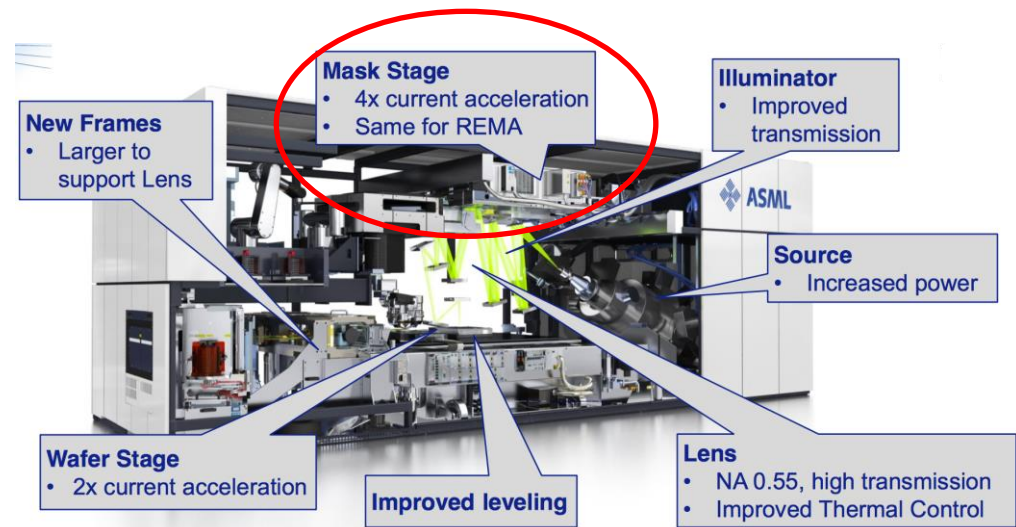
A: "Yes. **That's the failure of the qualification at TSMC—that's a good possibility. And from another source, I heard that they are still using optical and electron beam solution.**" -Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

In addition to a defective light source and optics, the key mechanical component of Lasertec's actinic mask inspection tool - the mask stage - is also fatally flawed. The stage is a critical component of both ASML EUV scanners that write the wafers as well as Lasertec's mask inspection machine, as it continuously moves and adjusts the wafer and mask into appropriate position in the x, y, and z planes during the writing or inspection process, requiring microscopic precision as EUV wavelength is only 13.5nm, a 10X higher resolution than previous DUV systems.

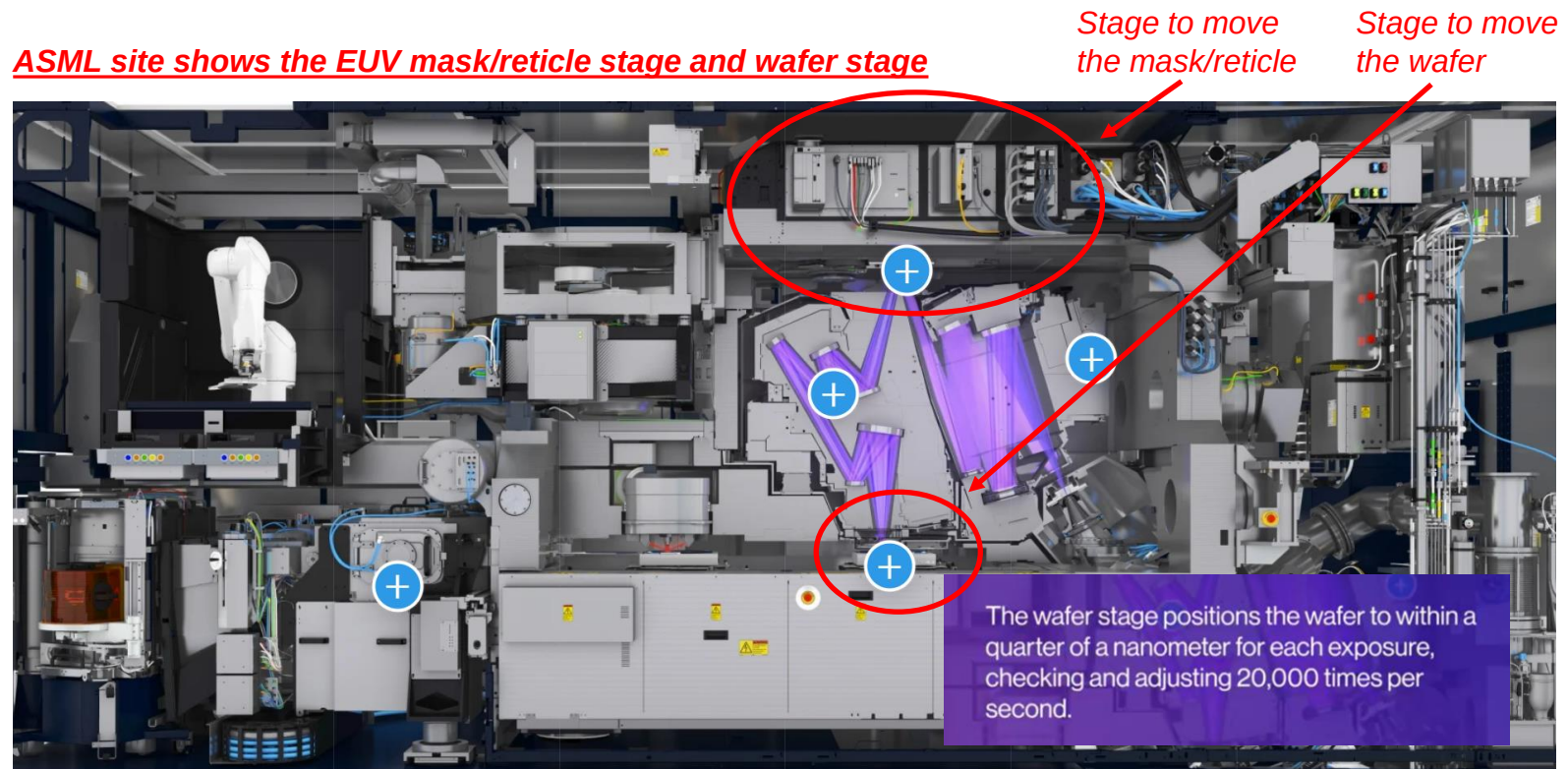
Lasertec diagram shows the mask stage



Mask stage shown in an ASML EUV stepper



The stage is a critical enabling technology and key factor in ASML's dominance: "to this day, the electric table, the superior alignment technology...are ASML's bedrock...its super-fast, super-precise stage and alignment have been unique selling points for decades." ASML's website states that the "stage positions the wafer to within a quarter of a nanometer for each exposure, checking and adjusting 20,000 times per second."



However, ASML developed a proprietary and unique magnetic levitation system for its stage, while Lasertec was forced to buy an off-the-shelf, defective stage from Sumitomo – the only supplier available. Unfortunately, the Sumitomo stage “is extremely complex and has a huge problem because of the contamination it produces...Lasertec chose the easiest route because it was available...you cannot buy a magnetic levitation stage...Lasertec cannot do it....that’s the second thing they compromised, in addition to the light source.”

The Sumitomo stage that moves the mask/reticle is highly flawed and produces contamination inside the inspection tool

*“The second key component is the stage that moves the reticle of the mask in the x and y plane. A regular stage can’t be used because the resolution increases from DUV to EUV, from 119nm to 13.5nm which is more than 10x higher resolution. The same goes for stability in the z-direction has to be maintained when you move the mask. So, you need to move the mask with extremely high stability in xy and in terms of the z motion, which is related to the focus. The only stages that you can use are either magnetic levitation or the **Sumitomo stage, which is extremely complex and has a huge problem because of the contamination it produces.** Lasertec chose the easiest route because it was available. You cannot buy a magnetic levitation stage - the only one is ASML’s and it is proprietary. Lasertec cannot do it. It takes ages to do something like that. It’s incredibly difficult. **That’s the second thing they compromised, in addition to the light source.**”* –Former KLA executive in a leadership role in their EUV mask inspection group

ASML’s site shows and describes its unique, proprietary magnetic levitation stage



Magnetically levitating wafer tables that accelerate faster than a fighter jet? Sensors that measure positioning 20,000 times per second with an accuracy of 60 picometers (less than the size of a silicon atom)? It’s all part of the extreme mechanics in ASML machines.

The Sumitomo stage uses a crude pneumatic system that operates inside a vacuum, using pressure to move the mask versus ASML's magnetic levitation. The pressure forces and releases particles, causing contamination that ruins the mask, in addition to being extremely complex and expensive with a cost in the millions per stage: "it's very complex and very, very expensive...multiple millions...it uses pressure and then they try to suck it back up...it releases particles into the cavity of the optics...that's the number one source of particle generation that they have."

Sumitomo stage requires a vacuum and 'releases particles into the cavity of the optics'; top source of contamination

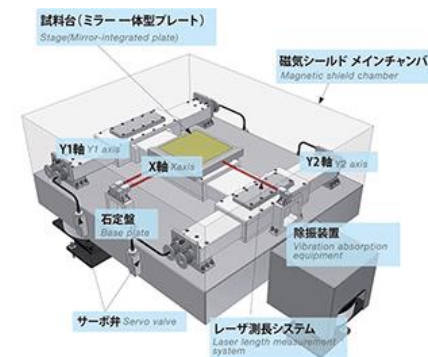
Q: "You said the stage needs a lot more precision in movement because the resolution is so high?"

A: "Correct. Because the depth of focus is a function of λ/NA^2 squared."

Q: "What is the issue with the Sumitomo stage?"

A: "The Sumitomo stage is a pneumatic stage that operates inside a vacuum and it has two issues. One, it's very complex and it's very, very expensive - millions of dollars they pay for that stage - millions, not one, multiple millions. And two, which is the most important - because **it uses pressure and then they try to suck it back up, it releases particles into the cavity of the optics. That's the number one source of particle generation that they have.**" –Former KLA executive in a leadership role in their EUV mask inspection group

Sumitomo "Vacuum Air Servo Stage" for semiconductor manufacturing



The stage creates a third source of contamination - compounding the problems from tin splatter from the plasma reaction used to emit EUV light, and off-gassing: “the stage has a problem with particulation...all of them contribute to the degradation of the imaging optics and also the particulation on the actual mask”; “the stage has two problems: significantly high cost and complexity and...the fact that it produces particulation into the entire area” of the cavity of the tool, which consists of a large, open vacuum chamber that can’t be partitioned with barriers as EUV cannot travel through air or glass.

Particulate contamination is a major problem as the EUV optics must be in an open cavity without barriers; can’t be partitioned with windows or barriers

“The stage has two problems: significantly high cost and complexity. and two, the fact that it produces particulation into the entire area. The problem with particles is two-fold. One, there is all the optics of the mask inspection system whether it's Lasertec or KLA is open optic. There is no window anywhere because you cannot put windows in EUV. So, it's a big, big chamber or two chambers are connected with a phalange.” –Former KLA executive in a leadership role in their EUV mask inspection group

The stage “makes particles” that cause contamination, degrading the imaging optics and mirrors, costs millions of dollars a year to clean

Q: “You said the optics are getting off-gassing and particles from the stage. So, you said they have to replace the optics or they have to clean those as well?”

A: “Together with the problem with the source of off-gassing, also **the stage has a problem with particulation and off-gassing, all of them contribute to the degradation of the imaging optics and also the particulation on the actual mask** of the inspection. And this one, you cannot clean. It coats the source of the EUV light until it develops a few mono-layers of coating of this organic composition, and gradually, you lose it, and you have to replace it.” –Former KLA executive in a leadership role in their EUV mask inspection group

3. The “new” ACTIS A300, announced with fanfare in Nov 2023 as Lasertec’s next-generation EUV product and driver of growth, is a hoax – a cover story to conceal fatal flaws with the Ushio-supplied EUV light source in its flagship tool, the A150, and the resulting, costly recall and replacement effort.

Lasertec announced the ACTIS A300 on Nov 24, 2023 with much fanfare, as the latest version of its actinic EUV mask inspection machine. It replaces the A150 - announced in 2019 - as its flagship EUV inspection product. The most important “innovation” in the A300 is a “new” EUV light source which Lasertec calls “Urashima.” The second upgrade is that the A300 purports to be High NA-compatible, which refers to next generation EUV lithography with high numerical aperture (“NA”) for printing at higher resolution, density, and smaller size.

Press release announcing ACTIS A300 with the “new” Urashima light source, Nov 24, 2023

Lasertec

PRESS RELEASE

Lasertec Corporation

2-10-1 Shin-yokohama, Kohoku-ku, Yokohama
(Tokyo Stock Exchange Prime Market, Code 6920)

Lasertec Releases ACTIS “A300” Series Actinic EUV Patterned Mask Inspection System

Yokohama, November 24, 2023 - Lasertec Corporation today announced the release of ACTIS “A300” series Actinic EUV Patterned Mask Inspection System for High-NA.

The semiconductor industry is expected to keep improving its manufacturing processes to pursue the production of smaller geometry devices to achieve higher performance, faster speed, and better energy efficiency. ACTIS A150 actinic EUV patterned mask inspection system provided by Lasertec has been facilitating the adoption of EUV lithography to high-volume manufacturing. It is renowned for its excellent inspection performance in the industry today.

The newly released ACTIS A300 is a next-generation model that meets the requirements of manufacturing processes using high-NA EUV lithography to enable the further miniaturization of device geometries.

The A300 series uses newly designed optics and high-brightness light source “URASHIMA”. It achieves a significant improvement in defect detection performance compared to the A150 series.

The anamorphic optics used in high-NA lithography have adopted different magnifications of projection in the X and Y directions. Inspection of EUV masks for high-NA lithography, therefore, requires different levels of resolution in the two directions. The A300 series can also be used to inspect EUV masks for the current NA lithography as well as EUV masks for high-NA lithography.

“The A300 uses newly designed optics and high-brightness light source ‘URASHIMA’. It achieves significant improvement in defect detection performance compares to the A150 series.”

“The A300 series can also be used to inspect EUV masks for high-NA lithography.”

Our research indicates that the entire A300 story is a fraud to conceal fatal flaws with the Ushio-supplied EUV source in the A150. Lasertec's stock would have imploded had the company revealed that the A150 source – the key component of an EUV inspection tool – was defective. Lasertec was forced to swap the Ushio source with one sourced from a Russian company called Isteq, which it simply renamed Urashima to make it appear as a Japanese-developed in-house technology. The A300 is nothing more than a cover story to mislead investors and spin a damage-control and recall effort as a “new” product. Even the photos of both products on Lasertec's site are identical except for different lettering.

Photos of the ACTIS A150 and A300



Rather than admitting that the Urashima light source is simply an old, re-packaged Russian EUV source no one else wanted, Lasertec appears to be misleading investors with a new name and claims that it is a Japanese-developed technology. We note the headline below which misrepresents it as “Lasertec’s In-house EUV Light Source: URASHIMA”; as well as similar representations by the CEO during recent quarterly Q&A.

CEO comments on recent Q&A

“...we announced ACTIS with High-NA, and in September, we also announced **an EUV light source developed in-house** with future needs in mind...” – Lasertec CEO, Jan 31, 2024 Q&A

“As for **URASHIMA, we developed it in-house** and leave the manufacturing to our partner companies as much as possible, just as we do with our other equipment.” – Lasertec CEO, Oct 31, 2023 Q&A

Press release announcing ACTIS A300 with the “new” Urashima light source, Nov 24, 2023

Lasertec's In-house EUV Light Source: URASHIMA

Urashima is designed for actinic mask inspection tools

Overview

- ✓ High brightness light source
- ✓ Stable EUV emission enhances reliability
- ✓ Small etendue to minimize heat load to pellicles

Inspection system verification

- ✓ Operating on site with multiple ACTIS
- ✓ Qualified by all customers

In-field development discussed in presentation 12751-3
Actinic inspection technology enhancement for n3 node and beyond

URASHIMA light source in ACTIS system

SPIE PHOTOMASK TECHNOLOGY + EUV LITHOGRAPHY 2023

While Lasertec represents Urashima as a “new” and improved EUV light source, it is simply an old Isteq LPP laser-produced plasma technology developed over a decade ago. Lasertec could easily have licensed the Isteq technology prior to the ACTIS A150 launch in 2019 but chose Ushio instead. Given that there are only two or three EUV source suppliers to choose from, it is obvious that Lasertec would have evaluated Isteq over five years ago prior to rejecting it for Ushio – and are now desperately trying to cover up the Ushio failure and trying Isteq as a Hail Mary. We note various Isteq papers going back to 2013 on their LPP system.

2013 presentation by Isteq’s parent entity



**Research review on plasma-based
EUV sources at RnD-ISAN/EUV
Labs**

V.Krivtsun, D.Abramenko, V.Bushuev, A.Chekmarev,
R.Gayazov, M.Spiridonov, O.Yakushev, G.Zukakishvili and
K.Koshelev

ISAN, RnD-ISAN, EUV Labs
Moscow, Troitsk

2013 EUV Sources Workshop Nov 4-7 Dublin, Ireland

2016 paper by the parent entity

RESEARCH ARTICLE **OCTOBER 31 2016**

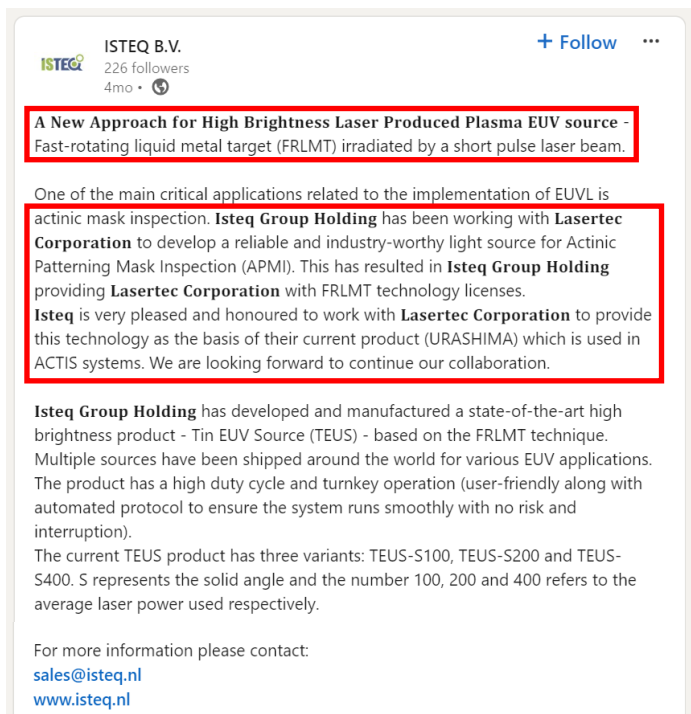
**Droplet-based, high-brightness extreme ultraviolet laser
plasma source for metrology**

A. Yu. Vinokhodov ; M. S. Krivokorytov; Yu. V. Sidelnikov; V. M. Krivtsun; V. V. Medvedev;
K. N. Koshelev

We report on the development of a high brightness source of extreme ultraviolet radiation (EUV) with a working wavelength of 13.5 nm. The source is based on a laser-produced plasma driven by pulsed radiation of a Nd:YAG laser system. Liquid droplets of Sn-In eutectic alloy were used as the source fuel. The droplets were created by a droplet generator operating in the jet break-up regime. The EUV emission properties of the plasma,

In a LinkedIn post coinciding with Lasertec's November 2023 announcement for its "new" ACTIS A300 with the Urashima source, Istec confirmed that Urashima is based upon a license for its TEUS LPP product. We note that both Istec and Lasertec misrepresent the technology as new – "a new approach for high brightness laser produced plasma EUV source" – despite licensing it to Lasertec over two or three years ago.

Istec LinkedIn post circa November/December 2023



ISTEQ B.V.
226 followers
4mo •

A New Approach for High Brightness Laser Produced Plasma EUV source - Fast-rotating liquid metal target (FRLMT) irradiated by a short pulse laser beam.

One of the main critical applications related to the implementation of EUVL is actinic mask inspection. **Istec Group Holding** has been working with **Lasertec Corporation** to develop a reliable and industry-worthy light source for Actinic Patterning Mask Inspection (APMI). This has resulted in **Istec Group Holding** providing **Lasertec Corporation** with FRLMT technology licenses.

Istec is very pleased and honoured to work with **Lasertec Corporation** to provide this technology as the basis of their current product (URASHIMA) which is used in ACTIS systems. We are looking forward to continue our collaboration.

Istec Group Holding has developed and manufactured a state-of-the-art high brightness product - Tin EUV Source (TEUS) - based on the FRLMT technique. Multiple sources have been shipped around the world for various EUV applications. The product has a high duty cycle and turnkey operation (user-friendly along with automated protocol to ensure the system runs smoothly with no risk and interruption).

The current TEUS product has three variants: TEUS-S100, TEUS-S200 and TEUS-S400. S represents the solid angle and the number 100, 200 and 400 refers to the average laser power used respectively.

For more information please contact:
sales@istec.nl
www.istec.nl

Isteq and its Russian parent RnD-ISAN jointly filed an international patent application for their LPP technology back in 2019/2020, and branded it as their “TEUS” EUV source product in sales presentation in 2021, if not earlier – once again illustrating that “Urashima” – TEUS in disguise – is far from new. The fact that the Isteq source was available prior to the ACTIS A150 launch in 2019/2020 suggests that Lasertec evaluated it years ago and concluded it was inferior to the Ushio source they first chose.

Old patent application by Isteq/RnD-ISAN

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(71) Applicants: **ISTEQ B.V.**, [NL/NL], High tech campus 9, 5656 AE Eindhoven (NL); **RnD-ISAN, LTD** [RU/RU]; Promyshlennaya St., 2B Troitsk, Moscow 108841 (RU);

(72) Inventors: **VINOKHODOV**, Aleksandr Yurievich, Sinezney Blvd., 10, apt. 116, Troitsk, Moscow 108840 (RU); **IVANOV**, Vladimir Vitalievich; Novatorov St., 36/3, apt. 336, Moscow 119421 (RU); **KOSHELEV**, Konstantin Nikolaevich; Troitskiy Blvd., 4, apt. 7, Troitsk, Moscow 108841 (RU); **KRIVOKORYTOV**, Mikhail Sergeevich; Miltseyskaya St., 62, apt. 114, Kirov region, Kirov 610027 (RU); **KRIVTSUN**, Vladimir Mikhailovich; Oktyabrskiy Prospekt, 23, apt. 85, Troitsk, Moscow 108840 (RU); **LASH**, Aleksandr Andreevich; Malyi Karetnyi per., 9-3, Moscow 127051 (RU); **MEDVEDEV**, Vyacheslav Valerievich; Fizicheskaya St., 12, apt. 50, Troitsk, Moscow 108840 (RU); **SIDELNIKOV**, Yuri Viktorovich; Oktyabrskiy Prospekt, 2, apt. 128, Troitsk, Moscow 108840 (RU); **YAKUSHEV**, Oleg Feliksovich; Sacco and Vanzetti St., 14, apt. 8, Ko-

WO 2020/216950

PCT/EP2020/061562

HIGH BRIGHTNESS LASER-PRODUCED PLASMA LIGHT SOURCE

FIELD OF INVENTION

[0001] The invention refers to high brightness laser-produced plasma (LPP) light sources designed to generate soft X-ray, extreme ultraviolet (EUV) and vacuum ultraviolet (VUV) radiation at wavelengths of approximately 0.4 to 200 nm, which provide highly effective debris mitigation to ensure the long-term operation of the light source and its integrated equipment.

2021 sales presentation for Isteq's LPP source



TEUS - high-brightness EUV LPP light source based on fast rotating target: product overview and specifications



2021 EUVL Supplier Showcase

A leading EUV lithography expert close to all major players including Lasertec, ASML, and KLA - as well as the industry's three key customers TSMC, Intel, and Samsung – mocked the Urashima source and nodded emphatically when we asked if Lasertec was simply trying to cover up a light source failure and spin it as new, advanced product by using an old Isteq source: “That’s right. That’s right. That’s right. You got it right. You got it...this is supplier talk from Lasertec and also from Intel because they are spending a huge amount of money, too, and can’t look bad.”

ACTIS A300 with Urashima is basically a fake “new” product announcement to cover up the light source failure with the A150

Q: “What is this Urashima ACTIS A300 announcement? They said that’s the Isteq source. They put out a press release in November saying it’s new, but if they started ripping out the light source ages ago, is that really a new product or just the same thing, and they’re trying to spin it?

A: “They took the technology, which is **exactly the same source from Isteq. They gave it a Japanese name** and then said we have licensed the technology; it is our source. But it’s not their source. They just essentially took it from Isteq, gave it a name. So, they’ve **taken the Russian stuff and gave it a Japanese name. That’s what they’re doing.**”

Q: “The big picture is that the A300 is not that different than the A150 - they’ve already swapped out the light source in a bunch of their existing machines, and they’re now just putting a new name on it. **It sounds like they’re trying to cover up a light source failure and they’re trying to spin it as a new, advanced product?**”

A: **“That’s right. That’s right. That’s right. You got it right.** You got it. This is supplier talk. So, you are right. This is **supplier talk from Lasertec and also from Intel because they are spending a huge amount of money, too, and they can’t look bad.** But Intel is Intel, and if it doesn’t work, you’re going to hear about it, but you won’t hear about it for two to three years when they’ll say, you know what? We found a better source from company B or X, and now that’s what we have.” – Leading EUV lithography expert close to Lasertec’s customers

Multiple other research sources revealed to us that far from being a “new” product announced in November 2023, Lasertec began quietly replacing the Ushio source with the Isteq one approximately 18 to 24 months ago, sometime in 2022. An Ushio field service engineer based at Intel’s fab in Oregon stated that Intel has only about ten Lasertec mask inspection machines in total in production – and that only three of these ten still have the Ushio source. The rest have the Isteq source now misleadingly labeled as “Urashima.” The engineer stated that of the 4 machines that were installed at the fab between 2020 and 2022, the first 3 had the Ushio source and then the Isteq/Urashima source starting in 2022: “...the last one was a different light source.”

Ushio states that Lasertec started using Urashima light source in 2022; only 3 of ~10 machines in Intel Oregon have the Ushio light source

A: “There might have been one. After the first 3, the rest of the machines, I don’t work on those.”

Q: “Is it a colleague at Ushio who works on those?”

A: “No.”

Q: “Who does the light source? They just handle it themselves on those other ones?”

A: “Yes.”

Q: “I see. So, you’re saying **they had a different light source for these other ones that they put in?**”

A: **“That’s correct.”**

Q: “So, you’re saying that they put in – how many machines between 2020 and 2022?”

A: “I think there were 4.”

Q: **“So, four between 2020 and 2022. And those were with the Ushio light source?”**

A: **“The first 3 were. The last one was a different light source.”** – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

The Ushio engineer confirmed that the remaining machines have already been “Urashima” machines since 2022: “they are not Ushio sources...I know they have a different light source...I know they announced the Urashima light source...that’s what I’ve heard it called...I’ve heard Isteq thrown around as the creator of the IP for that source.” We note that the engineer appeared nervous at times while answering our questions, in a manner that suggested to us a coordinated attempt between Lasertec and its key customers to conceal that the “new” A300/Urashima announcement is essentially fake.

Ushio engineer confirmed that the Isteq/Urashima source already powers most of the ~10 Lasertec machines Intel has at its fab

Q: “This Urashima light source, it sounds like they licensed it from a company in the Netherlands called Isteq, which is technology from Russia or the University of Moscow.”

A: “Those are all things that I’ve heard, yes.”

Q: “They have all these papers on the SPIE website. These other light sources, are they this Urashima light source, or is it something else that they tried?”

A: **“Yeah, I’m pretty sure it’s the same Urashima light source.”**

Q: **“So, you’re saying of the 10-ish machines up there, you say only 3 have an Ushio light source?”**

A: **“That’s correct.”**

Q: “Could it be that those other machines are actually the Ushio light source, and they’re just maintaining it themselves, or you’re pretty sure those are Urashima light sources?”

A: **“They are not Ushio sources...I know they have a different light source...I know they announced the Urashima source. That’s what I’ve heard it called. I’ve heard Isteq thrown around as the creator of the IP for that source.”** – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

A second research source - a longtime semiconductor industry advisor who consulted recently for Lasertec management - confirmed that the company already moved to the “new” Urashima source in 2022. Similar to the Ushio engineer, the expert was occasionally evasive in answering our questions and appeared uncomfortable when we asked bluntly if this means that the recent A300 product announcement is fake. He replied “That was a bit over a year ago” when we asked when Lasertec replaced the source – the interview was conducted in January 2024, and avoided commenting when we stated that the A300 is therefore not a new product: “Again, they are on a second-generation light source.”

Lasertec moved to the new Urashima light source “a bit over a year ago”

Q: **“When did they move to this new light source?”** *It’s from a Netherlands-based company called Isteq, affiliated with a company in Russia. It sounds like they just licensed it to Lasertec.”*

A: **“This was a bit over a year ago.”**

Q: *“Is this the Urashima light source? That’s what they refer to it as.”*

A: *“I really can’t comment on that because of a confidentiality agreement I have.”*

Q: *“This new A300 based on the Urashima light source - it seems to me that it’s not new, that it’s actually been out there for more than a year, maybe even more. Is it really a growth driver if they’ve already been selling it?”*

A: **“Again, they are on a second-generation light source...”** – Longtime semiconductor consultant recently working for Lasertec

Multiple sources told us that the Urashima light source is as defective as the Ushio one and fails to address its fatal flaws. The Ushio field engineer based inside the Intel fab works closely with Lasertec to service the machines, giving him day-to-day visibility into all aspects of the technology. He stated that while **“Lasertec tries to keep us in the dark”** about the performance of the machines with the Urashima source versus the Ushio one, that the Urashima tools **“are right next to our two”**: **“we can look...they can’t really stop us from looking.”** He stated that the Lasertec engineers he interacts with work on both EUV light sources, and that **“there’s as much downtime on those machines as there is on ours...my impression from being around them is that the production is about the same.”**

Lasertec tries to keep Ushio “in the dark” about Urashima light source problems; has just “as much downtime; Ushio engineers can see the basically two light sources side by side

Q: “Have you heard anything about how the machines with the Urashima source are doing versus the Ushio one?”

A: **“Lasertec tries to keep us in the dark about that.** The machines are right next to our machines and when we need to work with an engineer on one of the Ushio machines, we are getting that help from somebody that’s working on one of the other machines. It’s all the same Lasertec engineers. I get the feeling there’s **as much downtime on those machines as there is on ours...my impression from being around them is that the production is about the same.”**

Q: “Are some of the machines that you’re in charge of, with the other light source, right next to those machines, the Lasertec machines? Or are those in a different facility?”

A: **“Some of them are right next to our two.”**

Q: “Oh, so you actually see it? You see the one with the Urashima right next to yours?”

A: “Oh yeah.”

Q: “How does that work since they’re trying to keep you in the dark? Are you not allowed to touch it or look at it? That must be weird.”

A: “Yeah, I mean, we don’t obviously don’t touch it, but we can look. **They can’t really stop us from looking.**” – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

We asked the Ushio engineer whether the A150 and Urashima A300 machines look any different when they are open, except for the two different EUV sources: “it’s all the same, as far as I can tell,” replying affirmatively when we then asked if everything else inside the tools is identical. He stated that Lasertec’s field engineers “complain” about the Urashima source as much as the A150 and that they are skeptical of their own company’s technology: “I work a lot with the boots on the ground at Lasertec, and they complain about having to maintain their light sources...[they] feel that management made some compromises....”

Ushio sees no difference between the A150 and purported A300 machines with a new light source; “It’s all the same”

Q: “Do the Lasertec machines with the Urashima light source look the same as the A150 with the Ushio source? When you see the two machines open, do they look fairly similar?”

A: “Yeah, on the Lasertec side, **it’s all the same, as far as I can tell.**”

Q: “So, they just went with a different light source, but **everything else is pretty much identical.** It goes into the same space, the same housing - all that stuff?”

A: **“It seems that way.”** – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

Lasertec field engineers “complain” about the Urashima light source; “management made some compromises”

Q: “I imagine Ushio must feel like they got thrown under the bus if Lasertec is switching light sources. What do you guys think?”

A: “We’re not happy about it. **We are continuing on servicing the machines that are in the field, hoping that maybe somebody will come to their senses.**”

Q: “What do you mean “come to their senses?” Do you think they made a mistake with this Urashima product? Ushio would obviously have an interesting perspective on it.”

A: “Being boots on the ground, **I work a lot with the boots on the ground at Lasertec, and they complain about having to maintain their light sources,** and they kind of wish that Ushio was maintaining more of the light sources. I get the feeling that at least, the lower guys, **the field service engineers at Lasertec, feel that management made some compromises by reducing their costs, trading off cost for convenience.**” – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

The engineer indicated that the Urashima source is “down at least as often as the Ushio light source”; “they have the exact same inspection issues, so the Lasertec Urashima source has all the same problems”; “the light source has its own set of weaknesses” in addition to those that afflict the Ushio source; “I think that probably contributes to the downtime.”

Urashima source is “down at least as often as the Ushio light source”; “exact same inspection issues”; “all the same problems” and “downtime”

Q: “What are your impressions of that machine, being right there? You can literally see how much that one is down versus the other one.”

A: “Like I said, my impression is that **it's down at least as often as the Ushio light source**...they have similar issues...**they have the exact same inspection issues, so the Lasertec Urashima source has all the same problems. The light source has its own set of weaknesses. I think that probably contributes to downtime** on that machine. At the same time, the weaknesses of the Ushio source has sometimes caused issues on that machine.” – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

Although the Urashima source attempts to use laser-produced plasma (LPP) similar to ASML's technology, where the plasma is better controlled, versus the Ushio source's inferior laser-assisted discharge-produced-plasma (LDP) approach, Urashima is still plagued by the same inspection failures as Ushio. The plasma which emits EUV is just as unstable, leading to fluctuations and lack of brightness and power. The Ushio engineer provided details we find damning: "...the same results happen on their inspections" with Urashima; "they get a dark frame or they get a false defect...it's light source fluctuation"; "in these light sources, the brightness is just going to fluctuate"; "difficult thing to control"; "hard to get enough power...to get the brightness up."

Both Ushio and Urashima cause inspection problems such as dark frames and false defects due to light source fluctuation; problems to due to difficulties in controlling plasma; Urashima doesn't have enough power and illumination as "plasma is smaller"

A: "It's different issues, but the **same results happen on their inspections. They get a dark frame or they get a false defect...it's light source fluctuation. In these light sources, the brightness is just going to fluctuate** just as a function of the physics of the plasma. In the Ushio source there's a discharge between two electrodes. That means the **plasma has a lot of freedom to move. And so, that causes fluctuation and brightness. That's a difficult thing to control.** The other light source is laser-produced plasma. There's no discharge between electrodes. It's just the laser creates the plasma. That tends to be much more stable, but **it's hard to get enough power into that to get the brightness up. As a result, the plasma is usually smaller. That can create problems with not getting enough illumination into the tool."**

Q: "What happens when you don't get enough illumination in the tool?"

A: "It's similar. **If the plasma is moving, every once in a while, you'll get a frame where there's just not enough light and uniformity. If you're not getting full brightness all the way to the edge of the frame, then you're going to have dark areas, and that's going to cause a false defect to show up**...multiple things can combine, so if a light source is starting to dim and there's a component on the tool side that's dimming, a mirror or lens or something, then they combine and cause problems."—
Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

A few pages prior, we quoted a leading EUV lithography expert close to all major players including Lasertec, ASML, and KLA - as well as the industry's three key customers TSMC, Intel, and Samsung – who indicated that Urashima was simply a cover-up for the failure of the Ushio source. He described the Ushio source as an unmitigated disaster and ridiculed claims that the Isteq source is superior as coming from the “same people” – presumably Lasertec and Intel – who made similarly false claims about the Ushio source several years ago: “the Ushio source [had] degradation and they took it out...the same people who came to me and said, wow, we won the war and we got Lasertec...it's all good with Ushio...the same people two years later said yeah by the way everybody in the field has taken the source out.” He expressed similar doubt about the Isteq source: “Isteq has not quite done their job”; “Lasertec and Intel will...do a song and dance...well, how much data do you have...I have not seen long-term performance data at all.”

Ushio source had “degradation”; Lasertec/Intel did a premature victory lap and then had to rip out the light sources and replace with an Isteq source; new source also has problems and lacks performance data

“The Ushio source [had] degradation and they took it out. So, the same people who came to me and said, wow, we won the war, and we got Lasertec; it's all good with Ushio. There was this celebration. The same people came to me two years later and said, yeah, by the way, everybody in the field has taken the sources out, and we're replacing it with Isteq. Well, Isteq has not quite done their job. I don't think they have their sources in the field—if they have one or two already, there's not enough data. So, if you ask my opinion, Lasertec and Intel will come and do a song and dance and say, well, now we've got this new source supplier, and everything is good, it's all wonderful and all that stuff. My response will be, well, how much data do you have to say that because you were the same guys who came and said the same about Ushio? And I have not seen long-term performance data at all.” – Leading EUV lithography expert close to Lasertec's customers

He stated that an advisor to Lasertec confirmed to him that “Ushio’s source has been taken out everywhere.” He eviscerated Isteq’s source as an old, failed technology no one wanted before: “it didn’t use to work well before...so, I’m wondering what happened that all of a sudden this thing is working...this was an LPP source that nobody cared about...what happened in the last couple of years that all of a sudden they have improved their source?...Isteq’s source is a rotating tin target...you’re doing to drip the tin off it easily...it’s not a good situation.”

Isteq is an old “LPP source nobody cared about” that “didn’t used to work well”; implausible that it now suddenly works:

A: “I don’t know what exactly went wrong with Lasertec because Lasertec then bought that technology from Isteq, that Russian company, which is a refugee now in the Netherlands. Whatever the reason was, they have an exclusive license of 13.5 LPP for metrology and actinic inspection, which all seems good. But what if it doesn’t work, because **it didn’t use to work well before. So, I’m wondering what happened that all of a sudden, this thing is working very well? I am not so sure.**”

Q: “You’re saying the Isteq source didn’t used to work well in the past. Is that what you’re saying? The one they licensed.”

A: **“This was an LPP source that nobody cared about. So, the question is what happened in the last couple of years that all of a sudden they have improved their source?...the best source they could have bought would have been a Gigaphoton source. I heard that Gigaphoton wanted too much money, probably \$25-35 million because their source is made for the scanner. That source, in my opinion, has to be a lot more stable and better than Isteq’s source because Isteq’s source is a rotating tin target. That damned tin, you’re going to drip the tin off of it easily – it’s not a good situation.** I know the guy who made the decision—it could be that the source is about \$1.5 million.” – Leading EUV lithography expert close to Lasertec’s customers

Ushio source was replaced over a year ago; Ushio tried to cover it up and “pretended” nothing was wrong

Q: “When did they start ripping out the Ushio light source, and when did they replace them?”

A: **“This is from Lasertec.** When I was there last year at SPIE in February, **a good friend who is the advisor to Lasertec said, have you heard that Ushio’s source has been taken out everywhere?** I said I haven’t heard anything. I had just talked to the Ushio guy, and he didn’t say a word to me. He’s the head guy for the source. And I asked him what happened. He said nothing happened.”

Q: “You said you talked to a Ushio guy, and he pretended there was nothing wrong?”

A: “Yeah, **he just pretended.**” – Leading EUV lithography expert close to Lasertec’s customers

Av ex-KLA executive in a leadership role in their EUV mask inspection group corroborated the damning color on Isteq's source. He stated he has visited the company and is familiar with the management team, similar to the EUV lithography expert. He dismissed the Isteq source as simply a laboratory R&D project and not suitable for a commercial product: the "Isteq source will not be done in a professional way and it cannot work...you need to understand Isteq was done for research...research is done in bursts, not continuously...for inspection, you need something to work 24/7...it's really hard...it's totally different."

Ushio source "is terrible" and "creates a mess"; new Isteq source "cannot work"; just an R&D concept not suited for 24/7 production use; KLA Xenon-based source "works"

A: **"The KLA Xenon works.** The Ushio tin source failed. ASML is different. They're using tin and they're using IR laser to ignite it. That's ASML. The KLA technology is published, It uses a frozen Xenon gas with IR laser as the originator. **Ushio uses molten tin like ASML, but instead of using IR, they're using discharge electrodes, which is terrible. And the concept of maintaining two propellers creates a mess."**

Q: "It sounds like it takes ten years to figure it out. It sounds very complex."

A: "Correct...I think **that Isteq source will not be done in a professional way and it cannot work. You need to understand Isteq was done for research. Research is done in bursts, not continuously. For inspection, you need something to work 24/7, and that's it. And it's really hard. It's totally different.**" –Former KLA executive in a leadership role in their EUV mask inspection group

In addition to promoting the “new” and supposedly improved light source in the A300, Lasertec purports that the tool is High NA compatible, which refers to next generation EUV lithography with high numerical aperture (“NA”) to shrink the feature size. High NA is similar to regular EUV lithography but with larger optics with a 0.55 NA mirror vs. 0.33 currently, for better resolution and higher printing density, similar to changing the F-stop or aperture on a camera. ASML’s new High NA EUV scanners shrink chip features by 1.7X and triple the transistor density.

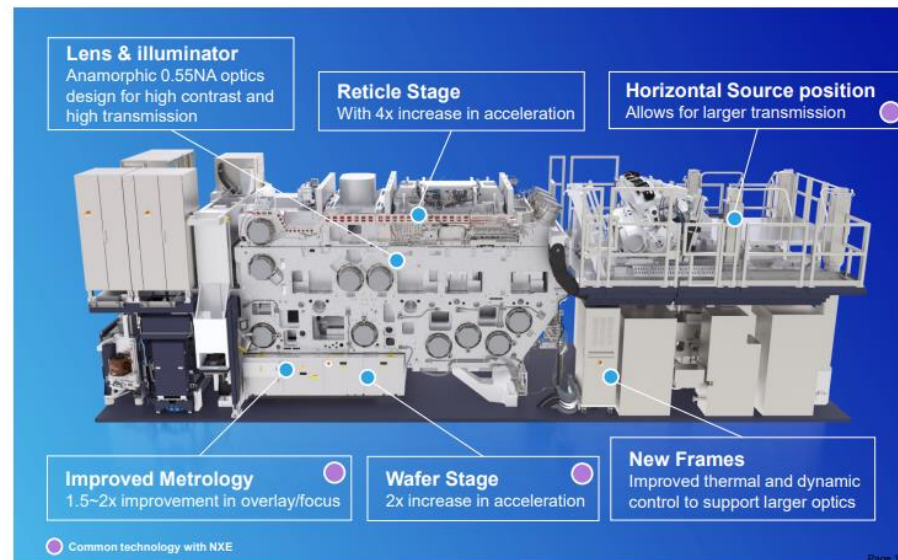
ASML presentation on its High NA EUV scanners, slide excerpt

High NA EUV

Increased NA enables shrink to 1.7x smaller features and 2.9x increased transistor density

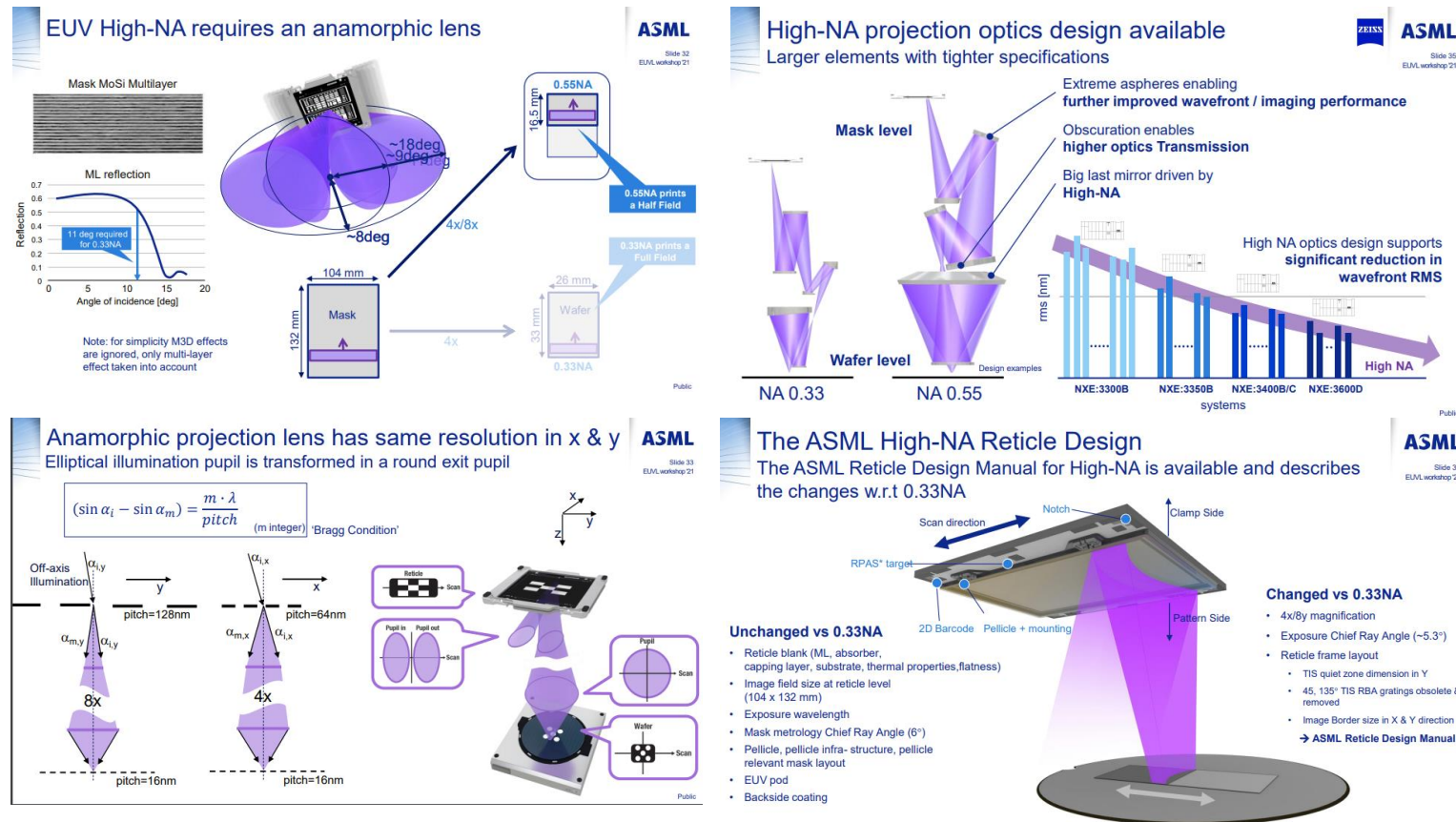
Increased contrast enables 40% improvement in local CDU at 1.4x lower dose

EXE:5000 product specifications	
Throughput at 20mJ/cm ²	≥185 WpH
Resolution	8 nm
Full wafer CDU & Focus	
• 10nm dense horizontal L/S	≤0.6 nm
• 14nm dense contact holes	≤1.0 nm
• Focus control budget	35 nm
On-product Overlay	1.7 nm
• Dedicated Chuck Overlay	≤0.9 nm
• Matched Machine Overlay	≤1.1 nm



However, changing the aperture to 0.55 for High NA requires completely overhauling the EUV source, optics, and other components required for actinic mask inspection – none of which Lasertec is capable of or appears to have done. ASML's materials explain the complexity of the upgrade, in particular the need for an anamorphic lens system and a significantly larger illumination system and projection optics versus current 0.33 NA EUV lithography and mask inspection.

ASML presentation on its High NA EUV optics illustrates the technical challenges, slide excerpts



We note that the German company Zeiss is the only developer of High NA anamorphic optics, as part of a strategic partnership with ASML, which precludes Lasertec from the intellectual property which is “secured by 2,000 Zeiss patents.” Zeiss states that its new High NA Mirror alone is twice as large and ten times as heavy as previous EUV mirrors, and required 1,500 dedicated staff - out of 7,500 total - to develop; and that each mirror takes an entire year to produce as they are “manufactured to atomic precision and consisting of more than a hundred layers...the mirrors are at the heart of High-NA-EUV technology.”

Zeiss website explains the complexity of new High-NA mirrors and optics - excerpts

The mirrors of High-NA-EUV lithography are unique in size and precision. Therefore we've developed a completely new system design.

Dr. Peter Kürz

Head of Field of Business High-NA-EUV lithography at ZEISS



The most precise mirrors in the world

Manufactured to atomic precision and consisting of more than a hundred layers, the mirrors are at the heart of High-NA-EUV technology. Compared to previous EUV lithography mirrors, they are significantly larger – and take around a year to manufacture.

The measurement technology for the mirrors of High-NA-EUV lithography is the most complex machine ZEISS has ever built. It was only possible thanks to the excellent collaboration with our strategic partner ASML and our network partners.

Dr. Thomas Stämmler

Chief Technology Officer ZEISS SMT

150 tons of absolute precision

Let's do a thought experiment: If the mirrors of High-NA-EUV lithography were to be enlarged to the size of Germany, the measurement technology would need to be able to measure accurately to 100 micrometers. Such measurement technology – precise in the subatomic range – did not exist until now. That is why ZEISS SMT has developed this measurement technology in parallel with High-NA-EUV lithography with its strategic partner ASML, to also be able to verify the quality of the manufactured mirrors.



Multiple industry executives in touch with Lasertec told us that the company's claims around the "new" ACTIS A300 and its purported High-NA capability are simply false and just vaporware – in contrast to KLA's upcoming machine which is "already ready for High NA." An ex-KLA executive in a leadership role in their EUV mask inspection group stated that he couldn't disclose details of Lasertec's technology given confidentiality agreements, but bluntly indicated that they do not have High-NA: "they don't have it now, without giving too much info on that – they don't have it...it will require replacement of their optics because...if they open up the aperture, the optics will block....they cannot do it right now...they need...a different curvature, different figure elements – it's a massive upgrade."

Unlike the upcoming KLA tool, Lasertec's machine is not capable of high NA – "they don't have it"

A: **"KLA's tool is already ready for high NA.** We didn't talk about high NA and low NA. Numerical aperture. ASML talks about the current ASML tool is a low NA tool. The numerical aperture is low, it's 0.2 or 0.something. They are going to go to 0.55. **It will more than double the resolution** because, remember, the resolution is equal to lambda over NA."

Q: "You're saying ASML's EUV tool is going to double the resolution as the aperture changes."

A: "Correct. The ASML High-NA tool they are saying will be next year. The HNA tool release, it says, 2026-2027. They delayed it. I knew that there was a delay. A year ago, I consulted on that and they said the end of the 2025 a year ago."

Q: "I'm looking. ASML to ship first High-NA EUV tool \$300 million per scanner."

A: "So, 300 million. They are almost doubling--the complexity of such a system is so high, it's incredible. I saw the actual system in [redacted]. I was there half a year ago for [redacted]. It's insane. But our KLA system, the mask inspection system has to be compatible for High-NA—**Lasertec will have to upgrade for High-NA and they don't have it now, without giving too much info on that—they don't have it. So, they will need to also invest money and do some fix to be compatible for High-NA. KLA, I'm mentioning because you asked about KLA pricing, KLA tool will always be High-NA compatible."**

Q: "How hard is it for Lasertec to make their tool High-NA compatible?"

A: **"It will require replacement of their optics because currently if they open up the aperture, the optics will block and... they cannot do it right now. So, they need to replace those models to a different curvature, different figure elements—it's a massive upgrade** or it's a new model."—Former KLA executive in a leadership role in their EUV mask inspection group

A longtime executive at Global Foundries, who deals directly with Lasertec and all aspects of EUV mask inspection, stated that Lasertec's November 2023 press release announcing the ACTIS A300 and its High-NA capabilities was "just a press release to match the shipment of the first [ASML] High-NA system to Intel, and they wanted to match that press release." He added that he has "not even heard of a single A300 system, a 300 system that shipped even...I don't know if it's pure vaporware." Intel ordered the first ASML high-NA EUV scanners in 2022, and ASML stated in Nov 2023 that the first pilot tools were about to ship. Lasertec's high-NA announcement was **simply a ploy to jump on the bandwagon of next-generation EUV lithography, despite lacking the technology to do so.**

ACTIS A300 is just a press release timed to coincide with Intel's High NA EUV announcement

*"I have not even heard of a single A300 system, a 300 system that shipped even. I don't know if it's pure vaporware, and it's **just a press release to match the shipment of the first High-NA system to Intel, and they want to match that press release.**" –*

Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

Dec 2023 press release announcing first shipment of ASML High NA system to Intel

ASML ships first "High NA" lithography system to Intel - statement

AMSTERDAM (Reuters) -Dutch semiconductor equipment maker ASML said on Thursday it is shipping the first of its new "High NA" extreme ultraviolet lithography systems to Intel Corp.

The new machines, which will cost more than \$300 million each, are expected to help computer chip makers produce smaller, faster semiconductors.

Three days before announcing its ACTIS A300 system in November 2023, several Lasertec engineers presented a paper at the SPIE photomask and EUV lithography conference, a professional society for optics and photonics. The 11-page paper focused on the A300's purported High-NA capabilities, yet was curiously vague on technical details in comparison to ASML's detailed presentations and contains only one short, 6-line paragraph on the projection optics key to High NA capability – without explaining how their optics could enable inspection of High NA anamorphic patterns which require different magnification in the X and Y directions. We are troubled that the paper is conceptual and written in the future tense as only a vision - in contrast to the press release days later that represents the High NA system as here and now.

Lasertec SPIE paper on ACTIS A300 High NA capabilities, Nov 21, 2023

Actinic Patterned Mask Inspection for High-NA EUV Lithography

Toshiyuki Todoroki¹, Ko Gondaira¹, Arosha Goonesekera², Hiroki Miyai¹

¹Lasertec Corporation, 2-10-1 Shin-yokohama, Kohoku-ku, Yokohama, 222-8552 Japan

²Lasertec USA, Santa Clara, CA 95054, USA

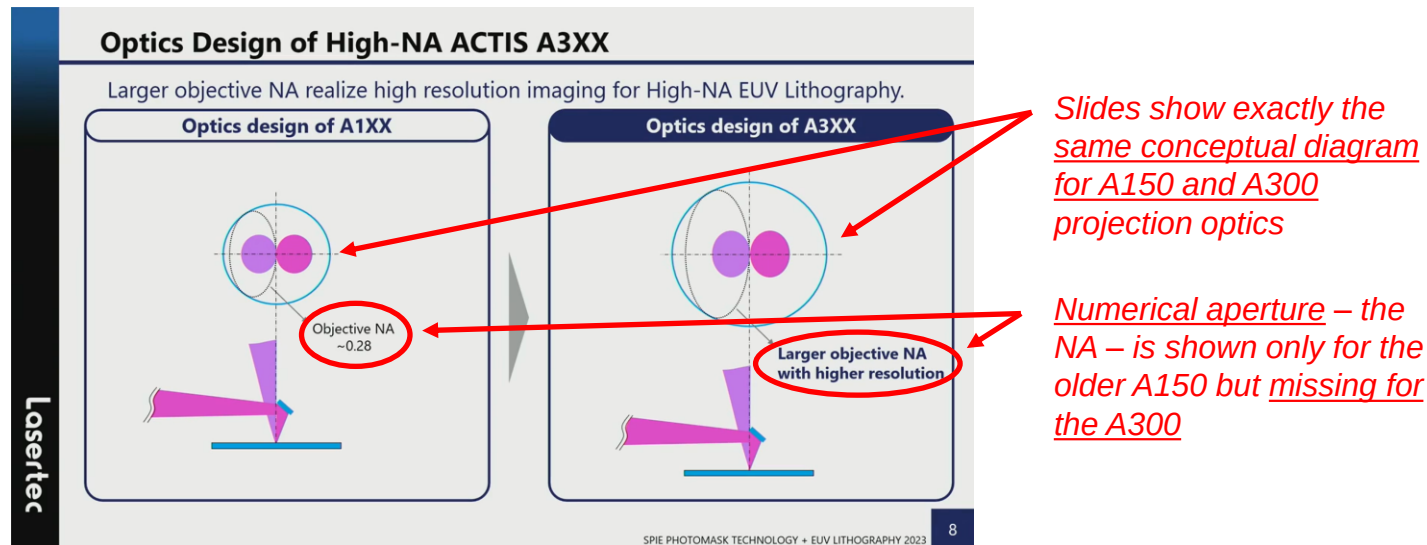
Paper contains only one vague, short paragraph on the system's optics, the heart of a High NA system

2.2 High NA projection optics of the next-generation ACTIS A3XX

One of the important aspects of the current ACTIS design is that it uses projection optics with asymmetric NA and resolution in the X and Y directions to maximize sensitivity and resolution. This feature is highly compatible with the inspection of anamorphic patterns, which have different sensitivity requirements in the X and Y directions. As shown in Figure 2, the next-generation ACTIS will follow the same design concept but with a higher NA, mainly to meet the high sensitivity requirements in the X direction for anamorphic patterns. This will provide optimal inspection for anamorphic patterns while allowing for further pattern width reduction in future technology nodes.

A diagram of the supposedly High NA optics in the A300 - shown in both the conference paper and accompanying presentation – is astonishingly lacking in technical specs or details. Instead of being transparent and showing photos of their High NA projection optics as ASML and its strategic partner Zeiss have done for years, Lasertec’s engineers show only a simple, crude drawing to compare the optics design in the A150 with the A300. Comically, both diagrams are identical except that the one for the A300 is just slightly expanded. As a major red flag, the diagram states the NA – numerical aperture – for the A150 but omits it entirely for the A300 and says only that it is a “larger objective NA.”

Slides accompanying Lasertec SPIE paper on ACTIS A300 High NA capabilities. Nov 21, 2023



As another red flag indicating that the A300 and its High NA capabilities are non-existent, we note that the paper fails to show its actual mask inspection capability. Instead, the paper resorts to a mathematical model versus actual results from using the machine, repeatedly using the word “simulation” or its variant: “imaging simulations using a rigorous optical simulator with electromagnetic calculations with conditions of circular illumination...in this simulation, A3XX shows a significant improvement in contrast....” If the A300 and its High NA technology were real, Lasertec engineers would have presented more credible information, similar to ASML, Zeiss, and others.

Lasertec SPIE paper on ACTIS A300 High NA capabilities, Nov 21, 2023 - excerpt

2.3 Resolution Simulation of High-NA ACTIS A3XX

To assess the inspection capability of the High-NA ACTIS A3XX, we performed imaging simulations using a rigorous optical simulator with electromagnetic calculations with conditions of circular illumination and a standard TaBn-based absorber with a thickness of 70 nm.¹⁶ Figure 3 shows the pattern contrast simulation of ACTIS A3XX with the high-NA objective mirror compared to the current ACTIS A1XX. In this simulation, A3XX shows a significant improvement in contrast, with more than 30% improvement for 30 nm half-pitch line-and-space and 64 nm contact holes.

*“Simulation” or “Simulator”
is used six times in the
short paragraph on the
A300’s High NA inspection
performance*

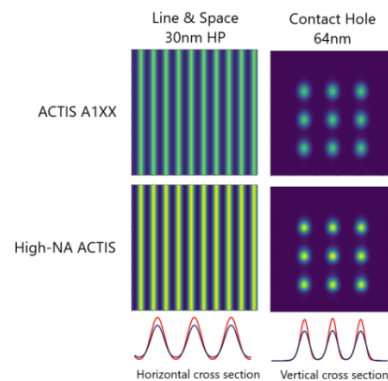


Figure 3. Simulation results of imaging capability of current ACTIS A1XX and High-NA ACTIS A3XX

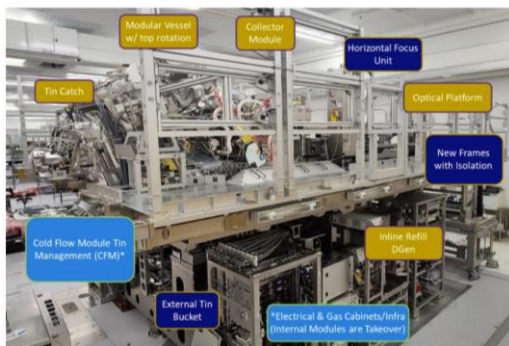
High NA EUV lithography and inspection also requires a new EUV light source with higher power, stability, and brightness than for conventional EUV at 0.33 NA, in addition to different optics given the higher aperture – an impossibility for Lasertec given its source is fatally flawed and doesn't even meet basic brightness and stability requirements for current EUV. A 2023 ASML paper presented at SPIE detailed the myriad new “requirements for source performance and geometry” for High NA, such as a new illuminator for “improved optical column transmission.”

ASML SPIE paper, April 2023 - excerpt

EUV light source for high-NA and low-NA lithography

Karl Umstadter, Matthew Graham, Michael Purvis, Alex Schafgans, Jayson Stewart, Peter Mayer, Daniel Brown

High NA (0.55NA) EUVL Scanners are in development for future nodes of device manufacturing, with new requirements for source performance and geometry. Our customers have all engaged in this program, some calling it the most important next step in lithography². In this paper we discuss our progress on the source integration, de-risking and acceptance testing for High NA.



Various changes and enhancements to upgrade the EUV source for High NA

- 5. Lasertec's EUV ambitions have always been vaporous given the impossibility of developing an EUV light source without investing USD billions. It is not - and will never be - a serious, credible player in EUV lithography/inspection, and investors who perceive it as "Japan's ASML" are misguided.**

As part of our investigation, we spoke to two of the most important and knowledgeable authorities on EUV light source technology for lithography and mask inspection: 1) Cymer, which developed the first working source and was acquired by ASML and became the foundational technology for ASML's EUV scanners; and 2) Isteq, a Netherlands-based affiliate of a Russian institute focused on plasma research and light source development, from whom Lasertec licensed their "new" EUV source, which they renamed "Urashima," misrepresenting it as a Japanese technology. We begin with the first expert, who spent over a decade working on EUV light sources at Cymer prior to its acquisition by ASML in 2013 and is one of the top scientists in the field. He recently led the EUV source development program at one of the largest semicap equipment companies, which we don't name to respect his confidentiality.

EUV plasma light source expert we consulted is one of the top scientists in the field; developed ASML's EUV source
"I spent 10 years at Cymer. I was hired actually for the EUV source. I spent 10 or 11 years working on all kinds of EUV sources. Around [year redacted], [semicap company name redacted] thought to do EUV mask inspection, and then they needed the EUV source guy, and they hired me. I was the leader of the EUV source development, then later EUV source and optic development."
– Leading figure in the development of Cymer/ASML's EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

The ex-Cymer scientist emphasized the extreme difficulty and cost of developing a reliable EUV source, highlighting that ASML spent \$4 billion to acquire Cymer in 2013 – the cost of entry just to have a light source to power their scanners. He stated that Cymer’s source still wasn’t working in 2013 and that the “scope and size of the investment” that was still required “was way above what Cymer could afford to make EUV technology happen” - “that’s why ASML had to buy Cymer.”

ASML had to invest \$4 billion to acquire Cymer for their EUV source; “scope and size of the investment was way above what Cymer could afford”

*“You can dig out in terms of the reason why ASML bought Cymer because Cymer said that they are not in a position to invest more because the **scope and size of the investment was way above what Cymer could afford to make EUV technology happen**. And they threatened ASML to quit because it's not sustainable, and that's why ASML had to buy Cymer. They had to buy a lot of basically infrastructure for EUV to make it happen. **They invested literally billions. \$3.6 billion to buy Cymer. They had a separate division and a separate building for that**. These are not unknown facts. You can google it.”* – Leading figure in the development of Cymer/ASML’s EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

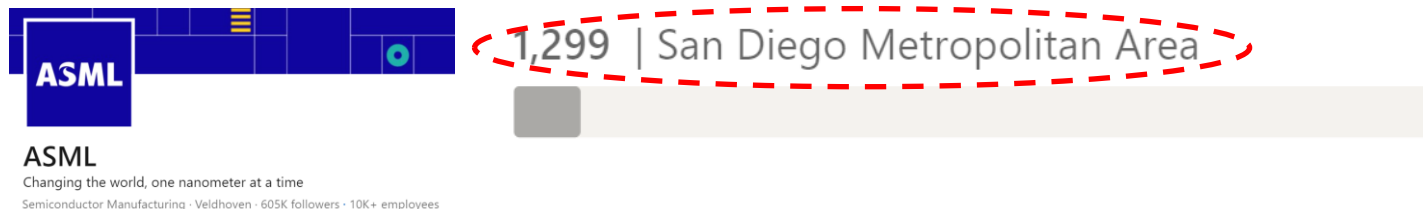
Press regarding ASML closing of Cymer acquisition, May 2013

ASML Holding NV Completes Acquisition of Cymer for \$3.7B

The Cymer scientist stated that it is impossible for Lasertec to replicate a tin plasma-based LPP source similar to ASML's, as it claims with its new “Urashima” technology. He stressed that it took Cymer and then ASML a combined 20 years of R&D to make it work, with Cymer having 300 dedicated staff just for the source; that ASML then spent “several billion dollars” more in R&D “on the source only,” after spending \$4B to buy Cymer; and that there are currently ~600-1,000 ASML staff in San Diego facility where Cymer was based, dedicated just to the EUV source – “that’s the scale”; “Lasertec is not in a position to throw 100 people working on a source.” We note that LinkedIn confirms that ASML has ~1,300 staff in San Diego, where its EUV source division is based.

Impossible for Lasertec to replicate ASML's source; took Cymer/ASML 20 years and required hundreds of people; required “several billion dollars on the source only”; ASML currently has ~1,000 staff working on their EUV source; “that’s the scale”
“Now, the ASML EUV source is tin-based. That's what I'm kind of telling, that yes, it's possible, but it would require hundreds of people, which are like 100 people, 10 years, and a lot of money. I don't think that, for example, Lasertec is not in a position to throw 100 people working on a source. **ASML and Cymer spent 20 years, and, on average, it was probably 300 people working per year, and then 20 years. ASML spent several billion dollars on the source only - maybe a couple of billion dollars in development. That's the scale.** I joined ASML at Cymer in 1999 or maybe 2000, and I worked there for 11 years. And it went into production basically in 2017. That tells you the scope. When I left Cymer, there were 250 or 300 people in the source development team engineering. **Right now, if you go to San Diego, I'm sure that there are 600-700, maybe 1,000 people working on it. It's a huge development.**” – Leading figure in the development of Cymer/ASML's EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

ASML's LinkedIn profile shows 1,300 staff in San Diego where Cymer EUV source group is based



In comparison to the 1,300 staff at ASML's EUV source group and the \$4B it spent on Cymer plus billions more in additional R&D, Lasertec is small with only 859 total employees in 2023, and historically invested little in R&D – generally \$9MM to \$15MM in R&D in the years preceding its EUV market entry in 2019. It strains credibility that Lasertec was able to develop a reliable EUV source with less than 1% of what Cymer/ASML spent. Ushio – its EUV light source supplier – spends \$80-90MM annually on R&D, but EUV sources are a small percentage of its revenue and EUV R&D expenditure would have been a tiny fraction of the total. Isteq, which developed Lasertec's "new" Urashima EUV source is little more than a Russia/Netherlands-based science project with only 12 employees on LinkedIn.

Lasertec and Ushio R&D expenditure was minimal prior to 2019 launch

\$ millions	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
<u>Lasertec</u>										
Revenue	134	124	149	155	192	267	395	633	665	1,057
R&D expenditure	10	9	15	13	25	33	31	51	63	76
<u>Ushio</u>										
Revenue	1,533	1,328	1,594	1,550	1,634	1,490	1,478	1,072	1,226	1,317
R&D expenditure	87	91	100	88	101	91	95	91	87	86

"New" Urashima source supplier only has 12 employees on LinkedIn

ISTEQ B.V.
 We make light brighter
 Nanotechnology Research · Eindhoven, North Brabant · 225 followers ·

[+ Follow](#) [Visit website](#) [...](#)

Home About Posts Jobs People **Insights**

PREMIUM

Total employee count ⓘ
 Based on LinkedIn data.

12
 total employees

The ex-Cymer scientist stated that ASML initially began with Ushio's source in 2009 but that it was a total failure: "it didn't work...they screwed it up"; the Ushio source "has a lot of problems" such as tin-based "discharge that...creates a lot of junk...nearby optics get literally covered in tin pretty quickly" – "still way more dirty than everybody wants it." He detailed the difficulty of developing a working source through ASML's failures for a decade, even years after spending billions on Cymer – illustrating Lasertec's insurmountable EUV light source problem.

ASML initially used the same Ushio source as Lasertec and "it didn't work...they screwed it up"; ASML struggled and failed repeatedly for a decade after in trying to get an EUV source to work, illustrating Lasertec's unsolvable light source problems
"ASML started a litho tool back in 2009 with an Ushio source. And it didn't work. And they screwed it up. They came to Cymer. They gave tons of money to Cymer, asked us, and they basically started a fully ramped EUV program. Two or three years later, they got their first EUV source from Cymer, and it didn't work. I was on the install team for the first couple of sources, and it didn't work. They screwed it up. They pushed it back by two years and bought Cymer. They hired people. They invested in infrastructure. And, again, in 2015, they failed miserably. They started to send those tools which produced the light but only for a very short period of time because the collector got contaminated. And then they spent another two or three years working on it. In 2017, they did not meet the requirements. ASML's margins on those were negative—they lost money. But they kept going, replaced them, improved them." — Leading figure in the development of Cymer/ASML's EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

Lasertec's Ushio EUV source "has a lot of problems" in "cleanliness and lifetime"; "lot of junk" covers "nearby optics" with tin "pretty quickly; methods to solve the problems "didn't really work"; "way more dirty than everybody wants"

Q: "What do you know about the Lasertec EUV light source?"

A: "They used Ushio. It's a problem. The source has a lot of problems, mostly in the cleanliness and lifetime. The lifetime, since it's a tin-based and there is some discharge that actually creates a lot of junk every time, and then nearby optics get literally covered in tin pretty quickly. They have to do something about it. It's difficult. There are several methods, but they didn't really work. They tried probably a combination of all of them...it's still way more dirty than everybody wants it." — Leading figure in the development of Cymer/ASML's EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

The Cymer expert also slammed the “new” Urashima source that Lasertec has licensed from Isteq, and we view his color as particularly damning given his intimate knowledge of Isteq and its technology. He stated that neither Lasertec nor Isteq has ever shown this source works and that it’s just a little lab experiment: “they did not show anything – I am really skeptical”; “as is, it would not work...it’s nothing but a lab prototype, slightly more than an idea.” He added that this source “is also tin-based...that’s why they would have similar problems” and that tin simply doesn’t work for EUV mask inspection: “requirements for cleanliness are ridiculous...I don’t think tin has any chance to do it.”

Lasertec nor Isteq have ever shown that the new light source works; “I am really skeptical”; just “a lab prototype” and only “slightly more than an idea”; “as it, it would not work”; “very difficult technology”

“They did not show anything - I am really skeptical about that source from Isteq or from the Russians. As is, it would not work. It’s nothing but a lab prototype, slightly more than an idea. There is some hardware around it, but I don’t think that it’s doable...it’s difficult. It’s not a surprise that they have a lot of problems with that source as well. It’s a very difficult technology” – Leading figure in the development of Cymer/ASML’s EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

New Urashima light source “is also tin-based” and has “similar problems”; tin doesn’t work for “EUV mask inspection”; “requirements for cleanliness are ridiculous”; tin has no “chance to do it”

“That’s why Lasertec was looking for a cleaner source or for a better source. They licensed the technology from Russia/Isteq, basically the idea and maybe bought a piece or two for internal development...that technology is also tin-based. That’s why they would have similar problems...I am personally not a big fan of any tin-based sources, especially for EUV mask inspection, because the requirements for cleanliness are ridiculous. So, I don’t think that tin has any chance to do it. But, again, then, you can basically physically block tin, but then it would mean that you lose a lot of light. That’s probably what they’re doing at this moment. And then, they still caught a lot of EUV photons and they get a lifetime out of that, and they hope to improve what they call the debris mitigation strategies to the point where they can either fully open or lens block or something like that.” – Leading figure in the development of Cymer/ASML’s EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

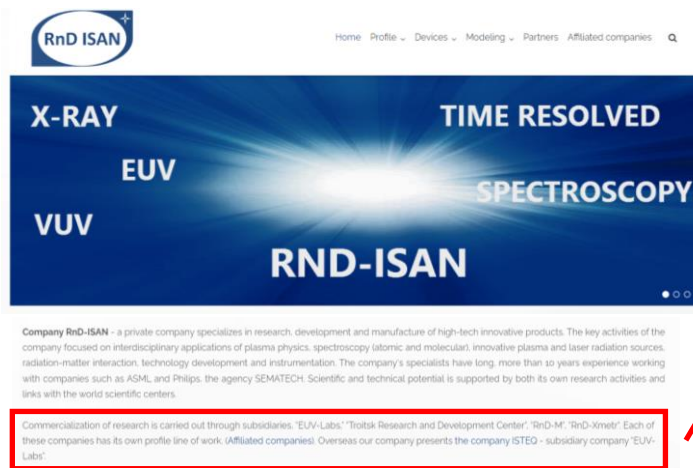
The expert detailed Lasertec's history of trying to use different light source suppliers and repeatedly failing, stating that they failed to scale from the simple Xenon-based EUV source they procured from Energetiq for their blank inspection tool, as actinic mask inspection is more complex: "the optics are much more complicated and they need significantly more light...that's why they are struggling...they moved to Ushio...they're still struggling...they have to move to something else." He stated that Lasertec **"cannot deliver" on their purported \$2 billion backlog due to their lack of a properly functioning EUV light source: "they have a \$2 billion backlog, which means that they probably have orders for 40 or 50 systems that they cannot deliver right now."**

Lasertec needs "significantly more light" from their source and "are struggling"; have never gotten any solution to work
"They started from a much simpler tool they call a blank inspector. The blank inspector - they use an Energetiq source, which is sufficient for that one. It's the only available commercial EUV source. It's not tin-based; it's Xenon-based. This company, which is Hamamatsu, which used to be called Energetiq, just outside of Boston. Now there are problems since the optics are much more complicated, and they need significantly more light. And that's why they are struggling. They moved to Ushio, tried to make it happen. They're still struggling. They have to move to something else." – Leading figure in the development of Cymer/ASML's EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

Lasertec "cannot deliver" on their backlog due to their lack of a functioning EUV light source
"In order for them to succeed and deliver all the - they have a \$2 billion backlog, which means that they probably have orders for 40 or 50 systems that they cannot deliver right now, basically something like that. I think it's publicly known that Lasertec has more than \$2 billion in backlog—they did announce it. Everybody suspects that 90% of that is EUV mask inspection; maybe it's a little bit of that is EUV blank inspection. If they have a few-plus billion dollars in backlog, it means that somebody placed orders for their EUV mask inspectors." – Leading figure in the development of Cymer/ASML's EUV light source; headed EUV source effort at one of the largest semicap equipment companies; one of the top scientist/experts in the field

The Cymer scientist's assessment of Lasertec's defective light source was corroborated by Isteq, the Russia/Netherlands company that developed the new "Urashima" source that Lasertec is now using. We spoke with a longtime, senior member of the technical team at Isteq, which is a Netherlands entity affiliated with a Russian firm which calls itself "RnD-ISAN" or "EUV Labs." Despite positioning itself as a non-Russian company given the current Russia sanctions, Isteq appears to be a subsidiary of RnD-ISAN. We note that the enterprise is extremely small and appears to be little more than a research institute, showing only 12 staff on LinkedIn. RnD-Isan's Russian website no longer appears to be working, but a cached version at the Internet Archive's Wayback Machine states that "overseas our company presents the company Isteq – subsidiary company EUV-Labs."

Isteq appears to be a subsidiary of Russia-based RnD-ISAN, per cached version of www.rnd-isan.ru



"Overseas our company presents the company Isteq – subsidiary company EUV-Labs"

The Isteq engineer stated that “we developed it,” referring to Lasertec’s new LPP source and revealed that they actually licensed it to Lasertec many years ago, making it far from “new”: “actually it was quite a long time ago...more than two or three years ago.” He emphasized that the EUV source is the heart of the entire machine: “the source part, the lamp, is crucial...absolutely – the source is the key.”

Isteq developed the actinic light source

*“Isteq specializes in making **light sources for actinic inspection...we developed it**...the history of Isteq is rather complicated, but Isteq has no relation to Russia nowadays. It’s not a secret that many of our employees are from Russia...our technical solutions are protected by patents that are officially registered in the U.S. and Europe, and in Japan and other countries.” – Key R&D employee at Isteq*

Isteq licensed the LPP source to Lasertec more than 2-3 years ago

Q: “Are you able to say when you licensed the technology to Lasertec?”

A: “Actually, it was quite **a long time ago. More than two or three years ago.**” - Key R&D employee at Isteq

EUV light source is the most critical part of the machine

“The inspection machine consists of two parts: the source itself and the inspection part where the inspection actually happens. The source is a sort of lamp, and this inspecting part is where you put the sample, and you try to find different defects. It’s not obvious to an observer, but **the source part, the lamp, is crucial**. So, the quality of the lamp determines the availability and the reliability of the whole machine. **Absolutely - the source is the key**. The source, I call it the place where we produce plasma. And this plasma radiates the light. We collect this light, and we transfer the light to the intermediate focus, where our inspection happens. So, our inspection happens either in the intermediate focus or very close to the intermediate focus.” - Key R&D employee at Isteq

The Isteq engineer was wary of revealing that they licensed the technology to Lasertec, perhaps because of the Russian connection: “Urashima...is a new source...for their newest A300 series, they are using a new light source, and it’s not an Ushio light source...this general concept was shared between the companies...now Lasertec and Isteq both have the same concept.”

Lasertec A300 is using Isteq’s light source technology, but Isteq appeared extremely wary of admitting it

Q: “Is Lasertec still using the Ushio light source, or are they using a different light source now?”

A: “From my information, **their new machine uses, you correctly called it Urashima. Urashima is the name of the light source.**

This is a new source. For their newest A300 series, they are using a new light source, and it’s not an Ushio light source.”

Q: “Are they developing it themselves or licensing it from somebody? Are they getting it from you?”

A: “This is their property. I just wanted to say that this is their property. They have full access to this technology.”

Q: “The only reason I’m asking is because they mentioned Isteq in their press release.”

A: “Yes, they mentioned Isteq. It looks like you know the materials that were introduced at a conference. It’s like this: there was the general concept of the source, and **this general concept was shared between two companies. Now, Lasertec and Isteq both have the same concept**, but they have different engineering solutions.”

Q: “Okay, so this is not a situation where they’re buying the light source from you like they were buying the light source from Ushio. They’re just doing their own thing.”

A: “No. Isteq has its own light source; Lasertec has its own light source... **We are independent companies. That’s all I can say.**” – Key R&D employee at Isteq

Shockingly, the Isteq engineer stated that the Urashima source they licensed to Lasertec has the same problems as Ushio's: "I don't see why it should be any different...no ultimate solution...there is no cure" – echoing the Cymer scientist's conclusion that Lasertec's tin-based approach is doomed no matter what. Isteq detailed how Urashima produces the same contamination as Ushio: "LPP machines are susceptible to different types of contamination...millions of micro-explosions...a lot of debris and vapor and tiny particles which have high speed and energy...like bullets...problem of almost all LPP laser-produced plasma machines...Lasertec's source...it's definitely not an exception...tin vapor everywhere...they ricochet...and they penetrate almost everything."

Isteq states the new Urashima light source has the same problems as the Ushio one: "no cure"; "no ultimate solution"; "different types of contamination"; "lot of debris and vapor and tiny particles"; like "bullets" that "ricochet"

A: "There are **potential risks that may come from the new light source, of course**. When we talk about the quality of the light source, we are talking about the **variability of the machine, the maintenance, and the lens. LPP machines are susceptible to different types of contamination**. We shoot with lasers on a tin target and just imagine **millions of micro-explosions and they produce a lot of debris and vapor and tiny particles which have high speed and energy. And these particles work like bullets**. This is a problem of almost all LPP laser-produced plasma machines. I don't think that Lasertec's source is an exception. So, it's **definitely not an exception. So, you have vapor everywhere, tin vapor everywhere**, so you have to protect the mirror from tin vapor and fast-moving tin particles. You have to protect the sample from tin particles because those tin particles are everywhere, and **they ricochet from the walls of the vacuum chamber and they penetrate almost everything.**"

Q: "You're talking about the old light source, or the Urashima light source has the same issue?"

A: "No, I'm talking about most light sources where laser-produced plasma exists."

Q: "Including the new one?"

A: "Yes. **I don't see why it should be any different from any others in this respect.**

Q: "It sounds like they have **all the same problems with the Urashima light source that they had with the Ushio light source.**"

A: "**Yes, of course. There is nothing that works like a charm in this world. There is no ultimate solution. There is no cure.**" – Key R&D employee at Isteq

Isteq bluntly dismissed Lasertec's Ushio source and its DPP discharge-based technology as a complete failure: “DPP...machines have no future...Ushio made a strategic mistake because they were somewhat deluded by the deceptive simplicity of DPP and they chose this way mistakenly.” He added that Lasertec's problems go far beyond the source and encompass other components: “the inspection machines are not in a mature state...this is the main problem in the industry...most of the designs...are plagued by different problems”; Lasertec's “technical problems stem from two different sources...the [light] source and the second is...the testing part of the system itself.”

Ushio made a massive error in using DPP technology for their source – “DPP...plasma machines have no future”

Q: “The Ushio light source has caused a lot of problems for Lasertec?”

A: “Partly because of it. There are two different types of sources: LPP and DPP, laser-produced plasma and discharged-produced plasma. Most sources introduced in the market today are LPPs. **DPP, discharge-produced plasma machines have no future.** I think that **Ushio made a strategic mistake because they were somewhat deluded by the deceptive simplicity of DPP machines, and they chose this way mistakenly.** So, now they have no choice but to develop what they have. They have to stick to it; they can't just throw it away.” – Key R&D employee at Isteq

EUV mask inspection machines are “not mature” and are “plagued by different problems”; Ushio “a strategic mistake”

A: “Most of the **inspection machines are not in a mature state. This is the main problem in the industry.** Literally, most of the designs introduced in the market, they are **plagued by different problems.** Different problems, like a child growing up.”

Q: “I think you're probably talking about Ushio? I talked to some of Lasertec's customers, and they've all said that the Ushio light source breaks down, they have to replace it, it gets dirty. It sounds like the light source is unstable.”

A: “Ushio is a strange scene. I have no doubt that they accumulate very good engineers and scientists. But from my point of view, **they made a strategic mistake.**” - Key R&D employee at Isteq

Lasertec product problem stems from both the light source and the rest of the system

“Their technical problems stem from two different sources. First, these are problems related to **the source.** And the second is the problem related to **the testing part of the system itself.**” - Key R&D employee at Isteq

When we asked Isteq if they are in touch with Lasertec's key customers – namely Intel and TSMC – they indicated that they were, but were vague due to NDA's. Nonetheless, Isteq indirectly indicated that both are unhappy with Lasertec's inspection machine: "it's under NDA so I cannot reveal the nature" of the communication with TSMC or Intel; "no company...is fully satisfied with their inspection product...absolutely every product now has different problems...no company...APMI, actinic patterned mask inspection is a huge challenge."

Isteq was vague due to NDA's but indicated that both Intel and TSMC are unhappy with Lasertec's product: no one is "fully satisfied"; "different problems"; "huge challenge"

Q: "Do you ever speak to TSMC or Intel?"

A: "Yes, but I cannot reveal the nature of my... I communicate with different people from different companies, but it's under NDA so I cannot reveal the nature of it."

Q: "Do you know if Intel and TSMC are happy with the Lasertec product?"

A: "At this moment, there is **no company that is fully satisfied with their inspection product. So, absolutely every product now has different problems...no company is fully satisfied...APMI, actinic patterned mask inspection, is a huge challenge.**" – Key R&D employee at Isteq

While Lasertec represents Urashima as a “new” and improved EUV light source, it is simply an old Isteq LPP laser-produced plasma technology developed over a decade ago. Lasertec could easily have licensed the Isteq technology prior to the ACTIS A150 launch in 2019 but chose Ushio instead. Given that there are only two or three EUV source suppliers to choose from, it is obvious that Lasertec would have evaluated Isteq over five years ago prior to rejecting it for Ushio – and are now desperately trying to cover up the Ushio failure and trying Isteq as a Hail Mary. We note various Isteq papers going back to 2013 on their LPP system.

2013 presentation by Isteq’s parent entity



Research review on plasma-based EUV sources at RnD-ISAN/EUV Labs

V.Krivtsun, D.Abramenko, V.Bushuev, A.Chekmarev,
R.Gayazov, M.Spiridonov, O.Yakushev, G.Zukakishvili and
K.Koshelev

ISAN, RnD-ISAN, EUV Labs
Moscow, Troitsk

2013 EUV Sources Workshop Nov 4-7 Dublin, Ireland

2016 paper by the parent entity

RESEARCH ARTICLE **OCTOBER 31 2016**

Droplet-based, high-brightness extreme ultraviolet laser plasma source for metrology

A. Yu. Vinokhodov ; M. S. Krivokorytov; Yu. V. Sidelnikov; V. M. Krivtsun; V. V. Medvedev;
K. N. Koshelev

We report on the development of a high brightness source of extreme ultraviolet radiation (EUV) with a working wavelength of 13.5 nm. The source is based on a laser-produced plasma driven by pulsed radiation of a Nd:YAG laser system. Liquid droplets of Sn-In eutectic alloy were used as the source fuel. The droplets were created by a droplet generator operating in the jet break-up regime. The EUV emission properties of the plasma,

In a LinkedIn post coinciding with Lasertec's November 2023 announcement for its "new" ACTIS A300 with the Urashima source, Isteq confirmed that Urashima is based upon a license for its TEUS LPP product. We note that both Isteq and Lasertec misrepresent the technology as new – "a new approach for high brightness laser produced plasma EUV source" – despite licensing it to Lasertec over two or three years ago.

Isteq LinkedIn post circa November/December 2023

ISTEQ B.V.
226 followers
4mo •

A New Approach for High Brightness Laser Produced Plasma EUV source - Fast-rotating liquid metal target (FRLMT) irradiated by a short pulse laser beam.

One of the main critical applications related to the implementation of EUVL is actinic mask inspection. **Isteq Group Holding** has been working with **Lasertec Corporation** to develop a reliable and industry-worthy light source for Actinic Patterning Mask Inspection (APMI). This has resulted in **Isteq Group Holding** providing **Lasertec Corporation** with FRLMT technology licenses.

Isteq is very pleased and honoured to work with **Lasertec Corporation** to provide this technology as the basis of their current product (URASHIMA) which is used in ACTIS systems. We are looking forward to continue our collaboration.

Isteq Group Holding has developed and manufactured a state-of-the-art high brightness product - Tin EUV Source (TEUS) - based on the FRLMT technique. Multiple sources have been shipped around the world for various EUV applications. The product has a high duty cycle and turnkey operation (user-friendly along with automated protocol to ensure the system runs smoothly with no risk and interruption).

The current TEUS product has three variants: TEUS-S100, TEUS-S200 and TEUS-S400. S represents the solid angle and the number 100, 200 and 400 refers to the average laser power used respectively.

For more information please contact:
sales@isteq.nl
www.isteq.nl

Isteq and its Russian parent RnD-ISAN jointly filed an international patent application for their LPP technology back in 2019/2020, and branded it as their “TEUS” EUV source product in sales presentation in 2021, if not earlier – once again illustrating that “Urashima” – TEUS in disguise – is far from new. The fact that the Isteq source was available prior to the ACTIS A150 launch in 2019/2020 suggests that Lasertec evaluated it years ago and concluded it was inferior to the Ushio source they first chose.

Old patent application by Isteq/RnD-ISAN

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(71) Applicants: **ISTEQ B.V.** [NL/NL]; High tech campus 9, 5656 AE Eindhoven (NL); **RND-ISAN, LTD** [RU/RU]; Promyshlennaya St., 2B, Troitsk, Moscow 108841 (RU);

EUV LABS, LTD. [RU/RU]; Promyshlennaya st., 2B, Troitsk, Moscow 108841 (RU).

(72) Inventors: **VINOKHODOV, Aleksandr Yurievich**; Sineyevy Blvd., 10, apt. 116, Troitsk, Moscow 108840 (RU); **IVANOV, Vladimir Vitalievich**; Novatorov St., 36/3, apt. 336, Moscow 119421 (RU); **KOSHELEV, Konstantin Nikolaevich**; Troitskiy Blvd., 4, apt. 7, Troitsk, Moscow 108841 (RU); **KRIVOKORYTOV, Mikhail Sergeyevich**; Militseyskaya St., 62, apt. 114, Kirov region, Kirov 610027 (RU); **KRIVTSUN, Vladimir Mikhailovich**; Otkryabryy Prospekt, 23, apt. 85, Troitsk, Moscow 108840 (RU); **LASH, Aleksandr Andreevich**; Malyy Karetnyy per., 9-3, Moscow 127051 (RU); **MEDVEDEV, Vyacheslav Valerievich**; Fizicheskaya St., 12, apt. 50, Troitsk, Moscow 108840 (RU); **SIDELNIKOV, Yury Viktorovich**; Otkryabryy Prospekt, 2, apt. 128, Troitsk, Moscow 108840 (RU); **YAKUSHEV, Oleg Feliksovich**; Sacco and Vanzetti St., 14, apt. 8, Ko-

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HIGH BRIGHTNESS LASER-PRODUCED PLASMA LIGHT SOURCE

FIELD OF INVENTION

[0001] The invention refers to high brightness laser-produced plasma (LPP) light sources designed to generate soft X-ray, extreme ultraviolet (EUV) and vacuum ultraviolet (VUV) radiation at wavelengths of approximately 0.4 to 200 nm, which provide highly effective debris mitigation to ensure the long-term operation of the light source and its integrated equipment.

2021 sales presentation for Isteq's LPP source



TEUS - high-brightness EUV LPP light source based on fast rotating target:

product overview and specifications



2021 EUVL Supplier Showcase

6. Lasertec's largest customers – notably TSMC, Intel, and Samsung, which are 77% of its revenue – are struggling, exasperated, and/or furious with its EUV mask inspection system, as well as its DUV systems, which we believe to still comprise the vast majority of its revenue. Key customers are “really struggling about what to do”; “it’s painful, extremely painful” to use Lasertec’s tool; “it would be stupid” to buy more machines.

Our research interviews indicate that Lasertec's largest customers – notably TSMC, Intel, and Samsung – are struggling, exasperated, and/or furious with its EUV mask inspection system, as well as its DUV systems, which we believe to still comprise the vast majority of its revenue. In this section, we shall quote detailed interviews with TSMC, Intel, Samsung, Global Foundries, and Marvell, in addition to former Lasertec employees based on-site at these customers, as well as experts in communication with these customers. We begin with an ex-KLA executive who described direct conversations with Lasertec and who was in a leadership role in their EUV mask inspection group. He stated that he began discussions “two years ago” with key customers who are “really struggling about what to do” with Lasertec: “the discussions are really hard.” Although Lasertec’s entire “monopoly” position in EUV mask inspection is based on its purported ability to inspect masks with a pellicle, the tool cannot reliably do so.

Customers “really struggling about that to do” with Lasertec tools failure to inspect pellicles: “discussions are really hard”

Q: “Got it. Are any of these customers—Samsung, TSMC, or Intel—have they had enough problems with Lasertec where they're just like we're going to now not use pellicles?”

A: “The **discussions are really hard**. I started to have them two years ago when I was there—**they're very hard with the customers. They're really struggling about what to do**. I'm not saying that the only reason why they delayed N3 but N3 right now is delayed.”

Q: “You're saying N3 is delayed because there's no good solution for pellicle inspection?”

A: “I'm not saying only because of that—that's part of the reasons. There are multiple reasons but definitely one of them is the ability to inspect the EUV pellicle.” -Former KLA executive in a leadership role in their EUV mask inspection group

While Lasertec's purported "monopoly" position is dependent on inspecting EUV masks with a pellicle – a thin membrane to protect the mask from debris – the problems with its tool are "too hard to fix and it takes too much time." Customers who bought it hoping to inspect with pellicles "have been paying a huge price" and now simply take the risk of inspecting without a pellicle rather than deal with the tool: "it's painful, it's extremely painful"; "they basically print without a pellicle"; "all these failures...slows down all the flow in the fab"; "it would be stupid to buy many more" machines; "I would do everything I can to avoid it."

Lasertec's EUV tool is only used because it purports to inspect masks with pellicles, but it can't: "they lied"

Q: "So, the only way you could use a pellicle is if you had a way to inspect it with a pellicle and Lasertec was the only game in town to inspect with a pellicle. Is that right?"

A: "Exactly. You got it."

Q: "Is that why they got all this market share, that they were the only game in town for pellicle?"

A: "Exactly. Basically, that's exactly what Intel told them. And they lied. They said, you know what guys? Others will buy this tool because you're the only ones and then you will have time to fix your shit together. **The problem is the shit is too hard to fix and it takes too much time."**

Q: "So, Lasertec is really just a P3 inspection tool."

A: "No, but I explained before that when you have pellicle, P2 and P3 are the same tool. So, it's P2/P3." -Former KLA executive in a leadership role in their EUV mask inspection group

"Stupid to buy" the Lasertec tool; "it's extremely painful"; "slows down all the flow in the fab"

Q: "Is anybody willing to buy an ACTIS A150 anymore with an Ushio tool or nobody's going to buy any because of the light source?"

A: "If I was a fab and I didn't have a P3 tool, I would buy it. I have no other choice. I would buy it. If I have to produce a mask, I have to buy it. Although, **it would be stupid to buy many more of those because I would do everything I can to avoid it. Some have been paying a huge price, they basically print without a pellicle because they don't want to inspect with a Lasertec tool. That's also a reason. But it's painful, it's extremely painful. And all these failures, it just slows down all the flow in the fab."** -Former KLA executive in a leadership role in their EUV mask inspection group

Customers are furious because they buy Lasertec's tool in order to inspect masks with a pellicle, but then can't use a pellicle given its inspection deficiencies, which compromises their entire inspection workflow given the risk of debris from not using a pellicle, resulting in yield issues, delays, and bottlenecks: "it's really a huge issue for the fabs"; "if you don't have a good actinic tool, you cannot use a pellicle"; "a particle will fall on the mask...every wafer will be zero yield"; "so what is the benefit?...don't use a pellicle but it's very painful." In the absence of pellicle inspection capability, customers just default to their legacy KLA mask inspection tools: "because you don't have a pellicle...so you just use the KLA tool" – negating the "EUV monopoly" thesis underpinning Lasertec's stock.

Lasertec tool causes its key customers huge delays, yield problems, and bottlenecks

Q: "You're saying they don't want to use Lasertec without the pellicles. Walk me through that again."

A: **"If you have an EUV mask and you decided not to put pellicle, so your P1, and P2/P3 inspection changes. You have only P1 and P2, and P2 now because you don't have a pellicle, so you just use the KLA tool.** After you finish, you clean the mask. You hope that there is nothing on top of it because you checked it with the KLA tool. You pattern it carefully. You move it to the fab. You put it on the scanner and start to print wafers. However, you print only 2000 wafers, not 10,000, and then stop to inspect them, almost all of them, to make sure that during printing, nothing was added and all the wafers are okay—and usually, more than 50% of the time, some of the wafers will be lost. Then you need to discard those wafers. **It's really a huge issue for the fabs."**

Q: "You're saying there are **delays, yield issues, bottlenecks in fabs because of the Lasertec tool?**"

A: **"Absolutely.** What happens is if you don't have a way to inspect EUV and use a pellicle, most likely, a particle will fall on the mask, and from then on, every print or every wafer will be zero yield. if you don't have a good actinic tool, you cannot use pellicle. So, **what is that benefit? Don't use a pellicle, but it's very painful.** That's what I'm saying."

Q: "But you're saying people are having enough problems with the Lasertec tool that they'd rather just do it without a pellicle?"

A: "With the Lasertec tool, yeah. So, TSMC has in the beginning slowing pellicle adoption because the pellicle was problematic, was exploding. Now, it's a point that TSMC is starting to adopt it but very slowly. Also **because the Lasertec tool is really problematic.**" -Former KLA executive in a leadership role in their EUV mask inspection group

Lasertec's customers can't reliably use it for P3 inspection, thus undermining the main reason for buying it and providing no benefit versus KLA's legacy DUV inspection tool that dominates P1 inspection. We note that P0 refers to inspection in the mask blank shop, done by an inexpensive tool; P1 is after the blank is printed with a mask pattern, and is dominated by KLA's Teron DUV line; P2 refers to re-inspecting the mask after repairing defect; and P3 is after adding a pellicle. Lasertec's only use case is P2/P3, but it introduces debris in P2 which requires redundant inspection with a backup KLA DUV tool, and it's too unreliable for P3 – eliminating any reason for its existence versus KLA.

KLA owns the P1 inspection in the mask shop with their Teron product line and no one else can compete

A: "P0 is done in the blank shop. That's a Lasertec tool that will remain a Lasertec tool. KLA is not there."

Q: "But that's a pretty cheap tool, right?"

A: "Yes, the source, buying from Energetiq, and that costs \$1 million. I think Ushio bought that company as well. From there, it goes to the mask shop. In the mask shop, the first tool is the EUV writer that writes the mask. Then you have P1, which is post-writing inspection. After that, it goes to Carl Zeiss repair tool. You can think about it as a multimillion-dollar tweezer that you can see all the problems that were discovered by KLA, and the KLA P1 tool is Teron 640."

Q: "You're saying KLA owns the P1 market, and nobody's going to beat them. They have a monopoly in the P1 market."

A: **"There's a monopoly there because KLA is the only tool that has Die to Database.** If you have a mask that has multiple dies, like small chips, Texas Instruments, stuff like that, there's a lot of chips. Then you can compare the right chip to the left chip, and you know that if three are the same, there is no problem."

Q: "So, KLA is going to keep the P1 market. Then P2, that's where Lasertec is used?"

A: **"P2 is before you put the pellicle after you repair. P3 is after you put the pellicle.** Let's start with P3 and go back to P2. After you put the pellicle, the concern is something will fold them up and you have to inspect P3 before you chip it with a stamp. However, in order to know—we talked about die-to-database, which is important to understand because Lasertec will claim that they are working on it. KLA developed this die-to-database. You compensate all the optical distortion and build a synthetic image. That is a huge task—more than 10 years of development. So, **that's perfected and that's what KLA is good at, so they dominate the P1.**" –Former KLA executive in a leadership role in their EUV mask inspection group

Due to “the struggle of the Lasertec tool both on the performance side and...on the stability or uptime side,” key customers are “begging” KLA to hurry their upcoming actinic mask inspection tool: “Oh my God...the amount of pressure, it’s inconceivable...the pressure is insane...from Samsung and TSMC combined...TSMC number one.” Customers complain that its lack of reliability requires them to purchase redundant tools: “that’s why sometimes there are two tools...they need to have a backup tool...it breaks everything...and that’s the number one problem they have.”

Lasertec customers are “begging” KLA to hurry their competing tool; “pressure is inconceivable...insane”; complained about performance, stability, and uptime of Lasertec tool; need backup Lasertec machines given lack of uptime

Q: “You said customers are begging KLA. When you were at KLA, you must have been talking to TSMC and Intel. What were they saying?”

A: **“Oh my God. The amount of pressure, it’s inconceivable...basically, the pressure is insane. From Samsung and TSMC combined. TSMC number one.”**

Q: “What were they saying?”

A: “The message was exactly in line with what I told you. **The struggle of the Lasertec tool, both on the performance side and both on the stability or uptime side.**”

Q: “When you say struggles with uptime, liability—what was the other thing that you said?”

A: “Cost of service.

Q: “Cost of service, uptime, and reliability.”

Q: “What happens if your collector has an issue and they need to fix the source and you need to inspect a mask. What do you do if there’s downtime? Aren’t you screwed?”

A: **“That’s why sometimes there are two tools.** That’s exactly the problem they have because if a tool is lousy, they need to have a backup tool; it breaks everything. **And that’s the number one problem they have.** If you have one tool that is a backlog to everything, you need to have a backup.” -Former KLA executive in a leadership role in their EUV mask inspection group

We asked if Intel and TSMC “hate” the Lasertec tool, and the expert replied “correct” and stated that their dilemma is to either not use a pellicle and avoid using Lasertec - TSMC’s approach - or to use a pellicle and suffer with the lack of reliability until KLA introduces its alternative. He stated that during his role in the KLA’s EUV mask inspection group, key Lasertec customers were coming “crying” to KLA and “begging us to expedite”: “I’m talking about customers coming to KLA... saying, guys, we need you to develop this tool...P2 can only be done by a clean tool...they were really begging us to expedite.”

Key Lasertec customers have been coming “crying” to KLA “really begging us to expedite” an alternative

Q: “You mentioned the customers were complaining. We talked about Intel, TSMC. You said you talked to people at an SPIE conference. You said the customers are crying. Explain what the customers are complaining about.”

A: “The customers were saying to us, even these days, not just at SPIE, this is in KLA, and that I can say because they said—”

Q: “Sorry, you said, not just SPIE but you said they were telling you at KLA?”

A: “Yeah, **I’m talking about customers coming to KLA and they were saying, guys, we need you to develop this tool, as I mentioned, P2 can only be done by a clean tool. And they were really begging us to expedite.**” –Former KLA executive in a leadership role in their EUV mask inspection group

Intel and TSMC customers hate the tool and only using it until KLA comes out with an alternative

Q: “Are Intel, TSMC, and Samsung able to get value out of the machine? What kind of uptime does it have? It sounds like it’s very painful to use or do they just not have an alternative?”

A: “Yes, they don’t have an alternative. The only way to inspect a mask after you pelliclize it—you cannot send a mask with a pellicle to the fab without inspecting after the pellicle. Why? Because the residue that can fall, the pellicle itself, especially because it’s made of nanocarbon nanotubes. They cannot degrade during pelliclization. You must do EUV actinic inspection after pelliclization. Although they have, let’s say, 70% uptime, they bite the bullet and say, I have to deal with that; otherwise, I cannot print [wafers].”

Q: “So, **you’re saying they have no option but to use Lasertec right now but it sounds like they hate it; they’re desperate for an alternative but they use it because it’s the only one right now.**”

A: “**Correct.**” –Former KLA executive in a leadership role in their EUV mask inspection group

6A. Struggles at Intel. Lasertec's EUV mask inspection effort is Intel's baby, which it sponsored with \$300-500MM in advances >5 years ago as its "founding" customer after failing to convince KLA. Intel's mask group is trying to save face and cover up that its investment has been a debacle, giving it supplier prizes for public PR while complaining behind the scenes, according to multiple research sources.

Lasertec - in its current incarnation as an “EUV inspection equipment” stock – is Intel’s baby. Intel’s mask inspection group seems keen to save face and cover up that its massive investment in and sponsorship of Lasertec has been a debacle. Circa 2018, the industry began planning to use ASML’s EUV scanners for 7nm and 5nm production. However, the lack of actinic patterned mask inspection (APMI) – inspecting EUV masks through a thin protective membrane called a pellicle – became a major gap in the EUV lithography supply chain. Our sources indicate that Intel pushed KLA to develop an EUV mask inspection tool, but KLA asked for too much money. **Intel then pressured Lasertec to rush a tool to market.** A 2018 trade publication stated that Lasertec “obtained funding for the technology” from “a large chipmaker” – asked if this was true, Lasertec replied “no comment.”

2018 trade publication article - excerpts

What’s Missing In EUV?

The gaps include mask inspection and pellicles.

Still, **the industry wants actinic pattern mask inspection (APMI).** APMI can supposedly find more defects than today’s inspection systems. **So far, though, APMI does not exist,** but several entities are developing the technology in R&D....While those efforts are ongoing, **Lasertec is reportedly developing an APMI tool, according to sources. Lasertec has also obtained funding for the technology from a large chipmaker, sources added**...Hiroshi Asai, general manager of corporate planning at Lasertec, in a recent e-mail. “We have not developed an actinic ‘pattern’ mask inspection system.” **Asked if Lasertec is developing an actinic pattern mask inspection system or considering to develop one, Asai said: “I have no comment** to make on your questions.”

A former KLA executive in a leadership role in their EUV mask inspection effort told us that Intel invested \$300 to \$500 million in Lasertec a few years ago via advances for future purchases but “at a significantly reduced price...they don’t have a lot of margin on those tools.” A longtime industry analyst who recently completed a consulting project for Lasertec and remains in touch with management stated that Intel “do have a significant say in the direction of Lasertec, and Lasertec views them as sort of their core customer or founding customer...they’ve obviously ordered and prepaid....”

Industry source believes that Intel gave Lasertec \$300 to \$500MM in advances about five years ago, to jumpstart the product

Q: “You said that Intel pushed them to make a product because you can’t do EUV without EUV mask inspection and that was a big hole in the infrastructure. You said ASML never focused on EUV mask inspection—that was a missing piece. When did Intel do the investment in Lasertec?”

A: “Roughly five years ago.”

Q: “Do you know how big of an investment it was roughly?”

A: “No, I don’t remember. \$500 million—between **\$300 to 500 million and that basically translates to the POs that they had from Intel but the POs are at a significantly reduced price. They don’t have a lot of margin on those tools, very low margins on those tools.**”

Q: “So, Intel basically gave an advance of \$300 to 500 million and low margin on those tools?”

A: “Correct.” –Former KLA executive in a leadership role in their EUV mask inspection group

Intel is “founding customer” of Lasertec with “significant say in the direction of Lasertec”

“I think that \$300 to \$500 million is a little high in my estimation. I’m sure they’ve invested some significant money in Lasertec, and **they do have a significant say in the direction of Lasertec, and Lasertec views them as sort of their core customer or founding customer per se...they’ve also obviously ordered and prepaid for some stuff...**” – Longtime semiconductor consultant recently working for Lasertec; close to KLA management

Multiple semiconductor executives confirmed that Intel is just as dismayed as TSMC and is similarly struggling with Lasertec's EUV tool. A longtime semiconductor equipment friendly with Lasertec's CEO stated that "I have met the VP of Intel's mask operation, a few times." He indicated **"they are very secretive"** and **replied "yes, of course" when we asked if Intel is having the same problems as TSMC.** The ex-KLA EUV mask inspection executive corroborated that **Intel is complaining behind the scenes** – without naming them specifically but leaving no doubt as to which **"major US customer"** he was referring to.

Intel is "secretive" but industry executives who interact with them believe they are also struggling with the Lasertec tool

Q: *"I heard Intel rushed Lasertec into coming up with a half-assed product and that's why you have all these problems because everybody was desperate. Is this consistent with what you heard?"*

A: *"Kind of, yeah. **I have met the VP of Intel's mask operation, a few times.**"*

Q: *"Did they say anything about Lasertec?"*

A: *"No. [chuckles] They are **very secretive.**"*

Q: ***"Is Intel having the same problems as TSMC is with the Lasertec tool?"***

A: ***"Yes, of course."** - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO*

KLA customer indicates a major customer, who we presume to be Intel, complained about the contamination and frequent replacement of the optics

Q: ***"Where did you see them replacing the optics?"***

A: *"In one of the customers."*

Q: *"I'm going to guess it was either Intel or TSMC, right?"*

A: *"One of the major - it's in the U.S."*

Q: ***"Was the customer complaining about it?"***

A: ***"Of course."** –Former KLA executive in a leadership role in their EUV mask inspection group*

We tried speaking with numerous Intel employees in its EUV mask inspection effort. One of Intel's most senior and tenured individuals, with a leadership role in EUV and the Intel/Lasertec relationship, agreed to a phone consultation but only at 300% of the usual expert rate – and then played games and revealed little. Intel's mask team clearly realized someone is investigating Lasertec and agreed to calls only to try and deduce our identity – another senior individual, from its mask shop and who we think spoke to the first executive, agreed to a call and then demanded an absurd \$5,000 expert fee immediately prior to the scheduled time (we declined). Nonetheless, the senior executive with whom we spoke did make a few damning admissions, saying that Lasertec's light source is “not a well-controlled process...it was painful but we would get some light...if you choose a bad technology you just can't just get there...DPP has inherent issues.”

Intel is highly critical of Lasertec's Ushio light source: “not a well controlled process”; “uncontrolled”; “painful”; “bad technology” with “inherent issues”

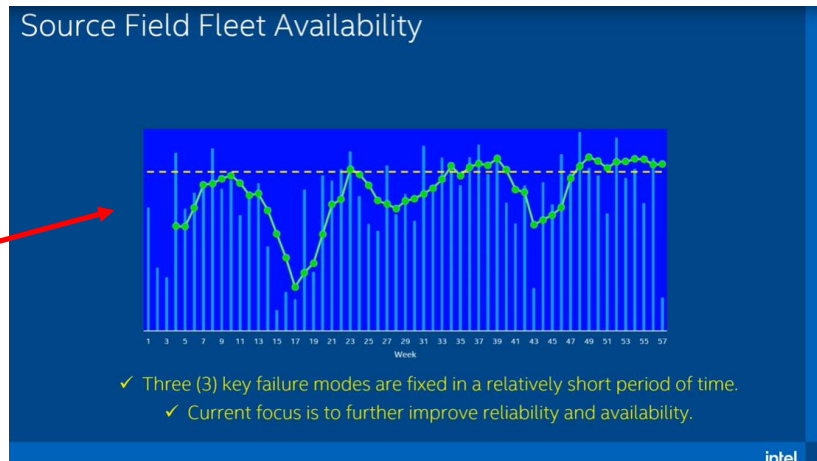
*“The Ushio light source is DPP. Discharge produced plasma. Think of the old days with a spark plug in a car with a gap where you apply voltage and you have a spark. Now imagine tin in the middle of the spark and using electrical energy to light the tin and get plasma. **It's not a well controlled process.** That's why people can't harness energy from lightning – **it's uncontrolled. It was painful but we would get some light. If you choose a bad technology you just can't get there. DPP has inherent issues.**”* – One of Intel's most senior and longtime engineers with leadership responsibility for EUV

We first realized that something is amiss as we read Intel's EUV mask inspection presentations at SPIE, the main industry association. We noted three presentations by Safak Sayan, a ~20-year Intel veteran per his LinkedIn profile and who we believe to be the head of Intel's mask shop. The presentations are typically co-authored with Lasertec employees, and seem rather eager to convey that the tools Intel purchased actually do work. For example, a 2021 presentation gives itself an A+ grade for EUV mask inspection, with the final sentence of the paper proclaiming that "No major development will be required to meet future needs such as higher sensitivity and throughput." Amusingly, however, the chart for the tool's uptime omits the Y-axis units entirely - a cliché case of "How To Lie With Statistics."

2021 SPIE presentation by Intel and Lasertec

23 March 2021
**Laser-assisted discharge produced plasma (LDP)
EUV source for actinic patterned mask inspection
(APMI)**
Safak Sayan, Gishore Chakravorty, Yusuke Teramoto, Takahiro Shirai, Shunichi Morimoto, Hidenori Watanabe,
Yoshimasa Sato, Kazuya Aoki, Ted Liang, Yoshihiro Tezuka, Marieke Jäger, Firoz Ghadiali, Frank Abboud, Steven L.
Carson

*Missing Y-axis for uptime
– units not even defined
and no numbers*



The same key Intel employee presented at SPIE in May and September 2023. Even the titles suggest ongoing struggles – the May paper is titled “Improvement of LDP EUV source performance” and the September one is “Progress report...” Once again, key performance charts in both presentations omit the Y-axis with no units or numbers, yet the presenter once again declares mission accomplished: “These improvements have enabled achieving a performance and reliability level that exceeds the HVM targets” – referring to high volume manufacturing.

2023 SPIE presentations by Intel/Lasertec

26 May 2023

Improvement of LDP EUV source performance for actinic patterned mask inspection

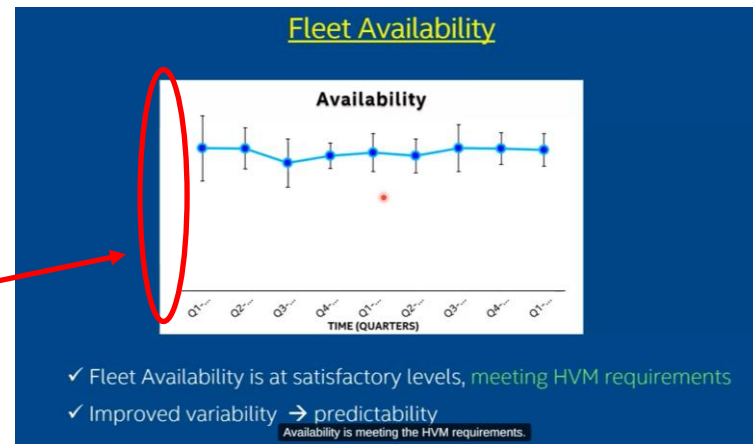
Safak Sayan, Kishore Chakravorty, Yusuke Teramoto, Bárbara Santos, Akihisa Nagano, Noritaka Ashizawa, Takahiro Shirai, Shunichi Morimoto, Hidenori Watanabe, Kazuya Aoki, Yoshihiko Sato

26 September 2023

Progress report on actinic patterned mask inspection for EUVL

Safak Sayan

Figure 12. Trend of continuous operation time.



The Intel SPIE papers confirm challenges with Lasertec's EUV tool, namely contamination and optics degradation from tin plasma ("ion induced damage to the optical components"); light source stability and brightness level; struggles in managing source brightness vs. source chamber pressure, as high pressure is needed to help mitigate tin debris but which reduces brightness; and unstable plasma which leads to dark pulses and inspection failures – "the plasma becomes unstable and brightness stability becomes worse...brightness may drop below the threshold required for mask inspection."

2023 SPIE presentations by Intel/Lasertec - excerpts

26 May 2023

Improvement of LDP EUV source performance for actinic patterned mask inspection

Safak Sayan, Kishore Chakravorty, Yusuke Teramoto, Bárbara Santos, Akihisa Nagano, Noritaka Ashizawa, Takahiro Shirai, Shunichi Morimoto, Hidenori Watanabe, Kazuya Aoki, Yoshihiko Sato

of the electrode design as well as other changes. Ion induced damage to the optical components such as downstream mirrors was mitigated by development of an effective debris mitigation approach. These improvements have

Fast moving ions originating from the plasma hit the collector surface and induce damage by sputtering. This ion induced damage limits the collector lifetime. It is necessary to reduce the velocity of the ions to minimize ion induced damage. Argon gas at high flow rate is injected in the path of the ions. The argon atoms collide with the fast-moving ions

If the discharge condition is not optimized, the plasma becomes unstable and brightness stability becomes worse. Brightness may drop below the threshold required for mask inspection. We term such EUV pulses that has intensity or brightness below the minimum threshold as dark pulses. Reducing and minimizing the number of dark pulses is crucial for quality and throughput of the inspection. Figure 8 shows the trend of the dark pulse count over time. Significant

An ex-Lasertec field service engineer provided us with particularly astonishing color on Intel's actual, behind-the-scenes view of Lasertec. The engineer was until recently based full-time, on-site at Intel's mask shop as part of nine field engineers servicing three Lasertec EUV tools. He was embedded with Intel staff and essentially their colleague. He stated "there was definitely a lot of frustration around" Lasertec's system, and that Intel's head of EUV mask inspection was so livid that "he got banned from a couple of meetings for blowing up on people" – we believe this to be the same person presenting the SPIE papers.

Head of Intel's EUV mask shop was allegedly so angry with the Lasertec tool and it's low uptime that he was banned from attending meetings "for blowing up on people"; "definitely a lot of frustration around it"

Q: "You must have become close with the Intel mask shop people. They're essentially your colleagues. What were they saying?"

A: "Their head of EUV - he seems pretty stressed out in general. I would give my health diagnostics reports to his direct reports...they would give us benchmark goals, and we would aim for them. But **there was definitely a lot of frustration around it.**"

Q: "The head of EUV mask inspection – was he stressed-out in general or because of the problems with the tool?"

A: "Hard to say. **He got banned from a couple of meetings for blowing up on people.**"

Q: "Blowing up at Lasertec?"

A: "Not necessarily us. He was often more **upset with the light source people** than us."

Q: "Yes, but he was blowing up in relation to the tool?"

A: "Yeah." – Former Lasertec field service engineer who worked in the Intel mask shop

When we asked the Lasertec engineer for more detail on the tension, he stated that the head of Intel's EUV mask inspection group was responsible for the tools – “the tools were his main area of management at Intel” – thus explaining the cover-your-ass tone Intel has struck in public with regard to Lasertec, given the size of Intel's investment and the ~\$50MM cost of each tool. Due to his anger around the tool, “he basically wasn't allowed in certain meetings anymore” by “his higher-ups”: “he wanted utilization closer to 85-90%, and being around 60-65% was not cutting it for him...he just tried to press, press, press to make it happen but there was only so much that could be done.” This, of course, explains the statistical omissions in the SPIE papers lauding the tool's purported uptime.

Head of Intel's EUV mask shop was allegedly frustrated at low utilization and banned from meetings due to his anger

Q: “You said that Intel's head of mask inspection was blowing up because of the tool, at the component supplier Ushio - that's what was causing the blowup? And then he got banned from meetings? I just want to make sure I understand this correctly. Intel's head of EUV mask inspection got so angry that he was told, hey don't come to the meeting until you cool down?”

A: “I wouldn't say it was Lasertec or Ushio banning him from the meetings. It would be more his higher-ups. Because **he was the head of the EUV-specific section of mask inspections. He basically wasn't allowed in certain meetings anymore.**”

Q: “The Intel people didn't allow him in a meeting because they thought that he was too angry to be productive, but was **his anger around the tool?**”

A: **“Yeah. I'd say it was centered around that. The tools were his main area of management at Intel.”**

Q: “He was responsible for the Lasertec tools?”

A: “Yeah, that was his main responsibility.”

Q: “Was it that he was stressed out because he was the champion for Lasertec, and it wasn't working, so he was stressed out after spending all that money? Is that what it was?”

A: “Yeah, I think that **he had some ideas of where he wanted utilization closer to 85-90%, and being around 60-65% was not cutting it for him. He just tried to press, press, press to make it happen, but there was only so much that could be done.**”

– Former Lasertec field service engineer who worked in the Intel mask shop

At the outset of our research, we were stumped as to why Intel includes Lasertec in its annual list of supplier awards – until we understood Intel’s role as Lasertec’s sponsor and anchor investor, and the natural incentive of its EUV mask inspection team to avoid embarrassment, particularly given Intel’s interest in promoting its EUV progress to investors and players like Apple. When we asked a senior Marvell employee about the award, he mocked it and laughed – “Marvell has also received that award...[it] is really sitting inside a cubicle that nobody uses...it’s not something that we write home about...yeah we’ll do a press release...but nobody talks about it”; “Intel is not a cutting-edge company in this day and age by any means...so if Intel is giving out awards...it’s not even worth its weight in salt.”

Marvell mocked Intel’s award as one they received as well – “sitting inside a cubicle nobody uses”; “not even worth its weight in salt”

Q: “It’s kind of funny Intel is giving Lasertec this annual supplier award. I know they give that award to a few dozen people.”

A: “**Marvell has also received that award as well.**”

Q: “Do they just give that award to everybody?”

A: “Yeah, I mean, we’ve gotten it [laughs]. That **award is really sitting inside a cubicle that nobody uses.**”

Q: “Yeah, I see a press release about it. Marvell Semiconductor receives Intel’s preferred quality supplier award.”

A: “It’s not something that we write home about. Yeah, we’ll do a press release for the publicity, but **nobody talks about it.** By the way, just for what it’s worth, **Intel is not a cutting-edge company in this day and age by any means at all. So, if Intel is giving out awards, nobody really—it’s not even worth its weight in salt.**” – Senior employee at Marvell Technology

Others indicated that the award is mainly to spite KLA, given tension over KLA's pricing, according to a former KLA executive – and that the grudge is Intel's main reason for buying Lasertec's tool versus anything special about their technology. A longtime industry consultant who we believe to be close to both KLA and Intel corroborated the dynamic: "KLA is genuinely arrogant because they have viewed that they control the wafer inspection market...particularly, there's an individual at Intel who has not been happy with KLA and their attitude about it and has directed business away from them on purpose."

Intel supplier award is just playing politics with KLA

Q: **"So, Intel is giving Lasertec the supplier award like a "fuck you" to KLA."**

A: **"Of course."**

Q: "Because they hate KLA's pricing. I see."

A: "Yeah, there's a long history with Intel." -Former KLA executive in a leadership role in their EUV mask inspection group

Intel only buying from Lasertec to spite KLA

"I think part of it is that KLA is genuinely arrogant because they have viewed that they control the wafer inspection market, the reticle inspection market for so many years, etc. **Particularly, there's an individual at Intel who has not been happy with KLA and their attitude about it and has directed business away from them on purpose**, looking for alternative sources of tools and inspection equipment." – Longtime semiconductor consultant recently working for Lasertec; close to KLA management

6B. Struggles at TSMC. The world's largest chipmaker's experience with Lasertec has been a disaster, not only with its ACTIS EUV systems but also with its outdated, previous-generation DUV tools.

Multiple research sources indicated that TSMC's experience with Lasertec has been a disaster, not only with its ACTIS EUV mask inspection line but also with its legacy DUV MATRICS X8 and X9 machines. We spoke with TSMC's former head of TSMC R&D, who played a key role in its 5nm transition and vendor/tool selection; a former longtime TSMC procurement executive, who stated that he originally introduced Lasertec's tools into TSMC and is in touch with current TSMC executives as well as with Lasertec management; a senior semicap executive in Japan who is friendly with Lasertec's CEO; and an ex-KLA executive in a key role in their EUV inspection group, who is also in contact with Lasertec. We begin with the ex-head of TSMC R&D, who stated that they still rely mostly on traditional KLA tools, not Lasertec actinic inspection: "Taiwan is very smart. They always wait for Intel and these guys to go first...they usually do not want to take the risk of being the first...that happened for EUV...were really slow...still used the traditional tools like KLA" which "has very strong inspection mask manufacturing for EUV."

TSMC is still relying mostly on traditional KLA tools, not Lasertec actinic inspection

Q: "Did TSMC buy a bunch of the Lasertec ACTIS A150 machines for actinic pattern mask inspection? Or did they use DUV and other ways of inspecting the EUV mask? What solution did TSMC go with in the end?"

A: "Initially in the first 7-nanometer or 5, it was a big phase for EUV...Taiwan is very smart. They always wants to wait for Intel and these guys to go first, and then TSMC really comes in big time to copy all the protocols...as we all know, TSMC has become very successful in 5-nanometer. So, they get a big Apple and AMD order. Taiwan is always very smart. They usually do not want to take a risk of being the first in the world to launch. **That happened for EUV. ...[T]hey were really slow to pick up because they are very careful about spending. So, they still used the traditional tools like KLA...there are second choices...KLA is not just sitting there. KLA also has very strong inspection mask manufacturing for EUV. So, KLA does offer some alternatives.**" – Former head of TSMC R&D; played key role in 5nm transition and vendor/tool selection

A longtime semicap executive in Japan - who speaks with Lasertec's CEO – said that he was “in charge of TSMC for about 15 years” at another tool supplier and “I was dealing with all the TSMC management people in the mask shop. I know how they function.” He confirmed the color from TSMC's ex-head of R&D that TSMC can't rely on Lasertec's tool. In particular, he stated that he recently met the head of TSMC's EUV mask shop - whose name we redact – and that he evaded questions about their struggles with Lasertec: “If they didn't have any [issues], he would happily talk about how things are going.” The expert conveyed that he knows the head of TSMC's mask shop well.

Senior executive who recently met with TSMC indicates they are having severe problems with Lasertec's tool

Q: “Your main source of information here is what you're hearing about TSMC, but you're hearing it from your contacts at [redacted] and [redacted] basically?”

A: “Yes. When I was at [redacted], **I was in charge of TSMC for about 15 years, so I was dealing with all the TSMC management people in the mask shop. I know how they function.**”

Q: “Have you talked to any TSMC mask shop people recently?”

A: **“In [a recent month, redacted], I met the top guy of their EBO, [name redacted], but he didn't want to talk about the issues with me, naturally. [chuckles].”**

Q: “He didn't want to talk about the issues with the Lasertec machine?”

A: “Yeah, any issues in the mask shop.”

Q: **“But was your sense that they had issues in the mask shop?”**

A: **“Of course. If they didn't have any, he would happily talk about how things are going.”** - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

The executive in Japan confirmed TSMC's desire to cover up its problems with Lasertec, after we asked if a 2023 SPIE paper co-authored by Lasertec and TSMC was an attempt by mask shop employees to save face after spending millions on tool purchases: “they placed an order, I think, for 10 tools, so they cannot come out with any negative papers [about Lasertec]”; “even upper management who accepted the purchase order will be put in an awkward situation...TSMC is a very hierarchical company...you don't want to make any superior people look bad.” We note that the TSMC mask shop executive that the expert spoke with is listed as one of the authors in the paper, shown below.

TSMC appears to be covering up problems with Lasertec tool in order to save face

Q: “**It sounds like TSMC might be trying to cover it up** because the KLA person told me to go to the SPIE website. I went there, and in November, a bunch of TSMC guys presented a paper talking about the early evaluation of this new Lasertec tool, the A300, with this new Urashima light source. They were very careful, but it sounds to me like they were trying to save face.”

A: “**Yeah, because they placed an order, I think, for 10 tools, so they cannot come out with any negative papers.**”

Q: “I see, because then the people in the mask shop look stupid.”

A: “Yes. Even **upper management who accepted the purchase order will be put in an awkward situation. TSMC is a very hierarchical** company. Hierarchy is very strong, so **you don't want to make any superior people look bad.**” - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

SPIE paper co-authored by TSMC and Lasertec staff

21 November 2023

Actinic inspection technology enhancement for N3 node and beyond

Hao-Ming Chang,¹ Hsin-Fu Tseng,¹ C. H. Lu,¹ S. C. Hsu,¹ Wei-Chung Hu,¹ Ajay Nandoriya,¹ Yi-An Huang,¹ Yung-Sheng Chang,¹ C. W. Wen,¹ Frankie F. G. Tsai,¹ Hiroki Miyai,² Masayasu Nishizawa,² Atsushi Tajima,² Hirokazu Seki²

¹Taiwan Semiconductor Manufacturing Co., Ltd. (Taiwan)

²Lasertec Corp. (Japan)

→ Semicap expert recently spoke with a TSMC mask shop executive listed as a co-author

The semicap executive was skeptical as “to what degree [TSMC] have succeeded in phasing in Lasertec tools to the production cycle” – confirming issues in both of its use cases in the mask shop as well as the fab.” Because TSMC is failing to get results from the ACTIS machine, he stated that they have to rely on redundant, backup inspection with traditional KLA and other tools: “so they need a combination of optical DUV and electronic beam inspection to inspect the mask.” He cited strong, recent adoption of a Holon e-beam tool by Intel, Samsung, and TSMC as evidence that they are struggling with Lasertec and resorting to backup solutions: “TSMC was the first one to adopt the Holon tool...other customers in the EUV space are also adopting” it.

TSMC struggling to phase in Lasertec; “hasn’t been very successful”; key customers relying on legacy DUV and SEM tools

Q: “You’re saying their using optical means that they’re **not getting the results from the Lasertec machine by itself. They have to use a second tool?**”

A: **“Correct.** Before Lasertec came out with their actinic tool, they needed to establish some way of inspecting the EUV masks. They used an optical tool from KLA – the Teron product line, an optical tool with a 193-nanometer light source. There’s a limitation to what defects it can detect. Very small defects, it cannot clearly distinguish. For something that looks like a defect, they record the coordinates, and then these coordinates are transferred to an electron beam inspection tool, which is only looking at these suspicious-looking defect locations. Is the defect on print or not? So, they need a **combination of optical DUV and electron beam inspection to inspect the mask.**”

Q: “So, they’re using the DUV device, and then who makes the e-beam device?”

A: “Mostly a company called Holon. **TSMC was the first one to adopt the Holon tool, and I think Samsung, Intel, and most of the other customers in the EUV space are also adopting the Holon tool.** The [redacted] of Holon is actually a good friend.”

Q: “Is TSMC still actively using Lasertec, or are they idling those machines because of all these problems with downtime?”

A: **“I’m not quite sure to what degree they have succeeded in phasing in Lasertec tools** to the production cycle, but from what I hear from this company, **Holon said that hasn’t been very successful so far. So, they are still ordering considerable quantities of Holon inspection machines.**” -Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec’s CEO

The executive stated that TSMC and “pretty much all customers are struggling in phasing in the Lasertec tools” due to “uptime, light source problems” and that “customers a) haven’t been successful with the qualification and b) the demand is low.” As a result, he indicated that TSMC is delaying Lasertec orders and that current sales are “probably pre-orders or initial orders” from several years ago.

“Pretty much all customers are struggling in phasing in the Lasertec tool”; current revenue likely previous pre-orders

A: “[W]hen TSMC delays orders, then all the tools, which are sitting in the fab, cannot move out, so the cash flow goes down.”

Q: “They have a billion dollars of inventory—at least, they value it at a billion dollars; I don’t know if they’re valuing it accurately. Most of that is disclosed as work-in-process inventory, not finished goods. It keeps going up every year, so the inventory is equal to revenue, which is unusual. There’s almost no semi-cap company where the inventory as a percentage of revenue is that high. It’s really weird.”

A: “Yeah, I totally agree. [chuckles] But I think it will clear up once they can start shipping what they have in the fab [chuckles].”

Q: “Have they had some problems shipping? Have they been sitting on a bunch of inventory - is that what you’ve heard?”

A: “Yeah, because customers a) they haven’t been successful with the qualification and b) the demand is low.”

Q: “Which customers have not been successful with the qualification?”

A: “TSMC. Pretty much all customers are struggling in phasing in the Lasertec tools.”

Q: “Because of the quality problems?”

A: “Yes. Uptime, light source problems.”

Q: “So, they are basically delaying the implementation of the tools.”

A: “Yes...there was a first wave [of orders], so the customer ordered, let’s say, 10 tools, and when the tools arrived, they started to see this and that problem, so they delayed the tools after that. So, there was a first wave, where they increased revenue, and now that they’re waiting for repeat orders, they find out that the phasing of the tools is not so easy.”

Q: “So, you think most of the sales that they recognize today are just pre-orders from a couple of years ago?”

A: “Probably pre-orders or initial orders.”

Q: “And now, people are holding back.”

A: Yes. And one is implementation takes longer than expected and the second one is the slow-down of the industry. So, the two factors work together to slow down the shipments. For example, TSMC Arizona is delayed by a year or two. So, equipment gets delayed.” - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec’s CEO

Echoing a theme we detail in the coming pages as well as a separate section, the expert stated that Lasertec's tool is so unreliable and has such little trust that TSMC has to run parallel, duplicate inspections with other tools to catch mask defects: "not so easy with a new technology like actinic inspection...they will have to run parallel inspections of the same masks...just to make sure there are no defects that go through without detection...if it's a fatal defect, it can cost them billions...they cannot ship wafers anymore."

TSMC has to run parallel/redundant inspections as Lasertec's tool is unreliable

Q: "What else have you heard about **TSMC's feedback**, whether through your connection or somebody else? I guess you said they're sort of struggling and making the transition. What are the reasons for that being? The optics or other problems?"

A: **"It's not so easy to change the process.** If they have an established process and an established yield number of, let's say, 19% or something, phasing in the new technology should improve the yield. So, everything needs to be fine-tuned to surpass the existing solution. And that's **not so easy with a new technology like actinic inspection. It's going to take time, so they will have to run parallel inspections of the same masks**, for example, every day, just to make sure that they get the same results..."

Q: "So, whenever they do use a Lasertec machine, they're having to kind of duplicate the inspection with other machines?"

A: **"Yes. Just to make sure there are no defects that go through without detection.** If that happens, the same defect will print on every single semiconductor. And if it's a fatal defect, it can cost them billions. They cannot ship any wafers anymore." - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

Some of the most damning color we received about TSMC's experience with Lasertec's tools is from a former longtime TSMC procurement executive, who stated that he originally introduced Lasertec's tools into TSMC and is in touch with current TSMC executives as well as with Lasertec management. He slammed the ACTIS A150's reliability: "the big thing I hear offline is it's not very reliable" and "needs a lot of support and it has a lot of issues." Referring to the "new" and purported A300, he stated bluntly that "TSMC is not buying the High NA Lasertec tool...there's no confidence in my opinion that it will be a reliable product."

TSMC is "not buying" the A300; "no confidence" that it's a "reliable product"; A150 is "not very reliable"; "has a lot of issues"

Q: "Is anybody buying less Lasertec, or have they essentially idled or just reduced their utilization of the Lasertec machines because of these frustrations?"

A: "Well, a recent example would be that **TSMC is not buying the High-NA Lasertec tool**, to my knowledge. And they're the company that does high volume, high-performance chips..."

Q: "You're talking about the recent ACTIS A300 that they announced in November, which sort of tied into the Intel High-NA announcement. Is that what we're talking about—the recent product announcement?"

A: "Yes, I'm talking about that...**there's no confidence in my opinion that it will be a reliable product...**"

Q: "You said you don't think this A300, this High-NA tool, is going to be reliable."

A: "I reviewed again the papers presented in the past year at the various conferences—there was the advanced lithography conference and a photomask conference. I reviewed the papers...**reliability, the big thing I hear offline is it's not very reliable.** I'm talking about the A150. A300 is just a product announcement. It's not shipped. **The A150...needs a lot of support, and it has a lot of issues.**" – Ex-KLA director of engineering focused on EUV mask inspection

He detailed TSMC's frustration and exasperation with Lasertec around a variety of issues, such as pitifully low mask inspection throughput and problems in inspecting with a pellicle, and described it as "kind of an endless negotiation...to try and convince Lasertec to improve their performance...to try to convince them to change their performance, the quality." He indicated that TSMC has no real interest in purchasing from Lasertec, and that both TSMC and Intel do so only to have pricing leverage with KLA: "TSMC will keep a minimal share for Lasertec...we believe [Intel] purchased Lasertec because they want to have some leverage power with KLA because KLA's price is even higher...TSMC spent a lot of money on KLA's particle inspection tools...that's very, very expensive."

TSMC is frustrated with Lasertec; "endless negotiation...to try to convince them to change their performance, quality"

"The Japanese guys, it is not easy to change their minds. We keep asking, you don't need to perform 100; you just perform 60 because I only needed the goal of 60. Okay, but I need to do it very fast. But the Japanese guys say, no, no, no. We care about the quality. We don't want to miss it. We don't want to miss any defects. But sometimes it's not necessary. And also come by another issue. The pellicle is also another issue. Some masks are with pellicle; some masks are without the pellicle. So, it's a kind of an endless negotiation. The negotiation is to try to convince Lasertec to improve the performance. The negotiation is to try to convince them to change their performance, the quality...You can decrease your quality but increase, improve your throughput."

– Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

TSMC and Intel mainly bought tools from Lasertec to have some pricing leverage with KLA

"TSMC will keep a minimal share for Lasertec...KLA is what we call opportunistic...they increased the price...they deal with a hard hand. For the past three years, Intel purchased some tools, maybe two or three from Lasertec. But we believe they want leverage with KLA. We mainly think they purchased Lasertec because they want to have some leverage power with KLA because KLA's price is even higher... TSMC spent a lot of money on KLA's particle inspection tools, like SP5, SP7, and SP6. That's very, very expensive." – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

The TSMC executive indicated that KLA's legacy DUV tools have superior economics, productivity, and throughput versus Lasertec's EUV systems, and that "KLA's share continues to increase" at TSMC: "it's already two-thirds, and I know it continues to increase"; "already two-thirds migrated from originally Lasertec to KLA...I have to say Lasertec has continued to decrease their share in TSMC." In particular, he criticized Lasertec's EUV source: "the big thing is the light...for production, for productivity, and for economic considerations, KLA performs better than Lasertec in TSMC."

TSMC indicates KLA DUV tools have superior economics, productivity, and throughput vs. Lasertec; using DUV inspection now vs EUV in production; TSMC has basically switched from Lasertec to KLA; "KLA's share continues to increase"

Q: "If Lasertec is still mainly selling DUV, that means they basically failed in EUV."

A: "The big thing is the light. **For production, for productivity, and for economic considerations, KLA performs better than Lasertec in TSMC. So, that's why I say already two-thirds of the business migrated from originally Lasertec to KLA already. I have to say Lasertec has continued to decrease their share in TSMC.**"

Q: "In fab 18, what happened?"

A: "**The economics, the throughput.** If I purchase a tool from KLA, probably for 3-nanometer, TSMC probably needs to purchase 20 mask inspection tools. But if I change to Lasertec, I have to purchase more than 30."

Q: "So, you're saying that TSMC tried using them for 5-nanometer, but it didn't really work because of the throughput; they only got two machines, and now they're not going to use them for 3-nanometer. Is that right?"

A: "Yeah, you can say that. Yes. **You can use the DUV light source** to inspect, no matter for a DUV mask break or EUV mask break, yes, because you're just trying to identify the same image and the defect."

Q: "Is TSMC using Lasertec in the new fabs? When you said they're reducing market share, who are they going with instead? KLA?"

A: "Yes. **Now, KLA's share continues to increase. Now, it's already two-thirds, and I know it continues to increase to get more from KLA.**" – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

The TSMC executive said that he recently spoke with a senior individual at Lasertec, who stated that things are “not well.” The executive stated that TSMC views Lasertec as merely a niche R&D tool that is not suitable in a fab: “we call it a lab tool, R&D tool...not a production tool...because you have a lot of limitations for productivity.” We did a second call with the executive after he checked with current staff at TSMC for any change in their negative opinion of Lasertec, and he indicated the situation is dire enough that TSMC holds a weekly meeting in Taiwan with Lasertec: “my colleagues in procurement and from the fab told me they have a weekly meeting to push Lasertec to improve...to improve their performance...their productivity.” In contrast, he stated that “KLA improved a lot” with their legacy DUV systems which they rely instead for EUV: “they have a very quick scan, and now their resolution is getting much, much better.”

Lasertec executive recently told TSMC contact that things are “not well”; TSMC views it as a niche “lab tool, R&D tool”

Q: *“When you talked to the Lasertec executive, what else did they say? Are things going well or not going well for them?”*

A: *“Not well.”*

Q: *“What did he say?”*

A: *“Not well. I think Lasertec will be kind of—we call it a lab tool, R&D tool.”*

Q: *“It’s an R&D tool? Lab tool, not a production tool.”*

A: *“Yes, a lab tool. Yes, not a production tool. Because you have a lot of limitations for productivity...even today, TSMC still has a weekly meeting with Lasertec. My colleagues in procurement and from the fab told me they have a weekly meeting to push Lasertec to improve.”*

Q: *“You said they have a weekly meeting with Lasertec to push them to fix their problems?”*

A: *“Push them to improve their performance, to improve their productivity...and the reason is if you increase the throughput, you cannot inspect the mask; it’s so detailed. But KLA improved a lot. They have a very quick scan, and now their resolution is getting much, much better.”* – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

He stated that TSMC views the Lasertec ACTIS EUV tool as just a niche R&D tool for their mask shop; that it is unusable for production in TSMC's fab; and that TSMC has stopped buying any more of Lasertec's flagship ACTIS A150 machines, replying "no, unfortunately no" when we asked if they're still willing to buy them. He emphasized that "it's not a tool to support production" and that "the Lasertec tool cannot meet demands in a production line."

TSMC stopped buying Lasertec EUV machines as it's only a niche mask shop tool; "not a tool to support production"

Q: "Has TSMC basically stopped buying A150 machines from Lasertec? **Are they still buying Lasertec A150 machines or no?"**

A: **"No, unfortunately, no."**

Q: "When did they stop buying them?"

A: "I will say something – even if TSMC continues to work with Lasertec, now it's mainly in the mask house, instead of the production house, production area. So, for the EPL, for the mask house..**it's not a tool to support production** because the production demands you have to quite frequently monitor and quite frequently check the EUV mask breaks. The demand requirement is quite serious for a production tool, and the mask house tool is different because, for a production tool layout, they have to identify the falling particles, the damage. But the damage is caused by the operation. For production, you are dealing with damage, the falling particles from the operation. Normally, the criteria are different. So, when you change to the production tool, **the Lasertec tool cannot meet demands in a production line.**" – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

He indicated that TSMC only purchased six of the ACTIS EUV machines at \$50 million each, mostly around 2019/2020 and don't appear to have purchased any more in years: "the pricing is so expensive." TSMC's only use case is in the mask shop, which requires only a small number of machines – we note that Intel's mask shop only appears to have three Lasertec machines. Hence, the TAM for EUV mask inspection machines is entirely dependent on their use in a production environment in a fab, where TSMC stated it is unusable: "it's not for production, it's for EPL," referring to their internal name for their mask shop.

TSMC only purchased six of the ACTIS A150 machines; don't appear to have purchased any in years

A: "TSMC probably purchased six ACTIS A150."

Q: "They purchased how many?"

A: **"In TSMC, six."**

Q: "Of the A150?"

A: "Yes."

Q: "And do you know when?"

A: **"2019. I think 2020 also...the A150, the pricing is so expensive, around \$50 million, and the purchasing price in TSMC is around \$50 million...yes, six tools. And they demand it for EPL. It's not for production; it's for EPL."** – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

The TSMC executive repeatedly and strongly emphasized that Lasertec's tool is a failure that cannot be used inside the fab to inspect masks during production, given its low throughput and inability to inspect more than a few masks per day: "No. TSMC is not happy with Lasertec...it's very expensive but the throughput is slow." He slammed both the current ACTIS A150 and the purportedly "new" A300 with the Urashima light source: "because of the throughput, TSMC will not consider the A300 or A150 for production...they will only exist in the...mask house."

TSMC "will not consider" Lasertec's EUV tool for production; only a niche lab/R&D tool in the mask shop; using legacy DUV inspection machines instead as EUV inspection "is not necessary"; only purchased six A150 machines

A: "Both the **LasertecX8 Ultra and X9 Ultra are the DUV light source, but they are inspecting the EUV mask. The main reason is for the throughput. And it is not necessary...**"

Q: "You're saying TSMC uses the X8 and X9 Ultra in the fab instead of the ACTIS?"

A: "Yes. **Because of the throughput. TSMC will not consider the A300 or A150 for production. They will only exist in the EPO mask house.** The EPO—we call it the EP Organization in TSMC. **TSMC has already purchased six sets of A150.** So, they already bought six tools. Six A150 tools. But last year, they also purchased the first A300. And the first A300 because they changed the light source, Urashima." – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

TSMC "is not happy with Lasertec"; "very expensive but the throughput is slow"; only need quick, lower resolution inspection in the fab but with higher throughput which Lasertec's tool cannot provide

Q: "It sounds to me like TSMC is not really that happy with Lasertec. Is that accurate?"

A: "**No. TSMC is not happy with Lasertec so far...It's the throughput. It's very expensive, but the throughput is slow.** TSMC has explained to Lasertec, the defects TSMC wants to identify in production are like falling particles. The ASML EUV tool has a very strong light source, and it can damage your pattern. So, **TSMC just wants to do quite frequent monitoring. Every day, they check the mask. A quick check. If you go through a quick check, the best resolution is not necessary. It's not necessary.**" – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

The Lasertec machine's throughput is so low that it can only inspect "five pieces per day" – referring to masks – while TSMC's 5nm Fab 18 needs 550 masks inspected daily in production. Of the six ACTIS EUV tools TSMC has purchased, only two are in the fab and the rest are in the mask shop: "the problem is the throughput...they only have two in the production area because the throughput is too slow...it's not affordable for production demands...it's not economical."

TSMC can't use Lasertec EUV tool for production in the fab; "problem is the throughput"; can only inspect 5 masks per day;

A: **"...yes, six tools. And they demand it for EPL. It's not for production; it's for EPL."**

Q: "For the mask shop?"

A: "Each for TSMC's mask shop."

Q: "Six machines for the mask shop?"

A: **"Yes, exactly..the problem is the throughput. The throughput is so long, only five pieces per day...for the mask house, because it is not for production."** TSMC has an in-house mask house. In production, if you migrate to the fab, the demand of the fab, every time you change the mask, or they have to regularly monitor the mask. The [fab] demand is much bigger than the mask house. So, the **throughput is another big issue."**

Q: "You said TSMC has six of the A150 in the mask shop. How many A150s do they have in their fabs?"

A: **"I think the A150 they only have two in the production area because the throughput is too slow...it's not affordable for production demands...if you care about the throughput, the production throughput, you have to use a 213nm light source...six or eight only. And the others are KLA.... If you care about throughput productivity, quality and productivity, it's kind of a dilemma...for the throughput, for the production side, it's not economical."** – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

Lasertec throughput of 5 masks per day is "not economical" as TSMC's 5 nm fab has to inspect over 500 masks per day

A: **"It's not economical."** Their competitor, KLA, their resolution, their quality is maybe not as good as Lasertec because they are using a 193-nanometer light source."

Q: "But you're saying the throughput is faster."

A: "Yes. Five hundred masks a day in fab 18. For 5-nanometer only."

Q: **"In fab 18, 5-nanometer, you said TSMC has to inspect 500 masks a day?"**

A: "550 per day." – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

The TSMC executive indicated that the “new” ACTIS A300 with the Urashima light doesn’t work and that it failed TSMC’s evaluation: “Yes. It failed” – laughing as he told us. He indicated that the evaluation occurred last year: “TSMC will not consider the A300 or A150 for production.”

Lasertec’s new A300 actinic tool “failed” evaluation at TSMC

Q: **“I heard the light source in the new Lasertec tool, the A300, doesn’t work, and TSMC evaluated it, but I heard that it failed at TSMC. Is that true?”**

A: **“[chuckles] Yes. It failed.”** – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

TSMC “will not consider” Lasertec’s EUV tool for production; only a niche lab/R&D tool in the mask shop; using legacy DUV inspection machines instead as EUV inspection “is not necessary”; only purchased six A150 machines

A: **“Both the LasertecX8 Ultra and X9 Ultra are the DUV light source, but they are inspecting the EUV mask. The main reason is for the throughput. And it is not necessary...”**

Q: **“You’re saying TSMC uses the X8 and X9 Ultra in the fab instead of the ACTIS?”**

A: **“Yes. Because of the throughput. TSMC will not consider the A300 or A150 for production. They will only exist in the EPO mask house. The EPO—we call it the EP Organization in TSMC. TSMC has already purchased six sets of A150. So, they already bought six tools. Six A150 tools. But last year, they also purchased the first A300. And the first A300 because they changed the light source, Urashima.”** – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

An ex-KLA executive who was in a leadership role in their EUV mask inspection group and who detailed his direct conversations with Lasertec confirmed that TSMC is highly disgruntled with Lasertec's EUV tool: "what I know is that TSMC is upset because the source is not working...TSMC is upset...the source does not have uptime, cost of service is too high, they are flickering." He corroborated that TSMC evaluated the A300 and that it failed and that Lasertec has to write off its inventory: "Lasertec probably must write off those tools."

TSMC is upset with Lasertec; problems with the source; A300 with new light source failed in the qualification period

A: "It's exactly what I told you. Lasertec convinced TSMC to take a new upgrade of the source, which is an Urashima source. There was a lot of wow around it. And when they released that source to TSMC, and that's it. We didn't hear anything else. **What I know is that TSMC is upset because the source is not working.**"

Q: "So, TSMC evaluated the Isteq source for six months and it failed."

A: "Yes. Intel doesn't return the tool. **TSMC is upset. The source does not have enough uptime, cost of service is too high, they are flickering.** Therefore, they upgraded to the new source one or two—I don't remember how many tools that have the new Urashima or Isteq source. Mid last year to the end of last year. **At the beginning of the year, I heard that they failed.**"

Q: "When they switched the source—they announced a product a few weeks ago in November 2023—they had the Actis A150. They announced the Actis A300 and they said it has the Urashima light source. What is Urashima?"

A: "Urashima is basically Isteq's source and then the product is called Urashima. This is Lasertec's name. Isteq is the company that manufactures this source and it's actually a Russian university."

Q: "You're saying the A150 used the Ushio source. The A350, the Urashima, is the Isteq source. So, you're saying Urashima has already failed after six months with TSMC."

A: "Correct. That's my speculation...now I know exactly what happened because I thought that the tools were shipped back. **Lasertec probably must write off those tools.**" - Former KLA executive in a leadership role in their EUV mask inspection group

The KLA executive stated that TSMC began pushing back Lasertec EUV orders over a year ago, after the new light source failed inspection in early 2023 after six months of evaluation: “it’s not stable...it’s just not stable...it doesn’t work all the time...it just dies.” He replied affirmatively when we asked if TSMC is angry with Lasertec.

TSMC is angry and began delaying purchases a year ago; new light source failed in the qualification period; EUV source “is just not stable” and “just dies”

Q: “You said Intel gave them this big advance. You said they tried to get TSMC to take it. That they over-ordered with too many PO’s?”

A: “What I heard is that **TSMC is delaying for N5—for 5 nanometer—delaying purchasing until they fix their problem, and that was the part of the Urashima project that they built, which is the new source** that I mentioned.”

Q: **“When did they start delaying?”** When did you hear this?”

A: “Yes, just **a year ago** now.”

Q: “You’re saying TSMC is pissed off at all their problems right now?”

A: “Mm-hmm....because of the source, because the original source was Ushio. **The Isteq source has been qualified in TSMC and failed.** And now, they are basically working on the next generation of it. And they’re hiring a lot of engineers.”

Q: “For the source?”

A: “For the source. I’m just connecting all the dots.”

Q: “You said they tried to change the source and failed. They have Ushio, then they went with Isteq. You said Isteq failed. When did Isteq fail?”

A: **“Roughly at the beginning of 2023. It was in evaluation for six months in TSMC.”**

Q: “So, Lasertec changed their source to Isteq. TSMC evaluated for six months, and it failed.”

A: “Mm-hmm.”

Q: “Do you know why it failed? Just the same reasons as the Ushio source?”

A: “What I heard is **it’s not stable.**”

Q: “Is it the flickering issue?”

A: “No, **it’s just not stable. It doesn’t work all of the time, it just dies.** I know the source. I’ve been in Isteq and my engineers live in Russia.” -Former KLA executive in a leadership role in their EUV mask inspection group

TSMC has concluded that EUV mask inspection is a failure given Lasertec's defective technology, and now simply uses legacy DUV tools instead with a ~200nm light source – stating that DUV is sufficient for EUV masks in production. As a result, TSMC's purchases are focused on KLA and Lasertec's old DUV tools called MATRICS X8 and X9 – not its ACTIS EUV line. The TSMC executive stated that “if you care about production throughput, you have to use a 213nm light source.” He stated that even though EUV is 13.5nm and can theoretically catch smaller defects, DUV is sufficient for the quick mask checks that TSMC needs to do in production: “TSMC just wants to do quite frequent monitoring...a quick check...the best resolution is not necessary.”

Using legacy DUV inspection machines instead, as EUV inspection “is not necessary”

A: **“Both the Lasertec X8 Ultra and X9 Ultra are the DUV light source, but they are inspecting the EUV mask. The main reason is for the throughput. And it is not necessary...”**

Q: **“You're saying TSMC uses the X8 and X9 Ultra in the fab instead of the ACTIS?”**

A: **“Yes.** Because of the throughput. TSMC will not consider the A300 or A150 for production. They will only exist in the EPO mask house.” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

The legacy DUV X8 is the only Lasertec inspection tool that TSMC uses for production, not the ACTIS A150

“Their X8 is the DUV light source. X8 Ultra. For production, when we talk about the production tool, it's mainly for production, for the production mask inspection tool. From Lasertec, they have two models. The first one is called X8 Ultra. Another one is the X9 Ultra. They are using the original light source. It's 213 nanometers. X9 Ultra, they are using a 193-nm light source...Originally, for Lasertec, their original light source for DUV is 213. But now, they promote 193, because KLA is using a 193 nm light source.” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

Only need quick, lower resolution inspection in the fab but with higher throughput which Lasertec's tool cannot provide

“TSMC just wants to do quite frequent monitoring. Every day, they check the mask. A quick check. **If you go through a quick check, the best resolution is not necessary.** It's not necessary.” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

However, although TSMC is forced to rely on Lasertec's old DUV tools versus EUV, its DUV line – known as MATRICS X8 ULTRA, and the more recent X9 – is plagued by light source defects as serious as those in its EUV source, in addition to other problems. An ex-Lasertec field service engineer who was recently based at TSMC's fab in Taiwan, whose main responsibility was the MATRICS DUV machines, provided color as damning as that provided by other field engineers for the ACTIS EUV line: "the laser source is unstable for the X8, X9 ULTRA"; "it has a lot of software bugs"; "bug is always causing random problems...they cannot find critical defects, or the resolution is very bad...the image processing is very bad." Although TSMC can get by with its current DUV machines, the engineer stated that Lasertec's product simply won't work for TSCM's 3nm node.

DUV laser source is unstable for both X8 and X9 ULTRA legacy machines; "a lot of software bugs"; doesn't work for 3nm

Q: "How well did the X8 and 9 work? How reliable was it?"

A: "The **laser source is unstable, for the X8, X9ULTRA.**"

Q: "You're saying the light source for the X9ULTRA is not stable?"

A: "Yes, and X8ULTRA is more stable than X9ULTRA."

Q: "Was TSMC happy with the X8ULTRA, or were they having problems with it?"

A: "The image calculation function is not very natural...it has **a lot of software bugs**...when the X8ULTRA scans a photo mask, when they find some defect, the software **bug is always causing random problems...they cannot find critical defects, or the resolution is very bad.**"

Q: "The resolution is very bad?"

A: "Yes. The image processing is not good...the **image processing is very bad.**"

Q: "This is for the X8ULTRA?"

A: "Yes, because the X8ULTRA for 3 nanometers has limitations. So, probably 5 nanometers is good, but **3 is not always good because of the laser source, the design is all for 5 or 7-nanometers**...it was very suitable for 5 nanometers, **not for 3 nanometers.**" – Former Lasertec field service engineer at TSMC fab in Taiwan

The Lasertec field engineer stated that the MATRICS DUV product suffers from particle contamination issues just like the ACTIS EUV line, leading TSMC engineers to complain: “...design is very bad because of the particle issue...the Lasertec machine isn’t designed very well...it’s suffering from so many particle issues...[the tool] is not very clean.” The tools require a shocking 10 hours of daily cleaning to address particle contamination, leading to a low uptime of less than 60%, with an onerous protocol including UV light to scan for stray particles.

Lasertec DUV tool “isn’t designed very well” and “suffering from so many particle” issues; TSMC engineers complain

Q: “What would TSMC say about the machine—their engineers?”

A: “They would probably say that the **software bug is too much**. And the design is very bad because of the **particle issue.**”

Q: “You said the design is very bad because of the particle issue?”

A: “Yes, because of the flow—you have high pressure to low pressure, and the **particles still happen.**”

Q: “You’re saying there are particles in the X8ULTRA?”

A: “Yes, from the FUU - the fan and filter unit... **The Lasertec machine isn’t designed very well. It’s suffering from so many particle issues...[the tool] is not very clean.**” – Former Lasertec field service engineer at TSMC fab in Taiwan

DUV machine requires 10 hours of daily cleaning to address particle contamination; low uptime of less than 60%

Q: “You said the tool is not very clean inside?”

A: “Yes, they clean the machine every day. They **swipe the machine and use UV lights to check for the particles.**”

Q: “How much downtime did the machine have every day? You said it’s working for four or five hours. And how many hours does it need maintenance?”

A: “Daily. They start cleaning at 9 am, and then they finish at about 15 or 16 pm. Then, they invite the customer to check it for about one and a half hours. So, it’s almost done by 18 pm.” Then they would start around two or three masks, and they would take the value. And if they catch a particle on the dummy mask, they will release it...the total working hours is about eight hours. At that checkpoint, they will create a job to verify the data. So, **it costs about almost 10 hours.**”

Q: “And then how many hours do they use it after that?”

A: “14 hours...when they have a lot of masks that need inspection, they will just find the time and take it down and take out the particles.” – Former Lasertec field service engineer at TSMC fab in Taiwan

The field engineer stated that Lasertec's DUV light source is unstable for not just the old MATRCIS X8 DUV product but for the new X9 as well, which was an attempt to address the X8's source problems – stating that he heard this a few months ago “from my colleagues who are X9 ULTRA team members” at TSMC's fab in Taiwan. The similarity between both the light source and particle contamination issues for Lasertec's DUV and EUV products suggests a fatally flawed and defective technology platform.

DUV tool has an unstable light source, based on recent color from the X9ULTRA field service team at TSCM

Q: “And you're saying the **X9ULTRA, the light source is unstable?**”

A: **“Yes, [I heard this] from my colleagues who are X9ULTRA team members.”**

Q: “When did they tell you this?”

A: “About three months ago.”

Q: **“So, they were having a lot of problems with it?”**

A: **“Yeah,** because TSMC wants Lasertec to move the new tool from R&D to onsite, and they are going to production.” – Former Lasertec field service engineer at TSMC fab in Taiwan

6C. Struggles at other customers, such as Samsung, Marvell, and Global Foundries, the latter even laughing at the quality of the Lasertec's EUV tool and ceasing new purchases in 2021, calling it "just unacceptable" with a "problematic light source."

Global Foundries, one of the largest semiconductor companies in the world, blasted Lasertec's EUV product, corroborating the damning color from TSMC and Intel with regards to a defective light source, contamination, and unacceptably low uptime. We spoke with a longtime, knowledgeable executive who has been directly involved in purchasing Lasertec systems and in all aspects of mask inspection. He stated that the uptime is only "around 60-70%...disappointing"; "the uptime is clearly an issue...not only uptime but also debris from the tin-based system that they use." He summarized the issues as a "problematic light source, uptime, and user interface not as friendly or capable as a comparable KLA system." He even described their "new" Urashima light source as "the risk for them, no question...it sounds sketchy."

Uptime and debris are major issues for previous light source as well as the "new" Urashima source

"The uptime is clearly an issue. The Urashima system or the plasma-based light source that they use to produce EUV—it always was going to be the challenge. **It's not only uptime but also debris from the tin-based system that they use.** That is my experience as well. Uptime is certainly an issue." – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

"Disappointing" and low uptime; "problematic light source"; inferior UI versus KLA

Q: "What do your colleagues at Global Foundries said about the uptime?"

A: **"Around 60-70%, disappointing** from a trial inspection system. We would expect it to be 80-90%, **problematic light source, uptime, and user interface not as friendly or capable as a comparable KLA system.**" – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

New Urashima is a light source is a key risk and "sounds sketchy"

"Urashima is the **risk for them, no question...it sounds sketchy.**" – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

Global Foundries actually laughed at the quality of Lasertec's tool during our research interview, describing it as merely a "niche player" that has "struggled" with low uptime and a poor user interface, stating that it is highly inferior to KLA. The executive slammed its uptime as "just unacceptable" for inspection equipment: "doesn't meet what we would normally expect"; "if it was not the only actinic inspection tool in the market, people would definitely go with another supplier"; "struggled with uptime concerns"; "not as sophisticated or as capable as the classic KLA mask inspection tools."

Lasertec tool doesn't meet minimum uptime standards – "just unacceptable"; "doesn't meet what we normally expect"; no one would use Lasertec if another actinic tool was available

Q: "I talked to a former field service engineer from Lasertec who was dedicated to the Intel mask shop in Santa Clara. He described major problems with uptime. He said the head mask person and others were getting really angry with Lasertec.."

A: **"I can confirm that. Uptime for the A150, uptime is an issue. If it was not the only actinic inspection tool in the market, people would definitely go with another supplier. It doesn't meet what we would normally expect.** We typically in the industry for look for inspection equipment to have availability to be used certainly in excess of 80%, and most are over 90% and up. And **the A150 didn't meet the 80% threshold [laughs]."**

Q: "This person said it was 60-65%."

A: "Right, which **for inspection equipment is just unacceptable.** It's not like it's process equipment, a vacuum system, a system where you'd expect there to be lower availability due to preventive maintenance or the nature of the process." – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

"Niche player" that has "struggled" with low uptime; legacy KLA systems are sufficient for EUV mask inspection

"I think KLA were reluctant as far as how fast EUV will actually be implemented. **We think the legacy systems we have are good enough for early EUV debug.** We didn't have EUV pellicles until recently...I think Lasertec has **struggled with uptime concerns.** Also, the user interface is **not as sophisticated or as capable as the classic KLA mask inspection tools,** which very much benefit from the learning and scale that KLA has had in the wafer front-end. Lasertec is kind of a niche player. They don't have the same install base and customer feedback, I would say, that has driven improvements like KLA has." – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

The executive indicated that Global Foundries only purchased two of the Lasertec EUV actinic inspection machines and haven't bought any since 2021, disparaging it exactly as TSMC did in our interviews as merely an "occasional R&D tool" that can't be used in a fab/production environment.

Global Foundries only purchased two Lasertec actinic machines; haven't bought one since 2021; merely an "occasional R&D tool"; not used in a production/fab environment

Q: "How many of the Lasertec actinic machines does Global Foundries have now?"

A: **"We have two."**

Q: "When was the last one purchased?"

A: **"They're both A150, so 2019 and 2021."**

Q: "So, you guys haven't bought any since then?"

A: "Correct."

Q: "Why is that? Unhappy with reliability and so forth, or didn't need it?"

A: "I would just say capacity needs. We have a very limited program. We're not using EUV in the way others are using it."

Q: "Are those two Lasertec machines being used a lot? Are they idle? What's the customer feedback?"

A: "The customer feedback is **they're not flat out in terms of pushing max capacity**. There's some engineering work done on them—used for engineering evaluations, particularly of new materials."

Q: "Is it fair to say they're an occasional R&D tool?"

A: **"They are occasionally an R&D tool."**

Q: "So, they're **not really being used in production right now?**"

A: "They're not ..." – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

The executive stated that Lasertec's customers are now delaying "shipment of systems" due to the rampant product problems and it's product "not even being adequate": "they're holding off near-term shipments until they resolve these quality issues." He indicated that its defective EUV system has badly damaged Lasertec's reputation and described its customers as "frustrated and disappointed": "prior to EUV, Lasertec was known as a reasonable supplier"; "[customers] are frustrated...frustrates people because their brand has definitely taken a hit as a result of their EUV inspection equipment"; in the past Lasertec "was adequate, and that's the frustration today – not even being adequate."

Customers delaying shipment of Lasertec systems due to "quality issues"

"I've heard of a **delay of shipment of systems** to have Lasertec fix some things or engineering changes...**they're holding off near-term shipments until they resolve these quality issues.**" – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

Customers are "frustrated and disappointed"; "brand has definitely taken a hit"; "not even being adequate"

"[Customers] are frustrated. In the past, prior to EUV, Lasertec was known as a reasonable supplier and high-quality because we used them in my prior employer. We had an internal mask house. They were a niche player but a very reasonable supplier...people are just **frustrated and disappointed** because it breaks with their tradition of being a mainstream supplier that would meet reasonable expectations for product quality. So, that's what **frustrates people because their brand has definitely taken a hit as a result of their EUV inspection equipment.** You knew, okay, it wasn't going to cost as much as KLA; it didn't have all the bells and whistles of a KLA, but **it was adequate, and that's the frustration today—not even being adequate.**" – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

The Global Foundries executive stated that “senior technical people” from Intel and Samsung - peers who he spoke with - recently visited Lasertec’s headquarters to convey concern about its product, indicating the severity of frustration among all three of its top customers: “they were visiting in part because of...quality concerns with the equipment, and to get a first-hand view of what is done in the factory to improve outgoing quality”: “it was more than somebody just going there to complain...these are senior technical people.”

Intel and Samsung recently visited Lasertec headquarters due to concern about the product’s problems

Q: “Out of curiosity, why did your colleagues visit Lasertec this fall? Because you’re thinking about more machines? Or do you periodically visit all of your major suppliers?”

A: “I should have been clearer. These are colleagues from other companies. Let’s say former colleagues at my company who now work for other companies.”

Q: “Are these customers Intel or TSMC?”

A: “Yes. **They were visiting in part because of some of the concerns we’ve talked about, quality concerns with the equipment, and to get a first-hand view of what is done at the factory to improve outgoing quality of the machines.**”

Q: “So, that’s like getting called to the principal’s office.”

A: “Correct.”

Q: “There are only three options for who that customer could be.”

A: “Correct.”

Q: “Are you able to say the company?”

A: “Samsung and Intel. Two of the three.”

Q: “So, **you heard the same feedback from both Samsung and Intel. They went there, and they saw finished goods inventory.**”

A: “Correct. **It was more than just somebody going there to complain. These are senior technical people that offer engineering insight and knowledge about, okay, here’s what you should be doing or here are what we think are engineering changes or improvements you need to make for the equipment to be better.**”

Q: “So, they paid Lasertec a visit a few months ago or whatever.”

A: “Correct.” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

Marvell, one of the largest semiconductor companies in the world, expressed the same dismay with Lasertec's EUV tool as TSMC, Intel, Samsung, and Global Foundries. They stated that they purchased one ACTIS A150 EUV inspection machine and lease two others, with no interest in purchasing any more. Marvell noted the same light source defects and technical problems as other large players. In particular, a Marvell senior employee stated that he spoke with ASML, and that ASML "echoed the same problems that we were seeing...ASML customers were also reporting the same things with Lasertec with respect to the light source."

Marvell stated they purchased one ACTIS A150

"We are a current customer of their A150 series. We use that for our 5-nanometer wafers. We purchased their tool in 2021, I believe. So, we've been using it for a couple of years now, and I'm, as well, familiar with their new A300 series.. **We only have one of these that we own, and then we're leasing two** ...In some cases, we buy the machines ourselves, and in other cases, we're leasing capacity." – Senior employee at Marvell Technology

Marvell stated they paid \$62MM for an ACTIS A150

Q: "The pricing I heard for the A150—I got a number of \$40 million from Samsung and \$70 million from Global Foundries. And then, I got an estimated of \$50-60 million from Intel, which buys a bunch. Where in the range did you guys fall for that purchase in 2021?"

A: **"\$62 million."** – Senior employee at Marvell Technology

Marvell indicates corroboration from ASML regarding extensive problems with Lasertec's tool

Q: "Have you had conversations about product quality with TSMC, Intel, Samsung and other companies with regard to Lasertec?"

A: "We have had some conversations with **ASML. They echoed the same problems that we were seeing. ASML customers were also reporting the same things with Lasertec with respect to the light source."**

Q: "ASML was hearing from their customers they were having problems with Lasertec's light source?"

A: "Yeah. With respect to ASML, **my conversations with my counterpart over there is that ASML customers have reported the same issues with Lasertec's A150 tool with respect to the light source that we were seeing."** – Senior employee at Marvell Technology

Marvell stated that it is “aware” of the “new” ACTIS A300 EUV inspection machine but that “it doesn’t make sense for us right now” to even evaluate it. The senior employee expressed significant skepticism: “there was no data that they had”; “no real-world case studies from independently verified semiconductor industry associations”; “no real way to discern that this actually has a value-add.”

The earliest Marvell would evaluate the A300 is in 2025 – “doesn’t make sense for us right now”

“We are aware of the A300 series, and that is something that we will be looking at evaluating, but it most likely won’t be until the beginning of 2025 only because there’s hardware demand softness that we’re seeing, and so it doesn’t make sense for us right now to move forward with a new evaluation..” – Senior employee at Marvell Technology

Lasertec had “no data” to show for the A300; “no real world case studies” or independent evaluation; “no real way to discern” any value

Q: “The A300 sounded like vaporware to me. Did they even have any evaluation data or benchmarks? What was the status?”

*A: **“No, there was no data that they had.** They essentially just had value-add of how this is different, namely in their light source...there’s **no real-world case studies from independently verified semiconductor industry associations** or a semiconductor consortium. And so, there’s **no real way to discern that this actually has a value-add** that would offset the overall cost of taking possession of one of these machines.”* – Senior employee at Marvell Technology

The Marvell senior employee stated that one of their executives recently visited Lasertec's headquarters in Japan, and that Lasertec was desperate to offload its ACTIS EUV inventory, offering up to ~75% discounts on the A150, as well as other inducements and concessions that acknowledged uptime and reliability problems: "they were trying to also offer additional A150's that they had on the books that weren't, let's say, being taken ownership of...we were getting a reduced price as well if we wanted to purchase additional A150's...in the range of \$15-30MM." Marvell stated they previously paid \$62MM for the A150.

Lasertec desperate to offload ACTIS inventory; offering up to ~75% discounts on A150; offering various inducements and concessions, including some that appear to acknowledge uptime and reliability problems

Q: "Do you remember any color around the inventory build-up color?"

A: "When our [Marvell senior executive – title redacted] went to go visit, **they were trying to also offer additional A150's that they had on the books that weren't, let's say, being taken ownership of by other competitors that had presumably forecasted that they needed them. And so, we were getting a reduced price as well if we wanted to purchase additional A150s and that price was closer to about, in the range of \$15 to 30 million.**"

Q: "So, they were willing to try to sell those machines at half or a third of their price?"

A: "Yeah, and also, again, that's why the comment that I gave earlier, which is that if we were to buy an A150—"

Q: "Yeah, they've got to write down all that inventory."

A: "Yeah."

Q: **"My question is, did that really happen? They offered you guys \$15 to 30 million to take A150's? This literally just happened a few weeks ago?"**

A: **"Yeah."** – Senior employee at Marvell Technology

The details that Marvell provided suggest that Lasertec is desperate to offload bloated EUV inventory: “they made a multitude of offers to try to get us to basically purchase more equipment.” Notably, Lasertec offered terms that indicate an attempt at damage control given customer frustration with product defects and low uptime: “other concessions were made like extended warranty service...at no additional cost...even guaranteeing service time like 60-70%...if not then we could take some kind of reimbursement...for the lost time of use” – making us question how Lasertec could retain any margin given the three field service engineers required to maintain each machine plus the millions to service, clean, and replace the light source and optics.

Lasertec desperation evident in contract concessions and inducements

A: “But we also declined it because we just don’t need it. We didn’t need that, and then we also weren’t looking for an evaluation on their A300. So, **they made a multitude of offers to try to get us to basically purchase more equipment.**”

Q: “Wow, I’m shocked.”

A: “What **they were also offering was that if we purchased an additional machine, we would get—there were other concessions that were made like extended warranty service that would be at no additional cost** or discounted warranty services or discounted repair costs or **even guaranteeing service time, like 60-70% time, for example, and if not, then we could take some kind of reimbursement from them for the lost time of use.**” – Senior employee at Marvell Technology

Marvell told us that a senior executive visited Lasertec in December, and that Lasertec offered an arrangement where “they would actually swap out the A150 that we have with an A300 at a discounted price...because they wanted to try to get early purchase volume.” The offer suggests not only desperation but an attempt to mislead investors by essentially fabricating sales of the “new” A300, which is simply the A150 with a different light source and a new name in an attempt to cover up a defective light source. Marvell indicated that they declined the offer and that “we haven’t seen real proof of concept and enough case studies to actually show the real-world application of it versus the A150.”

Lasertec offered to swap A300 for A150 “at a discounted price” so that they could show “early purchase volume”; no “real proof of concept”

Q: “What have you heard about the A300? Have you actually had a conversation or engaged with Lasertec on it?”

A: “We’ve had some introductory conversations with them about it, and actually, insofar as if we were to purchase—one of the conversations we had was that **they would actually swap out the A150 that we have with an A300 at a discounted price.** Again, it wasn’t something that we were looking to entertain just because the overall quote that we got on it is fairly significant. But I have heard that they used a new light source.”

Q: “Yeah, Urashima.”

A: “Urashima, yeah. The only thing is that **we haven’t seen real proof of concept and enough case studies to actually show the real-world application** of it versus the A150.”

Q: “When were these conversations? They announced it in November. When did you have these conversations with Lasertec about the A300?”

A: “This was about a few weeks ago, around December, because they did a press release and **[an executive at Marvell – title redacted to preserve anonymity] who actually went to visit their headquarters in Japan and had a conversation with them about it and one of the things that they had stated was that because they wanted to try to get early purchase volume, they were willing to potentially entertain a swap of the A150 for an A300.** We respectfully just said that we wouldn’t be looking at evaluating something like this until probably closer to 2025.” – Senior employee at Marvell Technology

7. Lasertec customers who have recently visited Lasertec's headquarters confirmed seeing highly unusual levels of inventory - “pretty scary...pretty weird” - and other red flags: their “factory floor for finished goods is full.. a substantial amount of finished goods inventory that's either not passing [evaluation]” or “customers have deferred shipment”; “literally no space left on their final inspection bays before the equipment ships.”

Marvell was one of several customers we interviewed who recently visited Lasertec's headquarters and reported seeing an unusual level of machine inventory – the Marvell executive “mention[ed] a build-up of inventory when they visited the Lasertec headquarters – and indicated skepticism of Lasertec's inventory valuation: “I can tell you right off the bat...they haven't taken any impairments against their balance sheet, so they're probably over-valuing their inventory far greater than it's actually worth...a lot of these companies are sitting on so much inventory....”

Marvell witnessed “a build-up of inventory” during Lasertec visit; “lot of these companies are sitting on so much inventory” due to 5N slowdown

Q: “What was the overall feedback from the [Marvell executive – title redacted] who visited Lasertec?”

A: “[T]hey did mention **a build-up of inventory when they visited the Lasertec headquarters**...I can tell you right off the bat that the reason why that is offload probably because they **haven't taken any impairments against their balance sheet, so they're probably over-valuing their inventory far greater than what it's actually worth.**”

Q: **“What was the inventory comment that [Marvel executive – title redacted] made?** It's very interesting because that's a second data point, so I'm just trying to see how credible it is.”

A: **“There's credibility to it because that's what I've also heard as well. A lot of these companies are sitting on so much inventory** because the 5 nanometer, the market size of it, has basically softened to a significant degree from 2022 to currently. I can give you some additional context. In 2021, when we forecasted a 5-nanometer wafer throughput for our needs to, let's say, TSMC, we were forecasting that by 2023, we would need about 15,000 to 18,000 5-nanometer wafers. And our total throughput was only about 4,000-5,000. Now, when we did that forecast in 2021, that's when—you're not just forecasting TSMC but you're forecasting to all your hardware needs, all of your hardware manufacturers that you work with...**everybody is sitting on additional inventory...that's an issue across all 5-nanometer architecture.**” – Senior employee at Marvell Technology

Marvell indicated that Lasertec's existing inventory has little value, given the offer Lasertec recently made to "take back the A150 and then give is an A300 at a reduced price...the value of the A150 that we have on hand wouldn't be worth that much." The senior employee noted that inventory mis-valuation in the semiconductor equipment industry "is a common thing, unfortunately, that I've seen across multiple companies in the space...to manipulate earnings: "they defer out...that impairment...because they don't want to record a hit on their gross margin."

Swap for A150 implies existing inventory of A150's has almost no value

Q: "When they said swap, what did they mean? You don't have to pay for it? Or you trade it in?"

A: "So, **they would take back the A150 and then they would give us an A300 at a reduced price** from the retail price."

Q: "And did they say what the price is and what the discounted price might be?"

A: "That I can't speak on because I think those are confidential terms. I can just speak to the fact that what **they were offering was to take the A150 back and give us an A300 and essentially give us a discounted value based on us returning the A150.**"

Q: "The swap thing is weird. How is there any trade-in value from that? It just makes no sense to me."

A: "Well, semiconductor equipment as a whole has a very high depreciation rate. The machine that we already have is almost around three years old, so they factored in the depreciation. Again, semiconductor equipment overall depreciates very quickly and so that's probably why we were getting the price that we were getting. **Effectively, companies would take the A300 and not use their A150s anymore; that would be the idea. So, the value of the A150 that we have on hand wouldn't be worth that much anyhow.**" – Senior employee at Marvell Technology

Marvell senior employee describes mis-valuing of inventory to defer an impairment as common in semiconductor industry

"If they haven't taken any reserves or impairments against their total balance sheet, then with the cost of their inventory - we've seen it before, where the **valuation that they place is not what it actually is, and they defer out when they're going to do that impairment because they don't want to record a hit on their gross margin.** This is a common thing, unfortunately, that I've seen across multiple companies in the space. That's what I've actively seen to be done to **try to manipulate earnings.**" – Senior employee at Marvell Technology

Other customers who recently visited their headquarters confirmed seeing an unusual level of inventory piling up, and speculated that they “recognized revenue for prepayments of systems that aren’t yet shipped.” A Global Foundries executive conveyed color “from colleagues that I’ve spoken with” – “their factory floor for finished goods is full...a substantial amount of finished goods inventory that’s either not passing [evaluation by customers] – customers have deferred shipment...but from colleagues who have visited, they said they have literally no space left on their final inspection bays before the inspection ships.”

Executives who recently visited Lasertec saw finished goods inventory piling up; premature revenue recognition

A: “I have not heard of any A300s actually shipping. I’ll just say, through colleagues who have visited, that there are **systems on the floor that customers are doing evaluations on, but I don’t know of any that have actually shipped to a customer site.**”

Q: “Lasertec’s financial statements look really weird, the kind that to me typically signal some kind of accounting games or accounting fraud.”

A: “I’m not an expert in financial treatment, but **I’d say recognized revenue for prepayments of systems that aren’t yet shipped.**”

Q: “Their financial statements are very opaque. It’s very unclear. But the one thing that is very clear is that the inventory over the last five years on their balance sheet has absolutely exploded where inventory now is about one times revenue. I wonder if product is piling up or they’re misvaluing the inventory to inflate earnings.”

A: “From colleagues that I’ve spoken with, **their factory floor for finished goods is full.** So, I’m guessing that they have a **substantial amount of finished goods inventory that’s either not passing [evaluation] - customers have deferred shipment - I don’t know. But from colleagues who have visited, they said they have literally no space left on their final inspection bays** before the equipment ships”

Q: “When did your colleagues visit?”

A: “It would have been this fall, in the September/October timeframe.”

Q: “Do you know roughly how many machines were in that space?”

A: “I didn’t inquire about specifics around how many bays it was. Realistically, an equipment provider probably - I don’t know - given about the size of Lasertec, it’s probably not more than 10 or 15 maybe. I don’t know.” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

Global Foundries indicated that customers are refusing to accept shipment of the tools: “I’ve heard of a delay of shipment of systems to have Lasertec fix some things or engineering changes...they’re holding off near-term shipments until they resolve these quality issues.” They commented further that misclassifying finished goods inventory as work-in-process is easy: “tricks that any supplier can play...you don’t have to place it ready for shipment...any number of ways you can say it’s work-in-process...if you don’t do that final quality inspection....any number of minor steps...you hold off doing them and you still get to call it WIP.”

Customers delaying shipment of Lasertec systems due to “quality issues”

*“I’ve heard of a **delay of shipment of systems** to have Lasertec fix some things or engineering changes...**they’re holding off near-term shipments until they resolve these quality issues.**” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection*

Misclassifying finished goods as work-in-process inventory are “tricks any supplier can play”; many ways to call it WIP

Q: “In the footnotes of their financial statements, they say almost all the inventory is work-in-process, and they have no finished goods because if they showed all that inventory as finished goods, their stock would collapse because then people would realize what’s going on. “

*A: “These are the **tricks that any supplier can play**. You don’t have to place it ready for shipment - there are **any number of ways you can say it’s work-in-process**. If you don’t do that final quality inspection - there are any number of these minor steps that can be done at the end, and you hold off doing them, and **you still get to call it WIP**.” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection*

Global Foundries noted “a huge risk to Lasertec, if machines are piling up and KLA is going around reassuring people that if you can hold off until mid-2025” – a perfect storm where customers are first refusing shipment due to product defects and then simply waiting for KLA’s upcoming system – “Lasertec could be caught with inventory they’re going to have to significantly mark down...they would end up having to impair...an asset write-down.”

Global Foundries sees KLA as “huge risk to Lasertec” that “obviously will” take share; Lasertec would have “significantly mark down” and “impair” its inventory; “would probably require an asset write-down”

Q: “How many more Lasertec machines are people going to buy in 2024? Because if finished goods inventory is already piling up, 3-nanometer delays, TSMC--?”

A: “I think **there's a huge risk here to Lasertec, if machines are piling up and KLA is going around reassuring people that if you can hold off until mid-2025**, we will have a system commercially available to you. And in fact, we'll give early access in the second quarter of 2025, you can come and evaluate...High-NA is going to accelerate. People are making these massive capital investments in inspection equipment. I do think, by the way, even if KLA came out with, let's say, an A150 competitor, yes, **they will take share from Lasertec. They obviously will. And Lasertec could be caught with inventory they're going to have to significantly mark down** even to move on the A150 side. But if the A300 remains only on paper...it could be pure PowerPoint capability that the A300 never really materializes. And then they would be in a lot of trouble at Lasertec...”

Q: “This is a scenario where their sales completely collapse.”

A: “Yes, for EUV, correct. Correct. And they **would end up having to impair** I don't know price point-wise. **KLA, if they wanted to...could put a lot of pressure on them, which would probably require an asset write-down on the part of Lasertec.**” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

Marvell also recently visited Lasertec's headquarters and reported seeing an unusual level of machine inventory – the Marvell executive “mention[ed] a build-up of inventory when they visited the Lasertec headquarters” – and indicated skepticism of Lasertec's inventory valuation: “I can tell you right off the bat...they haven't taken any impairments against their balance sheet, so they're probably over-valuing their inventory far greater than it's actually worth...a lot of these companies are sitting on so much inventory....”

Marvell witnessed “a build-up of inventory” during Lasertec visit; “lot of these companies are sitting on so much inventory” due to 5N slowdown

Q: “What was the overall feedback from the [Marvell executive – title redacted] who visited Lasertec?”

A: “[T]hey did mention **a build-up of inventory when they visited the Lasertec headquarters**...I can tell you right off the bat that the reason why that is visited probably because they **haven't taken any impairments against their balance sheet, so they're probably over-valuing their inventory far greater than what it's actually worth.**”

Q: **“What was the inventory comment that [Marvell executive – title redacted] made?”** It's very interesting because that's a second data point, so I'm just trying to see how credible it is.”

A: **“There's credibility to it because that's what I've also heard as well. A lot of these companies are sitting on so much inventory** because the 5 nanometer, the market size of it, has basically softened to a significant degree from 2022 to currently. I can give you some additional context. In 2021, when we forecasted a 5-nanometer wafer throughput for our needs to, let's say, TSMC, we were forecasting that by 2023, we would need about 15,000 to 18,000 5-nanometer wafers. And our total throughput was only about 4,000-5,000. Now, when we did that forecast in 2021, that's when—you're not just forecasting TSMC but you're forecasting to all your hardware needs, all of your hardware manufacturers that you work with...**everybody is sitting on additional inventory...that's an issue across all 5-nanometer architecture.**” – Senior employee at Marvell Technology

The Marvell senior employee stated that one of their executives recently visited Lasertec's headquarters in Japan, and that Lasertec was desperate to offload its ACTIS EUV inventory, offering up to ~75% discounts on the A150, as well as other inducements and concessions that acknowledged uptime and reliability problems: "they were trying to also offer additional A150's that they had on the books that weren't, let's say, being taken ownership of...we were getting a reduced price as well if we wanted to purchase additional A150's...in the range of \$15-30MM." Marvell stated they previously paid \$62MM for the A150.

Lasertec desperate to offload ACTIS inventory; offering up to ~75% discounts on A150; offering various inducements and concessions, including some that appear to acknowledge uptime and reliability problems

Q: "Do you remember any color around the inventory build-up color?"

A: "When our [Marvell senior executive – title redacted] went to go visit, **they were trying to also offer additional A150's that they had on the books that weren't, let's say, being taken ownership of by other competitors that had presumably forecasted that they needed them. And so, we were getting a reduced price as well if we wanted to purchase additional A150s and that price was closer to about, in the range of \$15 to 30 million.**"

Q: "So, they were willing to try to sell those machines at half or a third of their price?"

A: "Yeah, and also, again, that's why the comment that I gave earlier, which is that if we were to buy an A150—"

Q: "Yeah, they've got to write down all that inventory."

A: "Yeah."

Q: **"My question is, did that really happen? They offered you guys \$15 to 30 million to take A150's? This literally just happened a few weeks ago?"**

A: **"Yeah."** – Senior employee at Marvell Technology

The details that Marvell provided suggest that Lasertec is desperate to offload bloated EUV inventory: “they made a multitude of offers to try to get us to basically purchase more equipment.” Notably, Lasertec offered terms that indicate an attempt at damage control given customer frustration with product defects and low uptime: “other concessions were made like extended warranty service...at no additional cost...even guaranteeing service time like 60-70%...if not then we could take some kind of reimbursement...for the lost time of use” – making us question how Lasertec could retain any margin given the three field service engineers required to maintain each machine plus the millions to service, clean, and replace the light source and optics.

Lasertec desperation evident in contract concessions and inducements

A: “But we also declined it because we just don't need it. We didn't need that, and then we also weren't looking for an evaluation on their A300. So, **they made a multitude of offers to try to get us to basically purchase more equipment.**”

Q: “Wow, I'm shocked.”

A: “What **they were also offering was that if we purchased an additional machine, we would get—there were other concessions that were made like extended warranty service that would be at no additional cost** or discounted warranty services or discounted repair costs or **even guaranteeing service time, like 60-70% time, for example, and if not, then we could take some kind of reimbursement from them for the lost time of use.**” – Senior employee at Marvell Technology

Marvell told us that a senior executive visited Lasertec in December, and that Lasertec offered an arrangement where “they would actually swap out the A150 that we have with an A300 at a discounted price...because they wanted to try to get early purchase volume.” The offer suggests not only desperation but an attempt to mislead investors by essentially fabricating sales of the “new” A300, which is simply the A150 with a different light source and a new name in an attempt to cover up a defective light source. Marvell indicated that they declined the offer and that “we haven’t seen real proof of concept and enough case studies to actually show the real-world application of it versus the A150.”

Lasertec offered to swap A300 for A150 “at a discounted price” so that they could show “early purchase volume”; no “real proof of concept”

Q: “What have you heard about the A300? Have you actually had a conversation or engaged with Lasertec on it?”

A: “We’ve had some introductory conversations with them about it, and actually, insofar as if we were to purchase—one of the conversations we had was that **they would actually swap out the A150 that we have with an A300 at a discounted price.** Again, it wasn’t something that we were looking to entertain just because the overall quote that we got on it is fairly significant. But I have heard that they used a new light source.”

Q: “Yeah, Urashima.”

A: “Urashima, yeah. The only thing is that **we haven’t seen real proof of concept and enough case studies to actually show the real-world application** of it versus the A150.”

Q: “When were these conversations? They announced it in November. When did you have these conversations with Lasertec about the A300?”

A: “This was about a few weeks ago, around December, because they did a press release and **[an executive at Marvell – title redacted to preserve anonymity] who actually went to visit their headquarters in Japan and had a conversation with them about it and one of the things that they had stated was that because they wanted to try to get early purchase volume, they were willing to potentially entertain a swap of the A150 for an A300.** We respectfully just said that we wouldn’t be looking at evaluating something like this until probably closer to 2025.” – Senior employee at Marvell Technology

Marvell indicated that Lasertec's existing inventory has little value, given the offer Lasertec recently made to "take back the A150 and then give is an A300 at a reduced price...the value of the A150 that we have on hand wouldn't be worth that much." The senior employee noted that **inventory mis-valuation** in the semiconductor equipment industry **"is a common thing, unfortunately, that I've seen across multiple companies in the space...to manipulate earnings: "they defer out...that impairment...because they don't want to record a hit on their gross margin."**

Swap for A150 implies existing inventory of A150's has almost no value

Q: "When they said swap, what did they mean? You don't have to pay for it? Or you trade it in?"

A: "So, **they would take back the A150 and then they would give us an A300 at a reduced price** from the retail price."

Q: "And did they say what the price is and what the discounted price might be?"

A: "That I can't speak on because I think those are confidential terms. I can just speak to the fact that what **they were offering was to take the A150 back and give us an A300 and essentially give us a discounted value based on us returning the A150.**"

Q: "The swap thing is weird. How is there any trade-in value from that? It just makes no sense to me."

A: "Well, semiconductor equipment as a whole has a very high depreciation rate. The machine that we already have is almost around three years old, so they factored in the depreciation. Again, semiconductor equipment overall depreciates very quickly and so that's probably why we were getting the price that we were getting. **Effectively, companies would take the A300 and not use their A150s anymore; that would be the idea. So, the value of the A150 that we have on hand wouldn't be worth that much anyhow.**" – Senior employee at Marvell Technology

Marvell senior employee describes mis-valuing of inventory to defer an impairment as common in semiconductor industry

"If they haven't taken any reserves or impairments against their total balance sheet, then with the cost of their inventory - we've seen it before, where the **valuation that they place is not what it actually is, and they defer out when they're going to do that impairment because they don't want to record a hit on their gross margin.** This is a common thing, unfortunately, that I've seen across multiple companies in the space. That's what I've actively seen to be done to **try to manipulate earnings.**" – Senior employee at Marvell Technology

An ex-KLA executive in a senior role in their EUV mask inspection group stated that Lasertec's EUV tool shipments hit the wall about two years ago with delays due to product defects: "I'm telling you, that's where the problem started two years ago and they said that we cannot ship...they are not shipping because it's a liability...I definitely understand why because they are not shipping...because they have problems with the source...the customers are delaying orders."

Not shipping product because of problems with the EUV light source

Q: "And they have to do this every month, and it costs a couple of hundred thousand dollars?"

A: "Correct."

Q: "That's like \$2.5 million a year just to clean the collector."

A: "That was the beginning. How much it improves the tin, that's **exactly the reason why they went to the Urashima light source...I'm telling you, that's where the problem started two years ago and they said that we cannot ship - and that was why they are not shipping because it's a liability. They ship something and they know that it's consuming all the margin they have.**" –Former KLA executive in a leadership role in their EUV mask inspection group

Customers not taking possession due to problems with the EUV light source, causing inventory to pile up

Q: "Their **inventory is out of control**, like the same as revenue and there is no semi-cap equipment company where the inventory is the same as revenue."

A: "**It makes perfect sense**...I know because I faced the same issues. When you deal with large vendors, the only way they will invest so much in developing the EUV process, which takes forever, is when you commit to them a minimum numbers of system that you buy from them a year. That's the only way they'll guarantee a price. So, it's really a draconian contract. They need to buy inventory, or they'll be fined if they don't actually buy. We do the same thing. If you don't buy the minimum, we committed to pay fines and **it goes to our inventory. So, I definitely understand why because they are not shipping.**"

Q: "What do you mean they're not shipping? Walk me through that."

A: "**Because they have problems with the source. The customers are delaying orders**, that's one thing. The second thing is they themselves, until they're not fixing the problem, the cost of service and the cost of warranty consume the margins. So, they have a double interest to fix those things. And, as a result, they're pushing—" –Former KLA executive in a leadership role in their EUV mask inspection group

The ex-KLA executive described Lasertec's \$1.2 billion of inventory as "pretty scary...pretty weird...doesn't seem normal...that's pretty bad." He stated that most of the initial advances and purchase orders for the EUV tool were from Intel, their "founding customer" – "when it started, most of the slots were Intel slots" – but that they are now being "pushed, pushed, pushed out" and will likely never be filled, with the PO's just sitting on Lasertec's balance sheet as advances until the inevitable impairment charge and capitulation: "Intel...delayed those slots...they will never cancel those PO's...because of the investment they got...it's just there because they owe Intel money..."

Inventory level is "pretty scary" and "weird"; "doesn't seem normal":

"To have a billion dollars of inventory, that's **pretty scary...it's pretty weird. That doesn't seem normal. That's pretty bad** if they've got a billion dollars of inventory." - Former KLA executive in a leadership role in their EUV mask inspection group

Intel was most of the advances and are being "pushed, pushed, pushed out" and will likely never be filled

Q: "You said a lot of their advances in their inventory is Intel gave them too many PO's. You said they were trying to transfer some of them to TSMC. Can you talk about some of that?"

A: "What I said is the following: in the beginning, **when it started, most of the slots were Intel slots.**"

Q: "Most of the slots were Intel slots."

A: "And then they were basically promoting that to TSMC helping Lasertec to put the case in TSMC that's the only tool, and TSMC was just starting a PO. And **what Intel did, they delayed those slots, and you started to see that there were more and more PO's coming up from TSMC and after that from Samsung, and those Intel POs are pushing, pushing, pushing out. I think they will never cancel those POs because some of those POs are basically there because of the investment they got. It's just there because they owe Intel money, so that's what they did. Those POs—**"

Q: "You're saying they're advances. There are a bunch of Intel POs that will never get filled." - Former KLA executive in a leadership role in their EUV mask inspection group

The executive indicated that Intel gave Lasertec larger advances and more purchase orders than it could use and then tried to pawn them off to TSMC. However, TSMC – as detailed in a previous section – is fed up with Lasertec’s product defects and has ceased buying their EUV inspection machines – “TSMC is...delaying purchasing until they fix their offload problem” and that they started delaying “a year ago now.” He stated that TSMC evaluated the “new” ACTIS A300 with the Urashima light source in early 2023 and that it failed their qualification process.

Intel over-ordered machine tried to pass machines on to TSMC, which began delaying purchases a year ago

Q: “You said Intel gave them this big advance. You said they tried to get TSMC to take it. That they over-ordered too many PO’s?”

A: “What I heard is that **TSMC is delaying for N5—for 5 nanometer—delaying purchasing until they fix their problem, and that was the part of the Urashima project that they built, which is the new source** that I mentioned.”

Q: **“When did they start delaying?”** When did you hear this?”

A: “Yes, just **a year ago** now.”

Q: “You’re saying TSMC is pissed off at all their problems right now?”

A: “Mm-hmm....because of the source, because the original source was Ushio. **The Isteq source has been qualified in TSMC and failed.** And now, they are basically working on the next generation of it. And they’re hiring a lot of engineers.”

Q: “For the source?”

A: “For the source. I’m just connecting all the dots.”

Q: “You said they tried to change the source and failed. They have Ushio, then they went with Isteq. You said Isteq failed. When?”

A: **“Roughly at the beginning of 2023. It was in evaluation for six months in TSMC.”**

Q: “So, Lasertec changed their source to Isteq. TSMC evaluated for six months, and it failed.”

A: “Mm-hmm.”

Q: “Do you know why it failed? Just the same reasons as the Ushio source?”

A: “What I heard is **it’s not stable.**”

Q: “Is it the flickering issue?”

A: “No, **it’s just not stable. It doesn’t work all of the time, it just dies.** I know the source. I’ve been in Isteq and my engineers live in Russia.” -Former KLA executive in a leadership role in their EUV mask inspection group

Other research sources corroborated the mountain of inventory, such as a senior semiconductor executive in Japan – who stated he is friendly with Lasertec’s CEO and that customers are “holding back the orders, waiting for improvement light source-wise or whatever”; “pretty much what the reality looks like...the reason why they are sitting on top of a lot of inventory.” In the quarter ended September 30, 2023, Lasertec’s gross margin mysteriously collapsed from 56% to 36% with no explanation. Management was evasive when asked by analysts after its earnings release. The executive suspected that Lasertec had to replace defective light sources in some tools, which destroyed their margin: “my suspicion is it was the light source...[which] along costs \$10 million...if they are forced to replace the light source two or three times...that will cost them \$40 million.”

“Sitting on top of a lot of inventory”; “holding back orders”; replacing defecting light sources

Q: “The KLA person I talked to said that Lasertec customers were “begging and crying” for them to hurry up and come up with an actinic product to compete with Lasertec. **That’s how miserable it sounded like the customers were. Is that consistent with what you’ve heard?**”

A: **“Yes, I think that’s pretty much what the reality looks like. That’s also the reason why they are sitting on top of a lot of inventory stock.”**

Q: “The customers just don’t want to buy it?”

A: **“They’re holding back the orders, waiting for improvement light source-wise or whatever.”**

Q: “This last quarter, Lasertec made a very interesting announcement. They were very secretive and vague about it, but their gross margin collapsed this last quarter, and they said we had a few machines that had no margin on them, but now they’ve cleared and now our margins... it was strange. Do you know what that was about?”

A: **“No. My suspicion is it was about the light source. The light source didn’t perform, so the customer demands a new light source. And I heard the light source alone costs \$10 million. So, if they are forced to replace the light source two or three times until it functions properly, that will cost them \$40 million, almost as much as a new tool, as you mentioned, \$40 million being the price. It can be very costly to replace parts. That’s my suspicion.”** - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec’s CEO

8. Lasertec's small, one-time EUV market opportunity is already exhausted, with little to no growth left in its flagship ACTIS line. Aside from product defects, irate customers, and the lack of a viable EUV source, an analysis of its growth drivers in the last 5 years indicates they were ephemeral.

Product defects, customer frustration, and exploding inventory naturally indicate that Lasertec's small EUV market opportunity is exhausted, with little growth for its ACTIS line. We now add further evidence for why its EUV prospects are saturated. Lasertec has only two growth drivers - Intel and TSMC, with Samsung as a distant third. TSMC is frustrated with the tool's reliability and throughput. Its ACTIS purchases have hit the wall, not to mention the ACTIS A300 failing TSMC's evaluation. That only leaves Intel, but ACTIS tools have only two use cases: in the mask shop and in production in fabs. The mask shop needs only a few tools, with TSMC and Intel's shops at a total of a mere 5-6 systems and no additions in ages. In addition, Lasertec's growth since the introduction of its EUV product has been driven by the industry transition to 5nm. That transition is completed, with 3nm orders delayed, per KLA: "...in the past two years it was the population of the N5 node...that was most of it...now it's N3 and N2...they're delaying those PO's."

Lasertec's recent growth driven by EUV transition for 5N and is now over; N3 purchase orders will be delayed

A: "Yeah, definitely. It's exactly according to what I said that in the past two years, it was the population of the N5 node from R&D to production [that drove their growth]. All the vendors are now, every fab, they move to production. When they move to production, they bought ASML tool and you know how much Lasertec they bought...s most of those orders are TSMC orders."

Q: "You said people moved N5 R&D to production."

A: "To production, yeah. And the number of layers for N5 significantly increased. Thirty percent of the layers are EUV."

Q: **"Have Intel, Samsung, and TSMC already bought all the machines they need for this 5-nanometer transition, or are they still going to be buying a lot more ASML tools and Lasertec tools?"**

A: **"No, that was most of it. Now it's N3 and N2 if they want to go; N5 is in production. For the main ones, yeah, they were N3 PO's. Now, they're delaying those PO's for N3, and those PO's will be delayed.** That's my projection."

Q: "Lasertec's N3 POs will be delayed?"

A: "Correct." - Former KLA executive in a leadership role in their EUV mask inspection group

That leaves only one theoretical driver of growth: Intel's potential addition of a few more ACTIS tools in its fabs, as TSMC views them as a failure for production and eliminates its fabs as a growth opportunity. We note that masks are first inspected in the mask shop, and also periodically inspected in the fab shop as wafers are printed. An ex-KLA director of engineering in its EUV group bluntly framed the dilemma, which captures why inventory has hit \$1.2 billion: Lasertec's tool is simply not production-worthy, which makes two mask shops at Intel and TSMC, both now full, as the bulk of its market opportunity: "the Lasertec tool is not 95% reliable...the mask shop...that's where more of the money is...inspecting in process in the wafer fab – the importance is lower than the mask shop...that's why spending money on an EUV actinic inspection in a wafer fab seems unlikely to me...it would be lower priority overall."

Not a "production-worthy" tool; tool purchases for wafer fabs are less likely as the value is mostly in the mask shop

A: "Production-worthy, not so quick."

Q: "So, **you're saying it's not production-worthy?**"

A: **"Yeah, definitely. The question is, is it production-worthy? Is it reliable? People want a 95% reliable tool. The Lasertec tool is not 95% reliable...the mask shop, and that's where—to be blunt—that's where more of the money is.** The volume is in the fab. Making sure that the mask is good is super important because every mask is making so many chips. Inspecting in process in the wafer fab - the importance is lower than the mask shop, and the market value is lower. And that's why **spending money on an EUV actinic inspection in a wafer fab seems unlikely to me...it would be a lower priority overall."** – Ex-KLA director of engineering focused on EUV mask inspection

It is therefore no surprise that Intel and TSMC are both delaying Lasertec EUV tool orders. A semiconductor executive in Japan in contact with Lasertec's CEO stated "it has been slow...companies like TSMC were holding back...Intel was delaying their orders for more than a year." One of Intel's most senior and tenured engineers with leadership responsibility for EUV added that "TSMC is pushing out and delaying orders." An ex-Lasertec field engineer based on-site at Intel's mask shop stated that the shop has long been at full capacity with only three Lasertec tools: "there are just the machines in San Jose...it didn't sound like they had plans of implementing any more...."

Both Intel and TSMC have been delaying Lasertec orders

A: **"It has been slow**, maybe for the last year, because companies like **TSMC were holding back** investment because of a lack of orders for their leading-edge processes. **Intel was delaying their orders for more than a year**, so they were holding back."

Q: "They were delaying their Lasertec orders?"

A: "Not only Lasertec but the whole lineup for EUV photomask manufacturing, including the writer tools. They had expansion plans, especially Intel, but this didn't happen as soon as they expected, so they were holding back on investment." - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

TSMC is delaying orders, according to one of Intel's most senior EUV engineers

"TSMC is pushing out and delaying orders. The Arizona original plan is now on hold. That's what they are telling their suppliers." – One of Intel's most senior and longtime engineers with leadership responsibility for EUV

Intel has only 3 Lasertec machines in its mask shop and utilization appears to be decreasing; plans to deploy machines in Arizona appear delayed

"There are just the **three machines in San Jose. It didn't sound like they had plans of implementing any more in that location.** Some of my ex-coworkers, they relocated some to Oregon... first, **they moved some guys to Arizona, and then they moved the Arizona guys to Portland.**" – Former Lasertec field service engineer who worked in the Intel mask shop

Our research interviews indicate that Intel has purchased about 13 total ACTIS A150 EUV inspection tools. We believe 3 are installed at Intel's mask shop in Santa Clara, CA, according to an ex-Lasertec field service engineer who was based on-site there; and we think that 10 are in Intel's fabs in Oregon, according to an Ushio field service engineer based at the fab. The mask shop appears to have hit capacity years ago with no new installs, leaving only Intel's fabs as a potential source of growth. However, since 2020, Intel's average rate of adding ACTIS machines at its fabs is only 2.5 per year, by our math. At a slightly discounted price of \$40MM each, Intel's fabs could drive \$100MM/year revenue at most. A Samsung executive estimated Intel would buy even less – only one machine per year, based on Intel's ratio of Lasertec tools per ASML EUV system.

Ushio states that Lasertec has approximately 10 A150 machines in its Oregon fab; averaged about 2 a year

Q: "How many Lasertec machines are there up in Oregon?"

A: **"A150? It's around 10."**

Q: "Just in Oregon?"

A: "Yes. The first one went in in 2020. I think they just installed the last one."

Q: "The first one was in 2020. **Between those two years [2020 and 2022], I think maybe four machines went into Hillsboro. Average about 2 per year."**

Q: "And then **the other 6 went in in 2023?**"

A: "Yes, they put a lot in last year... **I think it was more towards the latter half of the year...there might have been one in the first half.**" – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

Samsung executive expects Intel to buy perhaps only one Lasertec tool per year going forward

Q: "How many more machines do you think Intel is going to buy from Lasertec?"

A: "The trend in Intel for receiving ASML EUV tools is about five tools per year. So, **maybe one tool for a year from Lasertec.**" – Longtime Samsung executive in leadership role in EUV; worked closely with Lasertec; decision-maker for inspection tools

An ex-TSMC purchasing executive - still in touch with colleagues - told us that TSMC has only purchased six ACTIS EUV tools since their introduction in 2019, with no new purchases in years. At about \$50MM each – the executive complained that “the pricing is so expensive” - that is a mere \$300MM in cumulative EUV revenue in five years. We believe two of the six are in TSMC’s mask shop, and the rest are in its fab where it has struggled with their pitifully low reliability and throughput. The executive was clear that its only use case is as a niche R&D tool: “it’s not for production.” An ex-Lasertec field engineer at TSMC’s fab in Taiwan corroborated by stating that “I want to emphasize that the A150 are going into R&D at the site...mostly R&D.”

TSMC only purchased six of the ACTIS A150 machines; don’t appear to have purchased any in years

A: “TSMC probably purchased six ACTIS A150.”

Q: “They purchased how many?”

A: **“In TSMC, six.”**

Q: “Of the A150?”

A: “Yes.”

Q: “And do you know when?”

A: **“2019. I think 2020 also...the A150, the pricing is so expensive, around \$50 million, and the purchasing price in TSMC is around \$50 million...yes, six tools. And they demand it for EPL. It’s not for production; it’s for EPL.”** – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

Only two actinic EUV machines in the TSMC mask shop, just an R&D tool, not for production in the fab

A: “I worked for TSMC in the field, at TSMC fab 18A and 18B...there were **two A150s in the EBO factory [mask shop]...I want to emphasize that the A150 are going into R&D at the site...**”

Q: “You’re saying they just used the A150 for R&D?”

A: **“Yes, mostly R&D** in the field.” – Former Lasertec field service engineer at TSMC fab in Taiwan

The ex-TSMC purchasing executive bluntly affirmed that TSMC is no longer purchasing the A150 EUV tool, stating “no, unfortunately no” when we asked. He repeatedly mocked it as a “lab tool, R&D tool” – “it’s not a tool to support production...the Lasertec tool cannot meet demands in a production line.” He stated that he recently spoke with a Lasertec executive, who admitted that things “are not well” for them.

TSMC has stopped buying Lasertec actinic EUV machines as it’s only a niche mask shop took; “not a tool to support production”; “cannot meet demands in a production line”

Q: “Has TSMC basically stopped buying A150 machines from Lasertec? **Are they still buying Lasertec A150 machines or no?”**

A: **“No, unfortunately, no.”**

Q: “When did they stop buying them?”

A: “I will say something – even if TSMC continues to work with Lasertec, now it’s mainly in the mask house, instead of the production house, production area. So, for the EPL, for the mask house..**it’s not a tool to support production** because the production demands you have to quite frequently monitor and quite frequently check the EUV mask breaks. The demand requirement is quite serious for a production tool, and the mask house tool is different because, for a production tool layout, they have to identify the falling particles, the damage. But the damage is caused by the operation. For production, you are dealing with damage, the falling particles from the operation. Normally, the criteria are different. So, when you change to the production tool, **the Lasertec tool cannot meet demands in a production line.**” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

Lasertec executive recently told TSMC contact that things are “not well”; TSMC views it as a niche “lab tool, R&D tool”

Q: **“When you talked to the Lasertec executive, what else did they say? Are things going well or not going well for them?”**

A: **“Not well.”**

Q: “What did he say?”

A: **“Not well. I think Lasertec will be kind of—we call the lab tool, R&D tool.”**

Q: “It’s an R&D tool? Lab tool, not a production tool.”

A: “Yes, **a lab tool. Yes, not a production tool.** Because you have a lot of limitations for productivity...even today, TSMC still has a weekly meeting with Lasertec. **My colleagues in procurement and from the fab told me they have a weekly meeting to push Lasertec to improve.**” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

An Ushio field engineer who services the light sources in Lasertec's machines corroborated the color we received from the TSMC purchasing executive, confirming that TSMC bought six ACTIS EUV tools in total. He indicated that the last one was purchased in 2022. We note that the Ushio engineer is based on-site and full-time at an Intel fab in Oregon and is in touch with colleagues in Taiwan who service TSMC.

Ushio states that TSMC has only six ACTIS A150 machines; last one was installed in 2022

Q: "Do you know how it's going with the Ushio light source at TSMC in Taiwan? Do you know how many Lasertec machines they have in Taiwan?"

A: "I think there are **six machines in total**. Last I was aware, as of the end of 2023."

Q: "When did they put those in? Did they buy a bunch in 2022/2023, or were these older?"

A: "I think they started going in just before we got the ones in the U.S., so maybe 2019. I think **the sixth one went in at maybe the end of 2022.**" – Ushio field engineer who maintains light sources on Lasertec ACTIS A150 machines at Intel Oregon fab

TSMC cannot be a growth driver or key customer going forward for Lasertec for other reasons – an alarming fact as it is the world’s largest chipmaker and one of the three anchors that every inspection vendor depends on. First, Lasertec’s actinic inspection system is only relevant for masks with a pellicle, and TSMC generally doesn’t use pellicles – a thin membrane to protect the mask from debris. Lasertec’s tool purportedly allows inspection through the membrane, although in reality it introduces debris thus defeating its purpose and requiring duplicative inspection with a KLA DUV system. Second, TSMC is not moving to ASML’s new High NA system, using multi-patterning instead – eliminating High NA as a growth driver despite its hype as the next big thing in EUV.

Intel is Lasertec’s only TAM for actinic inspection, as TSMC doesn’t use pellicles

“Intel uses pellicles. TSMC has had a mixed history of using pellicles or not using pellicles, but for the most part, hasn’t been.”

– Longtime semiconductor consultant recently working for Lasertec

TSMC isn’t pushing hard on High NA

“One other thing I would point out is on the High-NA side. If you’ve been reading the news reports, it looks like TSMC is not going to push very hard on High-NA. What they’ve decided to do, apparently, is do multiple patterning standard EUV rather than trying to do single pattern High-NA.” – Longtime semiconductor consultant recently working for Lasertec

Recent press on TSMC’s disinterest in High NA EUV lithography

TSMC won’t adopt advanced High-NA EUV chipmaking tools until 2030 or later — Intel just received its first tool this week: Report

<https://www.tomshardware.com/tech-industry/manufacturing/tsmc-to-adopt-high-na-euv-tools-in-2030-or-later-report>

TSMC doesn’t need ASML’s new High-NA EUV lithography machines for its next-gen A16 process

<https://www.tweaktown.com/news/97827/tsmc-doesnt-need-asmls-new-high-na-euv-lithography-machines-for-its-next-gen-a16-process/index.html>

TSMC in no rush to embrace ASML’s High-NA EUV technology

<https://www.digitimes.com/news/a20240202PD200/tsmc-asml-high-na-euv.html>

Given the failure of Lasertec's EUV tool in a production environment, TSMC simply relies on legacy DUV tools instead, stating that DUV is sufficient to inspect EUV masks in the fab – an ominous fact for the Lasertec EUV bull case. The ex-TSMC procurement executive stated that the only Lasertec tool they are willing to use for production is their MATRICS DUV system, not their ACTIS EUV line – “TSMC will not consider the A300 or A150 for production”; “Lasertec X8 ULTRA and X9 ULTRA are the DUV light source but they are inspecting the EUV mask...for the throughput...[EUV] is not necessary.”

TSMC “will not consider” Lasertec’s EUV tool for production; only a niche lab/R&D tool in the mask shop; using legacy DUV inspection machines instead as EUV inspection “is not necessary”; only purchased six A150 machines

A: “Both the **Lasertec X8 Ultra and X9 Ultra are the DUV light source, but they are inspecting the EUV mask. The main reason is for the throughput. And it is not necessary...**”

Q: “You’re saying TSMC uses the X8 and X9 Ultra in the fab instead of the ACTIS?”

A: “Yes. **Because of the throughput. TSMC will not consider the A300 or A150 for production. They will only exist in the EPO mask house.** The EPO—we call it the EP Organization in TSMC. **TSMC has already purchased six sets of A150.** So, they already bought six tools. Six A150 tools. But last year, they also purchased the first A300. And the first A300 because they changed the light source, Urashima.” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

The legacy DUV X8 is the only Lasertec inspection tool that TSMC uses for production, not the ACTIS A150

“Their X8 is the DUV light source. X8 Ultra. For production, when we talk about the production tool, it’s mainly for production, for the production mask inspection tool. From Lasertec, they have two models. The first one is called X8 Ultra. Another one is the X9 Ultra. They are using the original light source. It’s 213 nanometers. X9 Ultra, they are using a 193-nanometer light source...Originally, for Lasertec, their original light source for DUV is 213. But now, they promote 193. That’s because KLA is using a 193 nanometer light source.” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

Although TSMC uses a few Lasertec DUV machines in production, there is no growth story there. The procurement executive stated that **“TSMC will keep a minimal share for Lasertec”** only to have some purchasing leverage with KLA. He added that **“two-thirds of the business migrated from originally Lasertec to KLA...I have to say Lasertec has continued to decrease...their share in TSMC.”**

TSMC and Intel mainly bought tools from Lasertec to have some pricing leverage with KLA

“TSMC will keep a minimal share for Lasertec ...KLA is what we call opportunistic...they increased the price...they deal with a hard hand. For the past three years, Intel purchased some tools, maybe two or three from Lasertec. But we believe they want leverage with KLA. We mainly think they purchased Lasertec because they want to give some leverage power to KLA because KLA's price is even higher... TSMC spent a lot of money on KLA's particle inspection tools, like SP5, SP7, and SP6. That's very, very expensive.” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

TSMC indicates KLA DUV tools have superior economics, productivity, and throughput vs. Lasertec; using DUV inspection now vs EUV in production; TSMC has basically switched from Lasertec to KLA; “KLA's share continues to increase”

Q: “If Lasertec is still mainly selling DUV, that means they basically failed in EUV.”

A: “The big thing is the light. **For production, for productivity, and for economic considerations, KLA performs better than Lasertec in TSMC. So, that's why I say already two-thirds of the business migrated from originally Lasertec to KLA already. I have to say Lasertec has continued to decrease their amount of share in TSMC.”**

Q: “In fab 18, what happened?”

A: **“The economics, the throughput.”** If I purchase a tool from KLA, probably for 3-nanometer, TSMC probably needs to purchase 20 mask inspection tools. But if I change to Lasertec, I have to purchase more than 30.”

Q: “So, you're saying that TSMC tried using them for 5-nanometer, but it didn't really work because of the throughput; they only got two machines, and now they're not going to use them for 3-nanometer. Is that right?”

A: “Yeah, you can say that. Yes. **You can use the DUV light source** to inspect, no matter for a DUV mask break or EUV mask break, yes, because you're just trying to identify the same image and the defect.”

Q: “Is TSMC using Lasertec in the new fabs? When you said they're reducing market share, who are they going with instead? KLA?”

A: “Yes. **Now, KLA's share continues to increase. Now, it's already two-thirds, and I know continues to increase to get more from KLA.”** – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

Legacy DUV sales will not save Lasertec from its EUV failure, as TSMC indicated that at most they may purchase 15 Lasertec DUV tools in coming years versus 10 of MATRICS X8 that they have currently. They indicated that the new X9 is still in evaluation, with an ex-Lasertec field engineer at TSMC telling us that the X9 has light source problems – thus jeopardizing even the 15 potential tool purchases.

TSMC plans to buy only 15 more Lasertec DUV tools for 3nm vs. 10 currently; KLA has 2/3 market share in DUV at TSMC

A: **“At TSMC, the demand is around 500 masks per day for production, for the fab, for 5-nanometer. TSMC highly relies on KLA. I will say KLA's share, KLA's tool is around 20 in 5-nanometer. And Lasertec is around 10.”**

Q: “But the X8, X9, that competes with KLA, right?”

A: “KLA, yes. But why TSMC still prefers to—when I asked procurement, they told me TSMC's target is 50 and 50 in the future because they want to keep the competition. **TSMC is thinking around 30 tools in the future.** So, 3-nanometer, they will buy another 30. Two-nanometer, they will buy another 30 tools.”

Q: “From Lasertec or total?”

A: “Total, yes, total.”

Q: “So, they'll buy 15 from KLA, and they'll buy 15 from Lasertec.”

A: **“Yes, 15 from Lasertec.”** – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

TSMC has only ~10 Lasertec legacy X8 DUV inspection tools; the newer X9 is only a demo version under evaluation

Q: “They have six of the A150. And do you know how many of the X8 and X9 Ultra they have?”

A: **“The X8, I think they have around 10 in production.** Ten sets in the production fab. So, it's not only for 5-nanometer, even for the 7-nanometer fab, you have. And now, **TSMC is evaluating their X9. But X9 is not 100% qualified yet** because they changed the light source from 213 to 193. The reason to do that is, in the fab, the throughput is number one because **KLA is using a 193-nanometer light source, and the throughput is around 25 pieces per day. But now, Lasertec's is around 20.”**

Q: “They have how many of the X8 and how many of the X9?”

A: **“X9 so far, I think it's only a demo. It's under evaluation.”**

Q: “X9 is under evaluation. How many X8s do they have?”

A: “I asked them, and they told me the X8 is around 10.” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

Samsung and Global Foundries are the only remaining buyers after Intel and TSMC where Lasertec could salvage a bit of EUV tool growth. However, executives at Samsung, Global Foundries, and a major Japanese tools supplier told us that **only one Lasertec EUV tool is required for every five ASML EUV scanners** installed. ASML reported a bit over 10 EUV scanners in each of the last 4 quarters: 12 (Q2/23), 11 (Q3/23), 13 (Q4/23), 11 (Q1/24) – implying at most two ACTIS EUV tools per quarter. In reality, **the actual number is closer to one ACTIS tool per quarter at ~\$50MM each**, as TSMC is a major portion if not a majority of ASML EUV systems – and TSMC is no longer buying Lasertec’s EUV tool.

Samsung indicates ratio is 1 Lasertec tool for every 5 ASML EUV scanners

“I’ve been told that **roughly one Lasertec tool is required for five ASML EUV’s**. Intel started to buy EUV tools from 2019, and until now, Intel has 22 EUV tools. Intel had 12 tools until 2019, 14 until 2020, 17 until 2021, and 22 EUV tools by last year. There’s roughly one tool for five EUV tools—one Lasertec tool for five EUV tools.” – Longtime Samsung executive in leadership role in EUV; worked closely with Lasertec; decision-maker for inspection tools

Global Foundries indicates ratio of 1 Lasertec tool for every 4 or 5 ASML scanners

“Looking at TAM, the major TAM is on the foundry side, but you also have the mask shop TAM. Because typically, the IDMs are a little different like Intel because of the in-house mask. But if you’re buying from Toppan or others, an EUV mask, they have to have inspection, an outgoing inspection at the commercial mask hub and then the foundry, of course, as you mentioned, would have an incoming mask inspection. **It’s about 1:4 or 1:5. That’s a reasonable ratio**.” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

Ratio of 1 Lasertec tool for every 5 ASML EUV scanners confirmed by other semicap industry experts

Q: “TSMC, they bought a fair number of Lasertec tools.”

A: “Eight or 10, I think.”

Q: “I heard from the Samsung person today that they need about **one Lasertec machine for every five ASML machines**.

A: “**I think that’s pretty accurate.**” –Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec’s CEO

The Samsung executive indicated that they purchased about 10-15 ASML scanners in each of recent years, indicating a ceiling of 2-3 ACTIS EUV machines if that rate holds – which seems unlikely given the flat and now declining quarterly EUV tool shipments recently reported by ASML. Samsung stated they pay about \$40MM for each Lasertec machine, which implies \$80-120MM of revenue at most – insufficient to maintain Lasertec’s growth and valuation as Samsung is one of the three legs of the customer stool they depend on.

Samsung expects to buy about 15-20 ASML EUV scanners per year, implying 4-5 Lasertec ACTIS tools at \$40MM each

Q: “When did Samsung buy the first Lasertec machine? How many EUV machines does Samsung have?”

A: “Samsung actually bought the first EUV tool very early. They ordered some Lasertec tools for the first time in 2019.”

Q: “How many ACTIS A150s did Samsung buy per year in the last few years?”

A: “I think the number of EUV tools they have is 48 as of last year. It shows 25 by 2020, 34 EUV tools by 2021, and 48 by 2022. This year (2023), they are expected to buy 15 more EUV tools. **They’ve got 9 Lasertec tools right now at Samsung—9 ABICS and 9 ACTIS.**”

Q: “9 ABICS and 9 ACTIS for how many EUV machines?”

A: **“And 48 EUV tools until last year. But this year, they are adding 15 more EUV tools, so probably around three more Lasertec tools for each, ABICS and ACTIS.”**

Q: “Do you know how much Samsung was paying for the ACTIS A150? Was it \$40 million or a different price?”

A: “The number I have is the price for ACTIS, so that’s \$40 million.

Q: “That’s what you guys were paying at Samsung?”

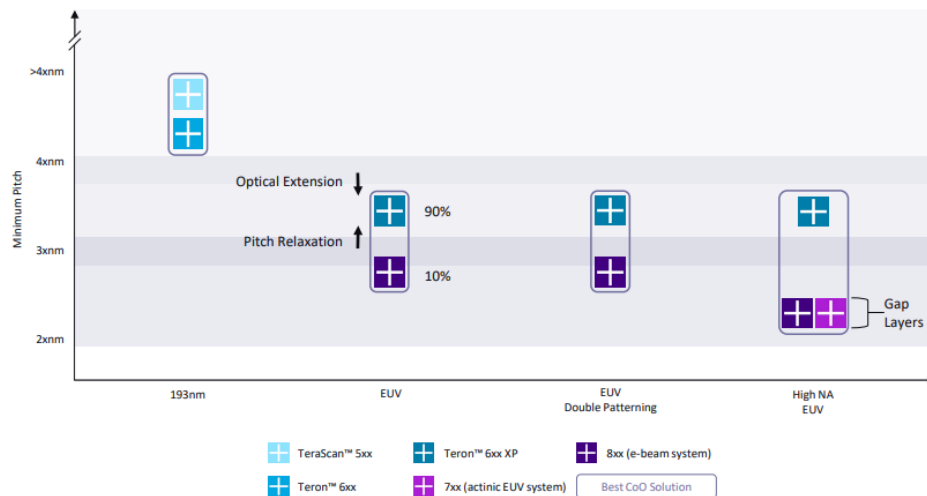
A: “Yes. That’s why I have those numbers. I know Intel—they paid \$40 million for one ACTIS tool.” - Longtime Samsung executive in leadership role in EUV; worked closely with Lasertec; decision-maker for inspection tools

9. KLA's impending launch of an EUV actinic patterned mask inspection tool will be catastrophic for Lasertec. We spoke with 3 KLA sources, as well as senior individuals at Intel, TSMC, Samsung, Global Foundries and others – all indicated that KLA's tool is likely coming in 2025, has a superior technology, and will destroy Lasertec's market share – hardly a surprise, given previous short-lived "monopolies" in semicap equipment that collapsed overnight when leapfrogged by a new entrant.

KLA's impending launch of an EUV actinic patterned mask inspection tool is an existential threat to Lasertec. We spoke with 3 KLA sources, Intel, Samsung, TSMC, Global Foundries and others – all indicated that KLA's tool is likely coming in 2025; has a superior technology; and will destroy Lasertec's market share. KLA has been tight-lipped about its technology and plans, but has publicly mentioned the tool is in development. KLA's last investor day was in 2022, where the EUV tool was hinted at several times under the name Teron "7XX."

KLA Investor Day, June 2022

EUV and Reticle Inspection Landscape



- Semi PC mission: release the best cost of ownership solution by leveraging the portfolio
- Optical reticle inspection is the workhorse of today's EUV in HVM
- Multiple 8xx e-beam systems shipped to address gap layers
- Actinic EUV tool in development, intersecting with High NA EUV

"Actinic EUV tool in development, intersecting with High NA EUV"

Semi PC reticle inspection strategy utilizes multiple technologies to deliver best CoO solution for customers

For anyone who doubts that KLA will enter the market, various statements by its management team underscore their seriousness. At a 2023 investment conference, KLA's CFO stated they have already invested "over \$1 billion" specifically for EUV mask inspection: "If you look at the reticle space, for EUV...at the end of this year, I was adding up the numbers as CFO, you kind of roll your eyes you go holy – wow, over \$1 billion across multiple products to solve the reticle challenge in terms of R&D investment." He noted their dominance and effective monopoly in inspecting EUV masks with their Teron DUV systems: "in the EUV space, 90%-plus of reticles are running through KLA systems" – once again illustrating the naivete of those who mistakenly think Lasertec is the monopoly.

KLA CFO Bren Higgins at Morgan Stanley, March 2023

*"And the economics of running the fab is playing to our optical solutions and our wafer inspection solutions to validate reticles in the fab environment. So clearly, that's playing out **in the EUV space, 90%-plus reticles are running through KLA systems**. We have new capability in the market that as they start to shrink feature sizes, that's an e-beam, electron beam-based capability that should meet those higher end post writing inspection requirements. And we feel pretty good about the strategy and the introduction of new capability, similar to what our competitor is offering and trying to engineer today when you move into a high NA environment [...] and so a lot of investment in the company to meet that timing. **If you look at the reticle space, for EUV, the company. I think at the end of this year, I was adding up the numbers as CFO, you kind of roll your eyes you go holly -- wow, over \$1 billion across multiple products to solve the reticle challenge in terms of R&D investment.**" – KLA CFO Bren Higgins*

KLA's actinic inspection tool was in development circa 2015/2017 but put on hold as they believed the market for such a tool was too small – a prescient view given the insignificant amount of Lasertec revenue that actually comes from EUV actinic tools versus its old DUV and other systems. A 2016 KLA presentation below details the “Teron 7XX” tool and its xenon-based light source – “Comprehensive multi-year investigation addressed and solved all significant technical risks.”

KLA presentation on EUV mask inspection tool program, 2016



Status of Actinic Patterned Mask Inspection at KLA-Tencor

Oleg Khodykin



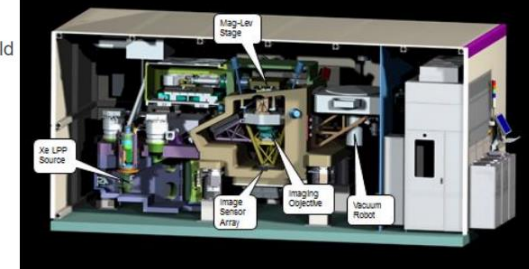
7xx program summary

EUV actinic patterned mask inspection

- System architecture defined
- EUV-specific image sensor designed and tested
- Optics concepts provide large field and high transmission
- Xe LPP source prototype shows required lifetime
- Ultra-clean vacuum prototypes tested
- Pilot production facility ready for build-out

710 System Concept – Cross-section

Rev 2 layout



Program ready for full-scale development

When KLA scaled back its EUV mask inspection program in 2015, the CEO's comments on its earnings call were devastating regarding the lack of a market for EUV mask inspection tools: "if we build that tool and there's not a market for it..."; "we're coupled to our customers on this and talking closely with them about their needs...to date, we have not seen compelling interest on their part to want to make that investment."

Industry article on KLA cutting its EUV mask inspection headcount, 2015

KLA-Tencor cuts back EUV investment

Workforce set to be cut by up to 10% amid lack of demand for KLA's extreme UV lithography mask inspection tool.

KLA CEO says no market for EUV mask inspection tools, earnings call excerpt, Apr 23, 2015

Patrick J. Ho, Stifel analyst

*"Maybe as a follow-up to Wes' question, for you Rick, in terms of **the mask inspection for EUV. How do you look at it from a KLA perspective, in terms of the risk/reward** the investments? Will you only do it if you're going to get some outside funding? Or is that something you can do, if the customer demand is there..."*

Richard P. Wallace, KLA CEO

*"No. This is one where we need to work closely with the customers -- the -- to make sure that: a, the demand is there and the commitment is there. Because right now, the uncertainty around the timing of high-volume EUV makes this a very different kind of question in terms of the investment profile. **Remember, if we build that tool and there's not a market for it, there's no other application for it. It can only be used for one thing, which is to inspect EUV masks.** And if you build an advanced wafer inspection tool, it really only matters that people are making wafers and doesn't really matter the technology they're making it with. So for that reason, we're coupled to our customers on this and talking closely with them about their needs. And we'll develop it in conjunction with them as we go forward. But **to date, we have not seen compelling interest on their part to want to make that commitment.**"*

KLA appeared to believe that actinic EUV mask inspection only made sense in conjunction with High NA EUV lithography – which is only beginning now, with ASML shipping the first High NA EUV scanner just a few weeks ago in April 2024 to Intel. At KLA's investor day, they stated that their actinic tool is “in development, intersecting with High NA EUV,” with an executive commenting that “we could have developed that earlier” but “we are confident that we are ready for the market at the right time...and be ready whenever high NA EUV will be available.” Given Lasertec's inspection tool lacks High NA capability, as detailed in an earlier section, KLA's patient entry into the market with High NA technology is an existential threat.

KLA investor day comments, June 2022

Ahmad Khan, KLA executive

“And then when high NA EUV comes, you can really shrink the pitch. That's why KLA says that for 5-nanometer, 3-nanometer, 2-nanometer, we think **our optical inspection 193 system, along with 8xx multi-beam reticle inspection system is sufficient**. And when high NA EUV comes with severe pitch reduction and mass pelliclization, you can bring in an actinic inspection system. **We could have developed that earlier**, but the complexity of the actinic system is much higher, and that's why we chose this path. And therefore, **we are confident that we are ready for the market at the right time. And we're giving actinic the appropriate time to do the development and be ready whenever high NA EUV will be available.**”

We spoke with 3 KLA sources who indicated that its actinic tool is coming in 2025: an ex-executive in a leadership role in its EUV group, who was directly involved in its development and is in touch with colleagues; a current executive with a key role in source development; and an ex-director of engineer in the EUV group. The first executive stated the timeline was originally end of 2024, but that he believes the beta version will ship in mid-2025 with KLA accepting purchase orders in Q3 2025. He stated it's "almost a billion dollar investment": "it's happening according to all indications I have...of course it's happening...it's not bullshit...they are investing tremendously...they have committed to Wall Street."

KLA is developing a competing tool that will launch in 2025 – "of course it's happening"

Q: "When do you think the KLA tool will launch?"

A: "They were saying last year the end of 2024. That's not going to happen. I believe **they will ship around mid-2025** the beta's. They will start ramping up production and when they see three months down the road installation goes well, they will start releasing or accepting PO's. I believe around the third quarter of 2025, they will accept POs and maybe start shipping in 2026-ish."

Q: "So, your job at KLA was to work on this tool?"

A: "**I was the head of [redacted] for that tool**. I had [redacted] engineers reporting to me, including the [redacted]."

Q: "Do you stay in touch with people at KLA?"

A: "I do."

Q: "But it's real, it's happening? It's just a question of exactly when. Is that fair to say?"

A: "**It's happening according to all the indications I have, it's happening. Of course, it's happening. It's not bullshit. They are investing tremendously. They committed to Wall Street. It's happening.**" –Former KLA executive in a leadership role in their EUV mask inspection group

KLA's may be investing roughly a billion dollars into their competing product

Q: "Do you know how much KLA is investing in their actinic product, roughly?"

A: "They were talking about **almost a billion-dollar investment**. Of course, I cannot disclose how much it eventually—you think about cleaning rooms and all this, and it adds up to roughly a billion-dollar asset." –Former KLA executive in a leadership role in their EUV mask inspection group

We note skepticism among some in the industry given KLA's secrecy and previous delays, which led us to press our research sources to verify that KLA's market entry is real. An ex-KLA executive stated that KLA spent years working on its actinic EUV tool and developed a prototype, and that the delays were not technological but due to a miscalculation as to when ASML's EUV scanners would launch, given that ASML had been delayed for a decade – without ASML EUV scanners, there would be no EUV masks to inspect and no market for such tools. A longtime industry consultant who is close to KLA and recently completed a consulting project for Lasertec stated KLA “made a conscious decision” to walk away from the market as no customers wanted to pay for the product.

KLA's delays simply due to miscalculating when ASML would introduce EUV; KLA already invested years in the technology

Q: “What were the delays at KLA? Was it just that Lasertec was kind of half-assed and rushed it, and KLA is trying to get it right so they don't have all the same problems? Why is the KLA thing taking so long?”

A: “KLA started 10 years ago, there was publication and a lot of presentations out there. They came up with a concept. They came up with some kind of a prototype for the source. They demonstrated to customers that guys, we have the technology. **Basically ASML delayed the EUV tool for 10 years until two and a half years ago.** When Intel came to Lasertec was just about or just before ASML started to talk about releasing, and KLA was already solving that problem for three years - and they just took their time. Basically, between the pellicle and the delay of ASML, **KLA was not carefully thinking when to start it and started too late. It's enormous, and it takes more than a regular program. It takes five to eight years,** not three to five years like a regular program.” –Former KLA executive in a leadership role in their EUV mask inspection group

KLA consciously let Lasertec have the actinic market and can easily re-enter it

“It **wasn't by accident that KLA missed the market. They made a conscious decision** to miss the market. They were working on an actinic inspection tool, and they decided to stop working on it unless they could get the customers to pay for it and probably, more specifically, Intel. They wanted something like \$500 million to continue the work. I think customers were frankly rather pissed about that and instead decided to work with Lasertec. **KLA went away from the actinic market for a few years** while others continued to work on it, including Lasertec.” – Longtime semiconductor consultant recently working for Lasertec; close to KLA management

A second KLA executive, with extensive experience in EUV source development, confirmed their technical ability to launch the tool given their “virtual monopoly for the DUV tool”: “KLA has all the ability to make it happen...that’s for sure...KLA is definitely it.” He replied “for sure” when we asked if he’s confident that KLA will solve any EUV light source challenges. A leading EUV lithography expert close to KLA as well as to Lasertec customers agreed: “KLA may have an interesting strategy...they will be ahead of Lasertec because they already have a better source.”

KLA has ‘all the ability’ to make a competing actinic tool; “that’s for sure”; can leverage their monopoly position in DUV

A: **“KLA has all the ability to make it happen. That’s for sure.”** They have the product, the team, the resources, they have the money. **If anyone can do it, then KLA is definitely it.** It may well be that the scope of the work is too large for Lasertec. All I can say is that the scope of work is huge, even with a source. Maybe it’s more than Lasertec can swallow. But since it’s possible to do and ASML showed that it’s possible, **KLA has all the necessary resources to make it happen,** to invest enough money, form the team, they know the market, they **still have a virtual monopoly for the DUV tool,** the same position as ASML”

Q: “You’re confident that KLA will solve the EUV light source problems?”

A: **“Yeah, that’s for sure.”** – Leading figure in the development of Cymer/ASML’s EUV light source; headed KLA’s EUV source effort; one of the top scientist/experts in the field

KLA “will be ahead of Lasertec” with “a better source”

“KLA may have an interesting strategy.” Let’s say they go with Gigaphoton. I don’t think they do, but they go with somebody else. Then, **in the long term, they will be ahead of Lasertec because they already have a better source.** So, they’re learning. KLA is a smart company...” – Leading EUV lithography expert close to Lasertec’s customers and to KLA

The “better source” the EUV lithography referred to is xenon-based. The ex-KLA executive corroborated that “the KLA xenon works,” using “frozen xenon gas with IR laser as the originator,” unlike Lasertec’s Ushio or Isteq tin-based sources which he stated are a failure. He predicted that the impact on Lasertec would be devastating, with KLA winning each step of EUV mask inspection, in addition to creating brutal price pressure as customers like TSMC used KLA’s entry to crush Lasertec’s pricing: “TSMC will play hard...KLA will win the P2 and P3 [inspection]...that’s for the mask shop.” As KLA’s Teron DUV tools already have a monopoly in P1, KLA would own the mask shop.

KLA Xenon-based source “works” while Ushio source “is terrible” and “creates a mess”

A: **“The KLA Xenon works.** The Ushio tin source failed. ASML is different. They're using tin and they're using IR laser to ignite it. That's ASML. The KLA technology is published, It uses a **frozen Xenon gas with IR laser as the originator. Ushio uses molten tin** like ASML, but instead of using IR, they're using discharge electrodes, which is terrible. And the concept of maintaining two propellers **creates a mess.”**

Q: “It sounds like it takes ten years to figure it out. It sounds very complex.”

A: “Correct...I think **that Isteq source will not be done in a professional way and it cannot work. You need to understand Isteq was done for research. Research is done in bursts, not continuously. For inspection, you need something to work 24/7, and that's it. And it's really hard. It's totally different.”** –Former KLA executive in a leadership role in their EUV mask inspection group

KLA will quickly replace Lasertec in the key inspection use cases in the mask shop at Intel, TSMC, and other customers; will also create price pressure

“There will be **price pressure, and TSMC will play hard,** and they'll play bold as usual. But **KLA will win the P2 and P3. That's for the mask shop.”** –Former KLA executive in a leadership role in their EUV mask inspection group

All three KLA sources we interviewed, as well as every major customer and other experts, stated that KLA's market entry will be catastrophic for Lasertec's EUV business. One stated "they will struggle...extremely hard to compete with KLA...I don't think Lasertec can pull it out in EUV [versus KLA]" – and that "they will disappear in five years" if they don't "re-engineer the tool [with] a simple source." A second indicated that "people would love to buy from KLA...there is no question about that...even if [their tool] were crap, then the customer would love to have a second supplier...they will get the orders, that's for sure."

Lasertec "will struggle" to compete with KLA; "extremely hard to compete"; Lasertec "will disappear in five year" is they don't fix their product problems

A: "There is KLA and KLA has proven in the past that they can pull it out eventually...my projection is **they will struggle**. There will be struggle there...I think it would be **extremely hard to compete with KLA**...I don't think Lasertec can pull it out in EUV [vs. KLA] because the cost structure is different."

Q: "You think they're just going to basically collapse as soon as KLA becomes a real competitor."

A: "I think that they will have to re-engineer the tool, a simple source, with low margin compared to KLA, for sure, significantly low to win or to sustain the fab. They will have to find engineering solutions, and I'm sure they work on it; otherwise, **they will disappear in five years.**" –Former KLA executive in a leadership role in their EUV mask inspection group

KLA customers are keen "to buy from KLA...no question...they will get the orders"; even "if it were crap" as customers want a second alternative to Lasertec

"People would love to buy it from KLA. There is no question about that. If KLA comes out with a product and it will be - even if it was only slightly better, people would do it - **even if it were crap, then the customers would love to have a second supplier.** There is no question about that. If KLA comes out with a product, **they will get the orders, that's for sure.**" – Leading figure in the development of Cymer/ASML's EUV light source; headed KLA's EUV source effort; one of the top scientist/experts in the field

One of the KLA sources noted that during his leadership role in their EUV mask inspection group, key Lasertec customers were coming “crying” to KLA and “begging us to expedite”: “I’m talking about customers coming to KLA... saying, guys, we need you to develop this tool...P2 can only be done by a clean tool...they were really begging us to expedite.” A second KLA source – an ex-director of engineering also in the EUV division – confirmed that “customers would really like KLA on this” – the Lasertec tool “has a lot of issues” and “doesn’t work well...that’s why customers want KLA to make a reliable tool.”

Key Lasertec customers have been coming “crying” to KLA “really begging us to expedite” an alternative

Q: “You mentioned the customers were complaining. We talked about Intel, TSMC. You said you talked to people at an SPIE conference. You said the customers are crying. Explain what the customers are complaining about.”

A: “The customers were saying to us, even these days, not just at SPIE, this is in KLA, and that I can say because they said—”

Q: “Sorry, you said, not just SPIE but you said they were telling you at KLA?”

A: “Yeah, I’m talking about customers coming to KLA and they were saying, guys, we need you to develop this tool, as I mentioned, P2 can only be done by a clean tool. And they were really begging us to expedite.” –Former KLA executive in a leadership role in their EUV mask inspection group

Customers “would really like KLA” to launch a Lasertec competitor

“I think there are a lot of issues with the Lasertec tool. It’s not reliable enough, and yes, I do think customers would really like KLA on this—if we could go back and have a time machine, KLA should have done this a long time ago, but people want a reliable tool with high performance...” – Ex-KLA director of engineering focused on EUV mask inspection

Lasertec tool “doesn’t work well enough for production”; “customers want KLA to make a reliable tool”

“Their tool doesn’t work reliably...the tool doesn’t work well enough for production. It has the capabilities that no one else has. So, I would say on the plus side, the capabilities are great. Production worthy, not great. So, you have to mix those things together...reliability is not so good. It’s actinic. It should do what needs to be done and that’s what customers think is that when it works. It doesn’t work well enough overall, and that’s why customers want KLA to make a reliable tool.” – Ex-KLA director of engineering focused on EUV mask inspection

Non-KLA sources reiterated the same doomsday scenario for Lasertec. A longtime semiconductor equipment executive in Japan – friendly with Lasertec’s CEO – stated that “everything could turn around in a second if KLA comes out with a superior tool...they’re out of business, basically.” The executive replied “correct” when we asked if it’s “game over” when KLA comes out with their product. He expressed skepticism of Lasertec’s stock and the average Japanese investor’s lack of understanding of EUV: “they were picked up by mass media newspapers and TV...that’s why they became popular...average people walking on the street have no idea what EUV is.”

Uninformed investors fueling the stock but “everything could turn around in a second” once KLA enters the market; one-trick pony that will be “out of business” with a new competitor

A: “When they started to enter the EUV market, they were picked up by mass media newspapers and TV, and that’s why they became popular, and that’s why their stock price increased so rapidly. **Average people walking on the street have no idea what EUV is, so they believe anything on TV or what the newspapers tell them. But everything could turn around in a second if KLA comes out with a superior tool.**”

Q: “Lasertec stock is hitting all-time highs but the people buying the stock don’t have any clue about these product problems.”

A: “Correct. **If someone invents a new light source or a new method to inspect EUV masks, they’re out of business, basically.**”

Q: “It sounds like these guys are basically a one-trick pony where people needed an EUV mask inspection tool. They did a bunch of pre-orders. Then they had a bunch of problems and **now it’s kind of like game over, basically.**”

A: **“Correct.”** - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec’s CEO

Intel and Samsung both told us that they believe KLA is soon entering the market. One of Intel's most senior and tenured engineers, who plays a leadership role in EUV and their Lasertec relationship, provided color that ought to make Lasertec investors quake in their boots, given Intel is Lasertec's "founding" and core customer: "it's a problem to be dependent on one supplier...we always want multiple choices...the timeline you heard from KLA is reasonable...I think KLA is serious...a year or eighteen months is reasonable." A longtime ex-Samsung executive, also in a leadership role in their EUV effort until recently and who works closely with Lasertec as a decision-maker and still speaks with colleagues, echoed Intel's view: Samsung "is always trying to diversify vendors...they don't want to just rely on one company because it's not good for negotiation power...they keep pushing KLA to produce the same tool as Lasertec."

Intel believes KLA is "serious" and 12-18 months from a tool; "problem" to be dependent on Lasertec

"It's a problem to be dependent on one supplier." As a customer we always want multiple choices. Hopefully another company will provide a solution. Competition is better...The timeline you heard from KLA is reasonable. I think KLA is serious. If they can develop the tool it will be good for the industry. A year or 18 months is reasonable." –One of Intel's most senior and longtime engineers with leadership responsibility for EUV

Samsung "pushing KLA to produce the same tool as Lasertec"

Q: "You said that Samsung has been pushing KLA to finish their actinic product?"

A: "Yes, of course, because Samsung is always trying to diversify their vendors, so they don't have to pay, they don't want to just rely on one company because it's not good for negotiation power. If KLA is promising the same actinic inspection tool, then it would be very good for Samsung, so they keep pushing KLA to produce the same tool as Lasertec." - Longtime Samsung executive in leadership role in EUV; worked closely with Lasertec; decision-maker for inspection tools

The Samsung executive stated that KLA's tool is coming soon: "KLA kept talking like it's very soon, just in a year...I think it's pretty soon...in a year or so...KLA is working very hard to develop those...." He predicted that KLA would get half of Samsung's business, and provided a particularly interesting piece of information – indicating that Samsung has already pre-ordered the tool from KLA: "they actually made pre-orders...for evaluation and testing...Samsung basically has confidence and trust in KLA...KLA is aggressively developing their own actinic tools...they're going to find a way...one way or the other, sooner or later."

Samsung expects KLA to launch a competing tool "very soon"; will give them at least half the business; have already placed a pre-order; has "confidence and trust in KLA" who is "aggressively developing their own actinic tools"

Q: "Do you know when KLA will finish it and launch it?"

A: "Until last year, when I was working at Samsung, **KLA kept talking like it's very soon, just in a year**. But it looks like they are not really complete yet on those actinic tools. I think it's pretty soon because people keep saying that they're going to produce those actinic tools very soon, in a year or so **KLA is working very hard to develop those kinds of tools**. It will come out very soon, maybe by next year or so."

Q: "When KLA launches a tool, is Samsung going to give KLA half the business, ¾ of their business? What will they do?"

A: **"Half the business"**. It depends on the performance."

Q: "Has Samsung already given them some kind of pre-order or investment, some advance or not?"

A: "Pre-order is not—actually, a demo tool. **They actually made pre-orders**. One demo tool but it hasn't come out yet."

Q: "So, Samsung has already paid KLA for a demo actinic tool?"

A: "Just one tool. That's for evaluation and testing. **Samsung basically has confidence and trust in KLA. So, KLA is aggressively developing their own actinic tools. They're going to find a way to do that one way or the other, sooner or later.**" - Longtime Samsung executive in leadership role in EUV; worked closely with Lasertec; decision-maker for inspection tools

Global Foundries view confirmed the color from Intel and Samsung, stating that “there’s a huge risk here to Lasertec...if machines are piling up and KLA is going around reassuring people that if you can hold off until mid-2025.” He predicted “they will take share from Lasertec...they obviously will” and that Lasertec would have to take an impairment charge on its mountain of inventory: “Lasertec could be caught with inventory that they’re going to have to significantly mark down...they would end up having to impair...could put a lot of pressure on them, which would probably require an asset write-down.”

Global Foundries sees KLA as “huge risk to Lasertec” that “obviously will” take share; Lasertec would have “significantly mark down” and “impair” its inventory; “would probably require an asset write-down”

Q: “How many more Lasertec machines are people going to buy in 2024? Because if finished goods inventory is already piling up, 3-nanometer delays, TSMC--?”

A: “I think **there's a huge risk here to Lasertec. If machines are piling up and KLA is going around reassuring people that if you can hold off until mid-2025**, we will have a system commercially available to you. And in fact, we'll give early access in the second quarter of 2025, you can come and evaluate...High-NA is going to accelerate. People are making these massive capital investments in inspection equipment. I do think, by the way, even if KLA came out with, let's say, an A150 competitor, yes, **they will take share from Lasertec. They obviously will. And Lasertec could be caught with inventory that they're going to have to significantly mark down** even to move on the A150 side. But if the A300 remains only on paper...it could be pure PowerPoint capability that the A300 never really materializes. And then they would be in a lot of trouble at Lasertec...”

Q: “This is a scenario where their sales completely collapse.”

A: “Yes, for EUV, correct. Correct. And they **would end up having to impair** I don't know price point-wise. **KLA, if they wanted to...could put a lot of pressure on them, which would probably require an asset write-down on the part of Lasertec.**” – Longtime executive at Global Foundries; directly involved with Lasertec and all aspects of mask inspection

TSMC appears to have stopped purchasing Lasertec actinic EUV machines several years ago, given their lack of reliability and throughput, as detailed in prior sections. However, KLA's actinic product is still likely to smash TSMC's purchases of Lasertec's legacy DUV machines, of which it still purchases a few for inspection in the fab, despite their lower resolution vs. EUV. TSMC would likely divert those purchases to KLA's actinic product versus trying to improvise with Lasertec's DUV system. An ex-TSMC procurement executive bluntly stated they already have a much higher opinion of KLA than Lasertec: "for productivity and for economic considerations, KLA performs better than Lasertec...two-thirds of the business migrated from originally Lasertec to KLA...Lasertec has continued to decrease...their share in TSMC...KLA's share continues to increase."

TSMC indicates KLA DUV tools are superior; "KLA's share continues to increase"

Q: "If Lasertec is still mainly selling DUV, that means they basically failed in EUV."

A: "The big thing is the light. **For production, for productivity, and for economic considerations, KLA performs better than Lasertec in TSMC. So, that's why I say already two-thirds of the business migrated from originally Lasertec to KLA already. I have to say Lasertec has continued to decrease their amount of share in TSMC.**"

Q: "In fab 18, what happened?"

A: "**The economics, the throughput.** If I purchase a tool from KLA, probably for 3-nanometer, TSMC probably needs to purchase 20 mask inspection tools. But if I change to Lasertec, I have to purchase more than 30."

Q: "So, you're saying that TSMC tried using them for 5-nanometer, but it didn't really work because of the throughput; they only got two machines, and now they're not going to use them for 3-nanometer. Is that right?"

A: "Yeah, you can say that. Yes. **You can use the DUV light source** to inspect, no matter for a DUV mask break or EUV mask break, yes, because you're just trying to identify the same image and the defect."

Q: "Is TSMC using Lasertec in the new fabs? When you said they're reducing market share, who are they going with instead? KLA?"

A: "Yes. **Now, KLA's share continues to increase. Now, it's already two-thirds, and I know it continues to increase to get more from KLA.**" – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

The TSMC executive continued that, unlike Lasertec, “KLA improved a lot...they have a very good scan, and now their resolution is getting much, much better.” A second TSMC source, who was previously head of R&D and played a key role in their 5nm transition and vendor/tool selection, stated that the current transition to High NA EUV mas inspection spells more bad news for Lasertec, as Intel is under a lot of pressure to use KLA instead. We note that Intel has made the biggest bet of any chipmaker on ASML’s new High NA scanners, recently taking delivery of the first High NA shipment: “Are they going to use Lasertec? I have some doubts because they were under a lot of pressure to use KLA...if they go to KLA...there will be more bad news for Japan.”

Lasertec executive recently told TSMC contact that things are “not well”; TSMC views it as a niche “lab tool, R&D tool”

A: “Yes, **a lab tool. Yes, not a production tool**. Because you have a lot of limitations for productivity...even today, TSMC still has a weekly meeting with Lasertec. **My colleagues in procurement and from the fab told me they have a weekly meeting to push Lasertec to improve.**”

Q: “You said they have a weekly meeting with Lasertec to push them to fix their problems?”

A: “**Push them to improve their performance, to improve their productivity**...and the reason is if you increase the throughput, you cannot inspect the mask; it’s so detailed. But **KLA improved a lot. They have a very quick scan, and now their resolution is getting much, much better.**” – Longtime ex-TSMC procurement executive who introduced Lasertec inspection tools into TSMC

High NA spells more bad news for Lasertec; Intel is “under a lot of pressure to use KLA” for high NA mask inspection

“Intel bought a High-NA version for EUV. They were the first ones. They spent \$500 million or whatever for a High-NA tool. Intel took a very high risk and then they said we are going to launch 3-nanometer, 2-nanometer EUV. So, the question is, what kind of inspection in mask engineering supplier are they going to use? **Are they going to use Lasertec? I have some doubts because they were under a lot of pressure to use KLA.** This High-NA version of the EUV is a very, very new thing...**we’ll have to see which inspection tool they’re going to get, Lasertec or KLA? If they go to KLA, then probably there will be more bad news for Japan.**” – Former head of TSMC R&D; played key role in 5nm transition and vendor/tool selection

TSMC's ex-head of R&D stated that President Biden's CHIPS Act is fast becoming a major threat to Lasertec, stating that it will force recipients of US government subsidies to purchase tools from US suppliers – the second that KLA launches its actinic EUV tool and creates a competing option, Lasertec as a Japanese supplier is in major trouble. The CHIPS Act became law in 2022 and authorizes ~\$300 billion in funding and subsidies for chip manufacturing on US soil. **Intel received \$8.5 billion in March 2024, the largest grant to date – illustrating the pressure it will be under to buy tools from US players like KLA; TSMC received \$7 billion the same month, to fund a fab in Arizona; and Samsung received \$6.4B. We note that these three chipmakers comprise almost all of Lasertec's revenue.**

CHIPS Act, recent press all from March/April 2024

The Chips Act has been surprisingly successful so far

Intel Receives \$8.5 Billion in Grants to Build Chip Plants

TSMC Will Receive \$6.6 Billion to Bolster U.S. Chip Manufacturing

U.S. gives Samsung \$6.4 billion to build chip factories in Texas

Source: Scorpion Capital consultation calls with experts; <https://www.nytimes.com/2024/03/20/us/politics/chips-act-grant-intel.html>;
[https://www.nytimes.com/2024/04/08/us/politics/tsmc-taiwan-chips-grants.html#:~:text=The%20CHIPS%20Act%2C%20which%20lawmakers,plants%20across%20the%20United%20States.](https://www.nytimes.com/2024/04/08/us/politics/tsmc-taiwan-chips-grants.html#:~:text=The%20CHIPS%20Act%2C%20which%20lawmakers,plants%20across%20the%20United%20States.;);
<https://www.washingtonpost.com/technology/2024/04/15/samsung-chips-act-taylor-texas/>; <https://www.ft.com/content/26756186-99e5-448f-a451-f5e307b13723>

TSMC's ex-head of R&D described the US CHIPS Act as a seismic event – “this is actually super, super big time for suppliers”; “Japanese suppliers are in a bit of trouble...KLA definitely has a super advantage...the Lasertec guys are going to be put into second choice...KLA is very, very strong...in terms of EUV...under the CHIPS Act...KLA becomes the favorite.” He continued that “Intel, Samsung, TSMC are under pressure from the US government to use US suppliers...Lasertec, I bet they will suffer....they may lose some of these things in the US because of the CHIPS Act.”

KLA will “become the favorite” and have a “super advantage” over Lasertec due to the CHIPS Act; Lasertec will be in “trouble because the CHIPS Act is for US suppliers”

Q: “What do you think will happen to Lasertec once KLA launches their competing product? It sounds like they are going to be in trouble at that point.”

A: **“That's right. That's right.** That's a very good question. I guess you are very familiar with the CHIPS program under the current Biden administration. This is actually super, super, big-time for suppliers, for foundries, and also universities like us. The CHIPS Act really focuses on U.S. manufacturing 5-nanometer and then so on to 3. So, EUV is one...KLA always brings up their option...the **Japanese suppliers are in a bit of trouble because the CHIPS Act is for U.S. suppliers.** So, Applied and KLA go first. The U.S. CHIPS Act right now, it's still at the very beginning because this is the first year it's going to launch. **I think KLA definitely has a super advantage...the Lasertec guys are going to be put into the second choice.** That's a problem. ASML is fine because nobody now can replace ASML. But **KLA is very, very strong. KLA apply in terms of EUV, metrology inspection, and packaging. But under the CHIPS Act, I think KLA becomes the favorite.** So, that is something you need to think about because it is starting to become very, very political.” – Former head of TSMC R&D; played key role in 5nm transition and vendor/tool selection

Lasertec's largest customers “are under intense pressure” to use US suppliers; Lasertec will lose business as a result “Intel, Samsung, and TSMC are under pressure from the U.S. government to use U.S. suppliers. That's for sure. So, whether Lasertec can really still be the choice - it's very political - unless KLA really screws up. If KLA screws up - which I don't think so - **Lasertec, I bet they will suffer a little bit...they may lose some of these things in the U.S. because of the CHIPS Act...** – Former head of TSMC R&D; played key role in 5nm transition and vendor/tool selection

10. Lasertec's actinic EUV tool is so defective and unreliable that TSMC, Intel, and Samsung have all quietly substituted non-EUV alternatives for inspecting EUV masks, improvising with KLA's legacy Teron DUV tool plus electron beam inspection - indicating that actinic EUV is useful if and when it works but is not essential and certainly not a "monopoly" that justifies Lasertec's absurd valuation.

Lasertec's actinic EUV tool is so defective and unreliable that TSMC, Intel, and Samsung have all substituted non-EUV alternatives for inspecting EUV masks, improvising with KLA's legacy Teron DUV tool plus electron beam inspection - indicating that actinic EUV is useful but not essential, nor worth \$50MM per tool. Industry sources further told us that all three of Lasertec's key customers have quietly built their own mask inspection systems as a workaround, using a very low power EUV source called High Harmonic Generation (HHG). One source close to these customers stated his surprise at learning that they were actually using Lasertec's old DUV tool while claiming they were using their EUV machine – we suspect Intel, whose EUV mask team we think is engaging in a face-saving cover-up, given their massive bet on Lasertec. A Japanese semiconductor executive close to both TSMC and Lasertec stated that TSMC is running parallel, redundant inspections with non-EUV tools: "...parallel inspections of the same masks...just to make sure there are no defects."

TSMC has to run parallel/redundant inspections as Lasertec's tool is unreliable

Q: "What else have you heard about **TSMC's feedback**, whether through your connection or somebody else? I guess you said they're sort of struggling and making the transition. What are the reasons for that being? The optics or other problems?"

A: **"It's not so easy to change the process.** If they have an established process and an established yield number of, let's say, 19% or something, phasing in the new technology should improve the yield. So, everything needs to be fine-tuned to surpass the existing solution. And that's **not so easy with a new technology like actinic inspection. It's going to take time, so they will have to run parallel inspections of the same masks**, every day, just to make sure that they get the same results..."

Q: "So, whenever they do use a Lasertec machine, they're having to kind of duplicate the inspection with other machines?"

A: **"Yes. Just to make sure there are no defects that go through without detection.** If that happens, the same defect will print on every single semiconductor. And if it's a fatal defect, it can cost them billions. They cannot ship any wafers anymore." - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

The Japanese semiconductor executive stated that TSMC has simply reverted to the EUV mask inspection process it used before Lasertec launched its actinic EUV machine in 2019/2020: 1) using KLA's Teron DUV tool with a 193nm light source; 2) recording the coordinates of potential defects it identified on an EUV mask; 3) transferring the coordinates to an electron beam inspection tool, also known as SEM for scanning electron microscope. DUV can't distinguish very small defects on EUV masks, so e-beam is used to then analyze "these suspicious-looking defect locations...so they need a combination of optical DUV and electron beam to inspect the mask."

TSMC relying on legacy DUV and scanning electron microscope tools

Q: "You're saying the fact that they're using optical means that they're not getting the results from the Lasertec machine by itself. They have to use a second tool?"

A: "Correct. But the second tool—before Lasertec came out with their actinic tool, they needed to establish some way of inspecting the EUV masks. **What they did was use an optical tool from KLA – the Teron product line. And this is an optical tool with a 193-nanometer light source.** So, there's a limitation to what defects it can detect. Very small defects, it cannot clearly distinguish. For something that looks like a defect, **they record the coordinates, and then these coordinates are transferred to an electron beam inspection tool,** which is only looking at these suspicious-looking defect locations. Is the defect on print or not? So, they need a **combination of optical DUV and electron beam inspection to inspect the mask."** - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

The semiconductor executive stated that Holon – a Japanese company – makes the e-beam inspection tool used by TSMC. He stated that he is friendly with Holon's CEO and that the tool is rapidly gaining adoption among all EUV players including Intel and Samsung, indicating that each of the three customers that comprise most of Lasertec's revenue have soured on its actinic EUV tool: "TSMC was the first one to adopt the Holon tool... Samsung, Intel, and most of the other customers in the EUV space are also adopting the Holon tool." The volume of Holon tools being ordered by Lasertec's EUV customers illustrate the failure of its flagship product: "I'm not quite sure... they have succeeded in phasing in Lasertec tools...they are still ordering considerable quantities of Holon inspection machines."

TSMC, Intel, Samsung are all using the Holon electron beam tool

Q: "So, they're using the DUV device, and then who makes the e-beam device?"

A: "Mostly a company called Holon. **TSMC was the first one to adopt the Holon tool, and currently, I think Samsung, Intel, and most of the other customers in the EUV space are also adopting the Holon tool.** The [redacted] of Holon is actually a good friend of mine. So, if I have some question, I can ask him."

Q: "Is TSMC still actively using Lasertec, or are they idling those machines because of all these problems with downtime?"

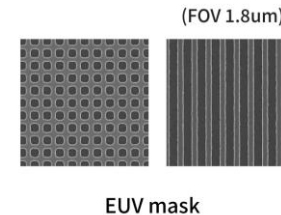
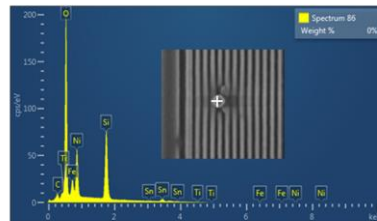
A: "**I'm not quite sure to what degree they have succeeded in phasing in Lasertec tools** to the production cycle, but from what I hear from this company, **Holon said that hasn't been very successful so far. So, they are still ordering considerable quantities of Holon inspection machines.**" -Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

Holon's technical literature actively promotes its scanning electron microscope tool for EUV mask inspection – “becoming indispensable for EUV lithography.” The semiconductor executive stated that Holon's tool only costs \$5 to \$7 million, versus \$40-60MM for Lasertec's actinic EUV tool. While KLA's DUV Teron tool is also expensive, its mask inspection throughput is vastly higher thus requiring a much smaller number of machines – TSMC complained to us that Lasertec's system could only inspect a pitiful 5 EUV masks per day, out of the 550 masks their fab needed inspected daily. Thus, the combination of KLA's DUV tool plus Holon's \$5-7MM SEM tool is far less expensive. TSMC would theoretically require 110 of the Lasertec tools at \$50MM each to cover their daily mask throughput of 550 – implying an absurd \$5.5 billion of spend.

Holon SEM tool is only \$5 to \$7 million

*“The KLA leading-edge optical inspection tool costs maybe \$40 or \$50 million. The SEM tool costs maybe \$5 or \$7 million...so the **financial setup of optical and the CD-SEM is less money.**” - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO*

Holon website - SEM tools for EUV inspection



An ex-KLA director of engineering in its EUV mask group – who worked on each of their DUV, actinic EUV, and e-beam projects - stated that non-EUV methods like KLA's Teron DUV tools are perfectly sufficient as “alternatives for the inspection.” While actinic mask inspection in theory “is the best way to do that” – using EUV so that the wavelength used in mask inspection is the same as that used to write the mask – it's only true “if the system is well-engineered,” which the Lasertec system is not. He pointed to “an award announced at last year's EUV mask event about the work that KLA and others have done about supporting EUV mask inspection with other methods” – “a well-engineered system is more important than the wavelength”; “the DUV tools actually do work [for EUV masks]...those tools work really well and have been well-engineered for a long period of time.”

Alternative methods for EUV mask inspection; KLA recently won an award for non-actinic method

*“I worked on the actinic project. I've worked on the DUV project, and I worked on the electron beam project. I worked on all three of those. I know them all very well...I would say that Lasertec definitely launched the actinic mask inspection capability sooner than KLA did; however, there are alternatives. In fact, there was an **award announced at last year's EUV mask event about the work that KLA and others have done about supporting EUV mask inspection with other methods. So, there are alternatives for the inspection.** Having said that, I do think that the actinic mask inspection is the best way to do that if the system is well-engineered. I do think there are current alternatives. KLA has an electron beam system, and they have a D2B system, and KLA is working on an EUV system.” – Ex-KLA director of engineering focused on EUV mask inspection*

KLA non-actinic DUV alternatives are “work really well” and are “well-engineered”; EUV wavelength is less important

*“The **current KLA systems are pretty good**, but they're not as good as actinic. Having said that, **a well-engineered system is more important than the wavelength**...they're using the current DUV tools, and it was announced at the most recent conference that people who are appreciating all the work that both KLA did on supporting EUV with the DUV tools. **EUV and DUV are different but for supporting EUV inspection, the DUV tools actually do work...Those tools work really well and have been well-engineered for a long period of time.** That's what people use. I think today, I think people are using the Lasertec actinic pattern mask inspection tool - they use that when it works...” – Ex-KLA director of engineering focused on EUV mask inspection*

The ex-KLA engineering director stated that if the Lasertec tool is “not working...people can get by” using DUV, exposing that Lasertec’s so-called “monopoly” in actinic EUV tools is irrelevant as DUV systems are sufficient – “KLA tool for DUV is the market leader...the mainstream tool...even for some of the EUV.” We note that KLA’s website actively showcases its Teron DUV systems for EUV mask inspection.

Customers “can get by” using the KLA DUV tool given reliability issues with Lasertec

A: “If the Lasertec tool is working, then great. If it’s not working, I think **people can get by, but it’s slow...**”

Q: “Are they using a DUV tool from KLA to get by? Is that the Teron?”

A: **“The KLA tool for DUV is the market leader,** especially for the original mask. You want to make sure that the mask has the design you want. There’s die-to-database. Lasertec has developed that, and I’m not sure how good it is. But, in general, people are using existing tools, in which case KLA is the market leader.**[The Teron] is the mainstream tool for everybody, even for some of the EUV.**” – Ex-KLA director of engineering focused on EUV mask inspection

KLA website - DUV tools for EUV mask inspection

Teron™ SL670e and Teron™ SL670e XP: EUV Reticle Quality Control

APR 12, 2022 3 MIN READ



A leading EUV lithography expert who interacts closely with all key players - Lasertec, Intel, KLA, and others - told us that all three of Lasertec's key customers have quietly built their own EUV mask inspection systems as a workaround given its defective tool, using a low power EUV source called High Harmonic Generation (HHG) - they "made essentially their own...in-house inspection system...so did Intel...Intel doesn't talk about it, but they did"; "they all ended up essentially using the same kind of technology...high harmonic generation...the power is very, very small...everybody was doing this, some quietly and some more publicly...Samsung still uses it...so does TSMC...most likely Intel does it too...but Intel is very quiet about what they do."

Intel, TSMC, and Samsung are quietly using their own in-house inspection system using an alternative HHG source given problems with Lasertec

A: "TSMC, but they did some things. **They generally made essentially their own inspection system and in-house inspection system, and so did Intel. Intel doesn't talk about it, but they did.**"

Q: "For EUV masks, they did their own system?"

A: "For EUV mask inspection so they can see what the defects are, and they all ended up essentially using the same kind of technology. They used a **high harmonic generation source** because that's the only other source that you can have that can give you close to 13.5 nanometers, but those are—the power is very, very small. Those are nanowatts. So, **everybody was doing this, some quietly and some more publicly. Samsung, we knew, but Intel and TSMC all did the same.**"

Q: "High harmonic generation—is that the term you used?"

A: "HHG. **They used an HHG source. Samsung still uses it, and so does TSMC, and most likely Intel does it too. But Intel is very quiet about what they do.** Intel has a mask shop in California. Frank Abboud runs it." – Leading EUV lithography expert close to Lasertec's customers

Tin-based plasma and HHG are the only two methods for producing EUV light to power actinic mask inspection. Only tin can produce EUV light at sufficient power and brightness for applications such as ASML's scanner; both the Ushio and Isteq sources Lasertec has tried are also tin-based, but defective. HHG is not a replacement for a proper actinic EUV source, but it illustrates that TSMC, Intel, and Samsung are able to improvise and get by without Lasertec's defective EUV inspection tool through a combination of HHG, KLA's DUV systems, and electron beam systems like Holon's. The EUV lithography expert stated that "they're doing a combination of these things" and expressed surprise that "some of the inspections that I was told were done with EUV machines from Lasertec were actually done with [their] 193nm machines" – that is, Lasertec's old DUV systems.

Intel and TSMC are using three different workarounds due to problems with Lasertec's EUV machine; customers are misrepresenting non-EUV inspections and claiming they are EUV

"I don't know how many HHG machines they are using because they're doing a combination of three things: they have the KLA machines, they have Lasertec's non-EUV machines. In fact, some of the inspections that I was told were done with EUV machines from Lasertec were actually done with 193nm machines from Lasertec. So, they have KLA machines, they have Lasertec machines, both with 193. And they have HHG machine, and then they have Lasertec EUV machine ." – Leading EUV lithography expert close to Lasertec's customers

TSMC, Intel, and Samsung's use of DUV, electron beam, and HHG systems indicates that Lasertec's market window for actinic EUV mask inspection is rapidly closing, as the next phase in EUV is High NA lithography, with Intel taking delivery of the first ASML High NA scanner last month. As detailed in an earlier section, Lasertec's technology cannot support High NA – and its claims around the purported High NA capabilities of the “new” ACTIS A300 lack credibility. High NA mask inspection requires an electron beam solution similar to Holon's. The Japanese semiconductor executive friendly with Lasertec's CEO explained High NA's requirement for optical proximity correct (OPC): “actinic is not going to be sufficient for mask inspection because sub-resolution assist features will become necessary...those features you cannot detect with actinic light...ultimately an electron beam solution is required...they need features on a mask that are smaller than 13.5nm...higher resolution than EUV is required...and people are working on an electron beam solution...KLA is one of them.”

Actinic/EUV mask inspection won't work for high NA lithography

A: “Just one more aspect. Ultimately, **actinic is not going to be sufficient for mask inspection because sub-resolution assist features will become necessary down the road. And those features you cannot detect with actinic light. So, ultimately, an electron beam solution is required.**”

Q: “Walk me through that. Are you talking about the High-NA EUV or something else?”

A: “Yeah, pretty much High-NA, what they call the inverted lithography—OPC technology, light correction on the mask. So, they need features on the mask that are smaller than 13.5 nanometers. So, these features **cannot be quantified using EUV anymore. A tool with a higher resolution than EUV is required. And people are working on an electron beam solution. And KLA is one of them.**” - Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

We note his comments that KLA is working on an electron beam solution, consistent with KLA's comments that it is timing its actinic EUV product to "intersect" with High NA. KLA appears ready to leapfrog and make Lasertec obsolete as the transition to High NA gains momentum in 2024 – Lasertec cannot support High NA, while KLA has patiently prepared for it as the next phase of EUV lithography. We find this particularly ominous as Intel is the first and most aggressive adopter of ASML's High NA technology – and Intel is Lasertec's "founding customer" and was its only hope for future actinic EUV purchases. The Japanese semiconductor executive stated that Lasertec's window is "maybe another two years": "KLA is working on it...Nuflare is working on a tool and ASML has a division where they're working on a multi-beam inspection tool...which means it has much better throughput compared to the Holon inspection tool."

Lasertec has a inspection two-year window before multi-beam mask inspection takes over; ASML, KLA, NuFlare and others will enter the market

Q: "How long do you think Lasertec's window is?"

A: **"Maybe another two years."**

Q: "And what happens then? E-beam or KLA, or what do you think will happen?"

A: "E-beam inspection will come out."

Q: "And are there any big companies working on it besides KLA for E-beam?"

A: "Yes, **KLA is working on it. NuFlare is working on a tool, and ASML has a division where they're working on a multi-beam inspection tool.**"

Q: "For EUV masks?"

A: "If they can establish the process, they can modify it for wafer inspection. And this is a **multi-beam solution, which means it has much better throughput compared to the Holon inspection tool.**"- Longtime, senior semicap equipment executive in Japan who is friendly with Lasertec's CEO

11. Lasertec's new "Innovation Park" in Yokohama is a fraud – a purported R&D and production campus with two supposed "fabs" that is 5X the size of its headquarters and its next driver of growth, announced in 2022 as the largest capital expenditure and expansion in its history. Multiple investigators visited the campus ~20 times from Dec 2023 and May 2024 and found a generally deserted and empty facility with no evidence of R&D, production, nor any indication that Fab 4A matches its promotional photos, leading us to wonder if the pics are fake.

In Feb 2022, Lasertec announced that it was paying ¥17 billion – \$150MM then and \$100MM at the current rate - for a new R&D facility called “Lasertec Innovation Park” or “InnoPa” for short. The complex is a 10-15 minute walk from Lasertec’s corporate headquarters in Yokohama, an hour south of Tokyo. An Aug 7, 2023, presentation shows a massive new cleanroom and fab called “Fab 4A”; claims the park is 5 times the size of headquarters; and that “operation started in July 2023.”

Lasertec presentation, Aug 7 2023

Our New R&D Facility: Lasertec Innovation Park (InnoPa)

We finished the construction of a cleanroom at the new site and started operation in July 2023.

Overview

Name	Lasertec Innovation Park (InnoPa)
Address	555 Toriyama-cho, Kohoku-ku, Yokohama, Kanagawa Pref. (10 min. walk from HQ) (16 min. walk from Shin-Yokohama station)
Site Area	15,830 m ² (4.8 times the area of HQ)
Status	- First round of construction completed in June 2023, and operation started in July 2023. - Second round of construction to begin in October 2023.



Fab 4A

Cleanroom in Fab 4A

11

We believe that Lasertec’s “Innovation Park” is a fraud and that key claims about any R&D or production activity, as well as its construction status, are false – as made by the company in press releases, presentations, earnings conferences, and during an in-person IR meeting at their headquarters, by a local consultant on our behalf. We hired multiple local investigators who visited the facility approximately 20 times between Dec 2023 and May 2024. Over many months, we found no evidence of R&D or production activity at the site; nor any evidence that Fab 4A looks like the photograph in Lasertec’s presentation – we wonder if it is a stock photo of a clean room. The investigators conducted extensive surveillance including: field visits and interviews; aerial drone surveillance over several visits, in particular of the purported “Fab 4A”; several weeks of 24/7 footage from motion detector cameras; and other photographic and in-person investigation.

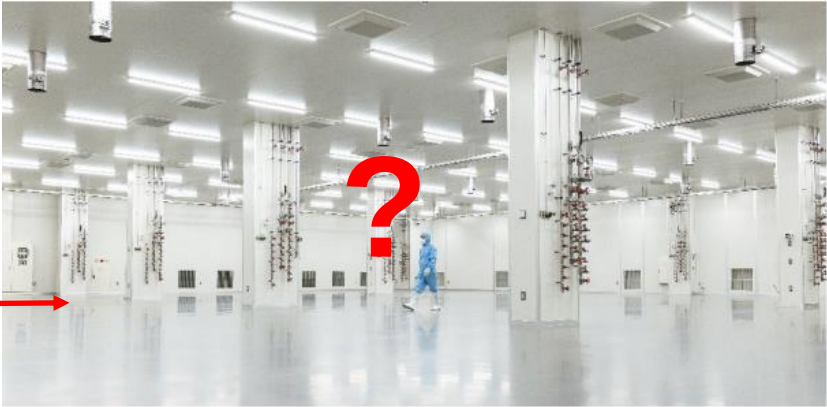
Lasertec presentation, Aug 7 2023 shows the purported “Fab 4A” cleanroom

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Across ~20 field visits, our investigators found that the 5-building complex was generally deserted and empty, except for one floor of a small 5-story office building that appeared to house low-level clerical or back-office staff, not R&D. Our investigators never saw any security or guards during any visit – not what one would expect from a facility with cutting-edge R&D, sensitive industrial secrets, intellectual property, and the production or components of \$50MM pieces of capital equipment. Our investigators saw no signs or gates forbidding entry onto the complex; wandered freely; were never asked who they were or asked to leave; and in fact were cheerfully welcomed by the occasional maintenance personnel, who were happy to answer questions about the status of the site.

Lasertec Innovation Park

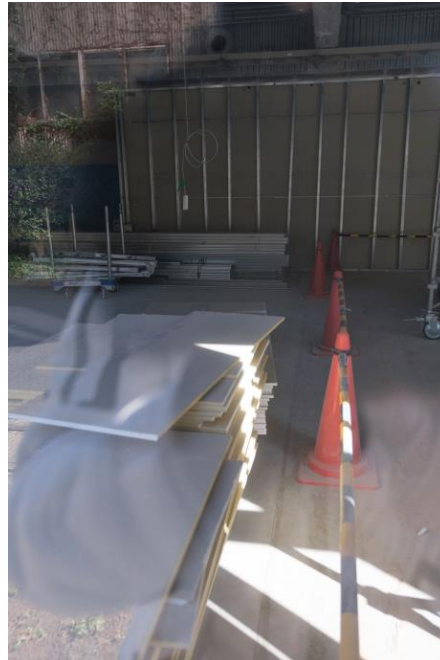


We were unable to be invited inside “Fab 4A,” and we conclude – as we show shortly via extensive photographs – that, at best, it is an overflow storage facility for the mountain of finished goods inventory overflowing at headquarters; but that no R&D, production, assembly, or testing takes place at the facility, even if the photos in Lasertec’s presentations accurately depict the interior. In addition to Fab 4A – which the company refers to as the “first round of construction completed in June 2023,” we saw no evidence of meaningful construction for Fab 4B, which it calls “the second round of construction” that supposedly began in Oct 2023 – just an office space with a ladder and few pieces of wood on the floor.

Fab 4A

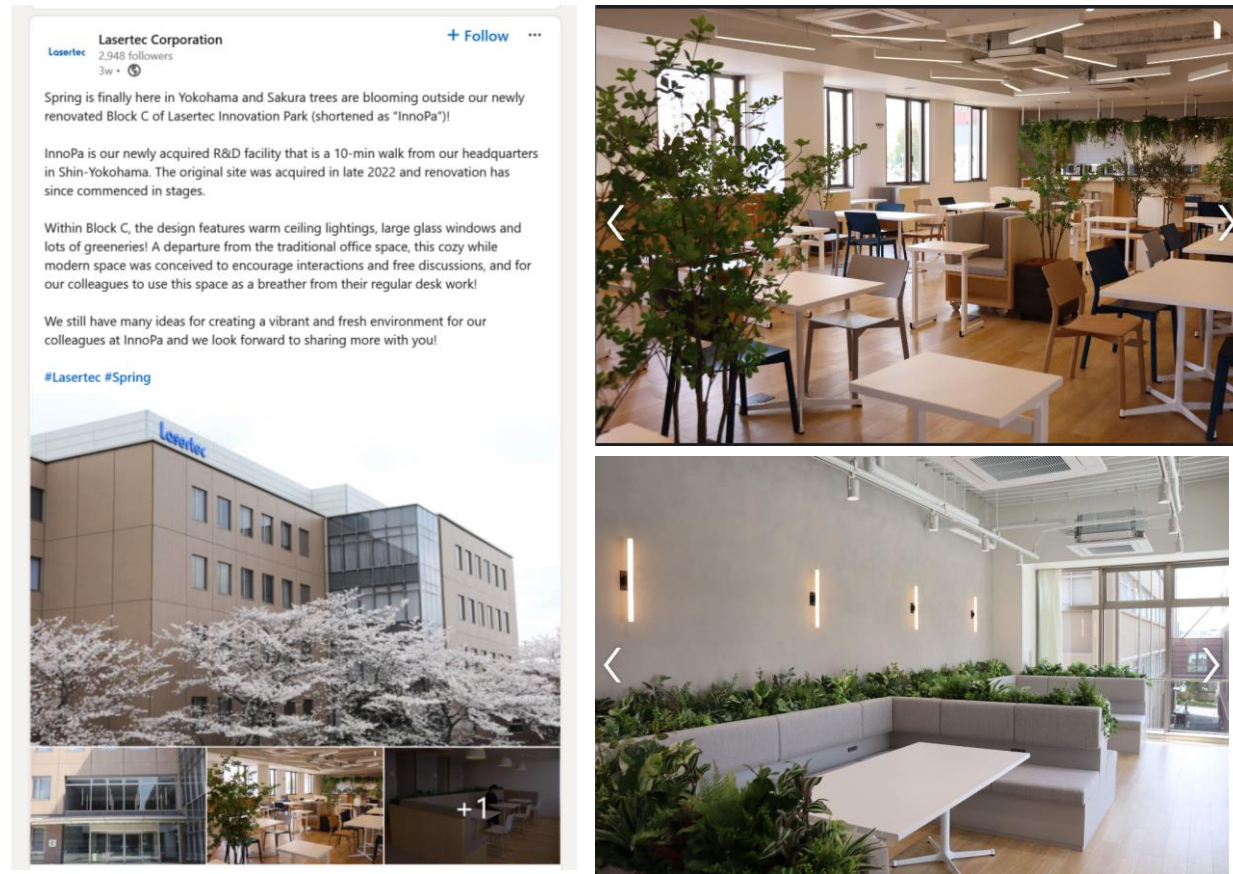


Fab 4B



Lasertec heavily promotes the Innovation Park to investors, even posting cherry blossom pictures in a LinkedIn post in early April 2024 for “our newly acquired R&D facility.” The restaurant and café look slick – but where are the employees?

Lasertec photos on LinkedIn post, April 2024



We hired a local investor relations/corporate access consultant who met with Lasertec's Investor Relations and Corporate Planning department at their headquarters on March 13, 2024. Prior to the meeting, Lasertec's IR representative agreed to a brief tour of their clean room at headquarters, but strongly refused multiple attempts asking for a tour of the clean room (Fab 4A) at the Innovation Park – we find it unusual that one clean room was offered but the other refused. We then asked for a brief, general tour of Innovation Park without entering Fab 4A – also refused. In a final attempt, we asked for simply a video clip of Fab 4A, which was also denied. We were told that a shareholder flew from Switzerland to see Fab 4A and was “extremely unhappy” at being similarly denied even a video. We quote from our consultant's write-up of the meeting:

Notes from meeting with Lasertec Investor Relations, March 13, 2024

““We took advantage of an opportunity that arose in context to ask if the photo released last year wasn't off the HQ cleanroom, and the IR Rep insisted that it was, in fact, of Fab-4A. He was generally vague about the progress of work on the other four buildings.”

“We asked about seeing a video of Fab-4A at Lasertec Innovation Park, since we couldn't see the place in person. The IR Rep seemed sincerely disappointed in explaining that no such video exists: ‘We in IR have been agitating,’ he lamented, ‘for the production of a DVD introduction to the facilities there but to no avail. I hosted an investor—a shareholder—who had come all the way from Switzerland to visit. He was extremely unhappy about being unable to get a look inside Fab-4A or to even see a video presentation.’”

Our request to see the finished goods bay at headquarters was also refused – given color we received from multiple customers that it was overflowing with inventory with no room left. However, in a telling comment, the IR rep “laughed” at the amount of storage space Lasertec needs, per our consultant’s write-up: “The IR Rep laughed that Lasertec has needed to rent space ‘all over the map’ to store tools.” The rep further confirmed Fab 4A’s role as a storage facility for excess finished good inventory, consistent with photographic evidence from our investigators, who saw no R&D or production activity at Innovation Park but did see shrink-wrapped machines occasionally being moved into storage.

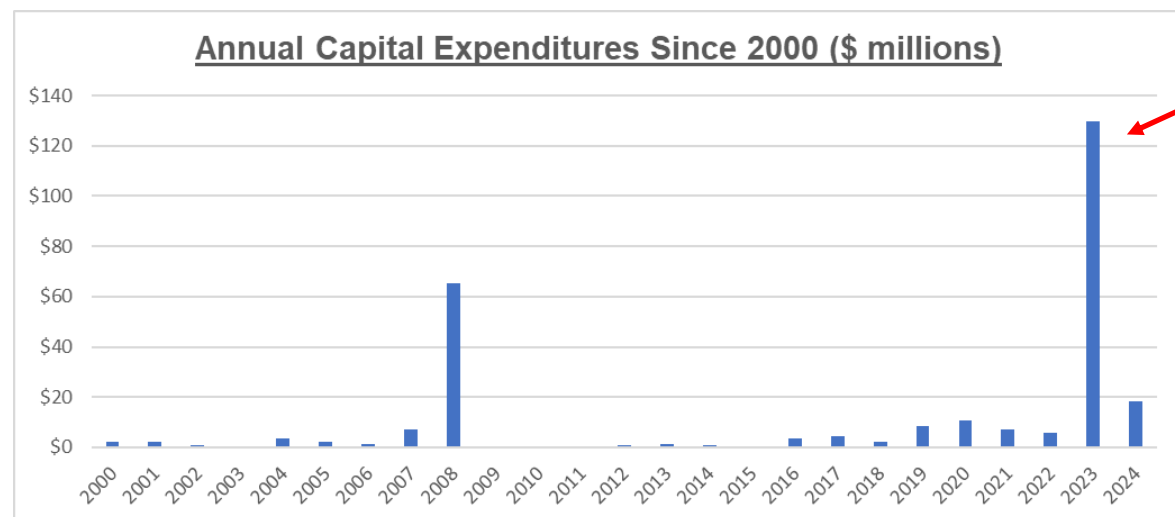
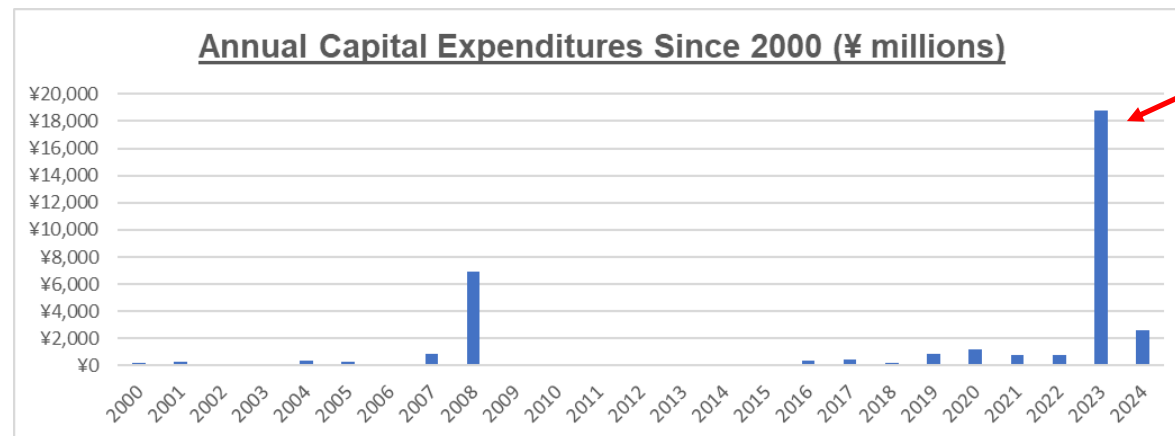
Notes from meeting with Lasertec Investor Relations, March 13, 2024

“The IR Rep **laughed that Lasertec has needed to rent space “all over the map” to store tools** before final tweaking and shipment. He says that **Fab-4A has been useful in relieving that pressure.**”

“We asked about getting a look at finished tools awaiting shipment. The IR Rep explained, however, that the **area in question was off limits.**”

Before we detail the photographic evidence, we note that the 2022 capital expenditure for the Innovation Park was by far the largest in Lasertec's history, dwarfing any other spending since 2000 – illustrating its level of importance. We show the annual spending in both JPY and USD.

Lasertec capital expenditures over time – JPY and USD millions



The timing of the announcement was suspicious. The stock had collapsed rapidly over 6 weeks, falling 40% from a peak of 36,000 in Jan 2022, to 21,000 on the day of the release on Feb 25, 2022 – a highly unusual moment to announce an expansion 5 times the size of headquarters, especially after sales plummeted that quarter. We suspect that management panicked and concocted a press release in a bid to halt the decline. Fake capital expenditure press releases are a trick we see often in corporate frauds, to create the illusion of customer demand, expansion, and growth. The chart below is what management saw on Feb 25, 2022 – anyone then could have feared a further 50-75% decline to 5,000 to 10,000. We note that trading volumes in the three weeks prior to the release were the highest ever in the stock – record volumes into a rapid decline are a sign of panic selling.

Lasertec stock chart as of Feb 25, 2022 – the day of the Innovation Park press release



Lasertec heavily promoted the investment as evidence of expansion and growth, typical of what we have observed when frauds push a capital expenditure narrative. The Lasertec press release on Sep 30, 2022, for the closing of the transaction hit the usual notes under “Purpose of acquisition”: “with the aim of **responding to business expansion**”; “we plan to increase our research and development and manufacturing capacity **in response to increased demand from customers...in line with the growth of our business.**”

Lasertec press release, Sep 30 2022



2022年9月30日

各位

会社名 レーザーテック株式会社
代表者名 代表取締役社長執行役員 岡林 理
(コード:6920 東証グライム市場)
問合せ先 執行役員 経営企画部長 三澤 祐太郎
(TEL:045-478-7111)

新研究開発拠点の名称決定と建設について

2022年2月25日付け「固定資産の取得に関するお知らせ」にて発表しております。新研究開発拠点の名称を「Lasertec Innovation Park（通称：InnoPa）」と決定いたしました。また、今後の計画を含む詳細について、下記のとおりお知らせいたします。

記

1. 拠点の概要

(1) 名 称	Lasertec Innovation Park（通称：InnoPa）
(2) 所 在 地	神奈川県横浜市港北区島山町 555-4 地 ※新横浜駅より徒歩16分、バスより徒歩10分
(3) 敷 地 面 積	約 15,830 ㎡
(4) 現 況	「新横浜東芝ビル」の名称で、東芝グループの研修所やグループ会社の拠点として使用 ※敷地内に5棟の建物：A棟、B棟、C棟、宿泊棟、機械実習棟

2. 取得目的

将来を見据えた顧客拡大への対応及び業務効率の向上、経営基盤の強化を目的として、お客様からの需要増に応じた研究開発および製造能力の増強を企図しております。
まずは、検査装置の新たな生産拠点として、クリーンルームを含む設備の建設を開始し、当事業の成長に即した研究開発、軌跡場所及び倉庫としての活用も検討しております。

3. 今後の計画

- 2022年9月30日を以て、弊社所有物件となりましたが、機械実習棟を除き、2023年9月30日を期限として、当面は東芝グループの利用が継続されます。
- ・ 第一期工事：機械実習棟のクリーンルームへの改修（完成予定 2023年6月末）
 - ・ 第二期工事：A棟、B棟、C棟、宿泊棟（東芝グループの利用終了後、工事開始予定）

2. Purpose of acquisition

With the aim of **responding to business expansion** with an eye to the future, improving operational efficiency, and strengthening the management base. We plan to **increase our research and development and manufacturing capacity in response to increased demand from our customers**. First, we will begin construction of facilities including a clean room as a new production base for inspection equipment. We are also considering using the space for research and development, as an office space, and as a warehouse **in line with the growth of our business**.

In addition to suspicious timing, we note three other unusual circumstances. First, the press release at closing stated that Toshiba used the site and its 5 buildings as a training center – but that Toshiba would continue to use the facilities for another year after the closing of the purchase. We have never seen a company make the largest capital expenditure in its history, and then agree to not use it for a year – especially if it was meant to “respond to customer demand.” If Toshiba needed a year to vacate the facility, Lasertec could have signed a purchase agreement with a future closing date – thus retaining ¥17 billion (\$150MM at the time) in capital for other uses, especially since just a few months earlier, Lasertec had to issue ¥10 billion in debt to fund an ¥8 billion dividend.

Lasertec press release, Sep 30 2022

2022年9月30日

各位
会社名 レーザーテック株式会社
代表者名 代表取締役社長執行役員 岡林 理
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“Used as a base for Toshiba Group training centers”

“There are 5 buildings on the site: Building A, Building B, Building C, Dormitory, and Machine Training Building.”

“Toshiba will continue to use the four buildings other than the machine training building until no later than September 30, 2023.”

“Lasertec will convert the machine training building to a clean room facility, the construction scheduled for completion by June 30, 2023.”

The second troubling circumstance: although it was a Toshina training facility, it was owned by Nomura Real Estate Holdings, which had purchased the facility from Toshiba in two tranches – and Lasertec explicitly declined to disclose it as a related-party transaction, stating “Nothing of note” for “Existing relationships between the seller and Lasertec.” Yet Nomura Asset Management is the second largest holder of Lasertec with 7%. While Nomura real estate and asset management companies may be separate legal entities, the web is murky: Lasertec shares a Supervisory Board member with Nomura Holdings Inc, which owns ~40% of Nomura Real Estate Holdings, the seller of the park; Nomura Asset Management is a top holder of both Lasertec and Nomura Real Estate Holdings; and so forth.

Announcement for research park purchased in 2022 - excerpt

3. Description of Seller

1) Company Name	Nomura Real Estate Building Co., Ltd.	
2) Head Office	1-1-1 Shibaura, Minato-ku, Tokyo	
3) Representative Director	Masatsugu Matsuzaki, President and Representative Director	
4) Description of Business	• Leasing • Development • Real Estate Consulting, etc.	
5) Capital	14,372 million yen	
6) Founded	December 22, 1932	
7) Major shareholders and shareholding ratio	Nomura Real Estate Holdings, Inc. 100%	
8) Existing relationships between the seller and Lasertec Corporation	Capital	Nothing of note
	Personal	
	Business	
	As a related party	

Nomura Real Estate Holdings Inc. owns 100% of the selling entity, Nomura Real Estate Building Co. Ltd.

“Existing relationships between the seller and Lasertec Corporation” – “Nothing of note”

We were unable to determine the details, financial flows, and potential circularity of the transactions between Nomura, Toshiba, and Lasertec, such as the actual value of the property versus the amount paid. We note only that both Nomura and Toshiba have an extensive history of episodes involving serious fraud. Nomura's real estate group, in particular, seems to be a compulsive repeat offender. In 2018, the US Department of Justice announced a \$480MM settlement with Nomura for "fraudulent conduct in connection with its Residential Mortgage-Backed Securities offerings." In August 2023 – less than a year ago – the DOJ announced a new \$35MM settlement with Nomura and the resolution of potential criminal charges after it "perpetrated a scheme...to defraud its customers...with the knowledge, encouragement, and participation of supervisors" – once again with respect to mortgage-backed securities.

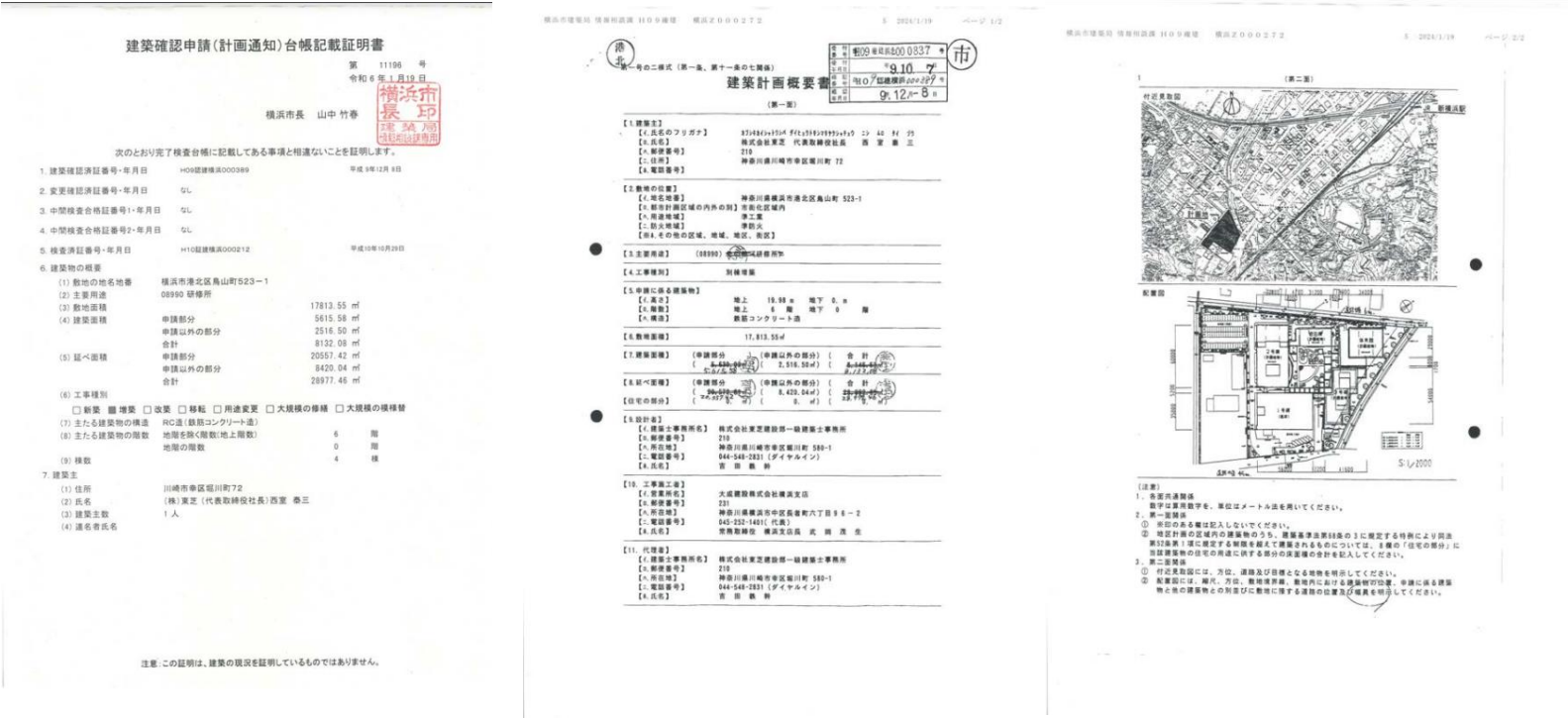
US Department of Justice press releases from Oct 2018 (top) and Aug 2023 (below)

Nomura Agrees to Pay \$480 Million in Civil Penalties
for Misleading Investors in Sale of Residential
Mortgage-Backed Securities

Nomura Securities International Agrees to Pay \$35
Million Penalty Stemming from Its Participation in
Securities Fraud Scheme

Third, we were unable to locate any construction or other permits for Lasertec’s purported build-out activity at the Innovation Park. Our investigators visited the City of Yokohama’s Housing And Architecture Bureau, and the only permits we could find were from 1997 for Toshiba’s construction of the four buildings, other than Building A, which dates from 1969.

Toshiba 1997 permit for what is now Lasertec’s so-called “Innovation Park”



The lack of any apparent R&D or production activity at the Innovation Park is troubling given Lasertec management's ongoing claims that it became operational in mid-2023. We noted previously its presentation on Aug 7, 2023: “we finished the construction of a cleanroom at the new site and started operation in July 2023”; “operation started in July 2023.”

Lasertec presentation, Aug 7 2023

Our New R&D Facility: Lasertec Innovation Park (InnoPa)

We finished the construction of a cleanroom at the new site and started operation in July 2023.

Overview

Name	Lasertec Innovation Park (InnoPa)
Address	555 Toriyama-cho, Kohoku-ku, Yokohama, Kanagawa Pref. (10 min. walk from HQ) (16 min. walk from Shin-Yokohama station)
Site Area	15,830 m ² (4.8 times the area of HQ)
Status	• First round of construction completed in June 2023, and operation started in July 2023. • Second round of construction to begin in October 2023.



Lasertec

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Management re-iterated these claims during the Aug 7, 2023 Q&A with analysts, stating that “a clean room has been newly established...and is being used as a research and development base from July 2023.” A BofA analyst asked “how much will production capacity grow” and the CEO deflected, but was otherwise specific: “the new InnoPa development site has been in use since July for product assembly and final adjustments” and that they would “further increase production volume” there “once the second phase of construction is completed.” The CEO was evasive when the analyst pressed for an answer on production capacity: “we have not announced...how much production will expand.”

Lasertec prepared comments and analyst Q&A, Aug 7, 2023

Management’s prepared comments on the slide showing the Innovation Park

“Next, I would like to talk about the new R&D center, Lasertec Innovation Park, InnoPa. **A clean room has been newly established at the new site and is being used as a research and development base from July 2023.** The first phase of construction was completed at the end of June and operation started in July. The second phase of construction is scheduled to begin in October 2023.”

Q&A with analysts - excerpts

Hirakawa [Q]: “This is Hirakawa from BofA Securities Japan. **I would like to ask you about production capacity**...when the first phase of construction was completed? Also, the second phase of construction is scheduled to begin in October of this year [...] when it is finished, **how much will production capacity grow?**”

Okabayashi [A]: “...we have been increasing production capacity every year. In the current fiscal year, although we are unable to provide figures, the **new InnoPa development site has been in use since July for product assembly and final adjustments, and we plan to further increase production volume once the second phase of construction is completed.**”

Hirakawa [Q]: “The second phase of construction is scheduled to begin in October, but, this is the same question, when will it be completed and **how much will production capacity grow?**”

Okabayashi [A]: “At this time, **we have not announced when it will end or how much production will expand.** We would like to discuss this when we are ready to make it public.”

Management continued to repeat these claims on the next two earnings calls on Oct 31, 2023, and Jan 31, 2024. In October, they suggested that more than one clean room has been installed: “some clean rooms have been installed.” In Jan, the CEO stated the park is “five times the space of our current headquarters”; that “production began in July”; that “equipment is now being assembled”; and that the second phase of construction which began in Oct 2023 “will contribute to production...we are working to increase production capacity...expanding our production space and workforce”; “working to increase production capacity.”

Lasertec management prepared comments Oct 31, 2023

*“...there was a very large investment in Q1 of last year due to the acquisition of a new base we call InnoPa, but since then, there has not been so much investment, although **some clean rooms have been installed.**”*

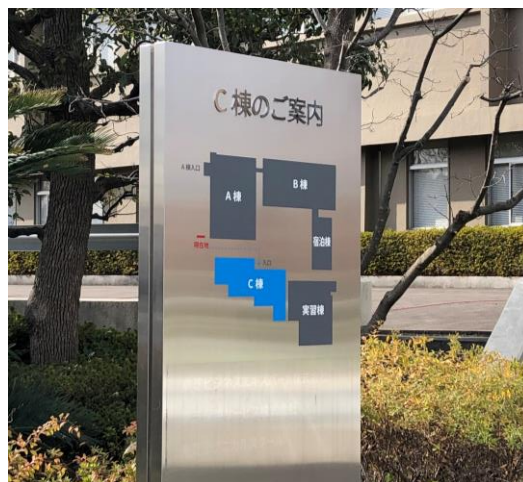
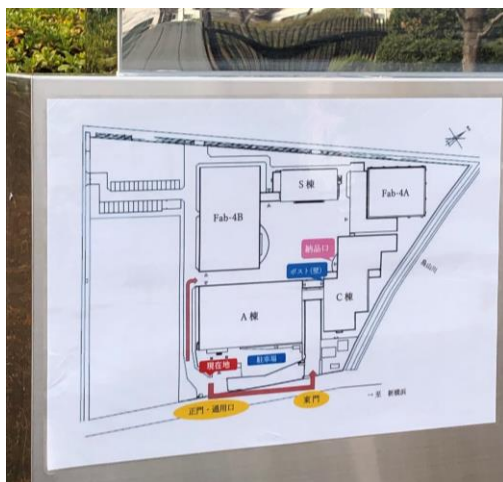
Lasertec analyst Q&A Jan 31, 2024

***Shibano [Q]:** “..in terms of lead time. I believe you explained that your company is **working on expanding capacity, such as with the new InnoPa**, and that a clean room was installed during the last fiscal year. I am wondering if the lead time is going to get shorter if you look to the future orders you may have...”*

***Okabayashi [A]:** “As you mentioned, we acquired property including land and buildings called InnoPa, which has about **five times the space of our current headquarters building**. The first phase of construction was completed last June, and **production began in July**. In addition, the **second phase of construction started last October, and when completed, it too will contribute to production. In this sense, we are working to increase production capacity** [...] We are also **expanding our production space and workforce**. As for the new InnoPa facility, as I explained earlier, the first phase of construction has already been completed and the **equipment is now being assembled and adjusted**, and the second phase of construction has started. As such, we are steadily **working to increase production capacity.**”*

The ~20 field visits our investigators made to Lasertec's Innovation Park indicate that management's claims are false with no evidence of R&D, production, assembly, testing; no evidence that Fab 4A is operational or even a completed clean room; no evidence of other clean rooms or a second phase of construction for Fab 4B. We begin with the entrance to Innovation Park which lays out a map of the complex, which consists of 5 buildings: 1) building A, an old three-story office building that is clearly not for R&D, and which appears mostly empty except for administrative personnel – not engineers – going in and out occasionally; 2) Building C, that appears empty except for a handful of clerical staff on the first floor; 3) Fab 4A, which looks like a gymnasium that is now supposedly a clean room and production facility that the company claims became operational in June 2023; 4) Fab 4B, which is the supposed “second phase of construction” but which has little such activity; and 5) Building S, which was a dormitory for the Toshiba Training Center and now appears unused.

Signs with site map at Lasertec Innovation Park



Our investigators noticed almost no activity when observing it from the perimeter – it looks old and not like a high-technology innovation park or R&D center.

Perimeter photographs of Lasertec Innovation Park



The first building one encounters after entering the complex is Building A, a dated 3-story office building. It is clearly not designed as an R&D facility. We saw almost no signs of activity except clerical and office staff on one of the floors, and no meaningful movement at typical peak times like morning, lunch, and end of day.

Building A at Lasertec Innovation Park



As they moved into the interior, it became clear that the complex is essentially deserted and a ghost town, except for some landscaping work in the courtyard. The building with the curved roof is Fab 4A, the so-called clean room where production is claimed to take place. It was previously a Toshiba training facility that appears to be a converted gymnasium and appears unsuitable as a large production facility for semiconductor capital equipment.

Interior photographs of Lasertec Innovation Park



As they moved even deeper into the interior toward the entrances and lobbies, no evidence typical of an R&D or production facility was evident.

Deep interior photographs of Lasertec Innovation Park



Source: Investigator field visits

“Fab 4A” from the exterior simply looks like a gymnasium, as it was previously a Toshiba training facility vs. being equipped for R&D or as a production fab.

Fab 4A from exterior



We monitored the entrance to Fab 4A with a 24/7 time lapse camera, for several weeks from several angles – we show photos from different days and times. We observed almost no activity, except for an occasional machine being delivered, as it serves only as a storage facility for finished goods inventory. We saw no staff entering or exiting at typical peak times such as morning, lunch, or after work.

Fab 4A – time lapse camera, angle 1



A second angle for the time-lapse camera monitoring the entrance to Fab 4A also showed that it is not staffed and is empty of people. We saw only an occasional maintenance worker or vehicle, or delivery of a finished machine.

Fab 4A – time lapse camera, angle 2



A third camera angle confirmed the lack of any personnel staffed at Fab 4A.

Fab 4A – time lapse camera, angle 3



Our investigators were allowed to wander freely around the complex without anyone asking them to leave. During one visit, they stood in front of Fab 4A as machines were being delivered. At that moment, we came to understand the only purpose that the so-called fab serves: as a storage facility for overflowing finished goods inventory. Note that the machines are already shrink-wrapped and sealed in plastic. The machines below appear far too small to be Lasertec's DUV or EUV systems – we assume they are their old, low-cost tools.

Fab 4A – loading truck at the entrance



We obtained numerous images of the loading area of Fab 4A, when the the vertical roller door was occasionally opened once or twice a week for delivery of finished goods inventory. We saw no movement at other times of parts, components, or staff as one would expect if any assembly, testing, production, or R&D was taking place.

Fab 4A – loading area



The machines going into storage at Fab 4A were hermetically sealed in extensive protective wrapping. This suggests that absolutely no testing or fine-tuning is meant to take place inside Fab 4A.

Fab 4A – loading area



The amount of wrapping is extensive and indicates it is finished goods inventory.

Fab 4A – loading area



Our investigators even looked at garbage containers around Fab 4A and around the complex, for evidence that the extensive plastic wrapping was being removed and discarded. They found no evidence, indicating once again that its only purpose is finished goods storage and that the wrapping is not being removed for any testing or fine-tuning.

Lasertec Innovation Park – example of garbage containers



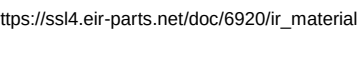
We were curious if any clean room exists at all inside Fab 4A, similar to the photo of the large, new, bright-white, extensively-lit room shown in Lasertec's presentations. We contracted an investigator for aerial drone surveillance, who took videos and photographs through the windows located at the top of Fab 4A on two separate days.

Lasertec presentation, Aug 7 2023 shows the purported "Fab 4A" cleanroom; drone surveillance

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From the top of Fab 4A, the drone could see straight through to sunlit windows on the other side of the structure, which still looks exactly like a gymnasium. The interior looks like a large open space with an exposed scaffold ceiling for HVAC equipment, as one would expect for a large gym-type space. We saw no evidence of a large clean room like the one in Lasertec's photo.

Drone surveillance of Fab 4A



Clean rooms for semiconductors and inspection equipment are the cleanest facilities on earth, where one microscopic particle discovered in a large facility every few weeks is considered a problem. Yet, the windows for Fab 4A are filthy and covered in dust. The HVAC equipment visible behind the windows appears caked in rust and/or debris.

Drone surveillance of Fab 4A



Close ups of mechanical and/or HVAC structures inside Fab 4A look like those in a ghost town gathering dust and spider webs for years.

Drone surveillance of Fab 4A



In addition to Fab 4A, where construction was supposedly completed June 2023 when it became “operational,” Lasertec states that the “second round of construction” began in Oct 2023 for Fab 4B – but investigator site visits indicate that these claims are also a wild exaggeration and fraudulent. The CEO stated in Aug 2023 that Fab 4B would “further increase production volume” but that “we have not announced...how much production will expand.” In Oct 2023, he added that this new phase of construction “will contribute to production...we are working to increase production capacity...expanding our production space and workforce”; “working to increase production capacity.” However, the entrance to Fab 4B indicates it is just an office space.

Exterior of the purported Fab 4B at Lasertec Innovation Park



Our investigators were able to easily look through the windows to “Fab 4B” to see the level of construction activity. They saw nothing but a ladder and a few pieces of wood on the floor, and no construction crews. No signs of progress for a fab are visible – not what one would expect if a major phase of construction began 8 months ago.

Interior of the purported Fab 4B at Lasertec Innovation Park

